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Hydraulic Systems

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Fluid Theory

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Pressure / Force / Area

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Introduction to Fluid Power

Machinery was invented to perform work and may use mechanical, electrical or fluid power to accomplish that work. The study of Hydraulics deals with the use and characteristics of a fluid and how it can be used to perform work. Hydraulics is the transmission of power through the use of a confined liquid. In industry, the use of fluid power is advantageous in some areas as a source of energy to power a machine. There are several advantages to using hydraulic systems, some are listed below;

- Variable speed control
- Reversible motion of prime movers
- System overload protection
- High power output in a small package
- Power components can stall out
- Motion can be transmitted without the slack or clearances needed in mechanical equipment
- Fluids don't wear out like mechanical machines
- Versatile, smooth, uniform action for many applications
- Can span great distances

Fluid power is so widely used as a source of energy we will begin by studying fluids and their characteristics.

Physical Properties of a Fluid

Different Types of Fluids

A fluid is a substance that can flow and possess no definite shape. A fluid may be a liquid or gas.

Molecular Energy

Any substance is made up of molecules. The molecular energy of any type of substance will define it and create the characteristics used in any application. When we consider the molecular energy you have to look at the three basic types of matter. The molecules of a solid have a very strong attraction force. They are so closely attracted to each other and barely vibrate creating the molecular energy of a solid. The molecules of a gas aren't attracted to one another, as a matter of fact the molecules will repel each other. Constantly expanding its boundaries and bouncing off other molecules or the container walls. This is the molecular energy of a gas. However unlike a gas, the molecules of a liquid are more closely attracted to each other. They are not in a rigid position nor do they have an attraction such as a solid.

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Liquid molecules are a combination of the molecular energy of a solid and a gas. Liquid molecules are continuously moving, slipping and sliding past one another, even when the liquid is at rest, creating the molecular energy of a liquid.

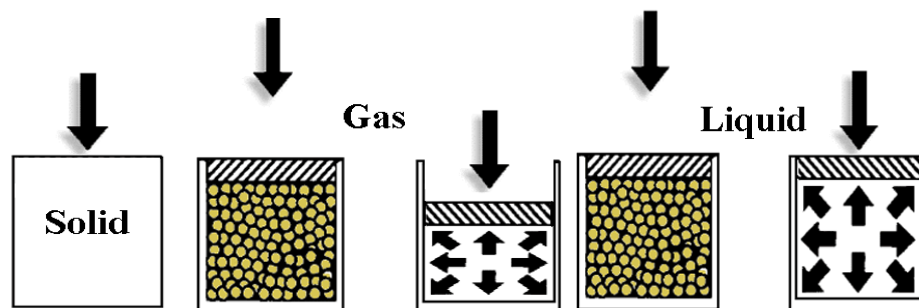
Since this slipping and sliding action of a liquid is continuously happening, it will take the shape of any container. It may or may not fill up the container depending on the volume of liquid. The liquids ability to accomplish this is what creates one of the first basic differences between a gas and a liquid.

Compressibility

The primary difference between a gas and a liquid is their compressibility. With the molecules of a gas trying to always expand to fill their container a gas is considered compressible. With the molecules of a liquid in close contact with one another, liquids will exhibit one characteristic of a solid. Liquids are considered non-compressible. Since liquids are relatively incompressible and can take the shape of their container, they possess certain advantages for transmitting a force.

Force Transmission

When a force is transmitted through a solid, the result will be a force in the same direction and the same magnitude (disregarding any frictional loss). This is an example of a mechanical source of energy; belts & pulleys, chains & sprockets, gear drives, etc. When a force is transmitted through a confined fluid it reacts in a special way. When a force is applied to a confined gas, the gas **will compress** and that same force (intensity) will be transmitted equally in all directions. When a force is applied to a confined liquid, the liquid **will not compress** and that same force (intensity) will be transmitted equally in all directions. The only difference between the two fluids would be their compressibility. The advantage fluid energy has over a mechanical source of energy is the mechanical energy is transmitted in one direction, where as the fluid energy is transmitted equally in all directions. Before we can discuss force transmission we have to define a force.



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The force (intensity) created inside the liquid is based upon the principle discovered in the 17th century by the French scientist Blaise Pascal. Pascal's Law states; the force applied to a confined fluid is transmitted equally in all directions in the form of fluid pressure.

Pressure / Force / Area

Pressure is a measure of a force's intensity. To measure pressure, the total force is divided by the area on which it is acting. The mathematical expression that describes Pascal's Law is:

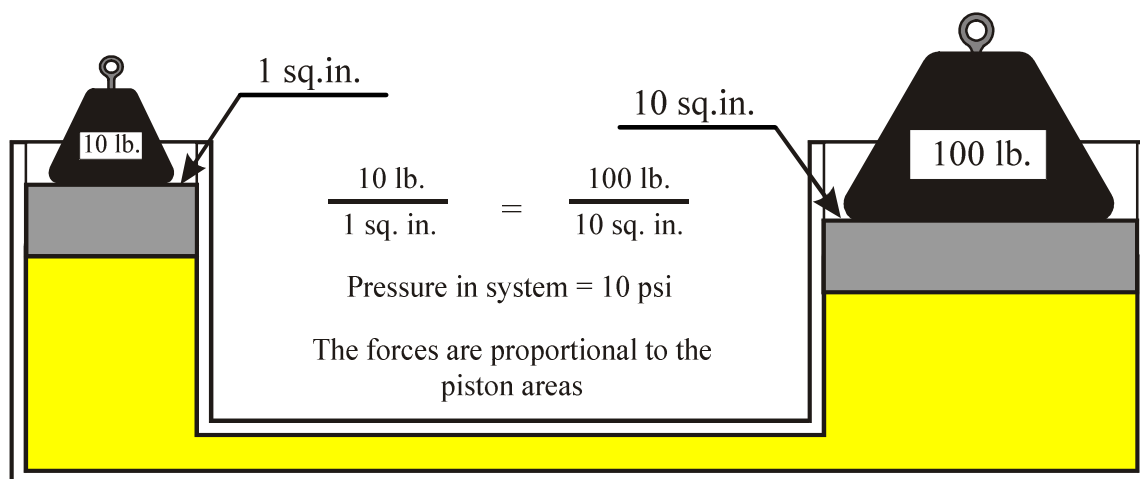
$$Pressure (psi) = \frac{Force (lbs)}{Area (in^2)} \quad or \quad Pressure (bar) = \frac{Force (kg)}{Area (cm^2)}$$

To solve for any other unknown:

$$P = \frac{F}{A} \quad or \quad F = P \cdot A \quad or \quad A = \frac{F}{P}$$

	<u>US</u>	<u>ISO</u>
<u>Force</u>	Pounds (lbs)	Kilogram (kg)
<u>Area</u>	Square inches (in ²)	Square centimeters (cm ²)
<u>Pressure</u>	lbs/in ² or PSI	kg/cm ²

To better understand the effect of pressure, we'll look at a more practical example of two pistons; one large, one small connected together by a fluid filled tube.

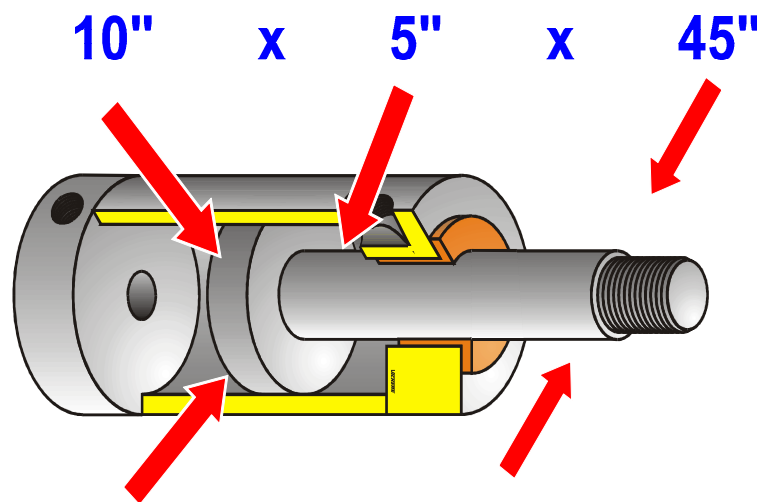


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The amount of surface area for the pressure to act on allowed the larger force to be supported. Since surface area was a key in supporting the larger force, then the amount of surface area (effective area) is just as important as the pressure or force values. Next we need to look at determining the surface area in an application.

The surface area of a particular application may or may not be a given value. Surface area in hydraulic actuators usually revolves around calculating the value from a given diameter. Typical cylinder call-outs (information) in industry, as illustrated below, use diameter as a basis for size. For example; the three most important values required from a cylinder application would be piston/cylinder/bore diameter, rod diameter and stroke length. If the rod diameter is of a standard size, for the cylinder size, then it may be omitted and the call-out may be simplified to piston/cylinder/bore diameter and stroke length.

Piston Dia. x Rod Dia. x Stroke Length



10" x 45"

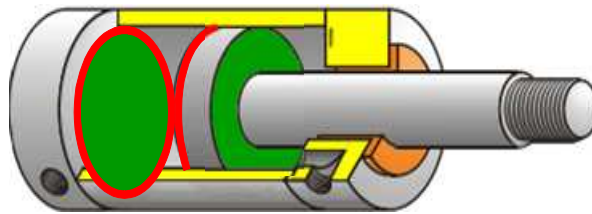
Piston Dia. x Stroke Length

Once the diameter is known the surface area can be calculated. The following are three of the most common area formulas used in fluid power.

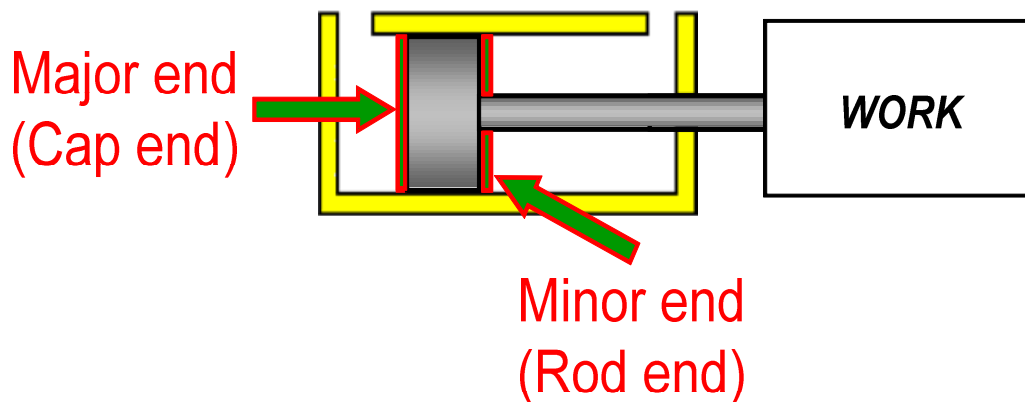
$$Area = \pi \cdot r^2 \quad or \quad Area = d^2 \cdot 0.7854 \quad or \quad Area = d^2 \cdot \frac{\pi}{4}$$

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The area relationship inside most hydraulic cylinder applications is not the same from one side to the other. The rod consumes some of the available (effective) surface area on one side of the piston (cylinder). A different approach is used to calculate the area on one side than the other. The difference in surface area from one side to another dictates that different terminology be used to distinguish the two sides. Major/Cap side is used where the full amount of surface area is fully utilized, whereas Minor/Rod side is used where the surface area is diminished.



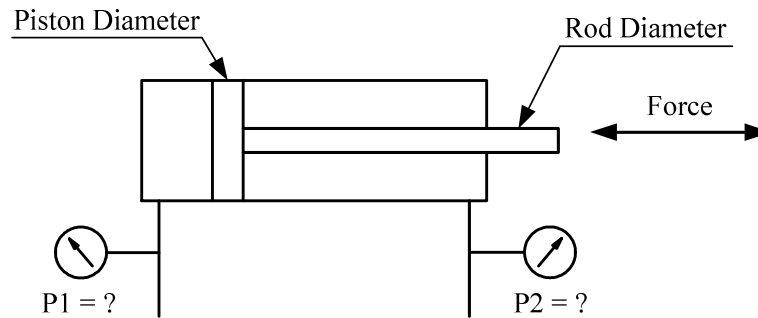
Minor area = Major area - Rod area



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Pressure / Force / Area Exercises

1. Calculate the missing information to extend **or** to retract the cylinder using the given information.

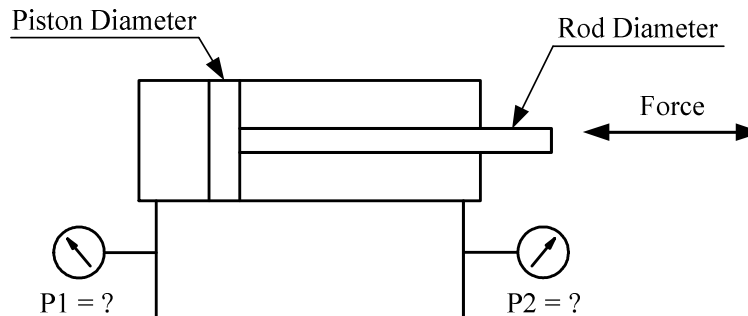


	<u>Pressure</u>	<u>Piston Dia.</u>	<u>Rod Dia.</u>	<u>Force</u>
a.	P1 = _____	2.0" (5.08 cm)		1400 lbs (635.03 kg)
b.	P2 = _____	2.0" (5.08 cm)	1 3/8" (3.49 cm)	1400 lbs (635.03 kg)
c.	P1 = _____	3 1/2" (8.89 cm)		2150 lbs (975.22 kg)
d.	P2 = _____	3 1/2" (8.89 cm)	1 3/4" (4.45 cm)	2150 lbs (975.22 kg)
e.	P1 = 1025 psi (72.06 bar)	_____		12880.56 lbs (5842.52 kg)
f.	P2 = 1025 psi (72.06 bar)	4.0" (10.16 cm)	_____	9660.42 lbs (4381.89 kg)
g.	P1 = 850 psi (59.76 bar)	5.0" (12.7 cm)		_____
h.	P2 = 850 psi (59.76 bar)	5.0" (12.7 cm)	2 1/2" (6.35 cm)	_____

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Pressure / Force / Area Exercises (Continued)

1. Calculate the missing information to extend **or** to retract the cylinder using the given information.



<u>Pressure</u>	<u>Piston Dia.</u>	<u>Rod Dia.</u>	<u>Force</u>
i. P1 = _____	1 1/2" (3.81 cm)		3180.87 lbs (1442.82 kg)
j. P2 = _____	1 1/2" (3.81 cm)	5/8" (1.58cm)	3180.87 lbs (1442.82 kg)
k. P1 = 1200 psi (84.37 bar)	2 1/2" (6.35 cm)		_____
l. P2 = 1200 psi (84.37 bar)	2 1/2" (6.35 cm)	1 3/4" (4.45cm)	_____
m. P1 = _____	6.0" (15.24 cm)		25000 lbs (11339.81 kg)
n. P2 = _____	6.0" (15.24 cm)	3" (7.62 cm)	25000 lbs (11339.81 kg)
o. P1 = 575 psi (40.43 bar)	_____		28902.72 lbs (13110.05 kg)
p. P2 = 575 psi (40.43 bar)	8.0" (20.32 cm)	_____	15241.67 lbs (6913.51 kg)

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Pressure / Flow

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The Relationship between Pressure and Flow in a System

Flow

Flow in a hydraulic system is created by the pump. Flow and pressure are the kinetic energy that allow a system to perform work. Flow from the pump, or system, is essential to cause movement of any actuator. There are two ways to measure the flow of a fluid; velocity and / or flow rate.

Velocity

Velocity is the average **speed** of the fluid passing a point or the average distance it travels per unit of time. It is usually measured in feet per second (ft/sec), feet per minute (ft/min), inches per second (in/sec), meters per second (m/sec), meters per minute (m/min), centimeters per second (cm/sec).

Flow Rate

Flow rate is a measure of the **volume** of fluid passing a point in a given time. It is usually measured in gallons per minute (gpm) or liters per minute (lpm). The actual speed of any actuator is dependent upon two factors; size of the actuator and the rate of flow (Volume) into it.

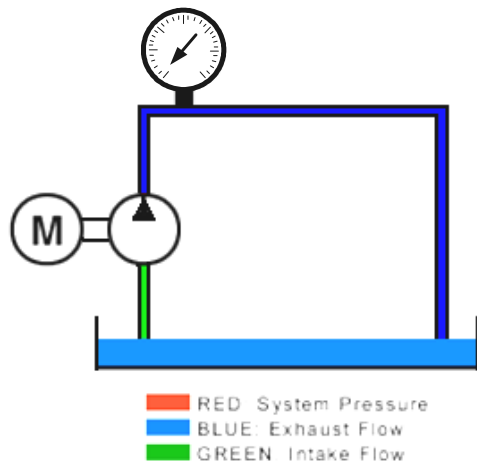
When relating to flow in a hydraulic system there are two fundamental principles;

- 1) “Flow makes it go”. For anything to move in a hydraulic system the actuator must be supplied with fluid.
- 2) Rate of flow determines actuator speed. Changes in pump, or system, output flow will change the speed of the actuator. Actuator speed will always be directly related to “how much you put into or how much you let out of the actuator.”

Pressure

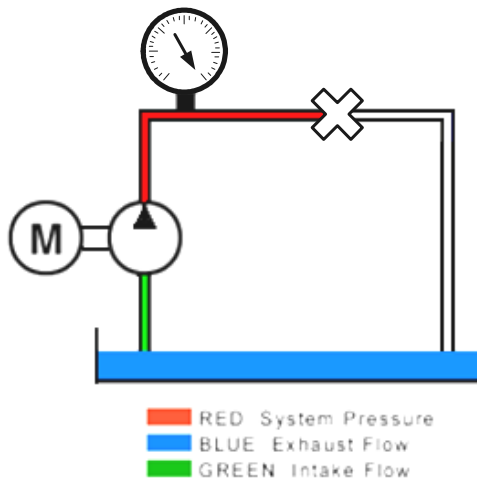
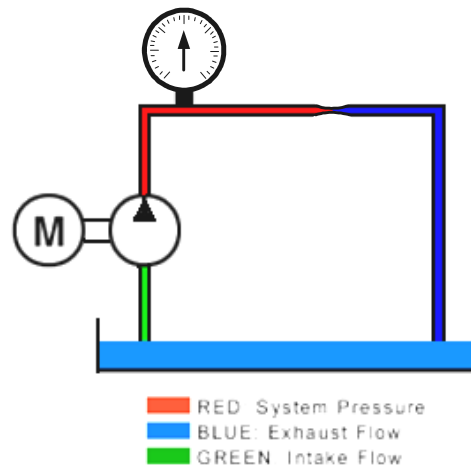
Pressure in a hydraulic system is directly related to the degree or amount of resistance to flow. Pressure gives the actuator it’s force, remember P/F/A relationships. The pump is producing flow, not pressure. To help illustrate this point;

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In this illustration pump flow is allowed to return back to tank/reservoir unrestricted. The amount of pressure created to push the fluid through the system is very low.

In this illustration pump flow is returning back to tank/reservoir but it encounters a restriction. The amount of pressure created is directly related to the degree of restriction. The pump has to create more pressure than before because of the extra resistance to flow in the system.



In this illustration pump flow is not allowed to return back to tank/reservoir but is completely blocked. Because of the blockage in the line system pressure will continue to rise as the pump continues to add fluid to the system. Pressure will continue to rise until it the system or a component fails.

All resistance to flow in a system creates pressure; including, transmission lines, components and fluid changing direction (such as elbows) are accumulative and contribute to total system pressure.

In summary; pressure relates directly to the force generated by an actuator, whereas flow rate relates directly to the speed an actuator operates at.

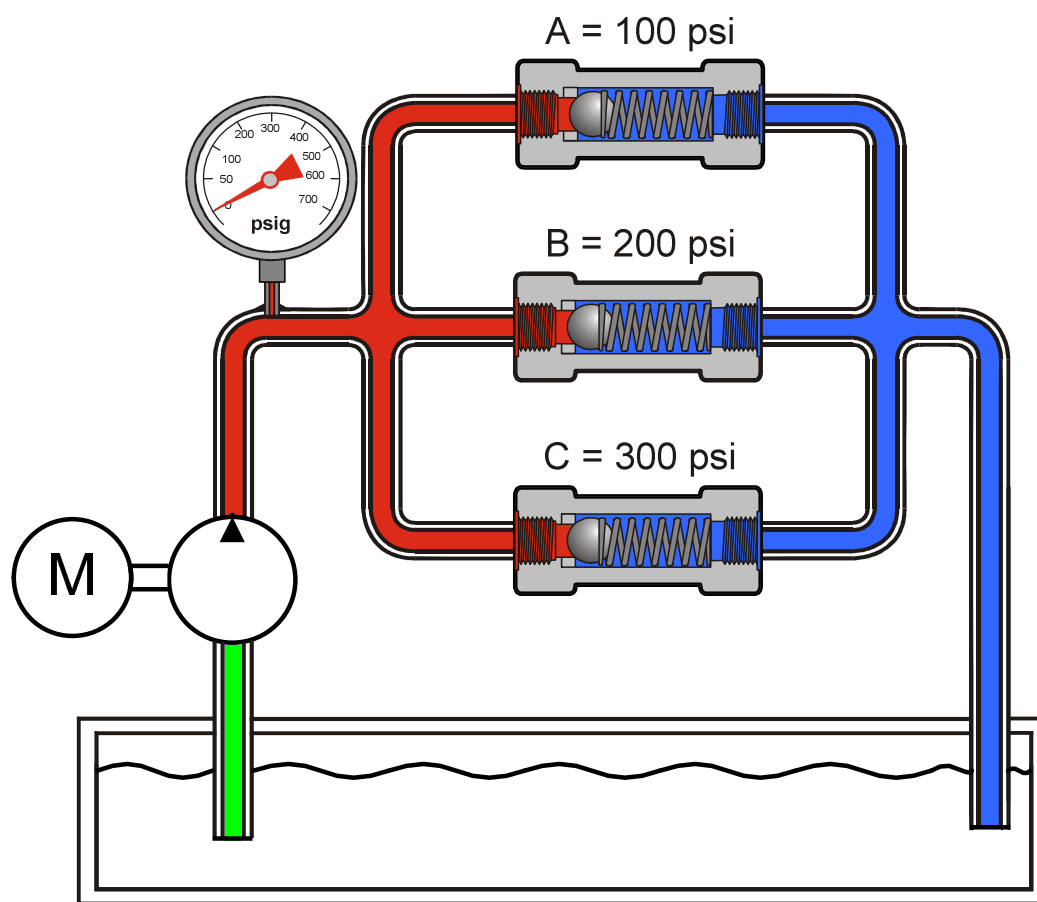
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Fluid Flow Paths

An inherent characteristic of a liquid is their ability to always take the path of least resistance.

Parallel Flow Paths

When two or more pathways exist the factor that determines the fluids flow path will be the degree of resistance. In the illustration below the fluid has three different flow paths.

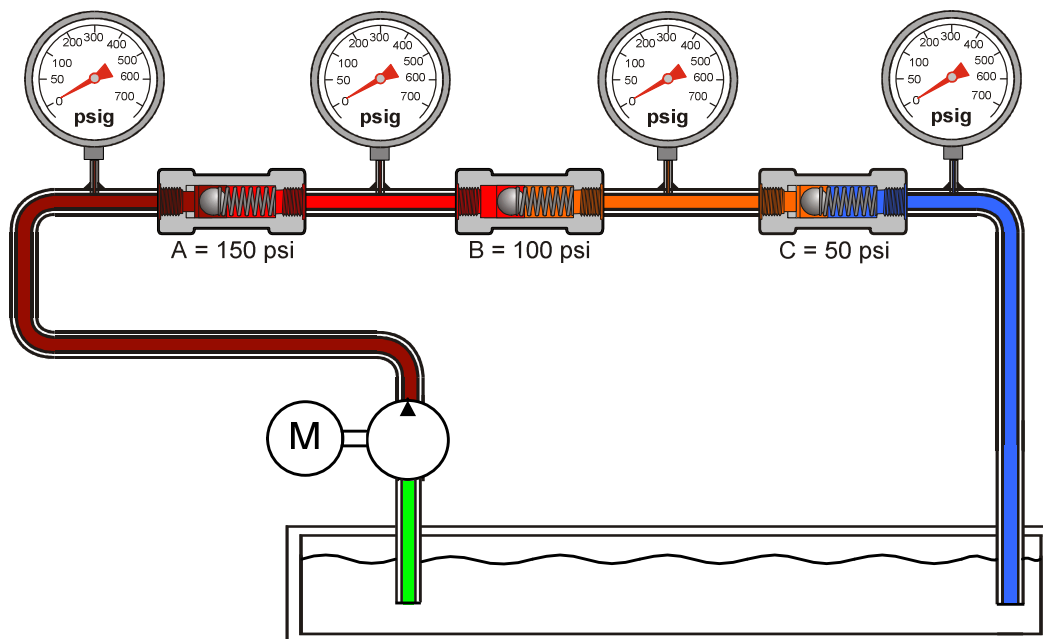


Parallel Flow Paths

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Series Flow Path

When the circuit components are connected in series the resistance to flow/pressures add up. In the illustration below the valves are connected in series. Pressure gauges are placed throughout the circuit to show the different pressure values within the different sections. The pressure gauges will show the pressure required to open each valve, plus the back pressure generated from the valves downstream. The pressure at the pump would be the total pressure required to open all the valves and create flow.



Series Flow Path

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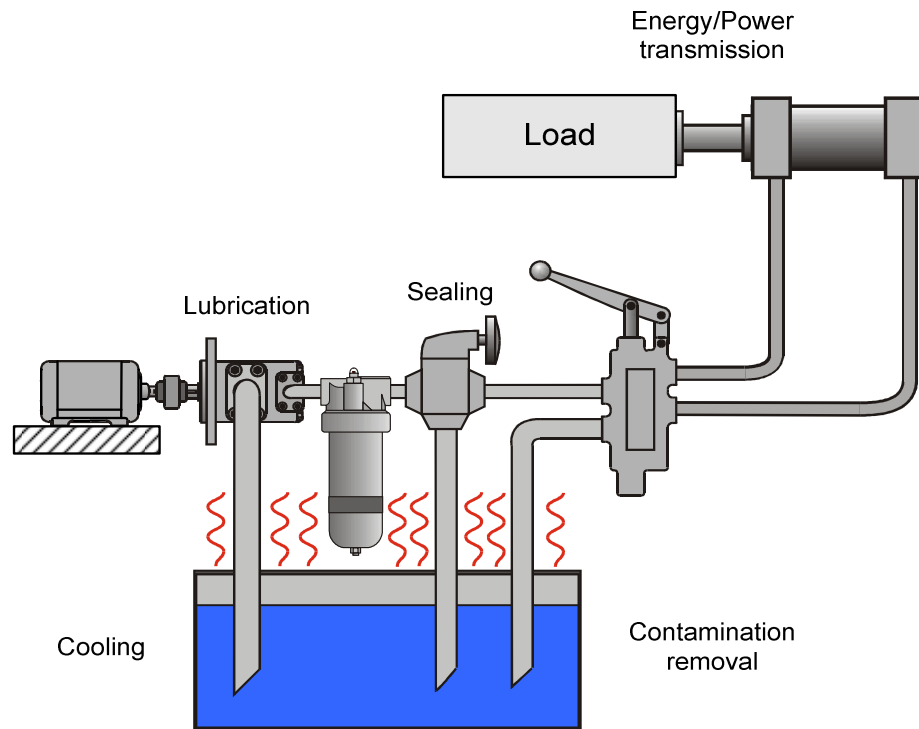
Viscosity / Temperature

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Petroleum Based Hydraulic Fluids

The most common type of oil used in an industrial hydraulic system will be a petroleum based fluid. A petroleum based hydraulic fluid has many primary functions it may perform in the system;

- Acts as a means of energy transmission
- Cools the system by removing heat
- Seals clearances between components
- Lubricates moving components
- Reduces friction



A petroleum based hydraulic fluid has many other secondary or quality type functions it may perform in the system;

- Cleans the system by washing away contamination
- Prevent rust, corrosion, pitting
- Prevent sludge, gumming, varnishing
- Reduce foaming
- Maintain stability over wide temperature ranges
- Separate water
- Be compatible with seals and gaskets

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Fluid Properties

The various properties of a hydraulic fluid that allow it satisfy its primary and some of the secondary functions;

- Viscosity
- Pour Point
- Lubricating ability
- Oxidation Resistance
- Rust & Corrosion Protection
- Demulsify

Viscosity

Of all the properties a hydraulic fluid may exhibit viscosity is, in most applications, the most important characteristic. Viscosity is the measure of a fluid's resistance to flow or ability to flow. If a fluid flows easily, it has a low viscosity or is a thin fluid. If a fluid has difficulty flowing, it has a high viscosity or is a thick fluid. The right viscosity for any application is always a compromise to satisfy the system requirements.

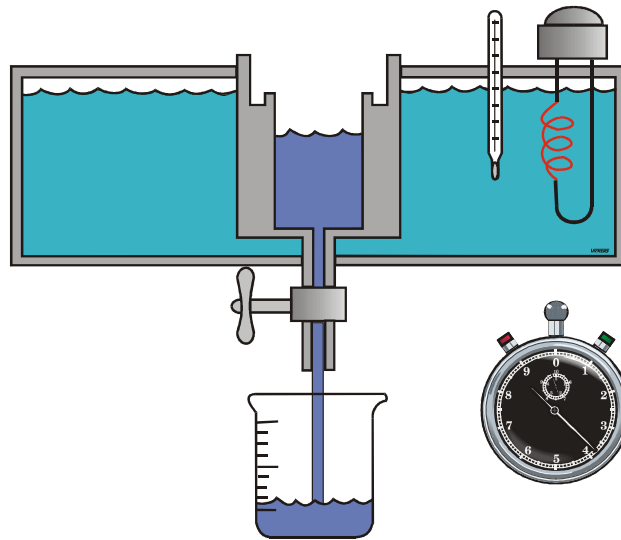
Measuring Viscosity

Many different methods exist to measure or define viscosity. Some of the most common are; Absolute (Dynamic) viscosity, Kinematic viscosity, Saybolt Universal Seconds and SAE values used for automotive oils. In the United States, hydraulic fluid viscosities are usually specified in Saybolt Universal Seconds or Kinematic viscosity.

Saybolt Universal Seconds

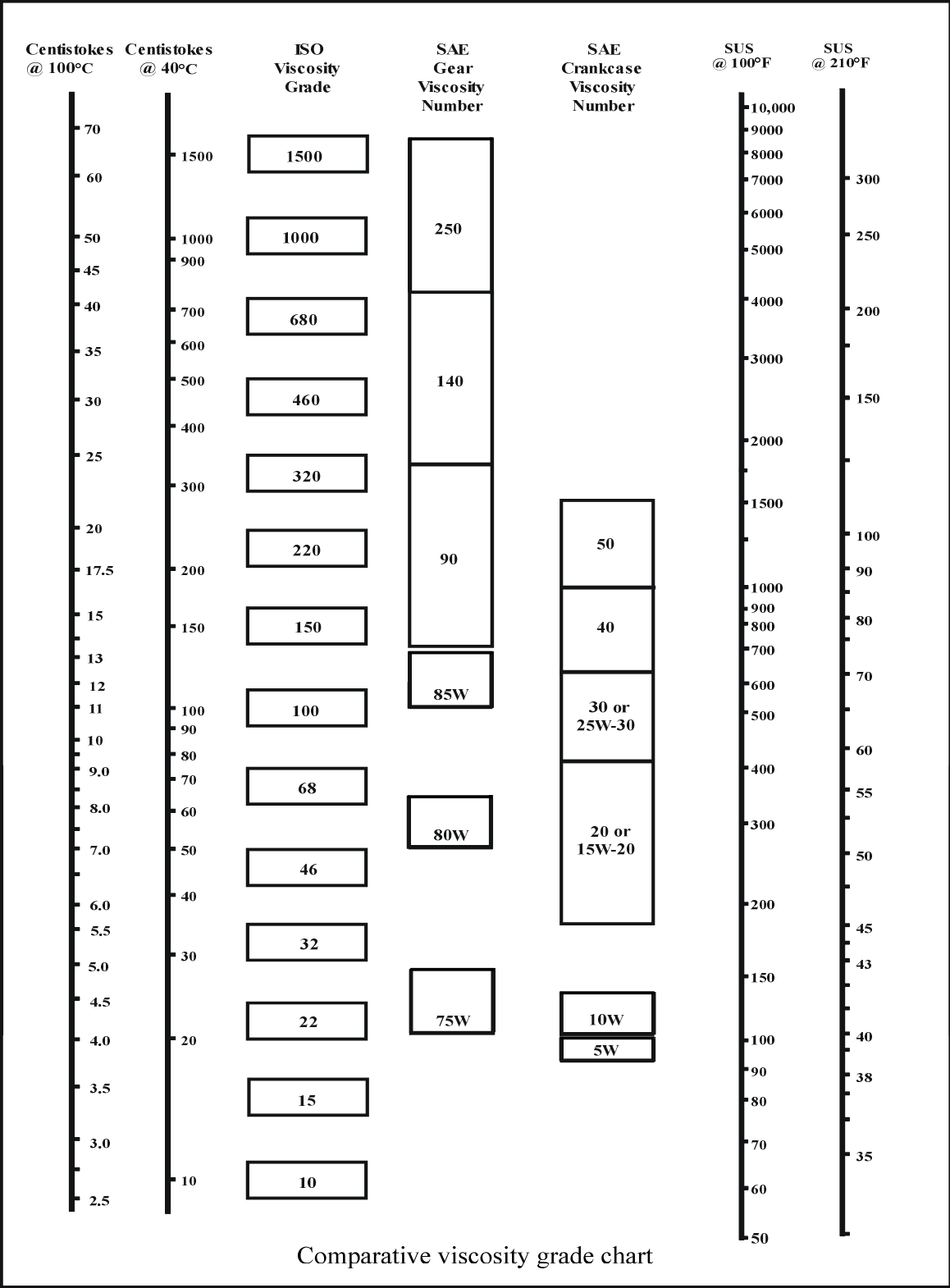
The Saybolt Universal Second system was created by George M. Saybolt in 1919 and is widely used in the United States. It is a measure of a fluids relative viscosity and uses SUS or SSU as an identifying abbreviation. A Saybolt Viscosimeter, invented by George M. Saybolt, is used to measure a fluids viscosity. The viscosimeter measures, in seconds, how long it takes for 60 ml of liquid to flow out of a restriction from a constant temperature bath. (Reference following picture)

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Kinematic Viscosity

The Kinematic Viscosity is the most widely used system worldwide. It is measured by the amount of time needed for a fixed volume of oil to flow through a capillary tube. The coefficient of absolute viscosity, when divided by the density of the liquid is called kinematic viscosity. Kinematic viscosity expresses the total resistance to fluid flow including internal fluid friction plus the effect of mass or weight of the fluid. The official ISO unit is meters squared per second (m^2/sec), but the centistoke (cSt) is commonly used in international fluid power industry. A centistokes is millimeters squared per second (mm^2/sec).



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Factors that affect viscosity

Several factors affect a fluids viscosity; listed below are some. Of all the factors that affect a fluids viscosity, temperature may have the most impact.

- Type of fluid
- Temperature
- Pressure
- Additives
- Contamination

Temperature

As with all liquids, hydraulic fluids are made up of molecules which are attracted to one another. The attraction force is not as strong as the molecules of a solid, but stronger than a gas. Heating any liquid will make the molecules travel faster and allow the fluid to flow easier, or have a lower viscosity. While cooling, any liquid will make the molecules travel slower and force the fluid to have more difficulty flowing, or a higher viscosity. In general, as temperature increases the viscosity of a liquid decreases; and as temperature decreases the viscosity of a liquid increases.

Not only can temperature have an effect on the fluids viscosity, but it can also affect other parts of the system.

Viscosity Index

Since the viscosity of a hydraulic fluid is an important factor and viscosity would change in relation to temperature. Any system which would have temperature changes or cannot maintain a constant temperature would need a fluid that could remain relatively stable over a given temperature range. Viscosity index (VI) is a measure of a fluid's resistance to viscosity change with temperature changes. A fluid with a stable viscosity over extreme temperature changes has a high viscosity index value. A fluid that does not have a stable viscosity over extreme temperature changes has a low viscosity index value.

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Conditioning

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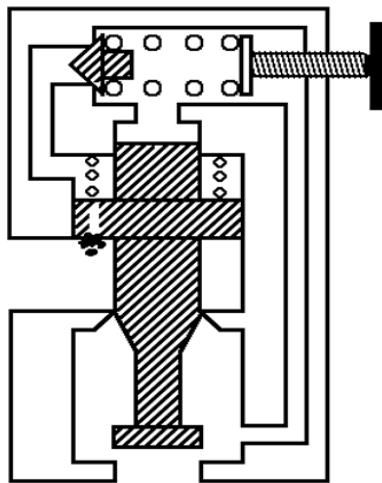
Conditioning

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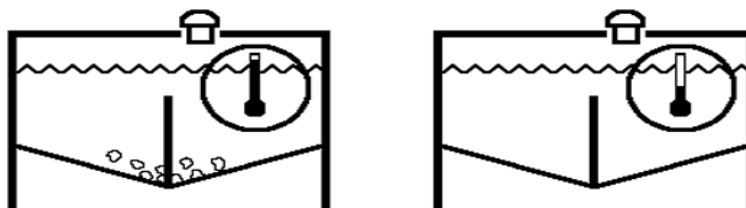
The Effects of Contamination on Hydraulic Equipment

Several factors are considered when dealing with contamination in hydraulic equipment. There are four primary functions of a hydraulic fluid; transmit power, cool or dissipate heat, lubricate moving parts and seal clearances between parts. Contamination can interfere or diminish the fluids ability to perform these functions.

Dirt interferes with the transmission of energy by blocking/plugging small orifices in hydraulic components; such as pressure controls, flow controls, pressure compensating pumps or any other component that depends on small orifices for control. Even air in the fluid can lead to damage from cavitation, reduced lubricity or erratic operation of components. The components then become non-productive, unpredictable and unsafe.



A hydraulic fluid's viscosity, system friction and changing fluid direction all generate heat during system operation. Dirty fluid interferes with the fluids ability to absorb heat from the system and a dirty reservoir will not allow the heat to dissipate from the fluid to the surrounding environment. **Clean hydraulic systems run cooler than dirty systems.**



The most serious effect that contamination can have on a system is when it interferes with the fluids ability to lubricate internal, moving parts. When lubricity is reduced or compromised excessive wear, slow response, erratic operation and premature component failure will be the result.

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Definition of Contamination

A contaminant is any foreign material that has a detrimental effect on the fluid's performance in the system. Contamination may exist as a gas, liquid or solid state in a system. A gas can either be in a dissolved state or entrained in the fluid as bubbles. A liquid, depending on its compatibility with the oil, can be in a free state or can be in a dissolved (emulsified) state. A solid, also may be known as particulate matter, can be characterized with it's density, hardness, ability to settle, ability to compact, ability to be transported, size, shape and concentration.

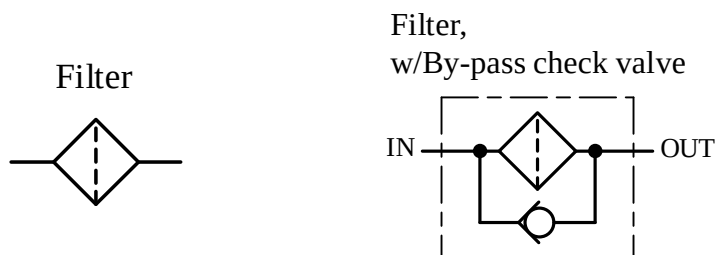
Sources of Contamination

Contamination has several ways to enter a hydraulic system;

- Built-in to the system
- Added to the system
- Generated within the system

Hydraulic Filters

All hydraulic fluids contain dirt to some degree and a large majority of component and system malfunctions are caused by particles of dirt. The need for a filter to reduce or remove contaminants is essential for system performance and dependability.

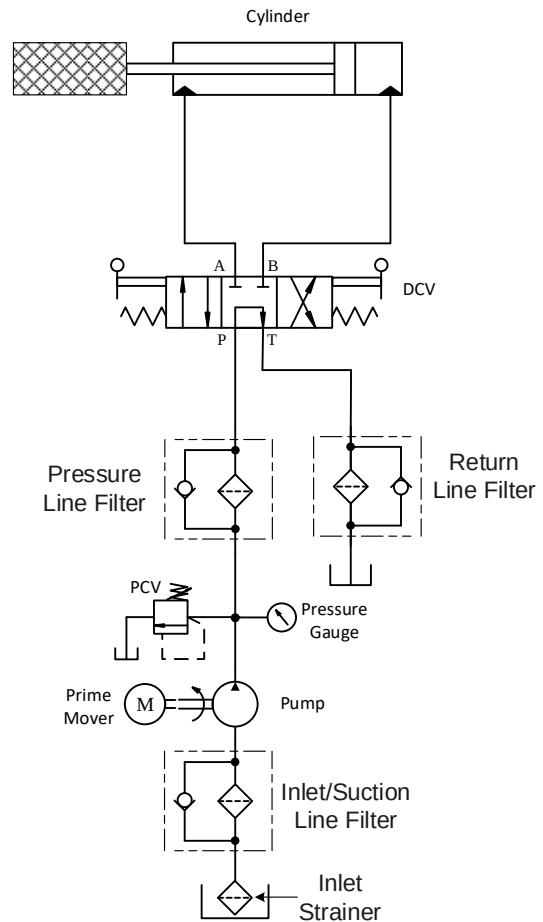
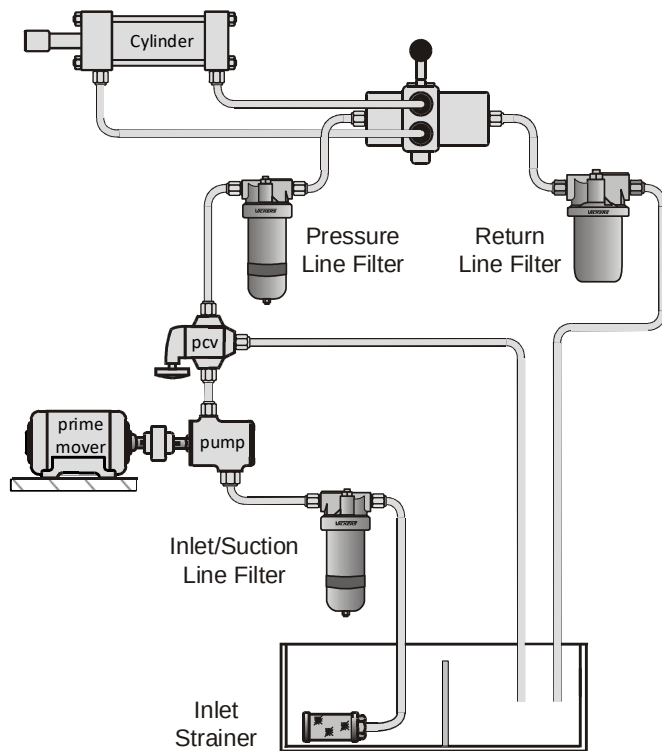


There may be two different types of filtration devices located in a system; filters or strainers. A filter is a device whose primary function is to trap, by some type of porous medium, contaminants from the fluid. A strainer is simply a very coarse filter. Each will usually have a different purpose in a system which usually will dictate a different position for each in the system. To put it in simple terms, whether it is a filter or strainer, their function is to trap contaminants.

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Filter Location/Position in a System

There are four typical areas for locating a contamination control component; inlet line (inlet strainer or suction filter), pressure line and return line.



Ideally each component should be equipped with its own filter, but this is usually impractical. To obtain the best results a balanced/strategic approach is used to locate the filters for necessary protection and maximum system efficiency.

Filter Ratings

Filter manufacturers commonly provide users with ratings/values that define the proper filter for any application. Flow ratings, in gpm or LPM, are used to size the filter properly for the required flow in the system. Pressure ratings are used to size the filter for the maximum pressure value experienced in the system. The filter element (medium) also receives rating to define the filters actual/approximate contamination removal ability or efficiency to provide adequate protection.

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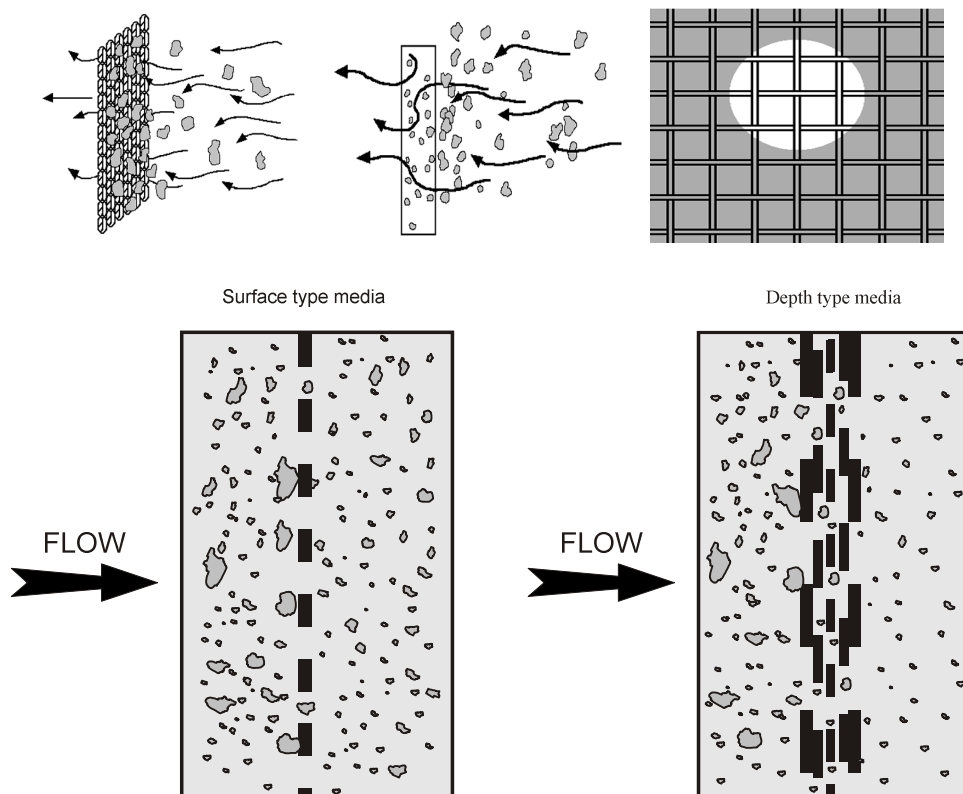
Filter Element Ratings

The function of the filter element is to remove contamination/dirt from the fluid. Depending on the location/position in the system, which determines purpose, there are two different numbers; a mesh number and a micron value.

A screen or strainer will use a mesh rating/number or a standard sieve number. The higher the mesh or sieve number the finer the screen/strainer, the lower the mesh or sieve number the more course the screen/strainer.

A filter will use a micron rating/number. The micron is referring to the measurable value in the metric system of the micrometre. One micrometre, one micron, is equal to one millionth of a meter or thirty-nine millionths of an inch. The micron rating/number refers to the hole/pore size created by the particular element material.

When a filter is specified with a micron rating/number, it usually refers to one of two different element/media rating systems, the nominal or absolute rating system. The filter element/media will be given a nominal rating if the material does not have consistent hole or pore size, thus the nominal rating is based on an average hole/pore size. The filter element/media will be given an absolute rating if the material does have consistent hole or pore size, thus the absolute rating is based on the absolute largest hole/pore size.



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Filter Efficiency Standards

Filters are made of a lot of different materials, such as Cellulose Media, Synthetic fibers, Wire-mesh, etc. All filters (surface or depth) are going to let some contaminant particles to go through them. The terms, absolute and nominal, have no standard definition regarding to filter performance and can vary from one manufacturer to another. Because of this, the filter performance should not be evaluated based on such ratings. To truly evaluate the performance of filters, the capture efficiency, at a given particle size, must be obtained.

Modern filters are engineered to meet the specific requirements of equipment manufacturers and are required to have exacting efficiency ratings.

Hydraulic filters are rated according to the size of the particles they remove and the efficiency with which they remove them. Filter efficiency can be expressed either as a beta ratio for a given particle size or as a percentage of particles captured.

Beta Ratio

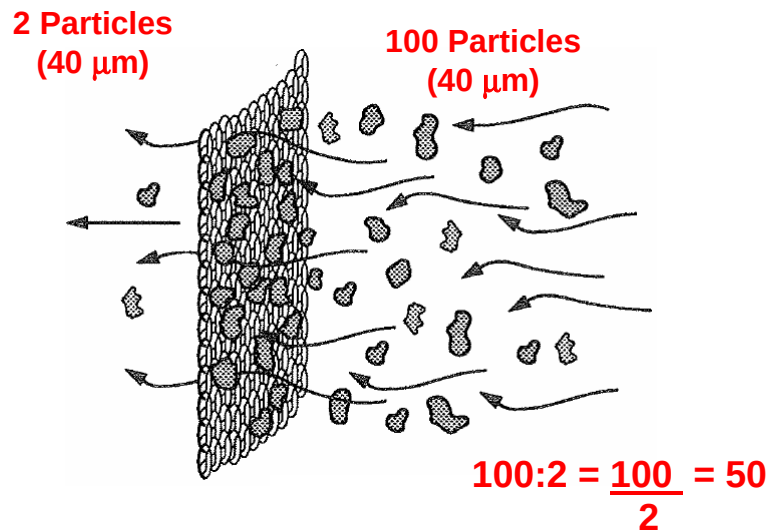
Until recently there has not been any universally accepted test method to measure or describe the media pore size or the size of particles a filter media can capture and hold. A test procedure called multi-pass testing or Beta ratio testing is a universally accepted test method that yields readily comparable test results. It was introduced with the object of giving both filter manufacturer and user an accurate and representative comparison amongst filter media.

Multi-pass testing uses a specified contaminate, of known sizes, added regularly in measured quantities to the fluid which is pumped continuously through the filter. Measured samples of the fluid are then taken at timed intervals from both the downstream and the upstream of the filter simultaneously, particles are measured and counted by electronic means using automatic particles counters.

Beta ratio is the number of particles, at the given size, that enter the filter divided by the number of particles that pass through the filter.

The bellow picture shows an example of a filter with a 50 Beta ratio

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Therefore, the beta rating of this filter is $\beta_{40}=50$

The beta ratio is an indicator of how well a filter controls particulate: i.e., if one out of every two particles in the fluid pass through the filter, the beta ratio is 2, if one out of every 200 of the particles pass through the filter the beta ratio is 200. Therefore, filters with a higher beta ratio retain more particles and have higher efficiency.

Efficiency for a given particle size can be derived directly from the beta ratio by the following equation:

$$\% \text{ Removal Efficiency} = \frac{\beta - 1}{\beta} \times 100$$

For the previous filter example $\beta_{40}=50$ the efficiency of the filter removing particles of 40µm will be:

$$\% \text{ Removal Efficiency} = \frac{50 - 1}{50} \times 100\% = \frac{49}{50} \times 100\% = 98\%$$

To remove confusion, manufacturers should describe the performance of the media used inside a filter by its 'beta ratio'. ISO 16889 (International Standard Organization) lists eight common Beta ratios used to report filter efficiency: Beta 2, 10, 20, 75, 100, 200, 1000 and 2000.

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Filter beta ratios and their corresponding efficiency percentages are shown in the following table.

Rating	Efficiency
2	50.0%
10	90.0%
75	98.7%
100	99.0%
200	99.5%
1000	99.9%

Even poor-quality filters can trap some level of 5 micron particles, but the Beta Ratio will be in the 2-10 range, which is not acceptable. Most experts agree a Beta Ratio for a given particle size should be at least 75 to achieve a reasonable level of effectiveness. High quality, efficient filter assemblies can achieve 200 to 2000 Beta Ratio for a given particle size.

ISO Cleanliness Code

Since filters are not 100% efficient, they can let pass some small particles that could not represent a big problem for some hydraulic components, but too many of these small particles could carry catastrophic failures due to sludge buildup and/or varnishing. While filter manufacturers publish beta ratings for filter media to describe efficiency performance levels, the ISO cleanliness code determine how clean the fluid is, because of this a direct connection between the beta rating scale and the ISO rating scale cannot be made.

The International Standards Organization standard ISO 4406 is the preferred method of quoting the number of solid contaminant particles in a sample. The code is constructed from the combination of three scale numbers selected from the following table.

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ISO Cleanliness Code 4406 for Hydraulic Fluid Contamination

ISO code number	Number of particles per ml	
	More than	Up to and including
22	20,000	40,000
21	10,000	20,000
20	5,000	10,000
19	2,500	5,000
18	1,300	2,500
17	640	1,300
16	320	640
15	160	320
14	80	160
13	40	80
12	20	40
11	10	20
10	5	10
09	2.5	5
08	1.3	2.5
07	0.64	1.3

ISO Cleanliness Level			
ISO Code	Particle/Milliliter		
	≥ 4μm	≥ 6μm	≥ 14μm
23/21/18	80,000	20,000	2,500
22/20/18	40,000	10,000	2,500
22/20/17	40,000	10,000	1,300
22/20/16	40,000	10,000	640
20/18/15	10,000	2,500	320
19/17/14	5,000	1,300	160
18/16/13	2,500	640	80
17/15/12	1,300	320	40
16/14/12	640	160	40
16/14/11	640	160	20
15/13/10	320	80	10
14/12/9	160	40	5
13/11/8	80	20	2.5
12/10/8	40	10	2.5
12/10/7	40	10	1.3
12/10/6	40	10	0.64

Note: The old ISO standard utilized 2μm, 5μm and 15μm instead.

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The following charts shows the suggested acceptable contamination codes for the most industrial hydraulic components and cross reference between the ISO and SAE Standards.

ISO code numbers	Typical components	Sensitivity
20 / 18 / 15	Flow control valves Cylinders	Average
19 / 17 / 14	Gear pumps/motors	Important
18 / 16 / 13	High pressure vane and piston pumps/ motors	Critical
18 / 16 / 13	Directional and pressure control valves	Critical
17 / 15 / 12	Proportional valves	Critical
16 / 14 / 11	Industrial servovalves	Critical

NOTE: The three figures of the ISO code numbers represent ISO level contamination grades for particles of $>4\mu\text{m}$, $>6\mu\text{m}$ and $>14\mu\text{m}$ respectively.

ISO 4406	SAE 749
14 / 12 / 09	0
15 / 13 / 10	1
16 / 14 / 11	2
17 / 15 / 12	3
18 / 16 / 13	4
19 / 17 / 14	5
20 / 18 / 15	6

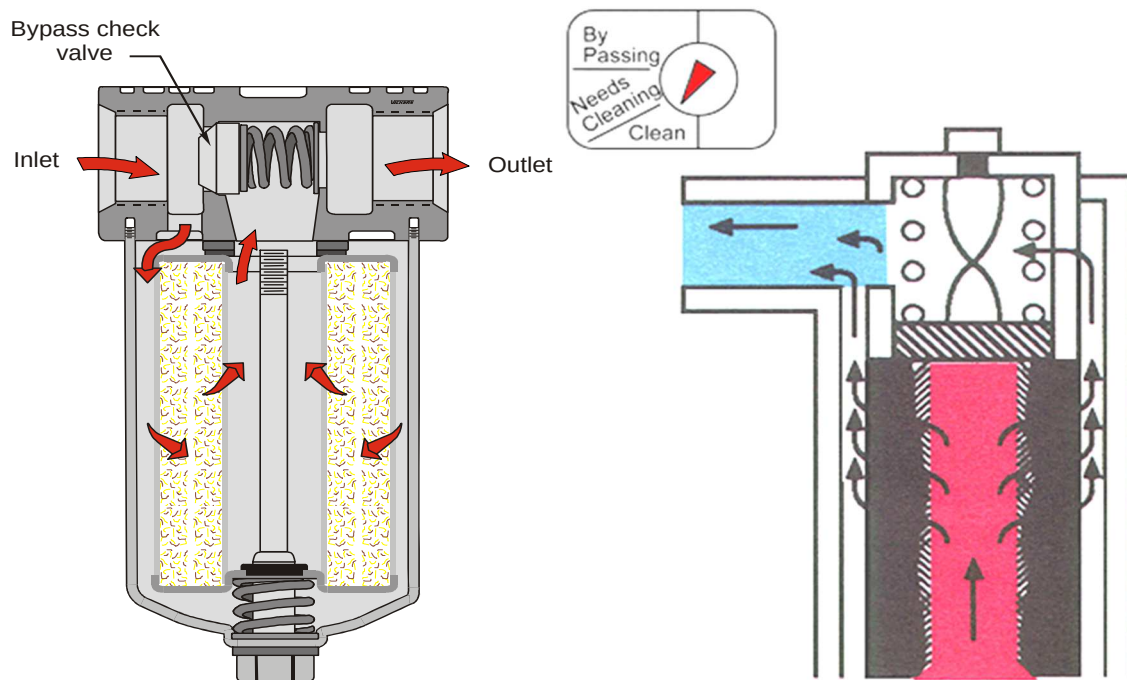
NOTE: The comparisons relate to particle count data only. To conform to any particular standard, reference should be made to the recommended experimental procedure.

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Filter Bypass Valves

Most, if not all, hydraulic filters have a bypass valve incorporated into the assembly to eliminate the possibility of element collapse or rupture when the element becomes clogged or when the fluid is very viscous. A bypass valve is basically a fail-safe device. As the filter element fills with dirt, the pressure required to flow through the element increases. When the pressure across the filter element is large enough to unseat the bypass valve and bypass the filter element completely. The bypass feature allows the fluid to bypass, instead of pushing the contamination through the filter element and re-entering the system or breaking down the element with higher than normal pressure which would destroy the element eventually. Although fluid should never be allowed to bypass the filter it does protect the filter and system components from excessive pressure.

The whole key to filter maintenance/performance revolves around cleaning or replacing the filter when needed. That's why most filters come equipped with filter bypass indicators to alert maintenance to a bypass situation.



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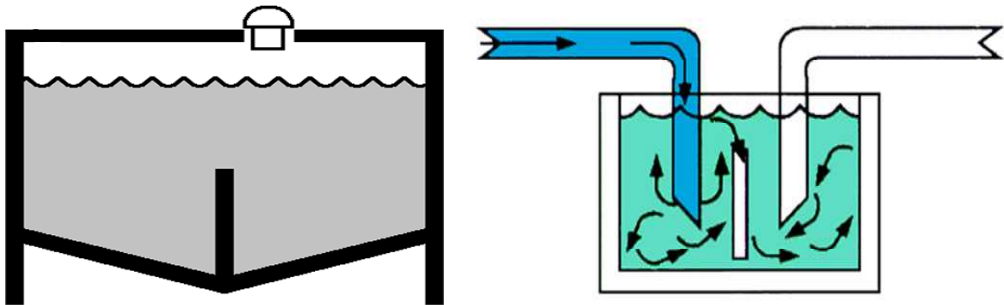
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Heat Exchange

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Hydraulic Reservoirs

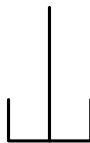
The hydraulic reservoir has many primary and secondary functions it serves. One function is to provide an opportunity for the fluid to transfer system heat from the fluid to the environment. By transferring heat through the reservoir walls it acts as a heat exchanger. Even-though the reservoir can be effective in removing heat from the system, sometimes it is incapable of removing all the heat. If this is the case, then a heat exchanger may be incorporated.



Reservoir (vented) Reservoir (sealed - pressurized)



Reservoir, w/drain line below fluid level Reservoir, w/drain line above fluid level Reservoir, w/main line above fluid level Reservoir, w/main line below fluid level



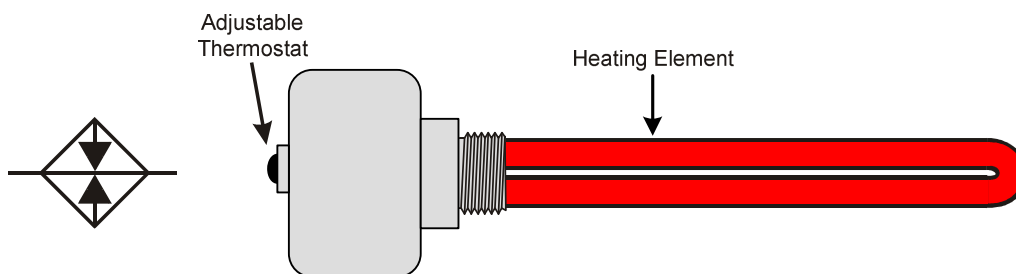
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Heat Exchanger

Even well designed hydraulic systems can be expected to turn some part of it's energy/power into heat, which contributes to the system inefficiency. Since no system is 100 percent efficient, heat is a major problem. When the fluid must be temperature conditioned, we typically think of cooling. In fact heat exchangers are commonly called coolers, because of their more frequent application – cooling the fluid. However there are some applications that require the fluid to be heated. The three types of heat exchangers are; heaters, air coolers and water coolers.

Heaters

A heater is installed in the reservoir, below the fluid level, and usually close to the pump inlet or pick-up line. Operates much like a heating coil or element in a household type water heater.

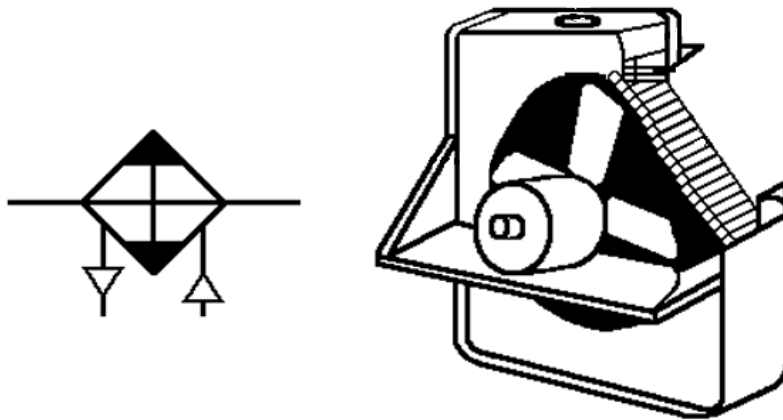


Air Cooler

An air cooler is typically used when cooling with water isn't practical or water isn't available. The hydraulic fluid is pumped through the tubes of the cooler and transfers the heat to the tubes/fins. Aluminum is frequently used as the basic material for the tubes because of aluminum's ability to absorb/transfer heat quickly. Then a fan or blower is used to move air across the tubes/fins to increase the heat transfer rate to the outside air. The operation is the same as an automobile radiator.

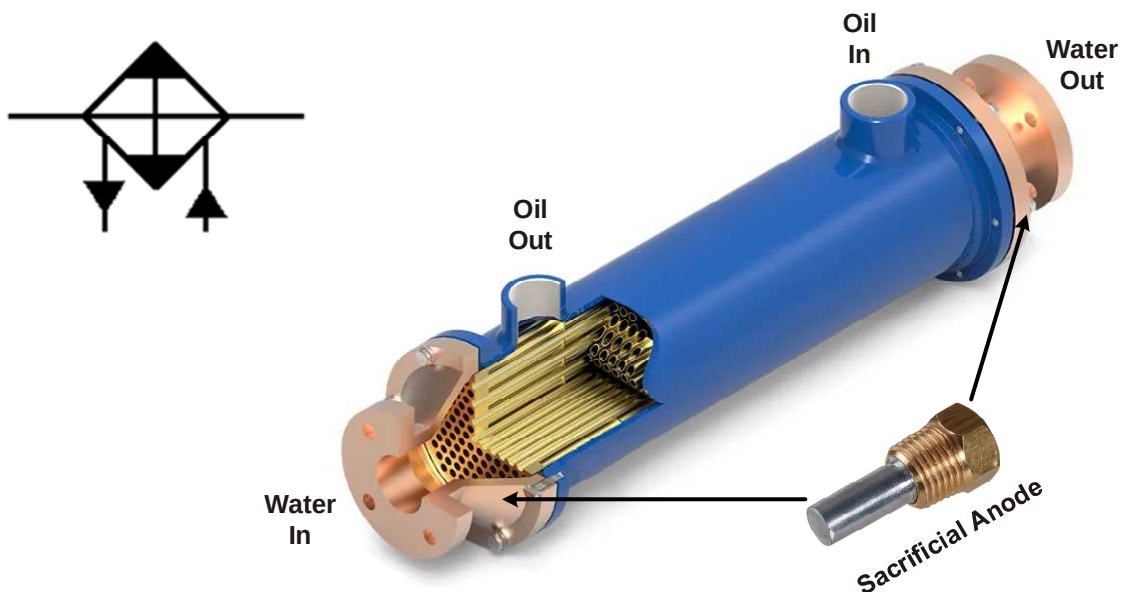
An air cooler isn't a true heat exchanger since it cools the fluid and usually doesn't heat the fluid. Generally it isn't practical to add heat to the fluid in this manner. Air coolers tend to be less efficient than water coolers and are ineffective in areas where a high ambient air temperature exists.

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Water Cooler

A water cooler is a true heat exchanger since it has the ability to cool or heat the fluid. The hydraulic fluid is pumped through the housing of the exchanger and heat is transferred to the tubes/fins. The heat is removed by the cooling water circulating through the tubes/fins. This type can be more efficient because the cooling water can be much colder than air used as a cooling medium. This type is also known as a shell and tube type heat exchanger.

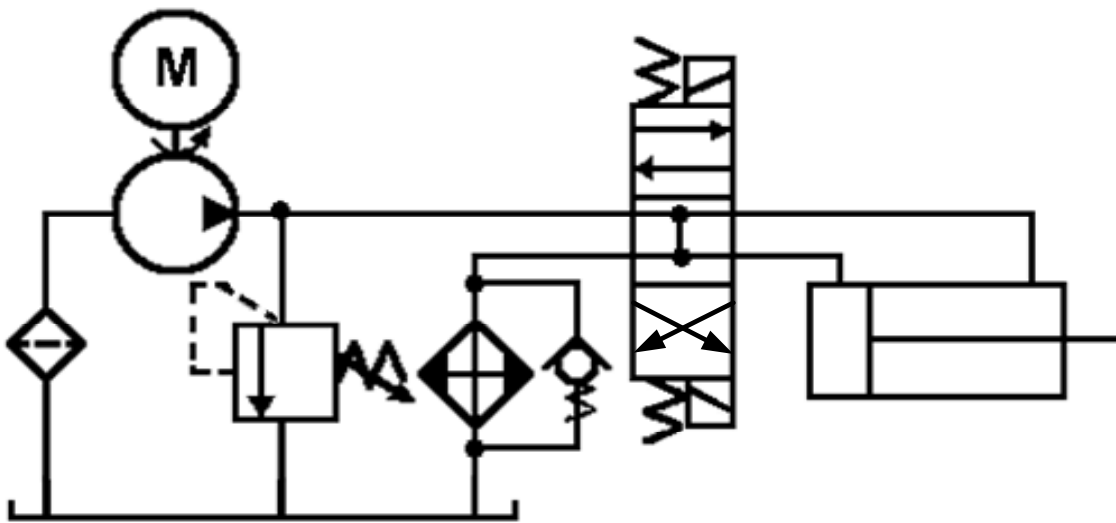


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Heat Exchangers in a Circuit

Heat exchangers generally have a low operating pressure, somewhere around 200 psi or less. This requires that they be located in a low pressure part of the system.

To ensure a pressure surge in the system does not damage the heat exchanger/cooler they are usually placed parallel to a low pressure spring loaded check valve.



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Components

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Pumps

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Introduction to Pumps

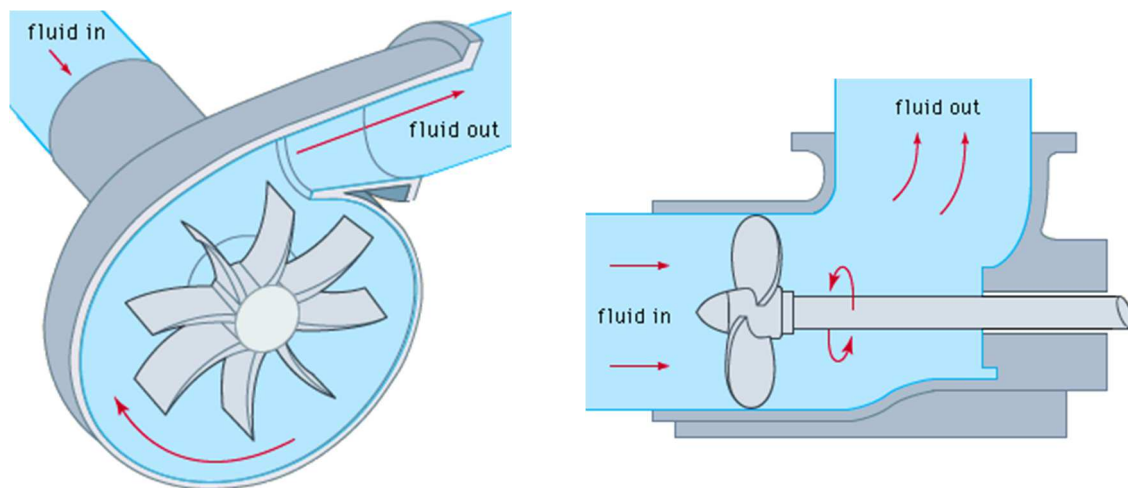
The function of a hydraulic pump is to convert mechanical energy, from the prime mover, into hydraulic working energy. It performs this function by pulling the fluid from the reservoir and then pushing the fluid into the system, creating flow (displacement) and pressure (force).

Displacement

The flow / volume capacity of a pump can be expressed as “displacement”, “displacement per revolution” or “output in GPM (or LPM)”. Displacement is the volume or amount of fluid moved in one revolution of the pumping (rotating) group.

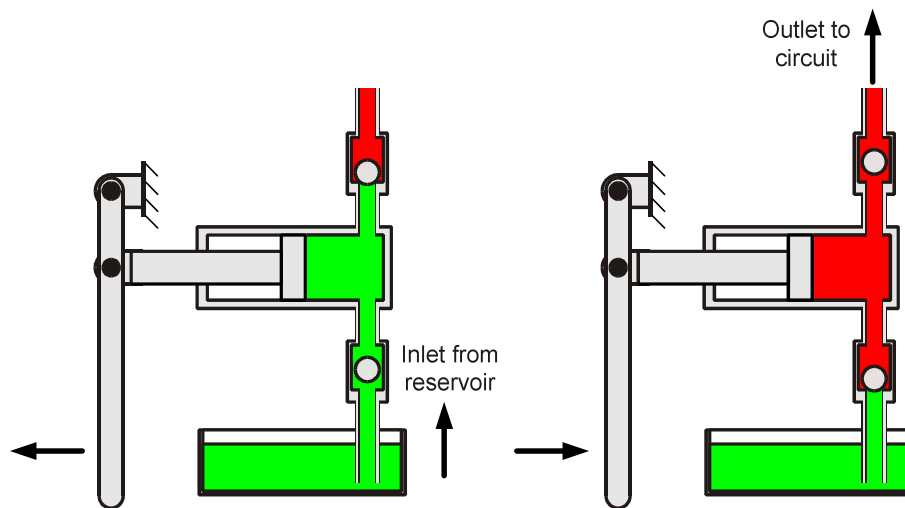
All pumps, no matter the type, fall into two basic categories; **positive displacement or non-positive displacement**.

Non-positive displacement pumps do not deliver a constant amount of fluid for every revolution of the pumping (rotating) group. The internal members (pumping group), which perform the pumping action, do not touch and/or seal against the inside housing. Since there is no contact or sealing between the internal members and the housing there is less drag or friction, but increased slippage or leakage during operation, especially at higher pressures. Non-positive displacement pumps are not useful in most industrial applications because of their limited pressure generation. Non-positive displacement pumps are primarily used to circulate or move fluids from one location to another. Some common examples would be; automobile water pump, dishwasher or washing machine pumps.



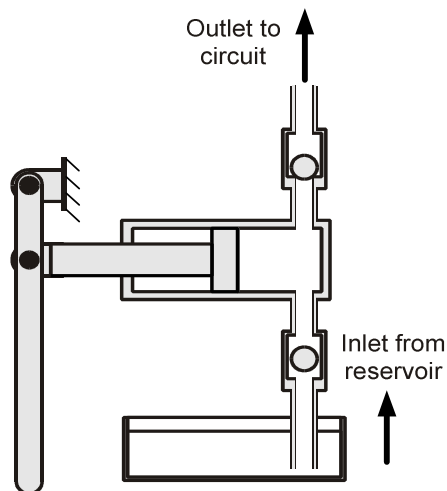
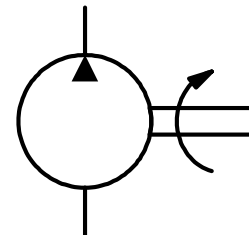
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The most common hydraulic pump used in systems today is a **positive displacement** type of pump. Positive displacement pumps deliver a constant volume / amount of fluid for every revolution of the pumping (rotating) group. The internal members (pumping group), which perform the pumping action, will actually touch and seal against the inside housing and provide consistent fluid flow, reduced slippage or leakage and higher pressure values.



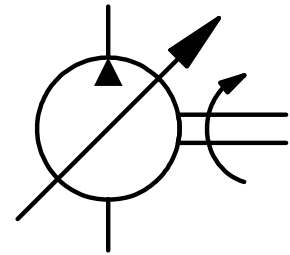
A positive displacement pump may be one of two different categories; **Fixed Volume (displacement)** or **Variable Volume (displacement)**.

A **fixed displacement** pump will deliver the same volume or amount of fluid for every revolution. Industrial pumps usually operate at 1200 to 1800 rpm, which means, as long as the rpm is constant the flow rate of a fixed displacement pumps flow rate will remain the same.



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In some situations it is required that the pump's flow rate be adjustable or variable. One way to accomplish **variable displacement** is to increase or decrease the speed of the prime mover. This method is usually impractical. The only other way to change the output volume/displacement is to change the pumps displacement internally by changing the pumping (rotating) group. Certain pump types have the ability to change their displacement.



Delivery in GPM (LPM)

A pump may be rated, for example, as a 20 gpm (75.7 lpm) unit, however it may actually pump more when under a no-load condition or less when under a loaded condition. Most manufacturers provide a table or graph showing pump delivery, horsepower, drive speeds and pressures under certain conditions.

Theoretically, a pump will deliver an amount of fluid equal to its displacement with each revolution. In reality, the actual displacement or output is less because of pump inefficiency. That inefficiency is in the form of internal leakage or slippage and will increase or decrease proportional to the system operating pressure. This difference between theoretical and realistic is termed as Volumetric Efficiency. Volumetric efficiency is equal to the actual output divided by the theoretical output and is expressed as a percentage.

Pump Ratings

A pump is commonly rated by its maximum operating pressure and displacement (output volume / flow) at a particular speed (RPM).

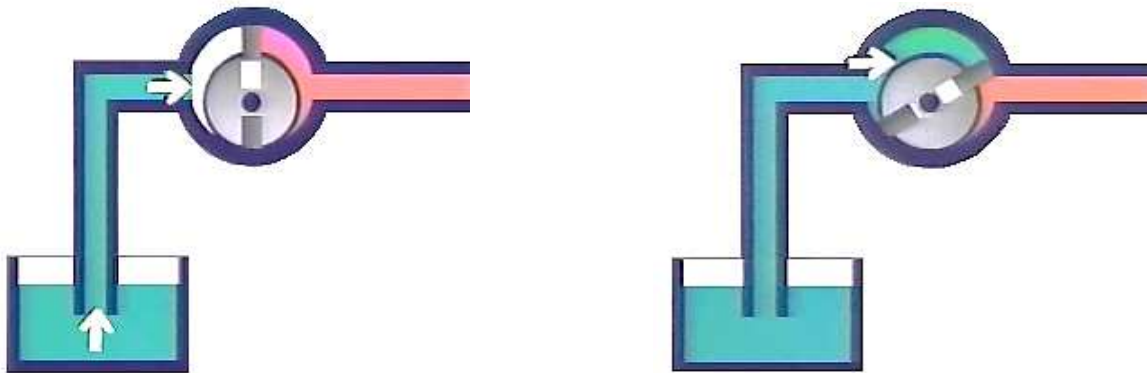
Operation at the Suction Side of a Pump

In hydraulic systems, pumps are used to pressurize the fluid and create flow. These pumps take hydraulic fluid from the reservoir, on the inlet or suction side, and force it out the pressure side to the rest of the system where it can be used to perform work.

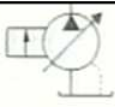
In hydraulic systems pressure is created at the outlet side of the pump, but the inlet side of the pump creates vacuum, the opposite of pressure. Pressure is normally measured in PSIG, which indicates how much pressure there is above atmospheric pressure. Vacuum is usually expressed in inches of mercury vacuum (in.Hg.), which means the gauge indicates how much less than atmospheric pressure (how much suction) is on the inlet side of the pump.

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Let's look at a simple hydraulic pump and see how pressure and vacuum affect its operation. When the rotor turns, the vanes (which are free to move in and out) are forced outward against the outer ring by centrifugal force. As the rotor turns, the space gets larger creating a partial vacuum on the inlet or suction side of the pump coming from the reservoir. Because there is less pressure in the inlet port, atmospheric pressure in the reservoir will push the fluid in the suction line and into the pump.



The pressure applied to the liquid by the atmosphere is used in two ways: first to move the liquid from the reservoir thru the suction line to the pump inlet port; and secondly it is used to force the liquid into the pump to fill the moving rotating group. If moving the liquid from the reservoir to the suction port requires too much of the atmospheric pressure (for example when the pump is been mounted too high above of the fluid level) there is not enough pressure remaining, to force the liquid into the rotating group fast enough.



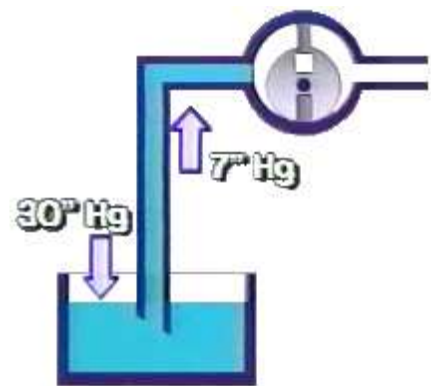
Installation Data

Inlet Conditions: 7 In. Hg. Max. Vacuum Condition
(At 1200 RPM)
5 In. Hg. Max. Vacuum Condition
(At 1800 RPM)

Operating Temperature Range: -40°F to 160°F

Filtration: SAE Class 4

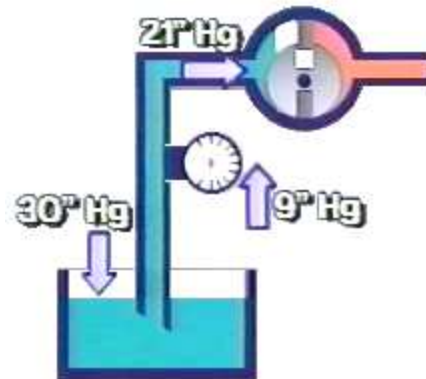
Installation Note: See page 29 of this bulletin for specific



Pump manufacturers give their suction specifications in terms of vacuum at sea level. If a pump manufacturer specifies that no more than 7 inches of mercury vacuum be present at the inlet port, this means that only 7 inches of mercury, of the available atmospheric pressure, should be used to get the fluid to the pump inlet port in order to have the remaining atmospheric pressure available to push the fluid into the rotating group.

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So, if we put a vacuum gauge on the suction side of the pump and find a vacuum to be more than 7 inches of mercury, we are likely to have erratic, noisy operation and this could damage the pump.



If the pump is to be used at an elevation higher than sea level (for example in Denver), where the atmospheric pressure is about 5 inches of mercury less than sea level, the lesser atmospheric pressure must be taken into account.

The vacuum reading should be approximately 2 inches of mercury; this is because we started out 5 inches of mercury short, due to the difference in elevation.



Installation Information

Fluid Recommendations

Premium quality hydraulic oil with a viscosity range (@ 100°F) of 150-250 SSU (32-54 cst.) for pressures up to 1500 PSI and 200-300 SSU (43-65 cst.) for higher pressures. Normal operating viscosity range between 80-500 (16-110 cts.). Maximum start-up viscosity is 2000 SSU (440cst.).

NOTE: Inlet or reservoir fluid temperature must be maintained between 40°F and 160°F. Recommended operating temperature is 100-110°F. Oil should have maximum anti-wear properties, rust and oxidation treatment.

Pump specifications are normally given for a pump operating at a certain speed and using a petroleum-based fluid, therefore the specification must be altered if a different speed or different fluid is used.

Most pumps get their maximum allowable inlet vacuum based on petroleum fluids, however fire resistant fluids can weight more than petroleum based fluids. Before changing fluids in the system be certain the fluid you choose will not exceed the pump manufacturers inlet vacuum specifications, otherwise a problem called cavitation could occur.

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Cavitation is the formation and collapse of cavities (holes within the fluid); these holes are created when the fluid boils, not from heat, but from low pressure. The pressure in the suction side of the pump is too low, in other words, too much vacuum is created and the fluid will actually begin to boil at normal operating temperatures.



When that happens, gas filled cavities form in the fluid. As the cavities in the fluid are carried to the outlet side by the rotating group, the pressure suddenly increases and the cavities collapse with tremendous force.

These gas filled cavities interfere with proper lubrication and destroy metal surfaces, especially on the outlet side of the pump. Continued cavitation will ruin the inside surfaces of the pump, even releasing metal particles that will be carried by the fluid to other components throughout the system.

Cavitation is the reason why when choosing or installing a pump it is very important not to exceed the pump's inlet vacuum specifications.

Cavitation vs Aeration

You have a noisy pump, cavitation or aeration, which is it?

What causes it?

Is it serious?

Absolutely.

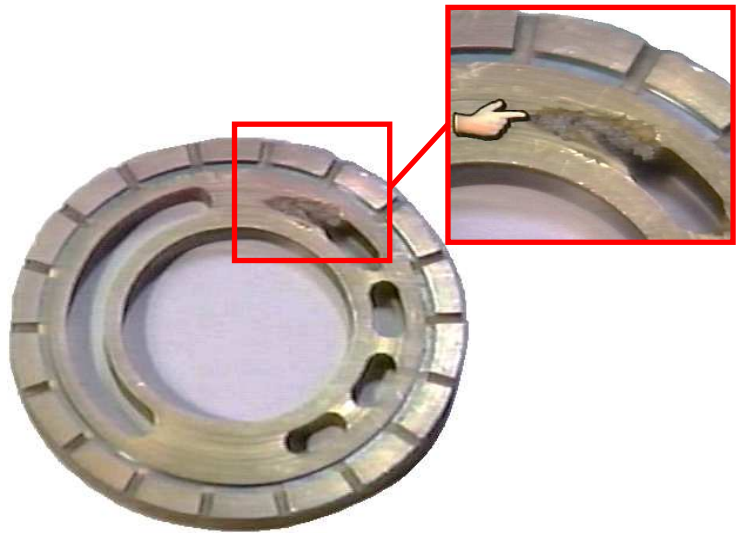
Cavitation (True-Cavitation)

Cavitation is very damaging to hydraulic components and usually manifests itself most dramatically in the pump. If it is not cured pump destruction will be the result.

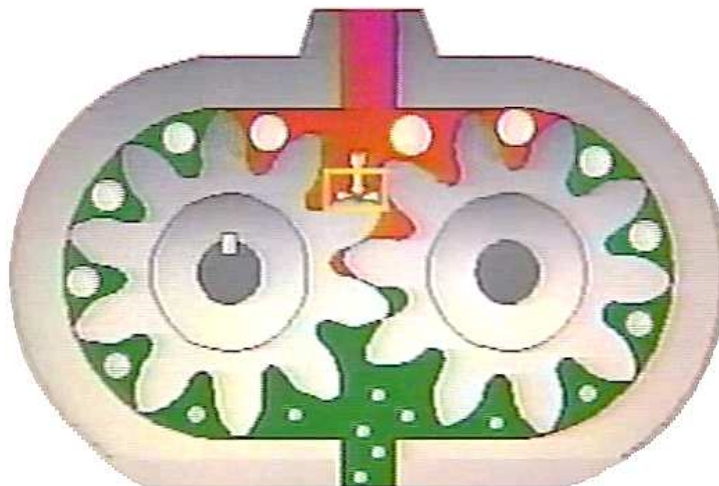
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This piston pump side plate shows the effects of cavitation damage. In this case the pump was still operating when this part was removed. Obviously, it would be only a matter of time, perhaps just a few hours, before the pump cease to operate.

Cavitation is a phenomenon that occurs when a fluids pressure becomes too low; that is why it is most common in pumps; although it can occur in other areas of the hydraulic system, whenever and wherever, pressure drops too far below atmospheric pressure.



In pumps, as shown in the picture, cavitation will occur when the fluid cannot enter the inlet side in sufficient quantity to prevent excessive vacuum.



This can be cause by a restriction in the inlet side, over-speeding of the pump, reservoir too far below the pump, inadequate reservoir breather, or anything else that tends to prevent the fluid from entering the pump. It is aggravated by elevated fluid temperatures, reduced fluid viscosity, and water in the fluid.

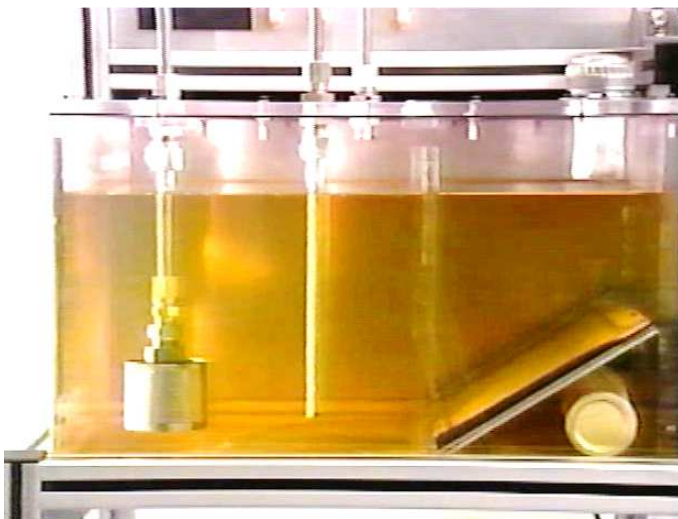
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Cavitation is evident because of the increase in inlet (suction) line vacuum, well beyond the pump specification. The noise is caused by the bubbles collapsing at the outlet side of the rotating group.

Because energy is absorbed in the cavitation process, the power output of the pump is reduced; there is a significant drop in the flow output of the pump as well as a slight drop in system pressure.



Vacuum Gauge



Reservoir



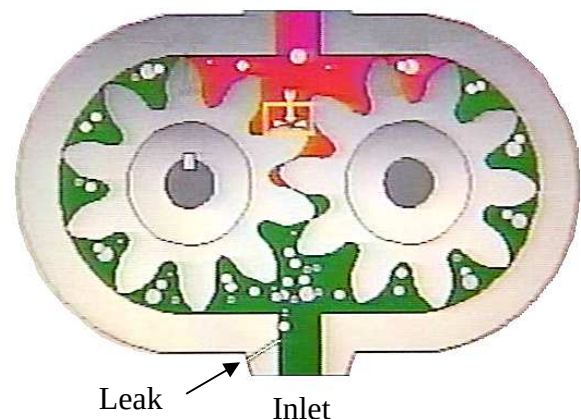
Flow Meter

Cavitation involves the vaporizing of a fluid and the vapor bubbles return to the fluid state at the outlet of the pump. There is little evidence of cavitation in the fluid stream or in the reservoir.

Aeration (Entrained Air or False-Cavitation or Pseudo-cavitation)

Aeration is caused by air mixing with the fluid. It can come from different sources:

- air mixed with the fluid.
- reservoir fluid level is too low.
- poorly designed reservoir, which allows fluid mixing with the air.
- hole or loose/leaky fitting in the inlet line of the pump.



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Air passing through the pump, behaves much the same as the vapor bubbles do; they expand at the inlet, and collapse at the outlet.

The collapsing bubbles cause erosive damage to the pump components in the same way as cavitation does. There is a slight difference in the sound aeration makes, although this is very difficult to distinguish, unless both aeration and cavitation sounds can be heard together.

The appearance is the same, bubbles appearing in the inlet side of the rotating group, although there is a marked difference in the inlet vacuum which is reduced due to the ingestion of air.

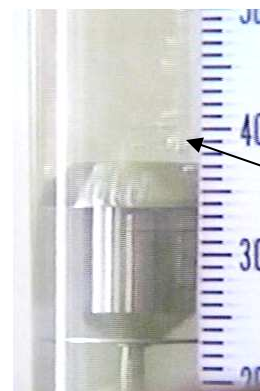
The pressure gauge at the outlet shows a slight pressure drop, similar to the experience with cavitation, because air is clearly mixed with the fluid as it travels through the pump, it cannot dissipate and it is still in the fluid as it travels to the reservoir. A reasonable amount of air that enters the reservoir will have a chance to flow to the fluid surface and be dissipated, but when the fluid becomes too aerated however, the air could make it through the reservoir and enter the pump again.



Vacuum Gauge



Reservoir



Flow Meter

The only correct solution to an aeration problem, is to find the source of the air and correct it.

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To determine which it is cavitation or aeration, look at the reservoir and check the inlet vacuum.

Cavitation will leave the liquid relatively air free, and is always coupled to a high vacuum. Aeration, will give the reservoir a white, cloudy or milky appearance, the air will be readily visible.

If cavitation appears to be the source of the noise, look for the inlet restriction, elevated temperature, and/or pour fluid that is causing it.

If aeration is the problem, look for the source of the air, loose/leaky fitting, low fluid level, and/or look at the design of the reservoir.

You have a noisy pump, cavitation or aeration, which is it?

Look at the reservoir, and check the inlet/suction line vacuum gauge. Is it serious?

ABSOLUTELY!

Types of Pumps

Pumps are made in many different sizes, shapes and types, however the pumping action is the same for each type. All pumps generate an increasing volume at the inlet (suction) side and a decreasing volume at the outlet (discharge) side.

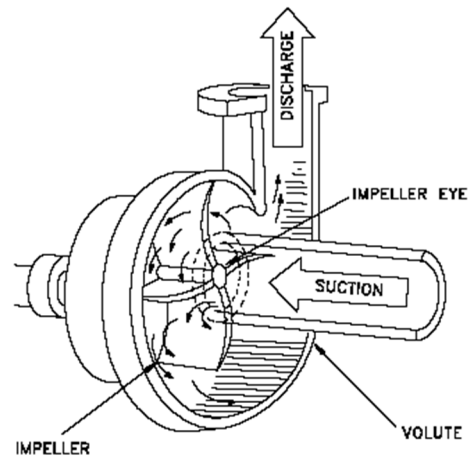
Centrifugal Pump

Centrifugal pumps, sometimes called hydrodynamic or turbine type, are non-positive displacement pumps. Centrifugal pumps are primarily used to circulate or transfer fluids from one location to another. Some common everyday examples would be; automobile water pump, dishwasher or washing machine pumps.



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Most centrifugal pumps operate by centrifugal force, thus the term centrifugal pump. Fluid enters the pump housing and is thrown outward by the impeller. Kinetic energy is transferred from the impeller and applied to the fluid forcing it out of the pump. Since the impeller does not actually touch or seal against the housing the amount of pressure that can be created is limited. The only resistance to flow created is the weight of the fluid and fluid friction. This type is usually impractical to use for most modern hydraulic systems.



Gear Pump

The simplest, and possibly the oldest, of all the common pump designs would be the gear pump. They are efficient pumps, capable of low to medium operating pressures (up to 2500 psi) and generally a medium discharge capacity (up to 50 gpm). They are available as single pump or double pump unit.



One gear (the driver) is rotated by the prime mover and is in mesh with another gear (the driven). A partial vacuum is created at the inlet (increasing volume) where the teeth un-mesh and fluid fills the cavity. A gear pump develops flow by the unmeshed teeth carrying fluid around the housing. As the gears rotate and try to re-mesh at the outlet (decreasing volume) fluid is forced out into the system.

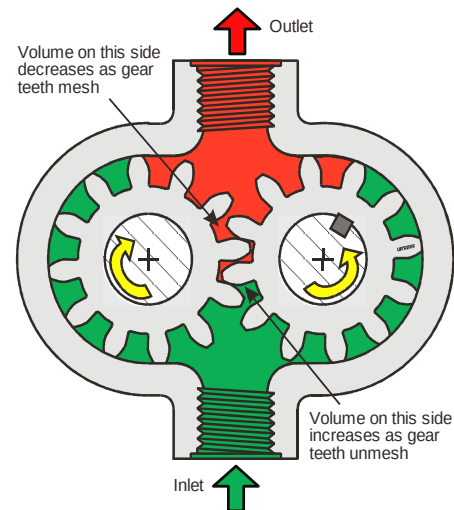
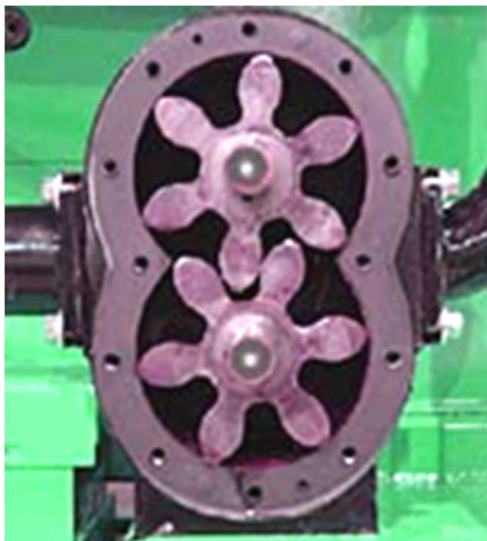
The gear pump is an unbalanced type of design. Low pressure at the inlet and high system pressure at the outlet creates an unbalanced load on the pumping group (rotating group). The gears, shafts, bearings, side-port plates, seals all feel the imbalance of force.

Gear pumps may consist of two different designs; external gear pump and an internal gear pump.

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External Gear Pump

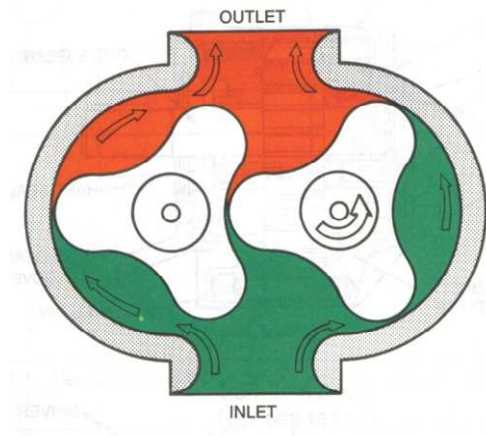
External gear pumps use the outer circumference of two external gears to perform the pumping action, increasing / decreasing volume.



External Gear Pump
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Lobe Pump

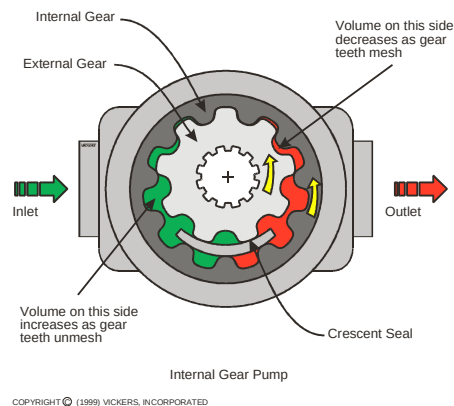
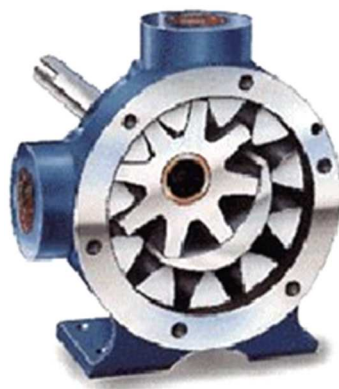
Also in the general family of gear pumps is the Lobe (Lobed) or Rotor pump. This pump operates on the same principle as the external gear pump, but can have a higher displacement than a conventional external gear pump. The higher displacement is possible because both gears will have less teeth, usually approximately three teeth each. Since the teeth are wider and more rounded than a conventional gear tooth more fluid can fit into the space for a higher output capacity.



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Internal Gear Pump

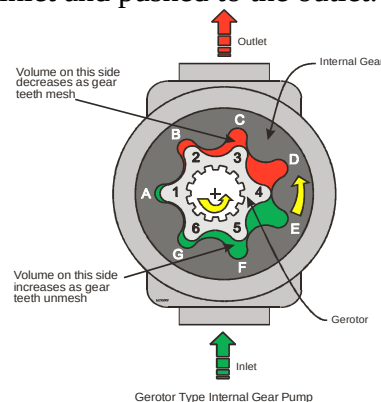
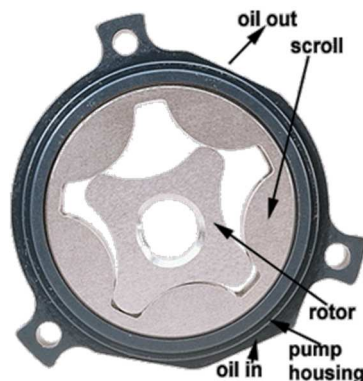
Internal gear pumps will use one external gear positioned inside an internal gear and the pumping action is the same, increasing / decreasing volume, because of the meshing and unmeshing of the teeth.



Gerotor Pump

The most common type of internal gear pump is the Gerotor Pump. A gerotor pump is an internal gear pump by design, however what defines the difference between an internal gear pump and a gerotor pump is the difference in the number of teeth. An internal gear pump can have differing ratios between the meshing gear teeth. The gerotor pump is unique in that it has only one tooth difference between both gears. The external toothed gear, which is the inner gear, has one less tooth than the internal toothed gear, which is the outer gear.

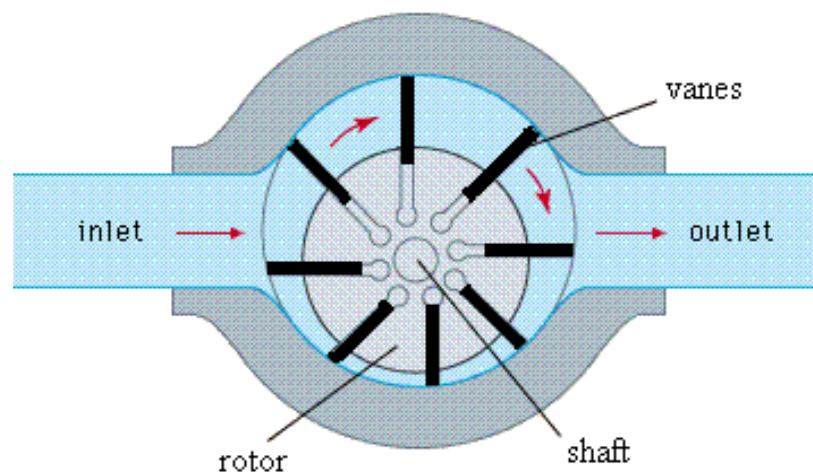
When the inner gear is turned by the prime mover, it rotates the outer gear. Just as in every other gear pump, an increasing volume is used as the teeth unmesh and decreasing volume is used as the teeth mesh. Fluid is then pulled from the inlet and pushed to the outlet.



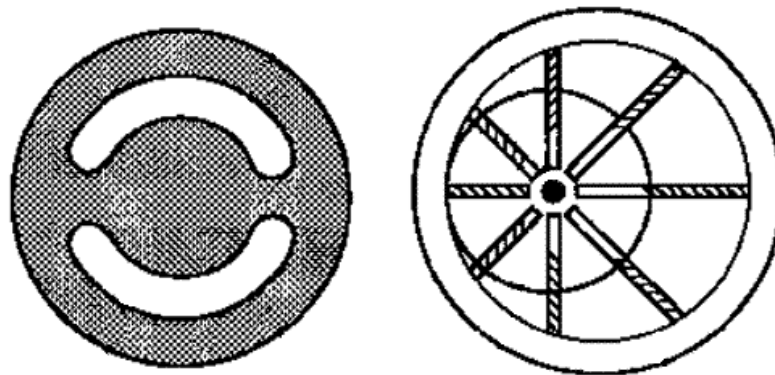
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Any gear pump design is considered to be a fixed displacement type. The displacement of a gear pump is determined by the fluid volume each gear displaces and the rpm that each gear is turning. The only way to change the discharge volume or displacement is by two methods. The first is to remove the existing gears and replace them with gears that have different teeth dimensions. The second method is to change the speed of the prime mover, more rpms would mean more volume displaced. Usually both of these methods are impractical in an industrial environment, thus gear pumps are not considered variable displacement.

Vane Pump

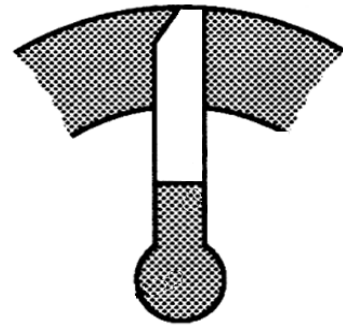


The vane type pump is widely considered to be the most common of all the hydraulic pump designs. They are very reliable, have a low operating noise level and are highly efficient pumps that are able to operate at low to medium pressure ranges (up to 3500 psi) and generally a medium to high volume capacity (up to 100 gpm). They are available as single pump or double pump unit.

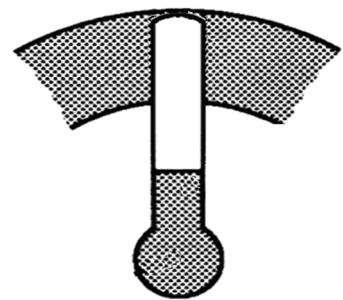


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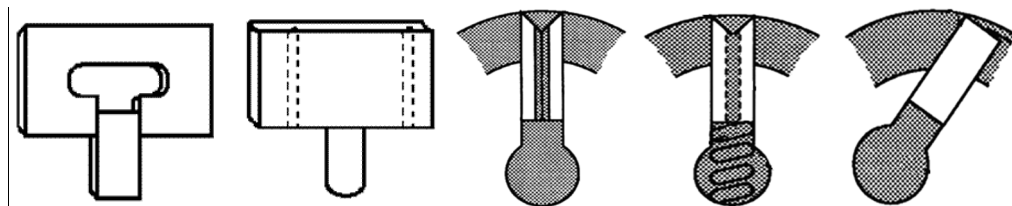
A prime mover drives a shaft which is connected to a rotor. The rotor is positioned inside the housing or pumping chamber. The rotor is slotted and fitted with different styles of vanes to make contact and follow the inner shape or contour of the housing or pumping chamber.



To ensure the vanes make contact and seal against the housing vane pumps generally require a minimum operating speed of approximately 600 rpm to allow centrifugal force to sling out the vanes and make contact. If the operating speed is slow or has a progressive start up or if a very viscous fluid is being used other means may be used to load the vanes and keep them in contact with the housing. Two other means are used to help load the vanes; one is to divert system pressure to the underside of the vane and secondly, light duty springs may be used to ensure the vane is extended or provides a good seal.



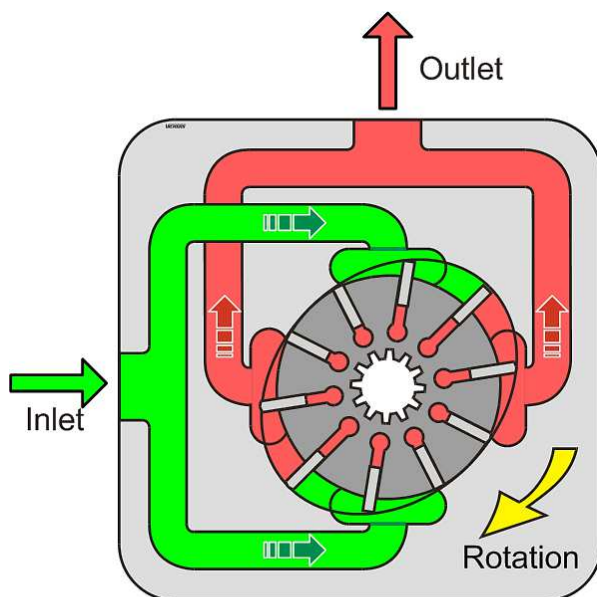
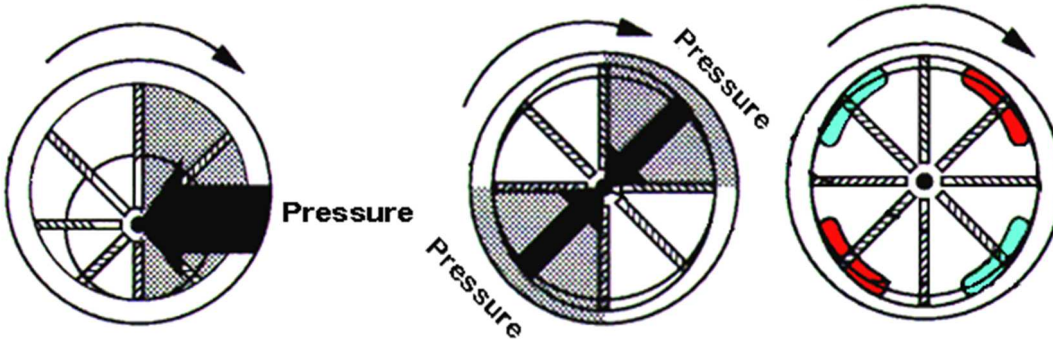
Loading a vane in any of the previously described manners may create too great of a force. If the force is too great there would be excessive drag, from friction, and too much wear on the vane itself. Manufacturers will compromise between creating the best seal and reducing the amount of friction and wear, so generally vanes are only partially loaded.



The rotor and pumping chamber are not aligned along the same centerline. Because the shaft and rotor are offset (eccentric) in relation to the pump housing, as the rotor rotates the distance between the vanes increases (increasing volume) and creates a partial vacuum. Fluid will fill the void or cavity. As the distance between the vanes decreases (decreasing volume) fluid is forced out into the system.

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Low pressure at the inlet and high, system, pressure at the outlet creates an unbalanced load on the pumping group (rotating group). The gears, shafts, bearings, side-port plates, seals all feel the imbalance of force. The vane pump previously illustrated is an unbalanced design. Unlike a gear pump, the vane pump may be **unbalanced or balanced** in design.



Balanced Vane Pump Design
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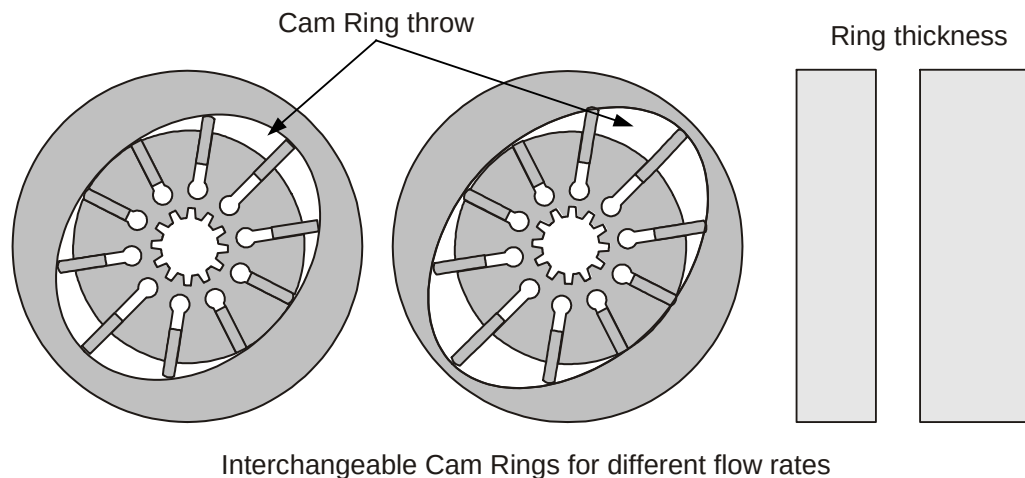
In a **balanced vane pump** the rotor and pumping chamber are aligned along the same centerline. With the two aligned along the same centerline there is no opportunity for the pumping (rotating) group to create an increasing / decreasing volume situation, thus there is no fluid flow or pressure generation. To overcome this, the pumping chamber (or cam ring) is elliptical (egg shaped, cam shaped), not circular as in the unbalanced design. With this type of design the two pressure quadrants oppose each other and cancel out or offset the shaft load (or side force) encountered in an unbalanced design.

Fixed volume, positive displacement vane pumps typically use a balanced design. Using a balanced design eliminates the shaft side loading issue, however it does present one other problem. With the pumping group (rotating group) running along the same centerline, the pump's displacement will be fixed and not variable.

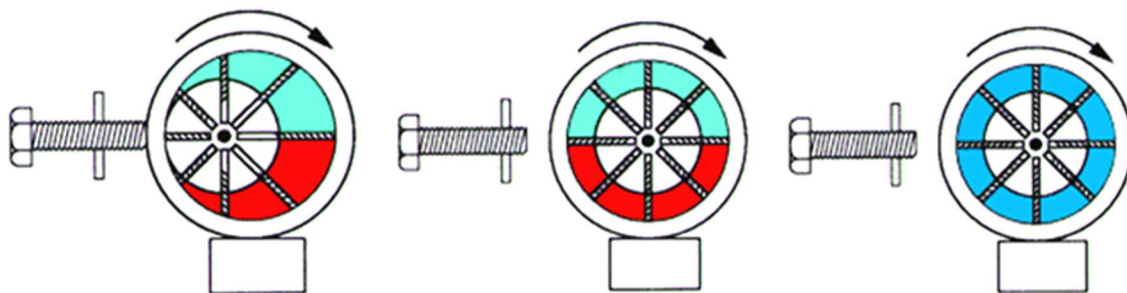
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Variable Displacement Vane Pump

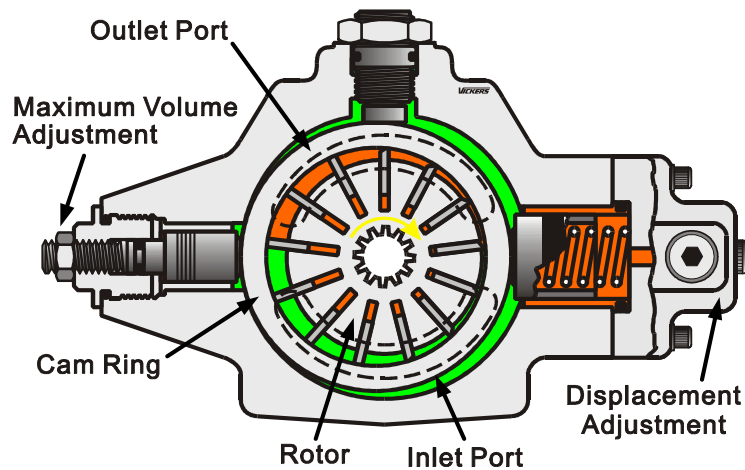
The displacement of any vane pump is determined by the fluid volume each vane (pumping / rotating group) displaces and the rpm it is turning. Many methods exist to change the discharge volume or displacement. The first is to remove the existing pumping (rotating) group and replace it with one of differing dimensions.



The second method is to change the speed of the prime mover, more rpm's would mean more volume displaced. Usually both of these methods are impractical in an industrial environment. If the vane pump is balanced in design it is considered a fixed displacement pumps, unless one of the two previously mentioned methods can be applied. If the vane pump is unbalanced in design it has the ability to change its displacement, while the pump is operating, thus making it very desirable in some applications. To change the volume of fluid discharged, while a pump is working, would require that something inside the pumping chamber would have to move to change the amount or volume being displaced. By using an unbalanced design and allowing the cam ring (inner housing) to move in a linear direction, the distance the vanes are extended can be changed, which in turn would change the displacement. This is known as a variable volume vane pump.



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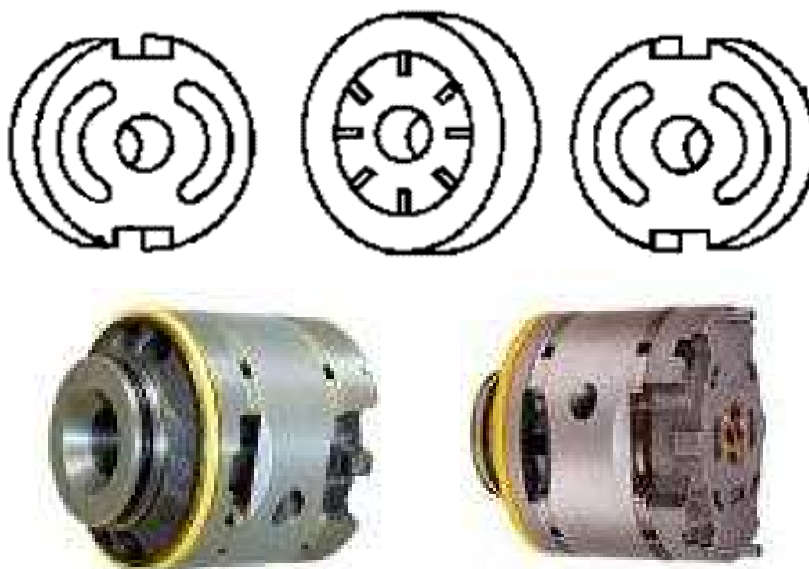


Variable Displacement Vane Pump

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Vane Pump Cartridge Assembly

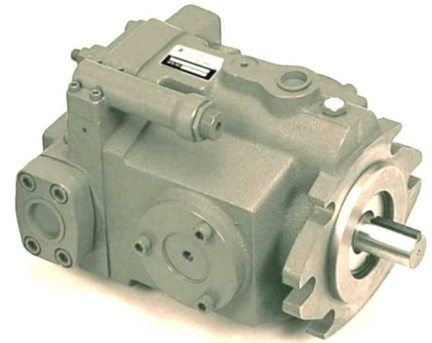
In addition to being variable displacement, vane pumps also set themselves apart from other pump designs by utilizing a cartridge assembly. Some vane pumps use, as their pumping (rotating) group what is called a cartridge assembly (preassembled cartridge). The cartridge consists of a complete pumping (rotating) group; ring, rotor, vanes, side port plates and any associated hardware required. Using a cartridge assembly type vane pump has two distinct advantages. One is easy pump servicing (fast replacement) of worn-out or broken parts. Second, if the pump output volume must be increased or decreased, a cartridge of the correct physical size and the appropriate volume can be substituted for the existing cartridge.



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Piston Pump

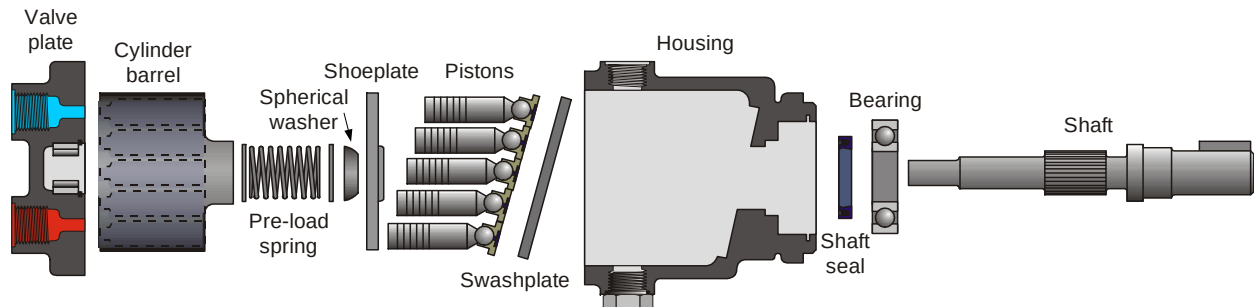
Piston pumps are highly efficient units available in a wide range of capacities. They are reliable and have a moderate (medium) operating level of noise. Able to operate in the medium to high pressure ranges (of 1500 to 3000 psi, some operating as high as 5000 to 10,000 psi) and generally a low discharge capacity of 10 gpm or less, but can be as much as 50 to 75 gpm. They are typically only available as a single pump unit.



Piston pumps fall into three basic categories; **Axial, Bent-Axis and Radial type**. The different names indicate the way the pistons move in relation to the drive shaft or center-line axis of the pump. No matter the type, all piston pumps operate on the principle of a piston reciprocating in a bore, much like a piston inside a cylinder.

Axial Piston Pump

In an Axial Piston Pump a prime mover rotates a cylinder (piston) barrel, positioned inside the barrel are pistons. The cylinder barrel and pistons are positioned axially in relation to the drive shaft or housing. As the barrel rotates the pistons travel with the barrel. To allow the pistons to generate a pumping action a spring loaded shoe-plate is used to keep the pistons pushed against a swash-plate. When the swash-plate is in a zero / neutral angle a no flow condition will occur because the pistons will not reciprocate inside the cylinder barrel and perform the basic function of a pump, increasing volume / decreasing volume. When an angle is created by the swash-plate the pistons will slide up (increasing volume) one side and slide down (decreasing volume) on the other side. The swash-plate position may be fixed or variable.



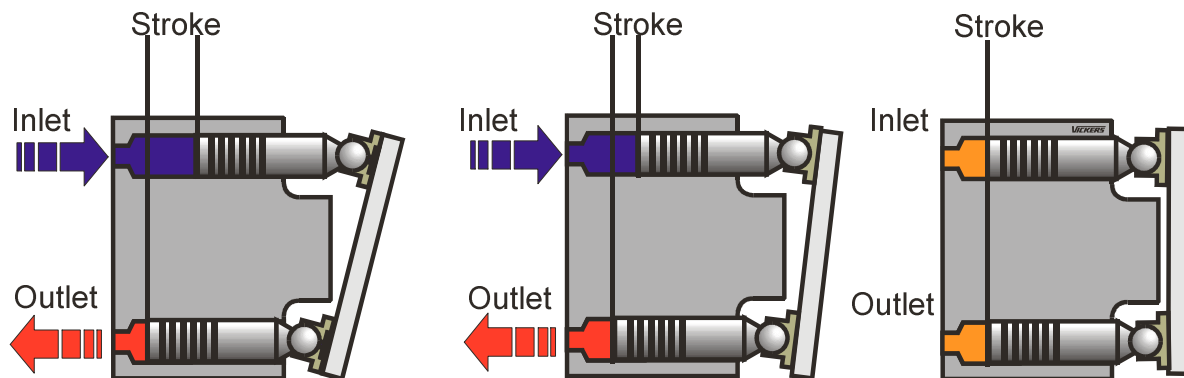
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Variable Displacement Axial Piston Pump

Much like the vane pump design, an axial piston pump may be fixed displacement or variable displacement. The amount of fluid an axial piston pump displaces, per revolution, is determined by the surface area of the pistons, number of pistons and stroke length of the pistons inside the barrel. During the pumping action nothing can be done to change the surface area of the pistons or the number of pistons. The only way to vary the displacement would be to change the stroke length of the piston in the barrel. Since the shoe-plate pushes the pistons against the swash-plate then the swash-plate becomes the key to pump displacement.

When the swash-plate is in a fixed position the pump displacement will be fixed. However changing the angle of the swash-plate will change the stroke length of the pistons, thus affecting the amount of fluid discharge.

The swash-plate operation may be manual, with servo control, with pressure compensation or by several other ways. Normally piston pumps are chosen because of how well they accomplish variable displacement.

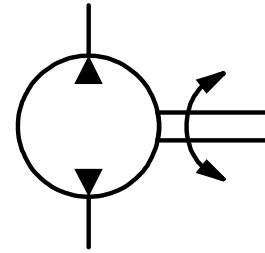


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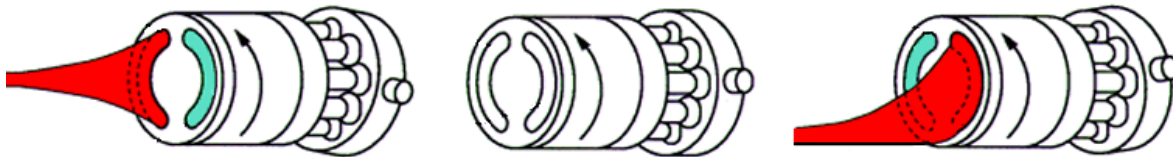
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Bi-Directional Axial Piston Pump

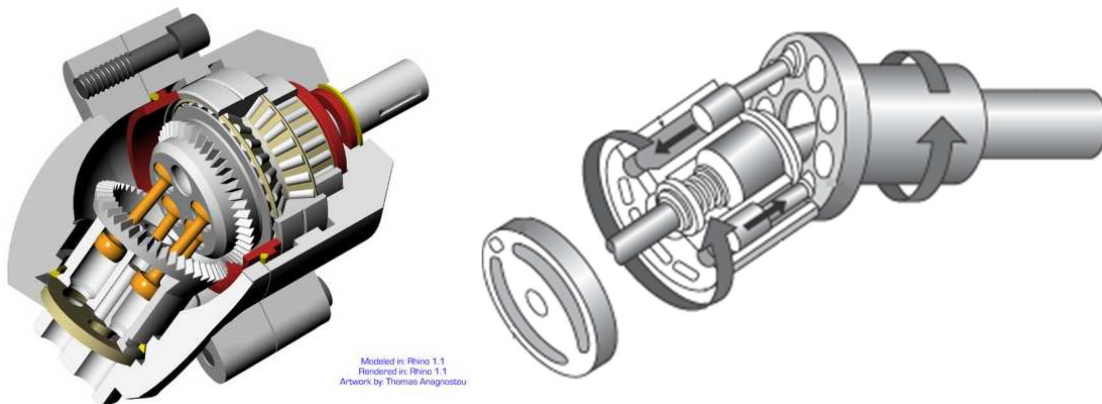
As discussed previously, the swash-plate is the key to the operation of an axial piston pump. In some versions the swash-plate has the capability to move over center, beyond the zero condition, in the opposite direction to reverse the direction of flow. It is called an over-center type of pump or bi-directional pump.



This allows the pump to change the direction of fluid flow without changing the direction of rotation of the prime mover (electric motor). The swash-plate operation may be accomplished manually, with servo control, or by several other ways.



Bent-Axis Piston Pump



In a Bent-Axis Piston Pump the pumping action or mechanism operates the same way as an Axial Piston Pump except the cylinder barrel is not in-line, or on the same center-line, as the drive shaft or the center axis of the pump. The angle formed by the bent-axis design creates the natural pumping action of the pistons reciprocating in the cylinder barrel. Fixed displacement versions are available with 23 degree and 30 degree angles.

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Variable Displacement Bent-Axis Piston Pump

In a variable displacement version an external control is used to change the angle, which changes the stroke length, thus changing the displacement of the pump.

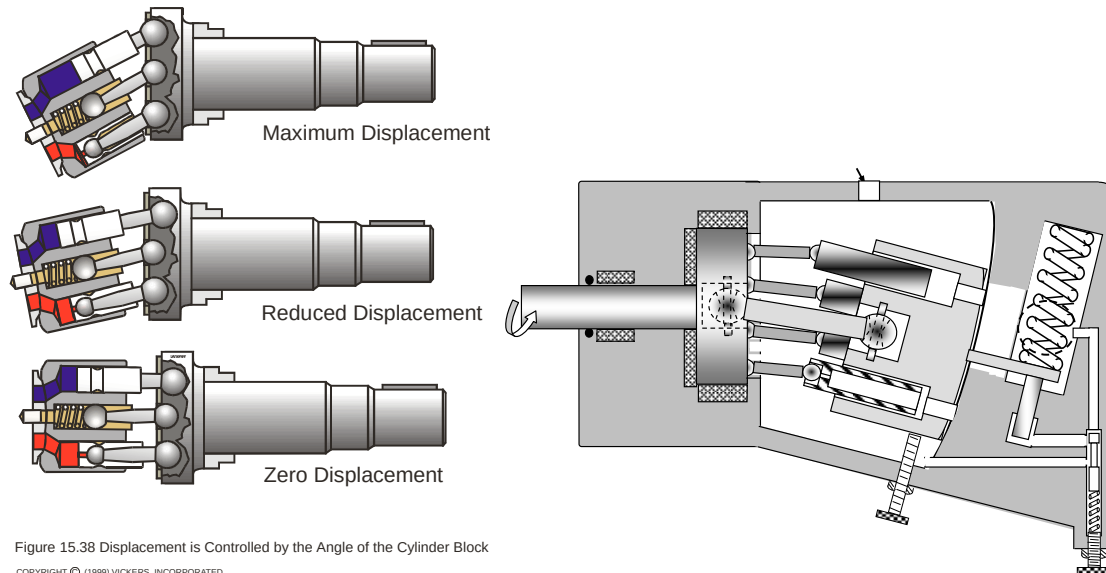
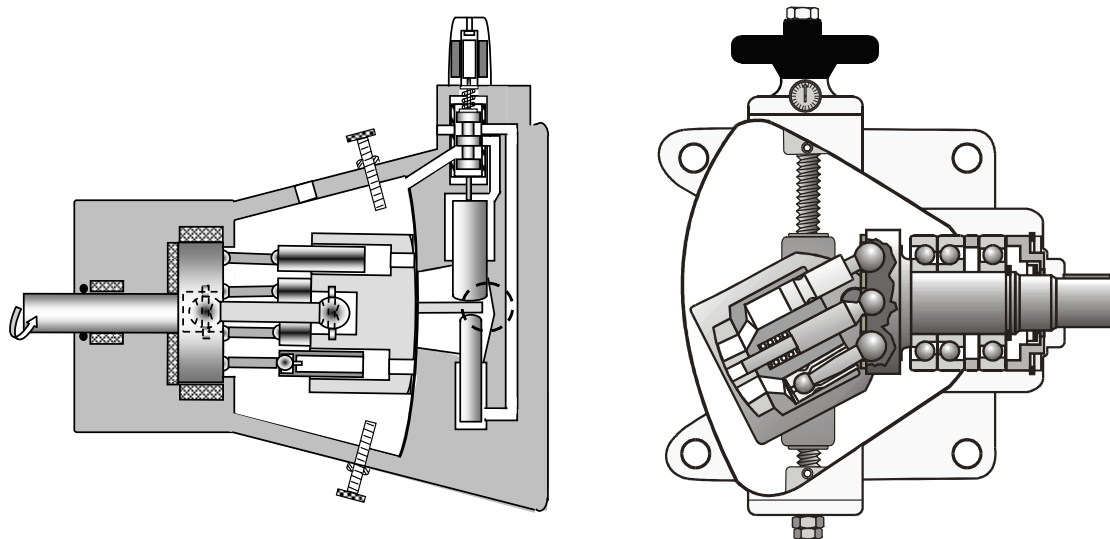


Figure 15.38 Displacement is Controlled by the Angle of the Cylinder Block
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Bi-Directional Bent-Axis Piston Pump

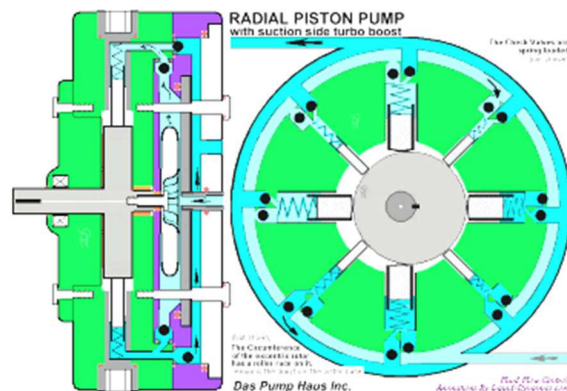
A bent-axis piston pump can also have bi-directional flow. It can intake or send fluid from either port while turning the same direction. The pumping (rotating) group or yoke can be moved over center to reverse the direction of flow, without changing the direction of rotation of the prime mover.



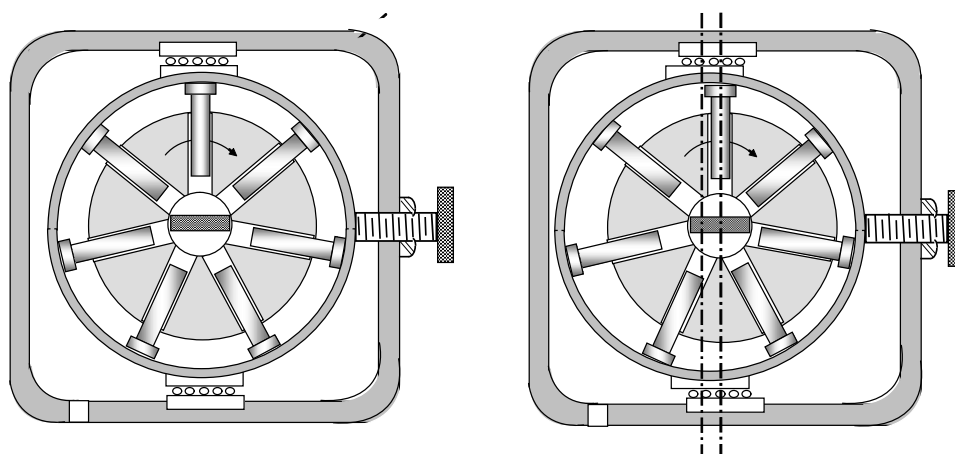
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Radial Piston Pump

In a radial piston pump a prime mover rotates a cylinder (piston) barrel, inside the barrel are pistons. The cylinder barrel and pistons are positioned radially inside the housing. As the cylinder barrel and pistons rotate the pistons are forced against the ring by centrifugal force, fluid pressure or mechanical force. The cylinder barrel is positioned off-center in relation to the center-line of the ring, which creates the opportunity for fluid to flow as the pistons reciprocate. Fluid enters (increasing volume) and is discharged (decreasing volume) from the pump through the cylinder barrel (valve block). The action is very similar to a vane pump's operation.



Variable Displacement Radial Piston Pump

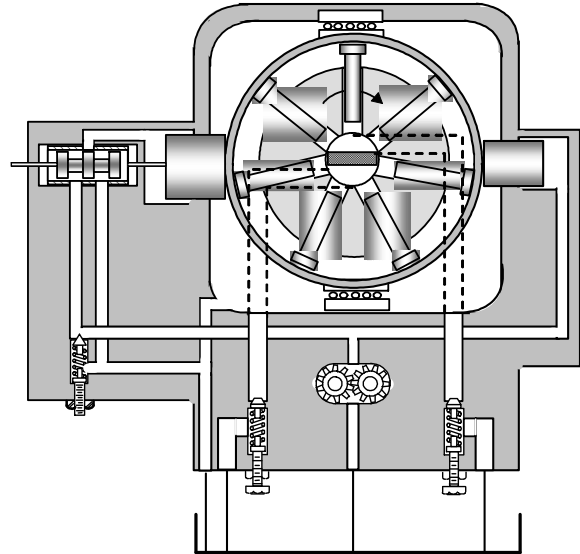


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Bi-Directional Radial Piston Pump

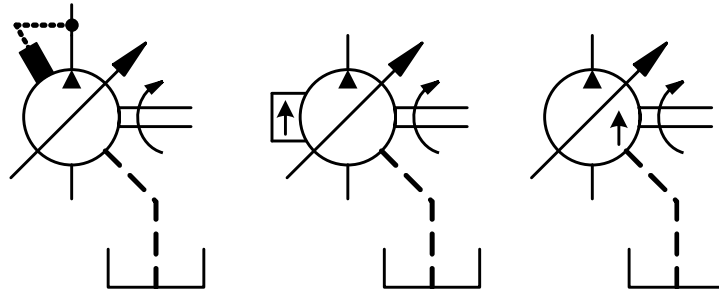
A radial piston pump can also have bi-directional flow. It can receive or send fluid from either port while turning the same direction. This pump design is used when it isn't practical to change the direction of the prime mover (electric motor) or is used in a closed loop circuit where all output goes to an actuator and return flow from the actuator goes back to the pump inlet.

A common circuit application for this is a Hydro-static Drive. Fluid from a bi-directional pump goes to a bi-directional motor to give an infinitely variable output speed and force in either direction of rotation without using a directional control valve.



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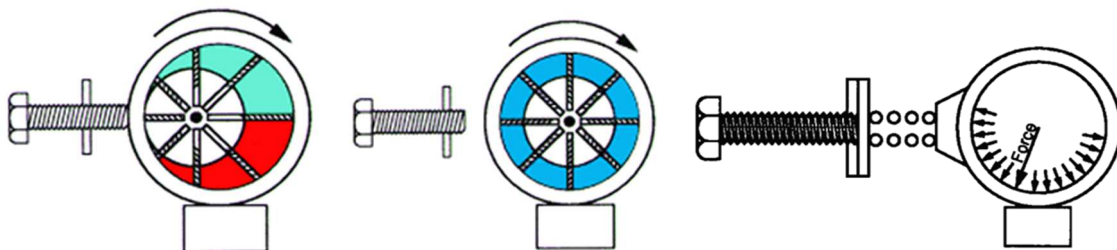
Pressure Compensated Pumps



In a hydraulic circuit power consists of gpm (lpm) and psi (bar). The amount of power generated by the pump and/or electric motor (prime mover) can be controlled (regulated) by reducing the generation of flow and pressure. This can be accomplished by using variable displacement, pressure compensated types of pumps. These pumps have the ability to reduce their displacement when the system pressure reaches a preset value. Vane and piston types of pumps are the only one's capable of accomplishing this concept.

Pressure Compensated, Variable Displacement Vane Pump

For a vane pump to be variable displacement / pressure compensated, the design will have to revert back to an unbalanced design, instead of the more desirable balanced design. Not only does the moveable cam ring provide variable displacement, but it can also be designed to allow pressure compensation.



The pumping action is the same as the fixed displacement unbalanced vane pump previously discussed. The difference is the Cam Ring is not fixed but can move in relation to the rotational group.

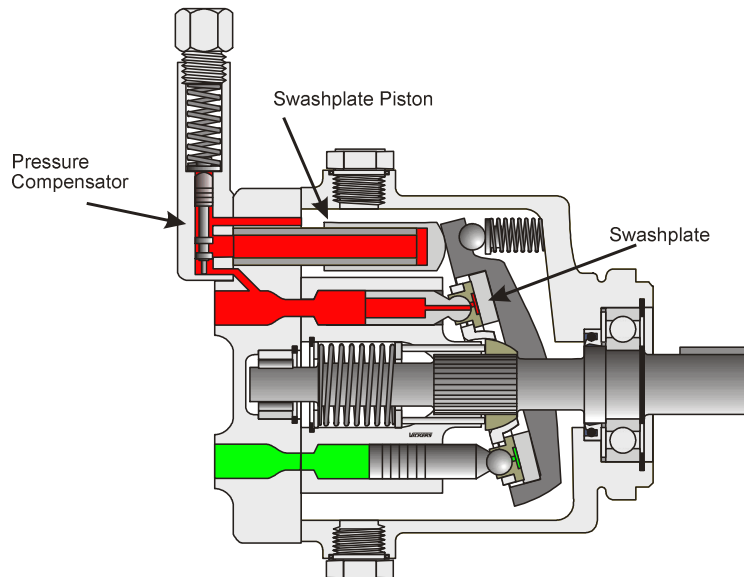
An adjustable force spring holds the cam ring in the offset (full displacement) position until enough pressure builds inside it to push against the spring and force it toward center. As the cam ring moves to the center output flow decreases until it finally stops. The cam ring never makes it all the way to center because some flow is always needed to make up for internal bypass.

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Pressure Compensated, Variable Displacement Axial Piston Pump

In a variable displacement, pressure compensated axial piston pump, the operation is the same as previously explained for the fixed displacement axial piston pump. The difference is this design has a moveable swash-plate that is held in place by a spring. The pump is always at full flow when pressure is below the compensator spring setting. When this pumps outlet flow meets resistant pressure from the output port it pushes against the spring loaded compensator spool. In the normal position this spool is allowing fluid behind the piston to go to tank through the case drain. When pressure is high enough to force the compensator spool to open it blocks fluid flow to tank and allows fluid into the chamber behind the piston. When enough fluid/pressure enters the chamber behind the piston, which pushes the swash-plate, and overcomes the spring the swash-plate moves to a position to stroke the pistons just enough to make up for bypass and control fluid used by the pump and any used in the circuit. This could be any amount of flow down to zero. The maximum displacement adjusting screw keeps the piston from retracting all the way even when pressure drops.

If the pressure drops off, the spool moves back and fluid is discharged from the piston and the spring returns the swash-plate to a greater angle and produces flow. The compensator thus adjusts the pump output to whatever is required to develop and maintain the preset pressure on the compensator. This prevents excess power loss by avoiding full pump flow when it isn't necessary.



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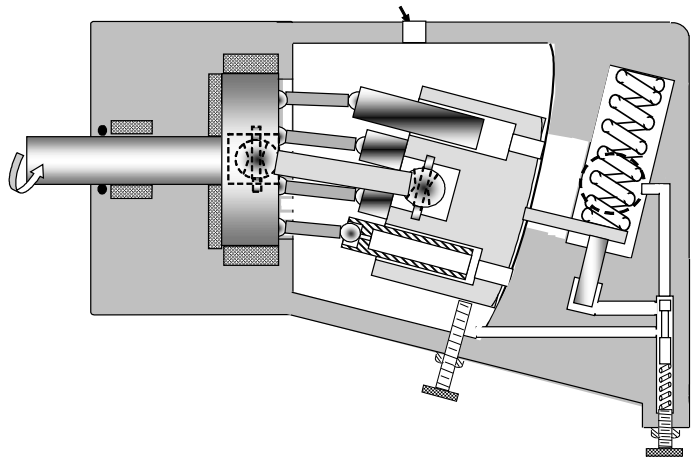
The piston shoe has recessed or hollowed out areas on its face that allow oil through an orifice in the piston. This oil lubricates the shoe and swash-plate. Always fill the case of a new, rebuilt or repaired piston pump before start-up. The pump needs lubrication and will have very little until it fills the case, also a filled case will seal clearances and make it easier for the pump to prime.

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Pressure Compensated, Variable Displacement Bent Axis Piston Pump

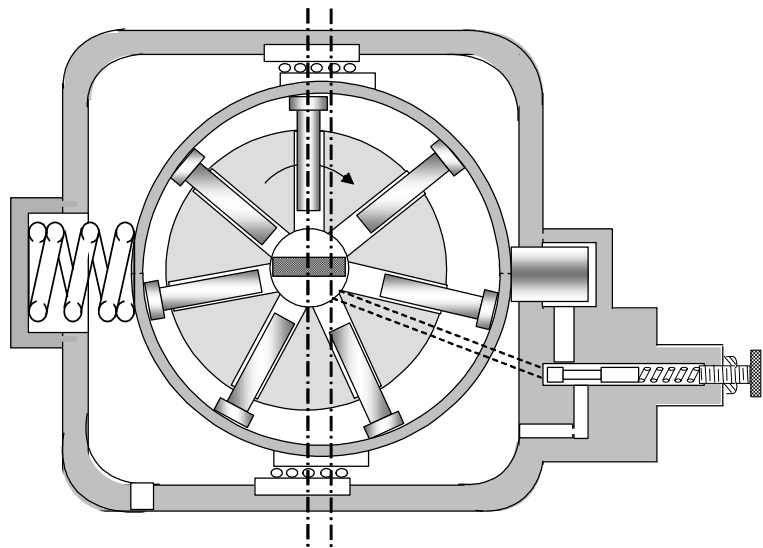
Variable Displacement, Pressure Compensated Bent Axis Pump operates in the same manner as a fixed displacement version, except the angle of the cylinder block containing the cylinder barrel and pistons can be any angle instead of fixed. This means pump output changes by a manual control or automatically changes as pressure increases to a predetermined setting by changing the angle of the cylinder block thus changing flow by using the compensator spring pressure setting. Also, the maximum angle of the cylinder block can be limited by the maximum volume screw to set maximum flow. This is an option on many manufacturer designs.

As pressure at the outlet increases to the setting of the compensator spring setting the spool is pushed back and forces the compensator piston to push the cylinder block to a smaller angle. When pressure reaches the preset spring setting the cylinder block stays in any position required to maintain the flow needed at the preset spring pressure.



Pressure Compensated, Variable Displacement Radial Piston Pump

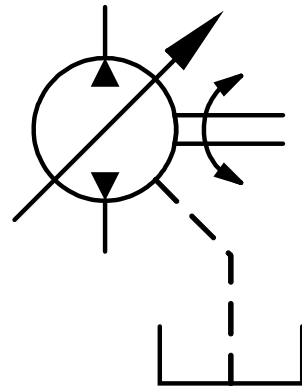
In a variable displacement, pressure compensated radial piston pump the operation is much the same as a fixed displacement type, except the reaction ring can move and change the amount of stroke length for the pistons. When pressure in the pilot line increases enough to push the compensator spool against its spring, pilot flow is connected to the compensator piston and its drain to the case is blocked. Pilot flow to the compensator piston forces the reaction ring to move against the return spring and reduce output flow.



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Case Drains

The internal leakage that occurs in a fixed displacement pumps generally goes into the housing and back into the inlet flow. In a variable displacement, pressure compensated pump internal bypass goes into the housing also, but has no passageway to the inlet line. All internal leakage must be drained from the housing directly to tank through a drain line. The drain line should exit from the highest point on the pump so the case stays full of fluid at all times, and make sure the drain line terminates below the fluid level inside the reservoir, so it cannot ingest air. Finally, always fill the case of a newly installed pump to make sure it has lubrication at startup.



Maximum Volume Adjustment

Some variable displacement and / or pressure compensated pumps have a maximum volume adjustment to keep the pumping (rotating) group from going to full volume or discharge. This feature makes it possible to adjust the maximum flow when pressure is below the compensator setting. This feature could be used to limit maximum horsepower when only a small portion of a higher flow pump is required. In most circuits this feature has no use since flow is usually controlled by flow controls or actuator size.

Multiple Pump Units

Multiple pump type units are typically available with gear and vane type construction. Most of the other pump designs are not practical to be used in a multiple configuration.



Double pump units, triple pump units, quadruple pump units, etc; can have a common inlet and common outlet or have different inlets and different outlets or a combination of inlets or outlets to satisfy many different circuit designs. Multiple pump units provide flexibility in a circuit, especially where circuit requirements may be differing displacement during circuit cycling or differing pressure values for work.

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Actuators

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Introduction to Actuators

The function of a hydraulic actuator is to convert the useful energy of a hydraulic circuit, in the form of pressure and flow, into mechanical work. Hydraulic actuators can be divided into two different types; **Linear Actuators (cylinders)** and **Rotary Actuators (rotary actuators and hydraulic motors)**.

Any actuator type may be **single acting** or **double acting**. A single acting actuator only uses fluid pressure to move the actuator in one direction and some other energy source to move it in the opposite direction. A double acting actuator uses fluid pressure to move the actuator in both directions.



Results Of System Changes On Linear Actuator Performance

System Change	Speed	Maximum Force
Increase pressure relief valve setting	No effect	Increase
Decrease pressure relief valve setting	No effect	Decrease
Increase system flow	Increase	No effect
Decrease system flow	Decrease	No effect
Increase piston diameter	Decrease	Increase
Decrease piston diameter	Increase	Decrease

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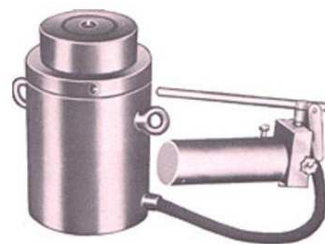
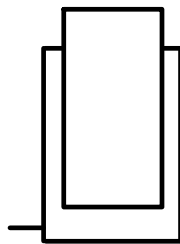
Types of Actuators

Linear Actuators (Cylinders)

Ram type Cylinder

A Ram type cylinder is probably the simplest of all the linear actuators and is considered to be single acting in operation. It has only one fluid chamber and uses the fluid pressure to move the ram in one direction. Most are mounted so as to use gravity, springs or the load/work to retract the ram. Common examples are; elevators, automobile lifts, floor jack, bottle jack.

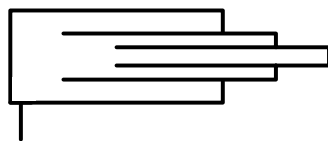
Ram type cylinder,
single acting,
single rod,
work/load return



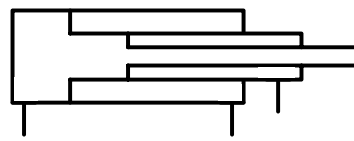
Telescoping Cylinder

A Telescoping cylinder is used when the existing space is limited, but a long extension of the cylinder is required. As many as four or five sleeves can be used to achieve the desired length. Telescopic cylinders may be single acting or double acting by design. If a single acting type is used, the retracting force may be gravity, springs or the load/work.

Single acting



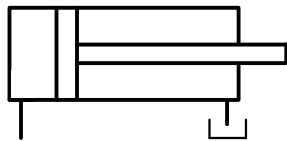
Double acting



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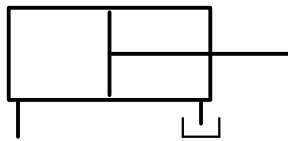
Single Rod Cylinder

The Single Rod cylinder is the most common cylinder application used in hydraulic systems. It may be single acting or double acting in design.

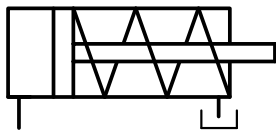


(Detailed symbol)

Single rod,
single acting,
work/load return

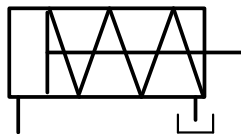


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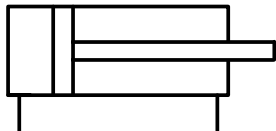
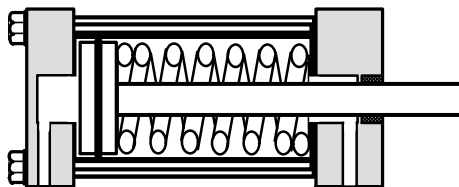


(Detailed symbol)

Single rod,
single acting,
spring return

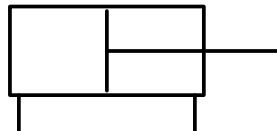


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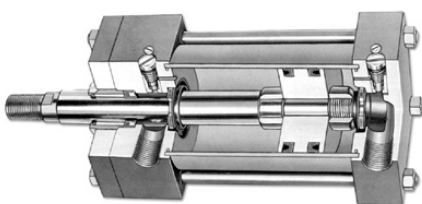
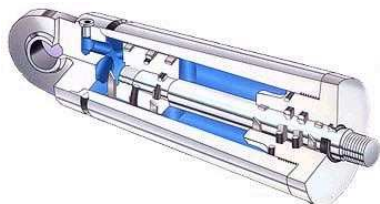


(Detailed symbol)

Single rod,
double acting



(Simplified symbol)

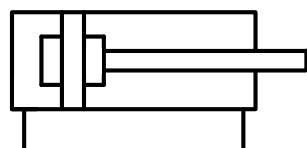


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Cylinder Cushions

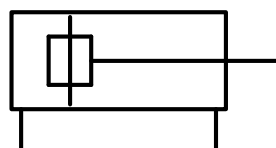
When the movement of a hydraulic cylinder piston is suddenly stopped at the end of its travel, the moving/inertial energy is changed into a concussion type of energy. This is also known as a “hydraulic shock” load to the cylinder. If enough energy is transmitted to the cylinder or prolonged shock loading occurs, the piston, rod, seals or end cap could be damaged.

Cylinders that experience excessive shock loading can be equipped with cushions. Cushions slow down a cylinder’s piston speed at the last moment of extension or retraction. Cushions can be utilized at both or either end of the cylinder. The piston speed is not affected while the piston is in the middle of its stroke or in essence between the cushions.

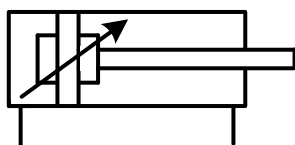


(Detailed symbol)

Single rod,
double acting,
w/ fixed cushions

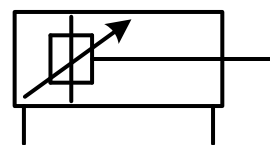


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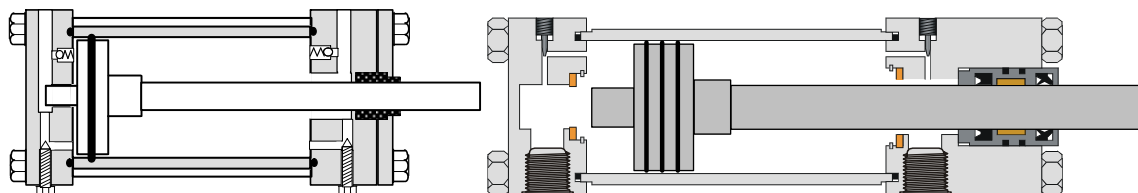
(Detailed symbol)

Single rod cyl,
double acting,
w/ adjustable cushions



(Simplified symbol)

As the cylinder is moving, full flow is allowed on both sides of the piston. However, as the piston approaches the end of the stroke, fluid is blocked from the large evacuation port and is diverted through the smaller evacuation port. The smaller port reduces the rate of flow, thus slowing down the piston speed. Most cushions include a needle valve to further control or slow down the fluid for better control.

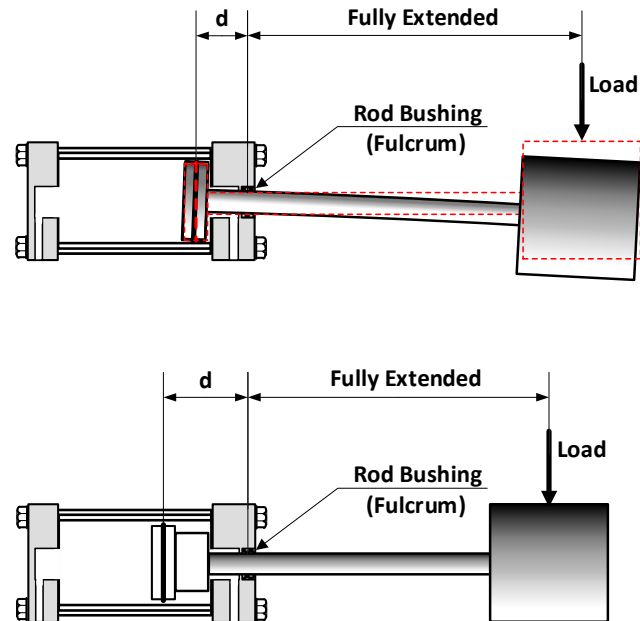


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Cylinder Stop Tube

A Stop Tube is a metal collar/spacer which fits over the rod. A stop tube is not used to decrease the stroke length of the cylinder, it will have that inherent effect. A stop tube is used to separate the piston (increase the minimum distance) from the rod bushing, in the end cap of the cylinder, when a long stroke cylinder is fully extended.

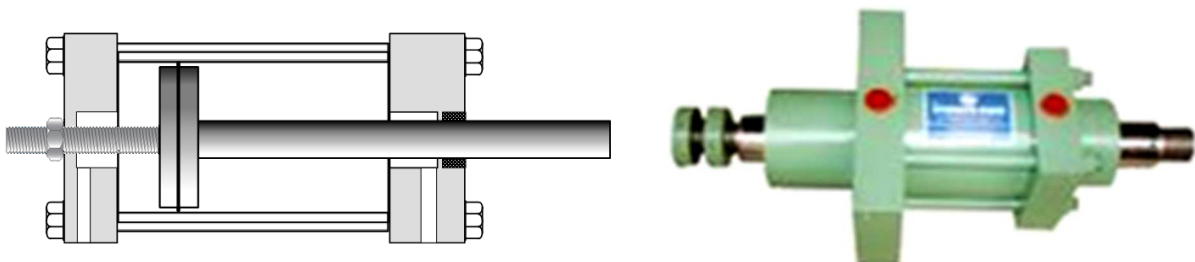
Most cylinders do not require stop tubes. Since it is a bearing, the rod bushing is designed to help the piston support the rod load. On long stroke, horizontally mounted cylinders at full extension, the piston and bushing are so close the rod does not have enough support. Since the piston and bushing are so close the rod only has one support point (fulcrum) and the rod basically becomes a see-saw or teeter-totter, will move back and forth significantly. This happens when the load is not supported or rigidly guided. Also long stroke cylinders will have significantly more rod weight than a standard length rod.



A stop tube is used to separate (increase the minimum distance) from the piston and bushing. By increasing the distance two support points are created and the rod has more rigidity, thus reducing the load on the bushing and the rod seals will have an opportunity to perform their function.

Cylinder Stroke Adjusters

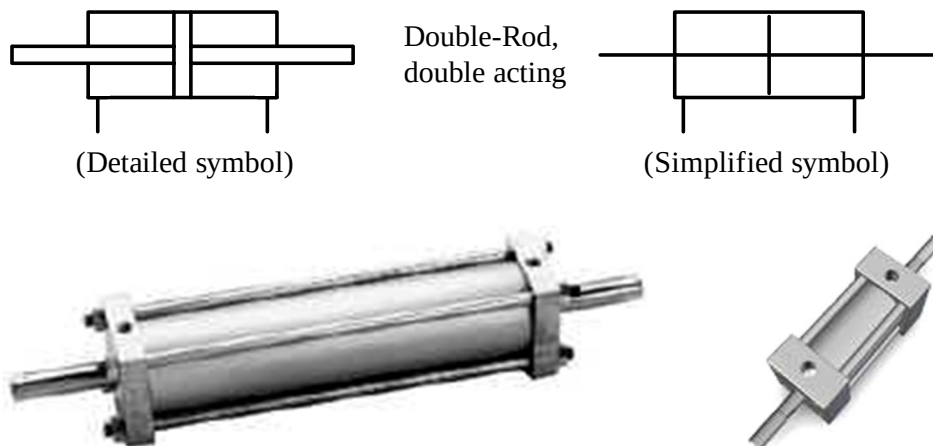
Sometimes the stroke length of a cylinder needs to be externally controlled. The external control/adjustment can be accomplished with a stroke adjuster, which is simply a threaded rod that can be screwed in or out of the end cap of the cylinder.



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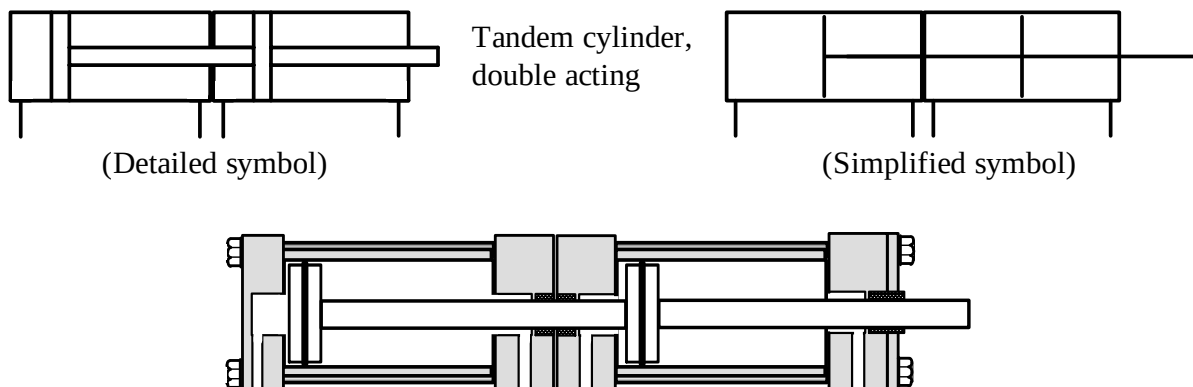
Double-Rod Cylinder

A Double-Rod cylinder is exactly that, a cylinder housing with a piston, but also with two rods opposing each other. This type of cylinder is used when equal rod speed or equal force is required in both directions, when it is desirable to attach two loads of the same value, or the opposing end may not perform work but is required to mount cams or limit switches/valves for control.



Tandem Cylinder

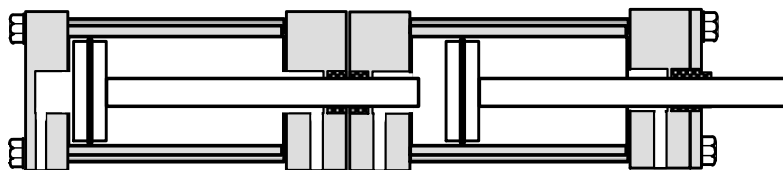
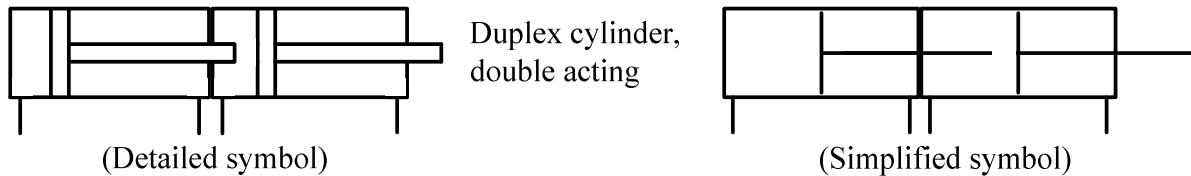
A tandem cylinder is a combination of two cylinders mounted in-line, with the pistons connected by a common rod. Tandem cylinders are used when mounting space is limited and increased output force is required.



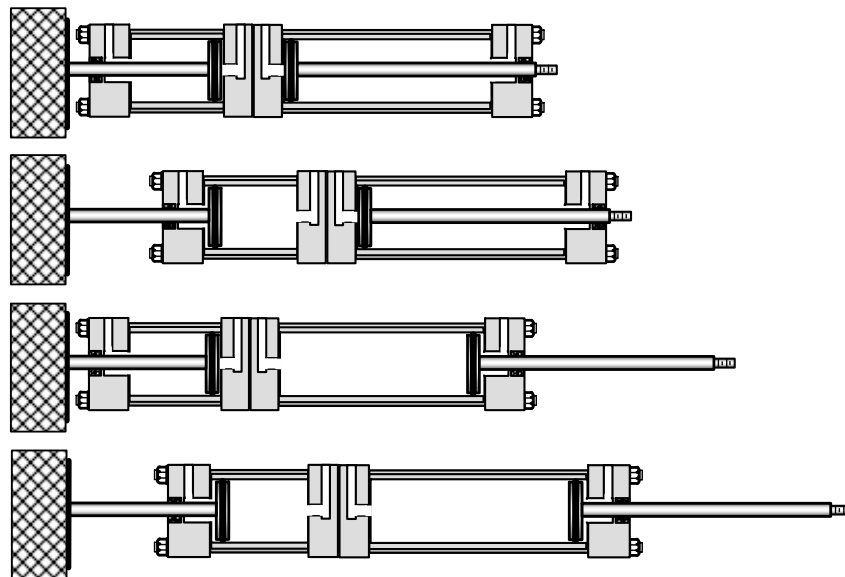
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Duplex Cylinder

A duplex cylinder is a combination of two cylinders mounted in-line, however the difference between a duplex and tandem is the pistons are not connected by a common rod. The cylinder assembly can operate just like a tandem setup or the cylinders can have independent operation. Duplex cylinders are used when mounting space is limited and increased output force is required.



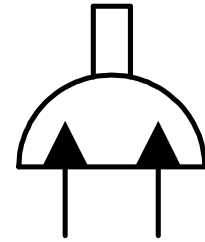
Other examples of duplex cylinder applications.



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Rotary Actuator (Limited rotation)

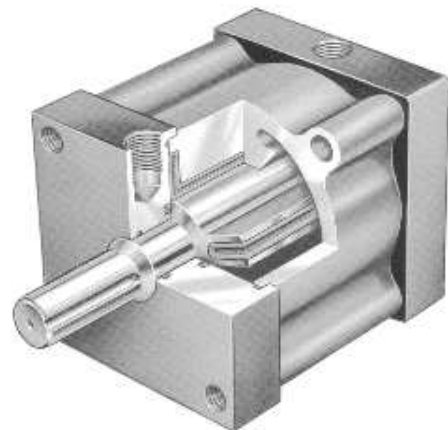
A Rotary type Actuator is basically a hydraulic motor, except it has one distinct design difference. A rotary actuator (Limited Rotation Motor or Torque Actuator) will have limited rotation in a full 360 degree circle. Just like a motor, it provides circular rotation and torque. However the circular rotation is not a continuous rotation, so it is usually not specified with an rpm rating.



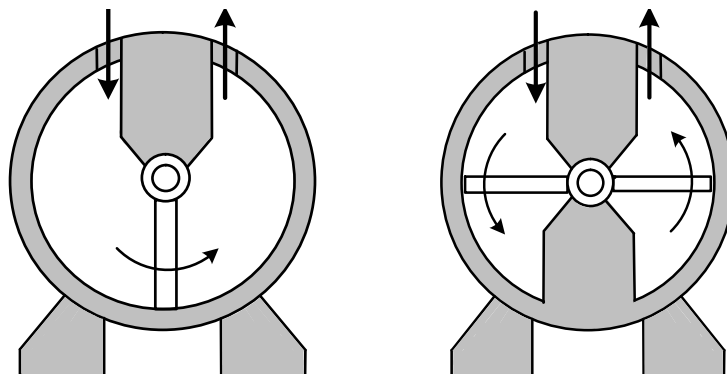
These actuators are limited to special applications or special motions and are limited in the number of types. The most common types are Vane type and Piston type.

Vane Rotary Actuator

The Vane type has a movable vane which separates the inner housing into two chambers. When fluid enters one of the two chambers, the fluid pressure acts on the vane creating a force imbalance. The force imbalance forces the vane to move inside the chamber. The vane is attached to a shaft and as the vane moves the shaft transmits the fluid force and flow into torque and rotational movement.



The vane type of actuator can be single vane or double vane by design. A single vane provides more rotation, as much as 270 degrees, but less torque. A double vane provides less rotation, usually around 120 degrees, but more torque. Vane type of actuators are chosen because they are very compact devices and are typically less expensive than any other variety.

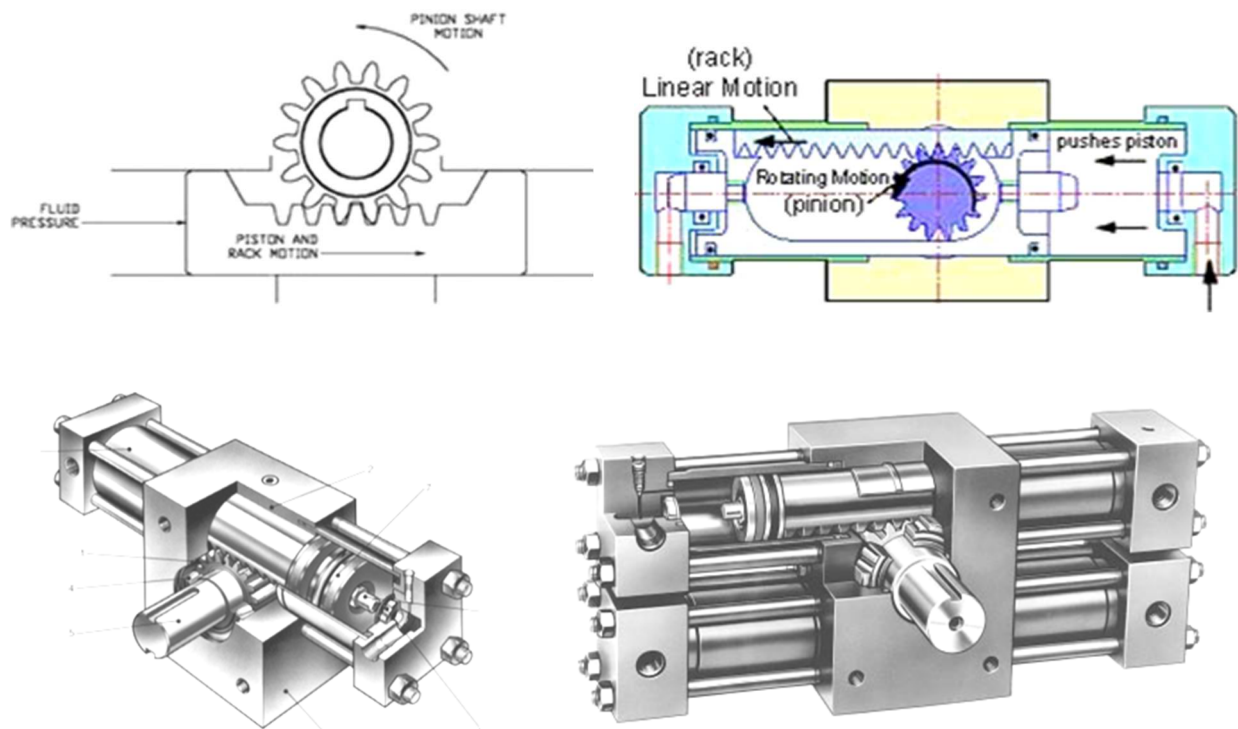


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Piston Rotary Actuator

In a Piston type, there can be several different types of mechanisms used to change the linear motion of the piston to rotary motion from the shaft. Some types can even offer rotary motion in excess of 360 degrees or more than one full rotation. Typically, piston type are classified as rack actuators (rack & pinion actuator), crank actuators or helical (screw type) actuators.

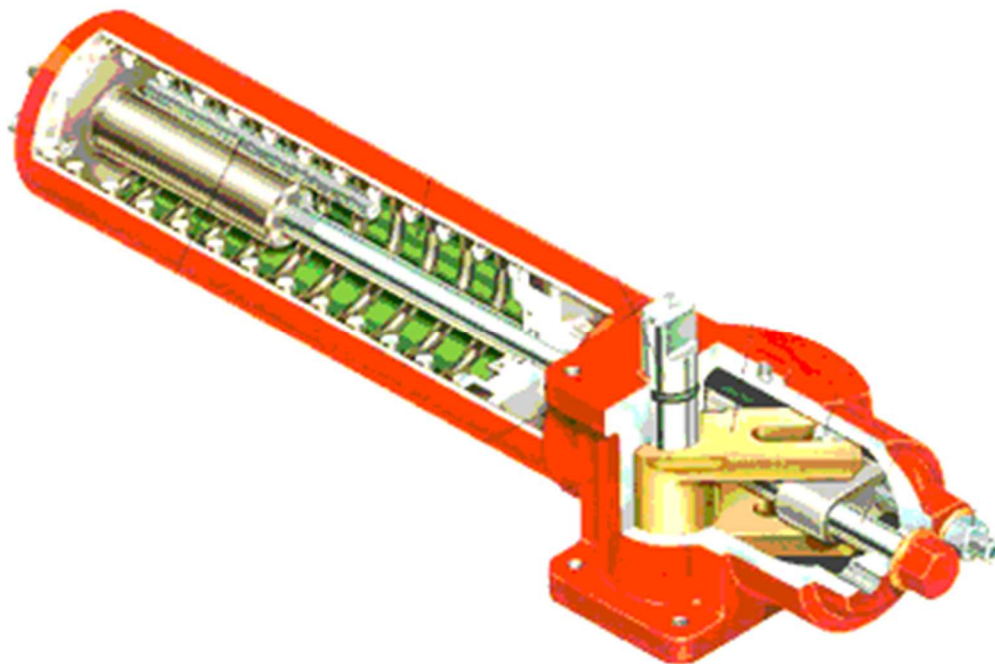
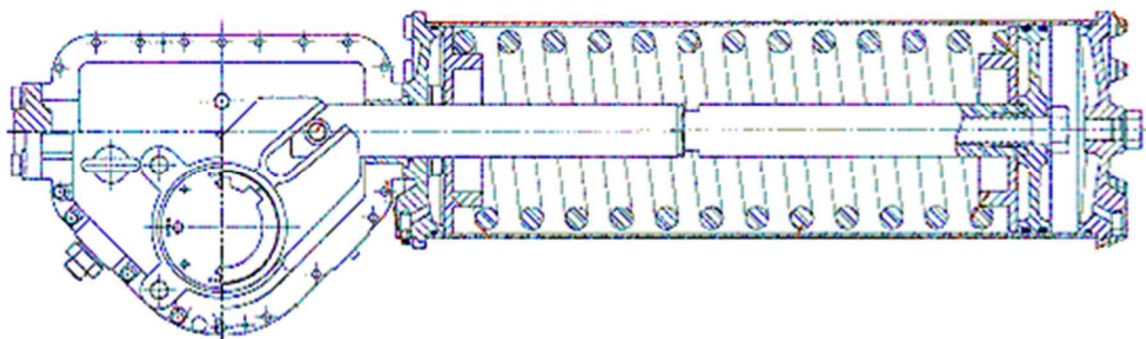
There are two basic types of rack actuators or rack & pinion actuators, a single rack and a double rack style. It may be single acting or double acting, and has the distinct advantage of providing more than one full rotation. Since the design and operation is very similar to a standard cylinder many of the same features (or accessories) can be used, such as cushions, stroke adjusters, proximity switches and special porting.



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A Crank type actuator utilizes the same principle of a standard cylinder also, in that system pressure move the piston inside the bore in a linear direction. When the piston moves a rod attached to the piston will move through a pivot point and reciprocate a crank which in turn rotates the output shaft.

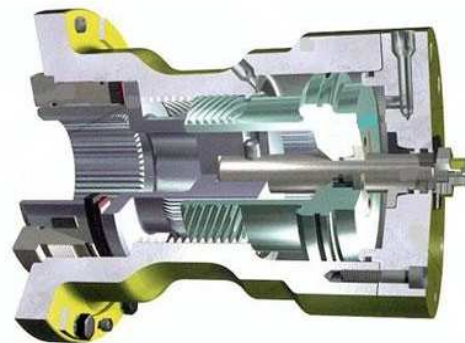
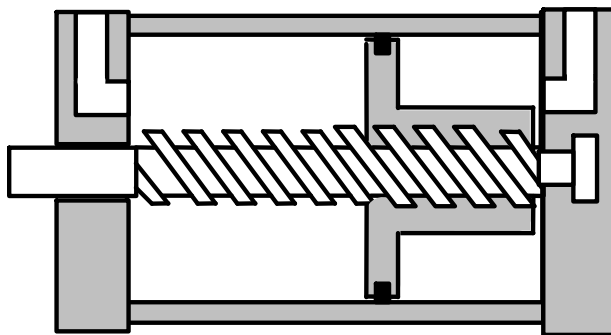
A crank type may be single acting or double acting. This type of actuator primarily used in industrial process control.



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A Helical (Screw) type actuator may be single acting or double acting. Like the rack & pinion type actuator it also can provide more than one revolution of the shaft depending on the size. It operates similar to a standard hydraulic cylinder in that system pressure is used to move a piston. Instead of a rod being attached to the piston, the piston is engaged with a helical screw that acts as the output shaft. As the piston moves in a linear direction, the helical grooves that engage the piston and shaft force the shaft to move in a circular direction.

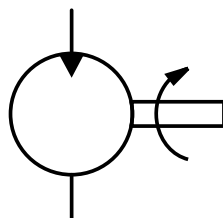
Helical designs provide the most torque output with the smallest cross section of all the rotary actuators. Double helix designs are also available to reduce the overall length of the actuator, but will double the torque output. Helical units are generally the most expensive of the rotary actuators, but they are also the most compact design.



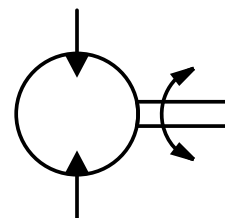
Rotary Actuator – Hydraulic Motor (Continuous rotation)

Hydraulic Motors are commonly referred to as rotary actuators and closely resemble a hydraulic pump in design. When in reality a hydraulic motor and hydraulic pump have different purposes in a hydraulic circuit. Instead of creating flow and pressure, like a pump, a hydraulic motor accepts flow and pressure and converts it into rotation and torque. Hydraulic motors may be unidirectional or bidirectional in design and may be fixed displacement or variable displacement. The most common types of hydraulic motors are gear, vane and piston.

Hydraulic motor,
fixed displacement,
unidirectional



Hydraulic motor,
fixed displacement,
bidirectional



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Hydraulic motors are rated according to; displacement (size), torque capacity, speed (rpm) and maximum pressure rating.

1. Displacement (size) - Displacement is the volume (amount of fluid) required to turn the rotating group (output shaft) in one revolution. Displacement is usually given as cubic inches per revolution (in³/rev) or cubic centimeters per revolution (cm³/rev).
2. Speed (rpm) – Motor speed is directly related to displacement and the volume (gpm) of fluid the system can supply.
3. Torque capacity – Torque is a turning or twisting effort (force) created by the fluid pressure acting on the exposed areas of the internal rotating group.
4. Maximum pressure rating – The maximum pressure rating of the motor will be a direct result of the torque load (resistance to movement) and surface area of the rotating group.

Results Of System Changes On Rotary Motor Performance

System Change	Speed	Maximum Torque
Increase pressure relief valve setting	No effect	Increase
Decrease pressure relief valve setting	No effect	Decrease
Increase system flow	Increase	No effect
Decrease system flow	Decrease	No effect
Increase displacement (size)	Decrease	Increase
Decrease displacement (size)	Increase	Decrease

Hydraulic motors can be classified by their application into three classifications.

1. HSLT (High speed, Low torque) – The primary types are gear, vane, axial piston and bent-axis piston.
2. LSHT (Low speed, High torque) – The primary types are internal gear, vane, axial piston and radial piston.
3. Limited Rotation Motors (Torque Actuators) – vane (single and double) and piston (rack, crank, helix/helical)

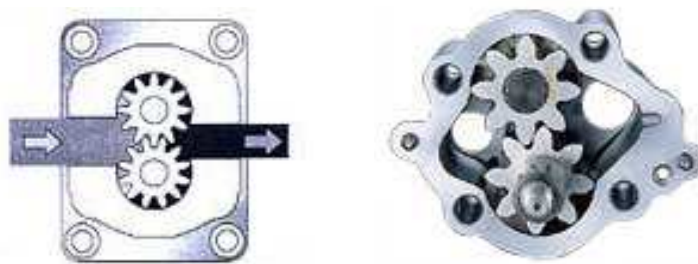
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Gear Motor

The simplest design of all the common hydraulic motors would be the gear. They are capable of low to medium operating pressures (up to 2500 psi) and generally a low rpm capacity (up to 2500 rpm). They have a high dirt tolerance and are moderately efficient.



A gear motor develops torque/rotation on the output shaft by the flow and pressure entering the body and acting on the gear teeth. The gears are in mesh and rotate together, with one gear attached to the output shaft.



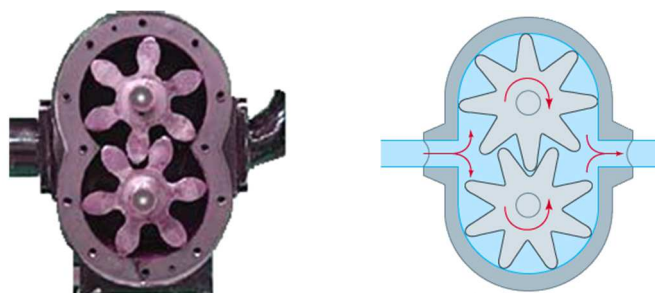
The gear motor is an unbalanced type of design. Low pressure at the outlet and high, system, pressure at the inlet creates an unbalanced load on the motor (rotating) group. The gears, shafts, bearings, side-port plates, seals all feel the imbalance of force.

The displacement of a gear motor is fixed and by the nature of its design it has no opportunity to be variable. The motor is reversible (bidirectional) only by reversing flow.

Gear motors may consist of two different designs; external gear motor and an internal gear motor.

External Gear Motor

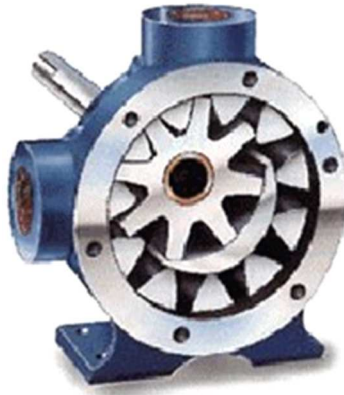
The previously described gear motor is an external gear type. External gear motors use the outer circumference of two gears to allow fluid to act and provide rotation.



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Internal Gear Motor

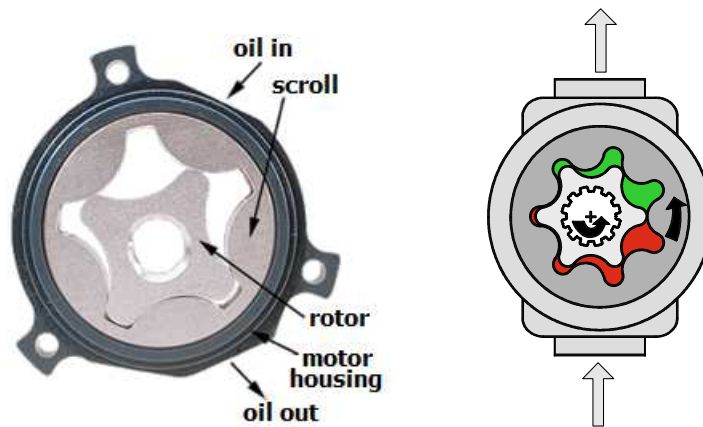
Internal gear motors will use one external gear positioned inside an internal gear, however the basic operation is the same. The pressure creates an imbalance of forces among the teeth of the gears, because of the meshing and unmeshing of teeth, rotation is created.



Gerotor Motor

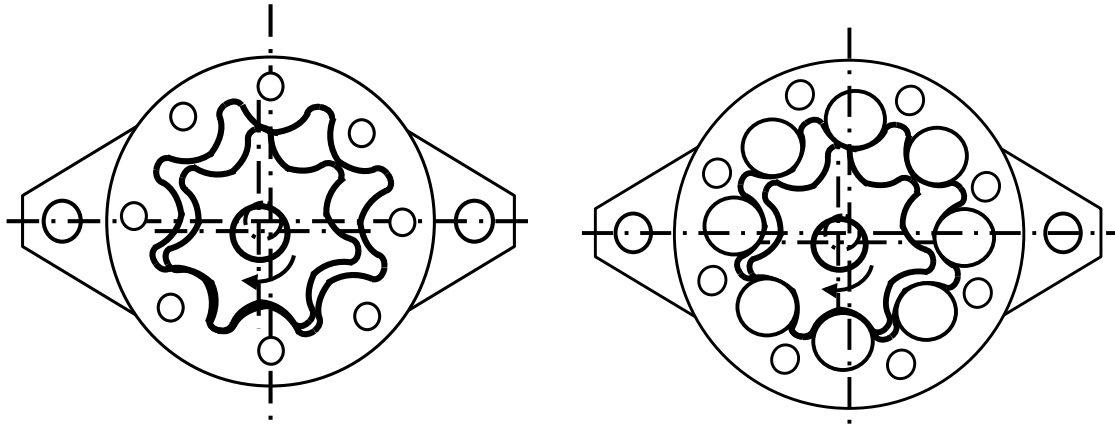
The most common type of internal gear motor is the Gerotor Motor. A gerotor motor is an internal gear motor by design, however what defines the difference between an internal gear motor and a gerotor motor is the difference in the number of teeth. An internal gear motor can have differing ratios between the meshing gear teeth. The gerotor motor is unique in that it has only one tooth difference between both gears. The external toothed gear, which is the inner gear, has one less tooth than the internal toothed gear, which is the outer gear.

Rotation is caused by the fluid pressure acting on the unequally exposed teeth, resulting in rotation and torque at the output shaft. Fluid entering the rotating group is separated from the fluid exiting the motor by the side port plate, causing the unbalance of force.



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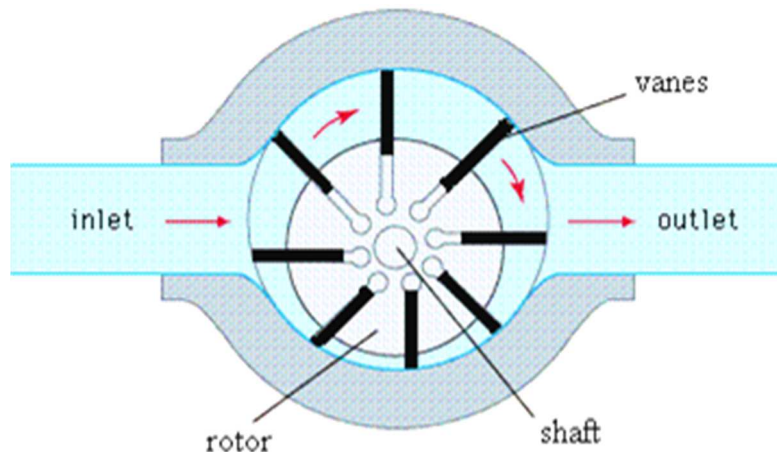
There are different variations of the gerotor motor design. Some variations include, additional gearing, which increases output torque while reducing rpm. Another variation is replacing the gear teeth with roller elements, called a Geroller, or Roller-vane gerotor motor. The roller design decreases friction which increases output efficiency.



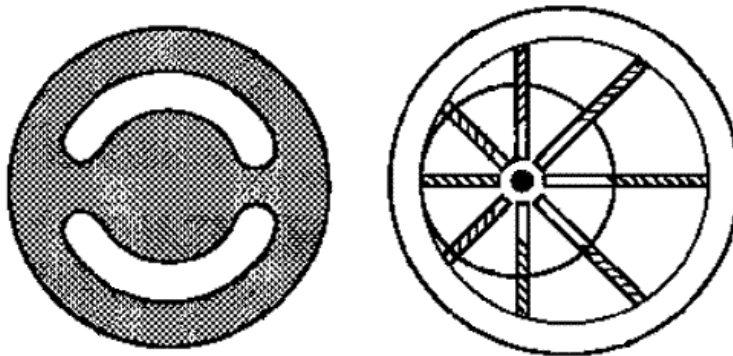
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Vane Motor

The vane motor is the most widely used of the common hydraulic motors. They have a high dirt tolerance and are moderately efficient. Depending on the type, vane motors are capable of low to high rpm (10 – 3000) and very high torque capacities (up to 5000 lb-ft). A vane motor is fixed displacement, usually isn't designed to be variable, and is reversible (bidirectional) only by reversing the system flow.



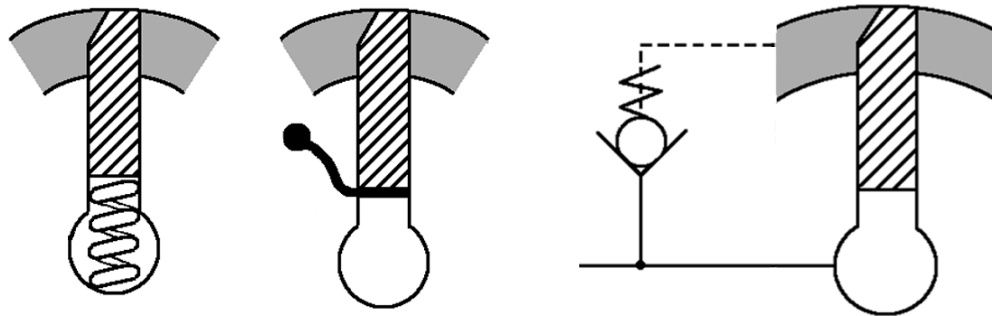
In a vane motor rotation and torque is created by system pressure acting on the exposed areas of the vanes positioned inside the rotor, which is inside the motor housing. As pressure acts on the vanes an imbalance of pressure/force is created. The vanes begin to move, as they move rotation is created. Rotation transfers from the vanes to the rotor and to the output shaft.



The overall operation of the motor is basically the same as a pump, however there is one key difference. A pump has a prime mover driving the input shaft, whereas the motor uses the shaft as an output for rpm and mechanical force. Which means the motor cannot depend on a prime mover for centrifugal force to keep the vane loaded/extended. Some other means must be used to keep the vane loaded/extended.

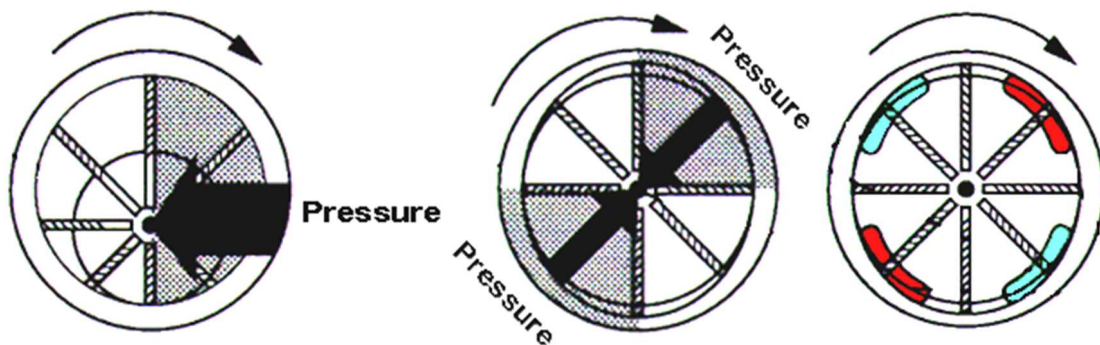
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To ensure the vanes extend, make contact and seal against the housing generally is done with two methods. One is to direct the existing fluid pressure, in the system, to the underside of the vane. Second is to use light duty springs on the underside of the vanes.



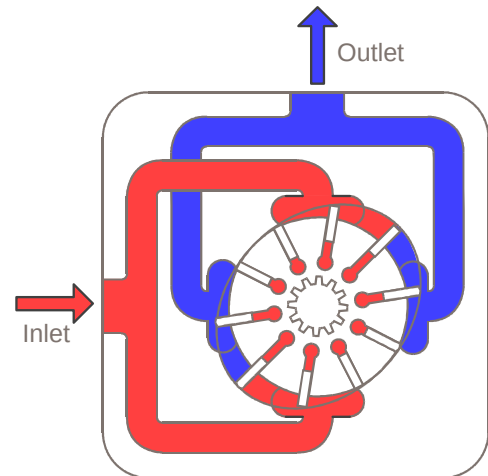
Just as in a vane pump, loading/extending a vane in any of the previously described manners may create too great a force. If the force is too much there would be excessive drag, from friction, and too much wear on the vane itself. Manufacturers will usually compromise between creating the best seal and reducing the amount of friction and wear, so the vanes are usually only partially loaded.

Just like in a vane pump, the rotor and inner housing are not aligned along the same centerline. Because the shaft and rotor are offset (eccentric) in relation to the motor housing, fluid will fill the void or cavity. High pressure at the inlet and low pressure at the outlet creates an unbalanced load/force on the rotating group. The gears, shafts, bearings, side-port plates, seals all feel the imbalance of force. The vane motor described up to this point is an unbalanced design. Vane motors may be **unbalanced** or **balanced** in design.



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In a **balanced vane motor** the rotor and inner chamber are aligned along the same centerline. With the two aligned there is no opportunity for the rotating group to create an increasing / decreasing volume situation, thus there is no rotation or torque. To overcome this, the inner chamber (or cam ring) is elliptical (egg shaped, cam shaped), not circular as in the unbalanced design. With this type of design the two pressure quadrants oppose each other and cancel out or offset the shaft load (or side force) encountered in an unbalanced design.



Fixed volume, positive displacement vane motors typically use a balanced design. Using a balanced design eliminates the shaft side loading issue, however it does present one other problem. With the rotating group running along the same centerline, the motor's displacement will be fixed and not variable.

Vane Motor Cartridge Assembly



In addition to being variable displacement, vane motors set themselves apart from other motor designs by utilizing a cartridge assembly. Some vane motors use, as their rotating group what is called a cartridge assembly (preassembled cartridge). The cartridge consists of a complete rotating group; ring, rotor, vanes, side port plates and any associated hardware required.



Using a cartridge assembly type vane motor has two distinct advantages. One is easy motor servicing (fast replacement) of worn-out or broken parts. Second, if the motor rpm or torque must be increased or decreased, a cartridge of the proper physical size and the appropriate displacement/volume can be substituted for the existing cartridge.



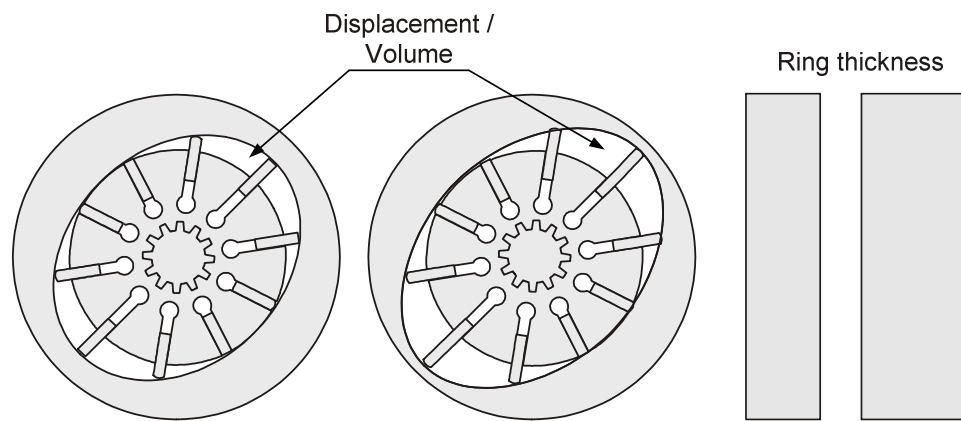
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The output rpm and torque of any vane motor is determined by the displacement/volume of the inner housing or rotating group.

By increasing the internal displacement/volume, the rpm of the output shaft would decrease. The opposite also holds true, by decreasing the displacement/volume, the rpm would increase.

By increasing the displacement, the amount of output force/torque would increase. The opposite holds true again, by decreasing the displacement/volume, the output torque would decrease.

The inverse relationship of speed and torque in a mechanical application is also true in a hydraulic motor.

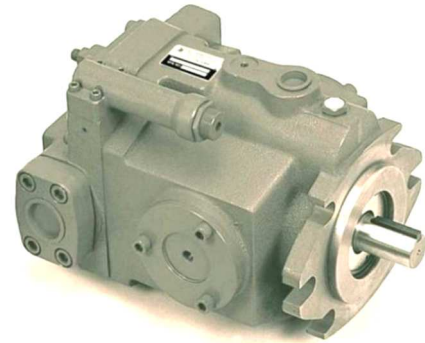


Usually it isn't practical to design the vane type of motor to be variable displacement, like a vane type of pump. Also it isn't practical to design the vane type of motor to be bidirectional. If the vane type motor is required to be bidirectional some other component in the system must change the direction of fluid flow to accomplish this; bidirectional pump or directional control valve for example. Systems that require variable displacement and/or bidirectional capability would probably use a motor that is more conducive to that type of application.

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Piston Motor

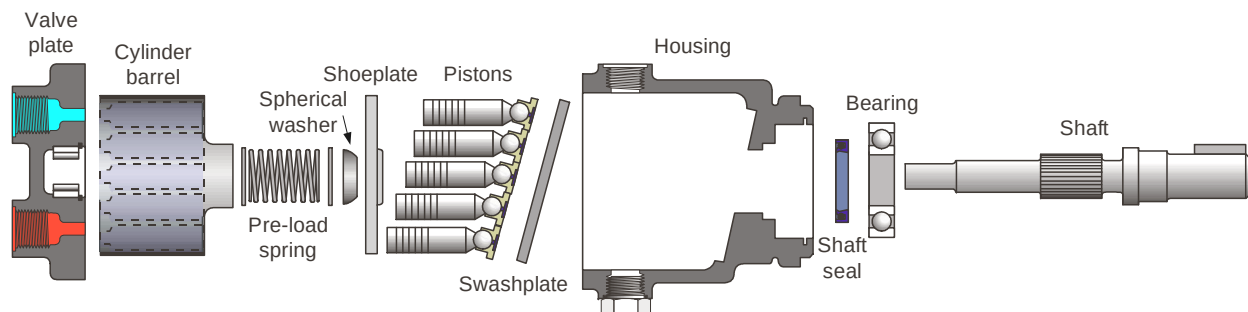
The most efficient and versatile of all the hydraulic motors is the piston type, this also makes them the most expensive. They operate in low to medium rpm ranges (10 – 2000 rpm) and with a low to high torque capacity. Their main use is low speed, very high torque applications. The largest benefit of the piston type of motor is the versatility.



Piston motors fall into three basic categories; **Axial, Bent-Axis and Radial type**. The different names indicate the way the pistons move in relation to the output shaft or center-line axis of the motor.

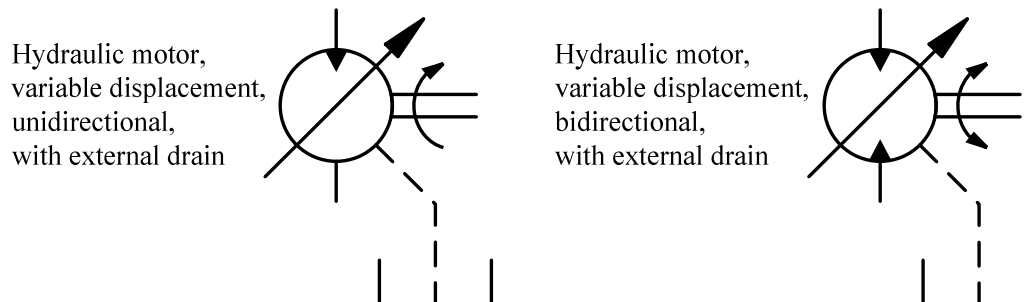
Axial Piston Motor

As fluid enters the motor and acts on the pistons, the force of the fluid will push the pistons against the swash plate. The position of the swash plate is the key to motor operation. When the swash plate is in a zero / neutral angle no rotation would occur. When the swash plate is positioned at an angle the pistons will slide in the direction of the incline. The pistons are situated in a circular barrel and when they slide along the swash plate they rotate the barrel which transfers the rotation to the output shaft. The swash plate position may be fixed or variable.



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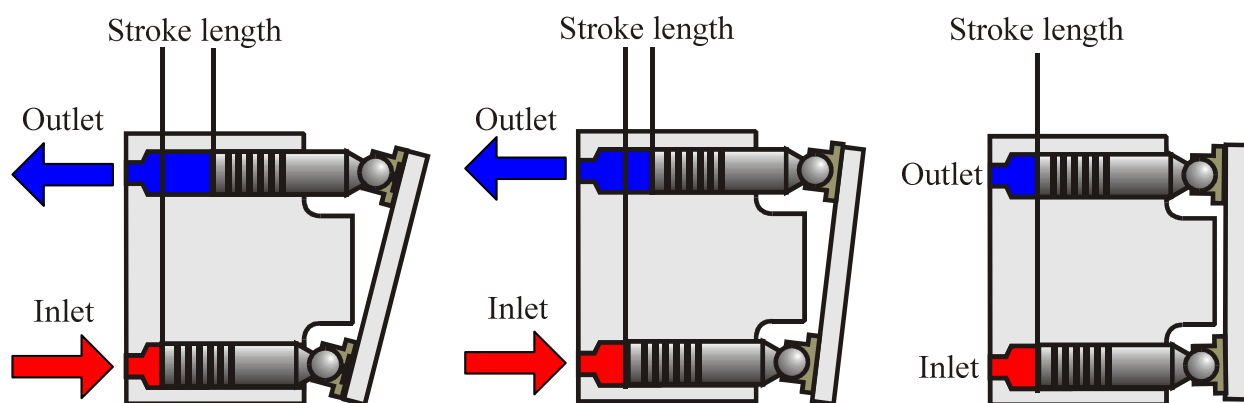
Variable Displacement Axial Piston Motor



The greatest advantage a piston motor may have over any other type of hydraulic motor is the ability to be fixed displacement or variable displacement. The amount of fluid an axial piston motor can accept, per revolution, is determined by the surface area of the pistons, number of pistons and stroke length of the pistons inside the barrel. During the operation, as the motor is accepting fluid nothing can be done to change the surface area of the pistons or the number of pistons. The only way to vary the displacement would be to change the stroke length of the piston in the barrel. Since the shoe-plate pushes the pistons against the swash plate then the swash-plate becomes the key to motor displacement.

When the swash plate is in a fixed position the motor displacement will be fixed. However changing the angle of the swash-plate will change the stroke length of the pistons, thus affecting the amount of displacement/volume. By changing the angle of the swash-plate the motor's displacement, consequently the shaft speed and torque output can be varied.

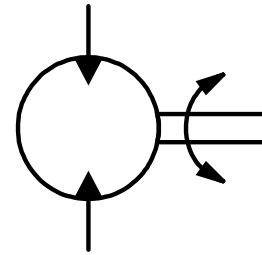
Operation of the swash-plate may be manual, with servo control, with pressure compensation or by several other ways. Normally piston motors are chosen because of how flexible they are when it comes to accomplishing variable displacement.



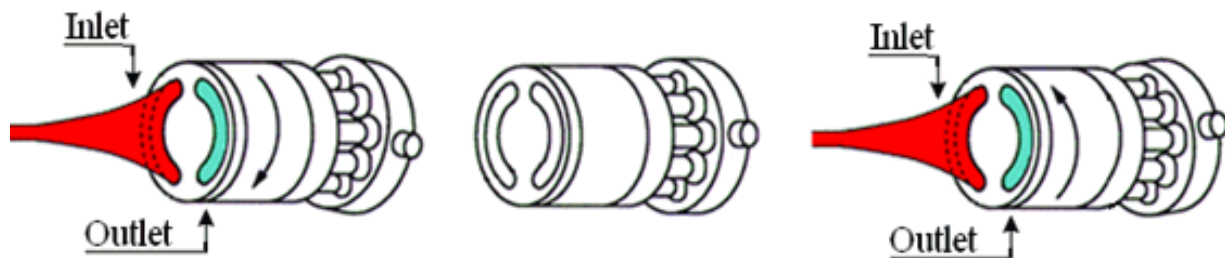
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Bi-Directional Axial Piston Motor

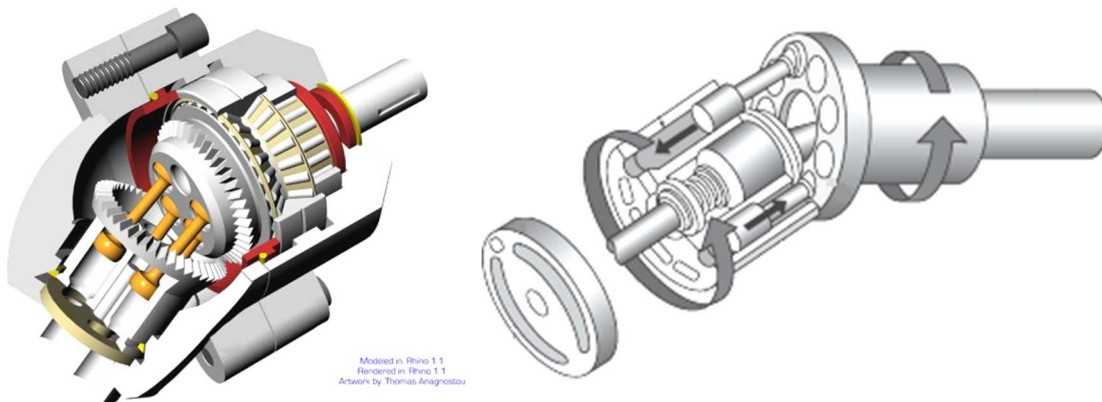
As discussed previously, the swash-plate is the key to the operation of an axial piston motor. In some versions the swash-plate has the capability to move over center, beyond the zero condition, in the opposite direction to reverse the direction of rotation. It is called an over-center type of motor or bi-directional motor.



This allows the motor to change the direction of rotation without changing the direction of fluid flow from the circuit. The swash-plate operation may be accomplished manually, with servo control, or by several other ways.



Bent-Axis Piston Motor



The operation of a Bent-Axis Piston Motor is the same as an Axial Piston Pump except the cylinder barrel is not in-line, or on the same center-line, as the output shaft or the center axis of the motor. The angle formed by the bent-axis design creates the natural movement of the pistons reciprocating in the cylinder barrel as system flow enters and acts on them. Both fixed and variable displacement models are available. Fixed displacement versions are available with 23 degree and 30 degree angles.

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Variable Displacement Bent-Axis Piston Motor

In a variable displacement version an external control is used to change the angle, which changes the stroke length, thus changing the displacement, which in turn changes the speed and torque of the output shaft.

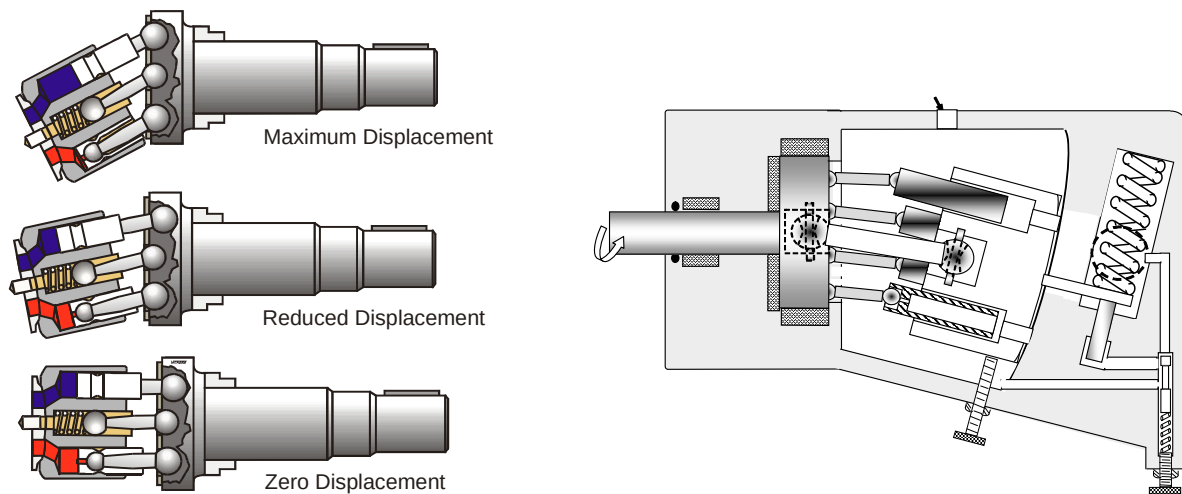


Figure 15.38 Displacement is Controlled by the Angle of the Cylinder Block
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Radial Piston Motor

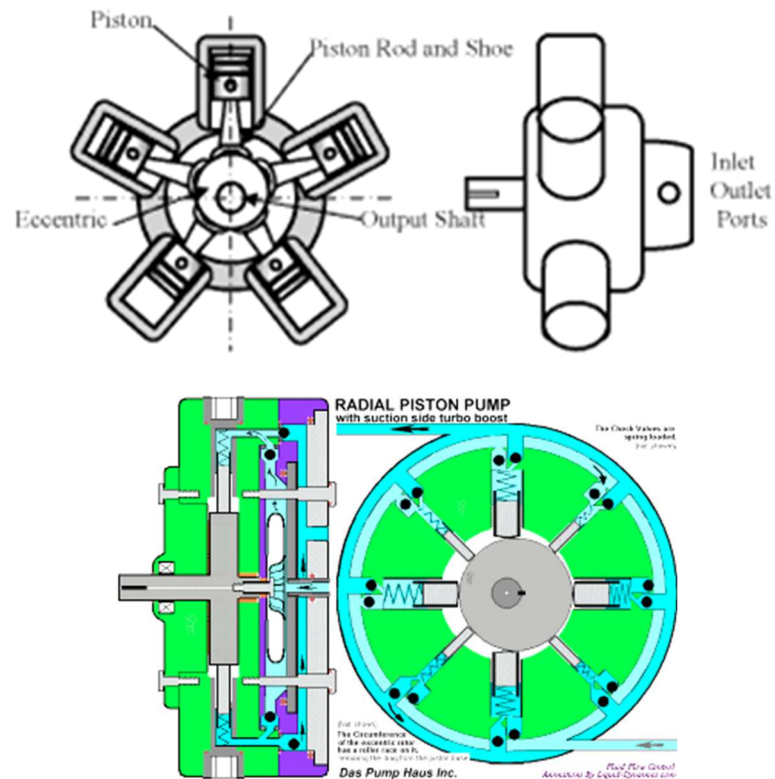
In addition to the Axial and Bent-Axis models, a Radial Piston Motor is also available. All piston type of motors operate on the same principle, it's the movement of the pistons in relation to the centerline of the motor that is different for each.

With the pistons positioned radially in the motor, as fluid enters and acts on them a downward force is created. The pistons push against an eccentric (cam type) shaft and cause the shaft to rotate. The pistons on one half of the motor are essentially pushing down, while the pistons on the other half of the shaft are exhausting their fluid and creating rotation.

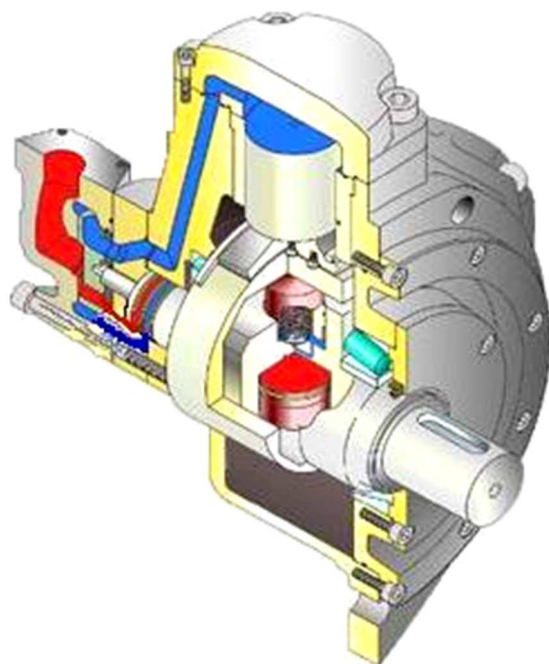


The radial piston motor can be fixed or variable in design. The one described previously is a fixed version.

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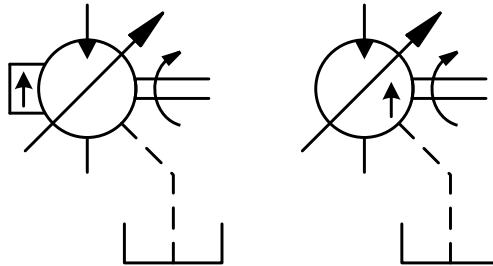
Variable Displacement Radial Piston Motor



A variable displacement radial piston type motor is similar to a fixed motor except the eccentric (cam throw) part of the shaft is movable. It can move in and out to change the stroke length of the pistons.

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Pressure Compensated Motors

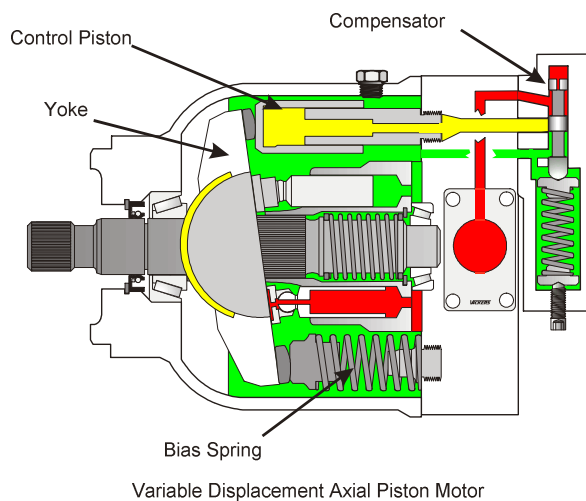


The compensator control used on hydraulic motors is used primarily to vary the motor displacement. A spring loaded piston is connected to system pressure and moves in relation to system pressure changes. The spring loaded piston is usually mechanically connected to the swash-plate (yoke) and will change the angle in response to pressure variations.

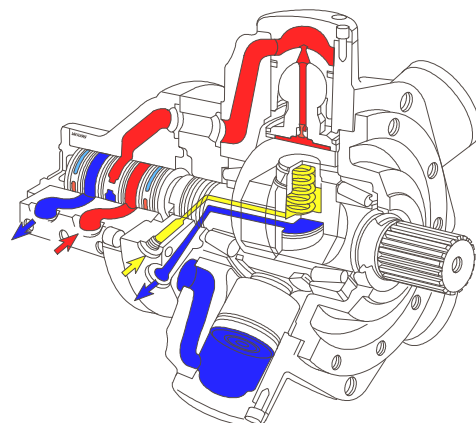
Increases in load force (torque) on the output shaft will be an increase in system pressure. The increase in system pressure will increase the swash plate (yoke) angle. An increase in swash plate angle will increase motor displacement, thus decreasing rpm. Normally, the motor is set to run at minimum displacement until system pressure builds up to a predetermined value. As the system pressure rises above this value, the displacement increases and the speed drops until the maximum displacement (highest torque) are attained.

Ideally a pressure compensated pump would regulate the displacement better to increase efficiency for changing load conditions, up to the pressure relief valve setting.

Various examples are illustrated below;



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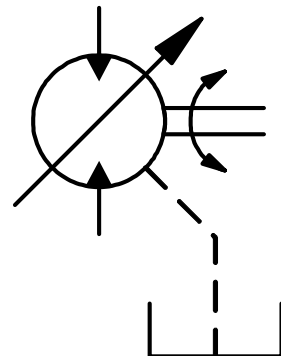
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External (Motor) Drains

Hydraulic motors are usually designed to be bidirectional, even motors that are only used in a unidirectional application are probably bidirectional in design.

All internal leakage (slippage) must be drained from the housing directly to tank through a drain line to protect the shaft seal. Bidirectional motors are typically externally drained.



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Hydraulic Formulas

Pressure, Force & Area

U. S. Units

$$\text{Pressure(PSI)} = \frac{\text{Force(lbs)}}{\text{Area(in}^2\text{)}} \quad \text{Force} = \text{Pressure} \times \text{Area} \quad \text{Area} = \frac{\text{Force}}{\text{Pressure}}$$

Metric Units

$$\text{Pressure(bar)} = \frac{\text{Force(kg)}}{\text{Area(cm}^2\text{)}} \quad \text{Force} = \text{Pressure} \times \text{Area} \quad \text{Area} = \frac{\text{Force}}{\text{Pressure}}$$

Hydraulic Cylinder Speed & Flow Rate

U. S. Units

$$\text{CylinderRodSpeed(in / min)} = \frac{\text{GPM} \times 231}{\text{PistonArea(in}^2\text{)}}$$

$$\text{CylinderRodSpeed(ft / min)} = \frac{\text{GPM} \times 19.25}{\text{PistonArea(in}^2\text{)}}$$

$$\text{GPM} = \frac{\text{CylinderRodSpeed (in / min)} \times \text{PistonArea (in}^2\text{)}}{231}$$

Metric Units

$$\text{CylinderRodSpeed(m / sec)} = \frac{\text{Flow(LPM)}}{6(\text{ factor}) \times \text{PistonArea(cm}^2\text{)}}$$

$$\text{Flow(LPM)} = 6 \times \text{CylinderRodSpeed(m / sec)} \times \text{PistonArea(cm}^2\text{)}$$

Hydraulic (Circuit) Horsepower

U. S. Units

$$\text{Horsepower} = \text{GPM} \times \text{Pressure(psi)} \times 0.000583$$

Metric Units

$$\text{Horsepower(Kilowatt)} = \frac{\text{FlowRate(LPM)} \times \text{Pressure(bar)}}{600}$$

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Hydraulic Formulas (Continued)

Hydraulic Motor Shaft Speed, Flow Rate, Torque & H/P

U. S. Units

$$RPM = \frac{GPM \times 231}{Displacement(in^3 / rev)}$$

$$GPM = \frac{RPM \times Displacement(in^3 / rev)}{231}$$

$$Torque(lb.in.) = \frac{Pressure(PSI) \times Displacement(in^3 / rev)}{2 \times \pi}$$

$$MotorHorsepower = \frac{RPM \times Torque(lb.in.)}{63025}$$

Metric Units

$$RPM = \frac{LPM \times 1000}{Displacement(cm^3 / rev)}$$

$$LPM = \frac{RPM \times Displacement(cm^3 / rev)}{1000}$$

$$Torque(N.m.) = \frac{15.9 \times Pressure(bar) \times Displacement(cm^3 / rev)}{1000}$$

$$MotorHorsepower(Kilowatt) = \frac{RPM \times Torque(N.m.)}{9543}$$

Hydraulic Pump Torque

U. S. Units

$$Torque(lb.in.) = \frac{Pressure(PSI) \times Displacement(in^3 / rev)}{2 \times \pi}$$

Metric Units

$$Torque(N.m.) = \frac{15.9 \times Pressure(bar) \times Displacement(cm^3 / rev)}{1000}$$

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Hydraulic Actuator Exercises

Instructions: Solve the following problems: (Disregard all internal friction)

1. What diameter hydraulic cylinder would be required to give a piston speed of 331 in./min. (0.14 m/sec) with a flow of 18 GPM (68.14 l/min.)?
2. What displacement hydraulic motor would give a torque of 42 lb.ft. (56.94 N.m.) at 300 psi (21.09 bar)?
3. A hydraulic cylinder with a 6.25-inch (15.88 cm) bore, a 20 inch (50.8 cm) stroke, and with a flow of 3.5 GPM (13.3LPM) will extend in what time?
4. A hydraulic motor turns 1200 RPM and uses 6.5 HP (4.85kW). How much torque would it develop?
5. A hydraulic cylinder with a 4-inch (10.16 cm) bore and a 20 inch (50.8 cm) stroke must move a load of 18,175 pounds (8244.04 kg) through its stroke in 6 seconds. How much hydraulic horsepower must be delivered to the cylinder?

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Hydraulic Actuator Exercises (Continued)

Instructions: Solve the following problems: (Disregard all internal friction)

6. How much flow is required to move a hydraulic cylinder with a 5.0-inch dia. (12.7 cm) bore, a 2.75-inch (6.99 cm) dia. rod, a 48 inch (121.9 cm) stroke, 120 in/min (0.051m/sec) on extension and 120 in/min (0.051m/sec) on retraction?

7. A hydraulic cylinder with a 4.0-inch dia. (10.16 cm) bore, a 2.25-inch (5.72 cm) dia. rod, and a 30 inch (76.2 cm) stroke, and with a flow of 20 GPM (75.71 l/min) will extend in what time?

8. What extension speed would be achieved with a 8 inch dia.(20.32 cm) bore cylinder with a 24 inch (60.96 cm) stroke receiving a flow rate of 50 GPM.

9. A hydraulic cylinder with a 2-inch (5.08 cm) bore and a 24 inch (60.96 cm) stroke must move a load of 2500 pounds (1133.98 kg) through its stroke in 5 seconds. How much hydraulic horsepower must be delivered to the cylinder?

10. What is the required displacement for a hydraulic motor is needed if the maximum system pressure is limited to 2000 PSI (140.61 bar) and the output torque has to be 75 lb.ft. (101.7 N.m.) to overcome the resistance?

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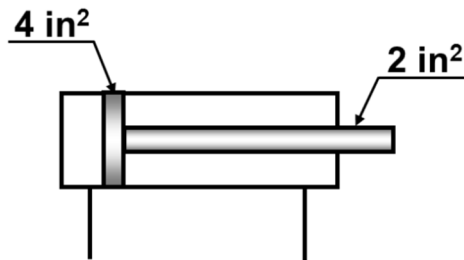
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Regenerative Circuit

Introduction

The speed at which a cylinder extends and retracts is frequently of great concern.

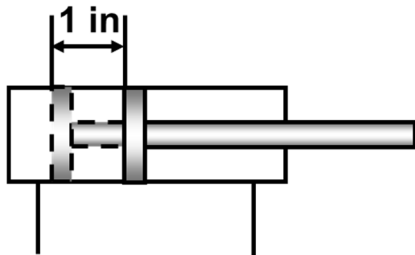
Let's analyze the extension and retraction of the following double acting, single rod cylinder.



As shown, the minor area is equal to 2 in^2 ($4\text{ in}^2 - 2\text{ in}^2$), then the area ratio is **2:1** ($4\text{ in}^2 / 2\text{ in}^2$).

$$mA = MA - RA$$

$$A_{\text{ratio}} = MA/mA$$

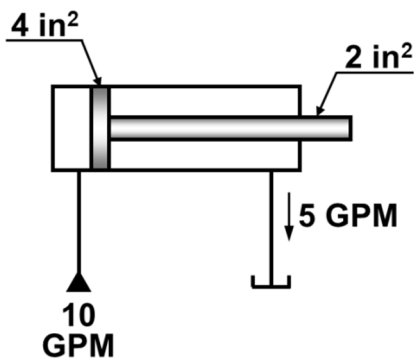


If the rod extends 1 in; the expanding volume in the major end is 4 in^3 ($4\text{ in}^2 \times 1\text{ in}$), and reducing volume in the minor end is 2 in^3 ($2\text{ in}^2 \times 1\text{ in}$). Then the volume ratio is **2:1** ($4\text{ in}^3 / 2\text{ in}^3$).

$$\text{Volume} = \text{Area} \times \text{Stroke}$$

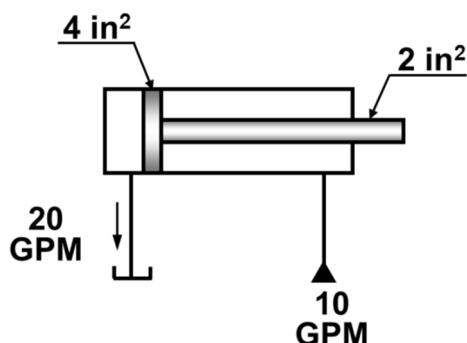
$$\text{Flow} = \text{Volume} / \text{time}$$

$$V_{\text{ratio}} = \text{Major End Volume} / \text{minor end Volume}$$



If 1 inch of rod extension is done in 1 min, then the flow going into the major end is equal to $4\text{ in}^3/\text{min}$ and the flow coming out from the minor end is equal to $2\text{ in}^3/\text{min}$. Therefore the flow ratio is **2:1** also.

$$\text{Flow}_{\text{ratio}} = \text{Flow IN} / \text{Flow OUT}$$



As a conclusion, for linear actuators, we can say that the **$A_{\text{ratio}} = V_{\text{ratio}} = \text{Flow}_{\text{ratio}}$** .

Then, if the pump's flow capacity is 10 GPM, and that flow is sent to the major end; the flow coming out from the minor end is 5 GPM, but if the pump's flow is sent to the minor end, the flow coming out the major end will be 20 GPM (flow intensification).

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The common way to control a double acting cylinder is to change the flow and pressure of the fluid to and from the cylinder when extending or retracting.

To extend the cylinder, a directional control valve is used to send fluid flow and pressure to the major end; fluid is allowed to flow from the minor end back to reservoir. There is little or no pressure on the minor end at this time.

To retract the cylinder, the directional control valve sends fluid flow and pressure to the minor end, fluid is allowed to flow from the major end back to reservoir. There is little or no pressure on the major end at this time.

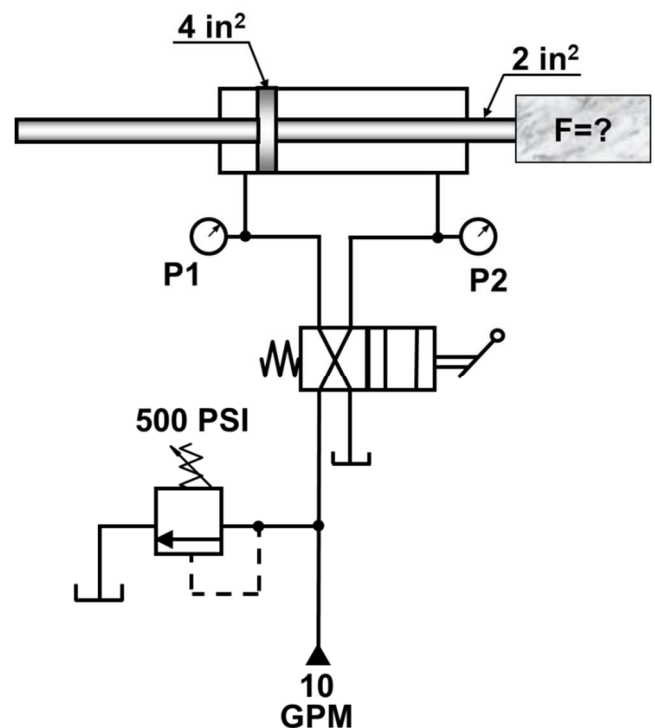
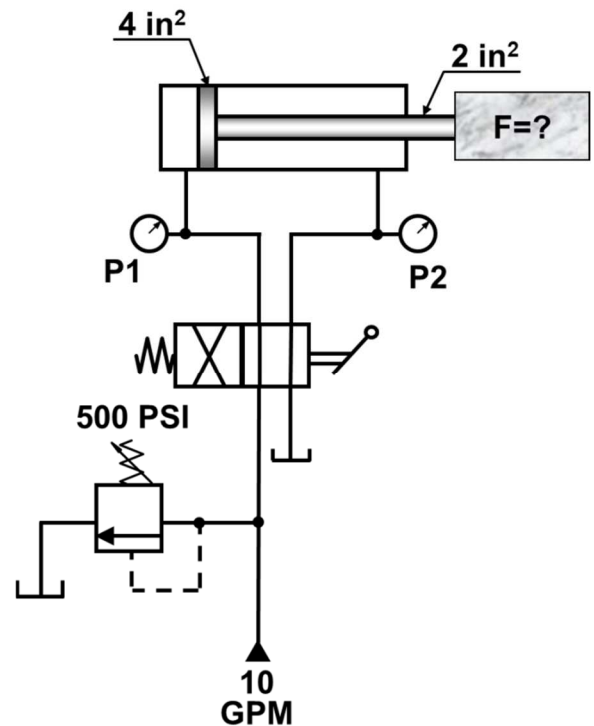
For the cylinder shown, the maximum extension force and speed will be **2000 lb** and **577.5 in/min**; the retraction values will be **1000lb** and **1155in/min**.

As shown previously, a double acting, single rod cylinder retracts faster than it extends. In some applications, it is required that a cylinder extends and retracts at the same maximum speed. One way of accomplishing this is with a double rod cylinder.

Since a double rod cylinder usually has the same rod diameter on both sides of the piston, piston areas exposed to system flow are equals. With the rate of flow to each side remaining constant, rod speed and forces are the same for extension and retraction.

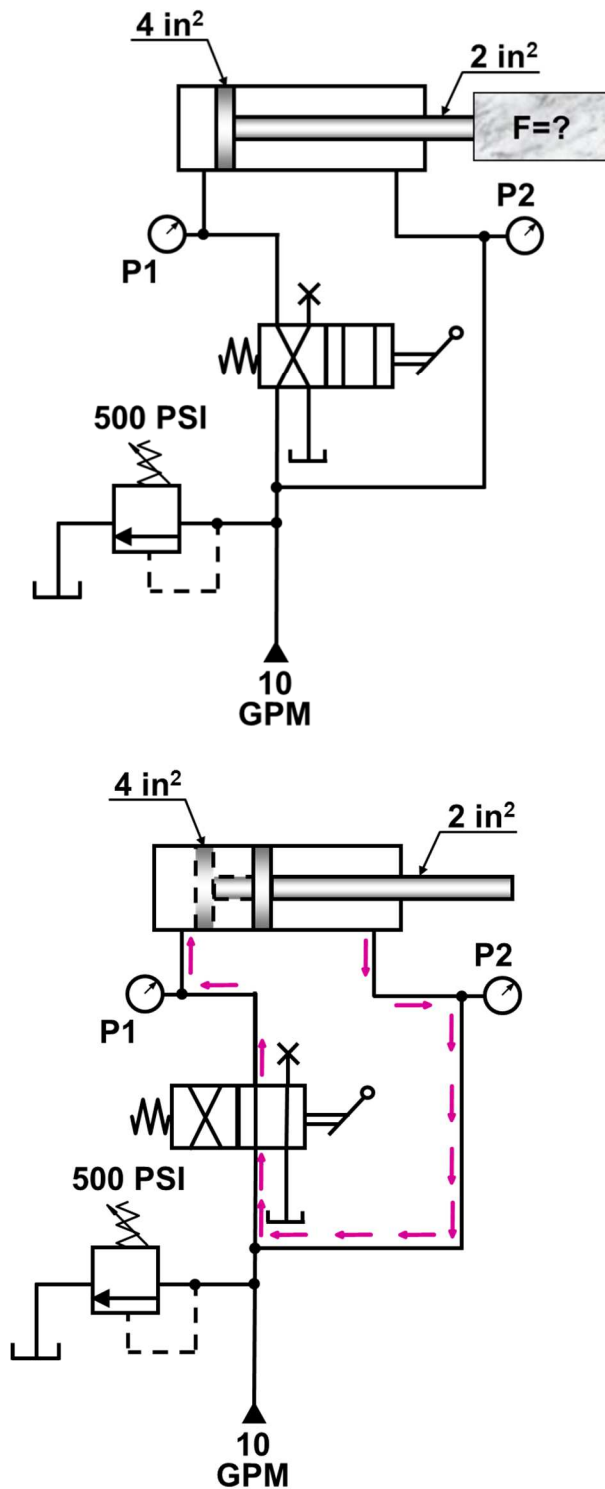
Some disadvantages of using double rod cylinders are:

- The rod speed is faster during extension, but the force is reduced.
- These cylinders cost more than a single rod cylinder.
- The extra rod needs space in which to move.
- The second rod seal is another potential leakage source.



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Regenerative Circuit



To eliminate the double rod cylinder problems; a regenerative circuit, as shown, can double the extension speed of a single-rod cylinder without using a larger pump or a double rod cylinder. That means, this circuit can save money because a smaller pump, motor, and tank can produce the desired cycle time. It also means that the circuit costs less to operate over the life of the machine.

In this example, the area ratio is 2:1, in other words, the rod side of the piston has one half the area exposed to pressure as the major end side. Actually, the rod area is not precisely half of the major area; however, we will consider it so in order to facilitate our calculations.

With the cylinder in the circuit, flow and pressure are directed to both sides of the piston at the same time. It may appear that the cylinder would be hydraulically locked. But, the force acting against the major and minor ends are 2000lb and 1000lb respectively. The unbalanced forces generated make the piston extend.

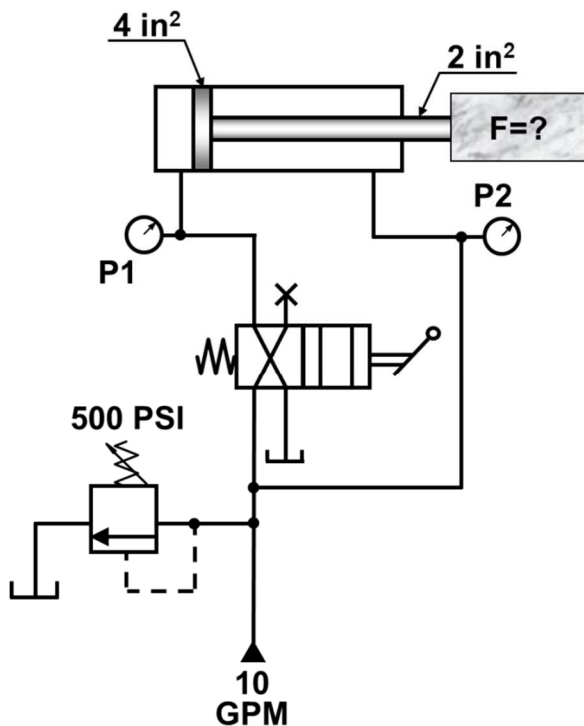
As the piston moves out, fluid which is displaced from the rod side must find a way to move; the least resistant flow path is the cap end (when the volume expands the pressure drops).

Let's analyze the flow, as a step by step process:

- First, as was shown before, if the pump sends 10 GPM to the major end, then 5 GPM will come out from the minor end (2:1 Area Ratio).

- Second, the flow coming out from the minor end is added to the pump flow; now the major end receives 15 GPM and will be 7.5 GPM leaving the minor end (2:1 Area ratio).

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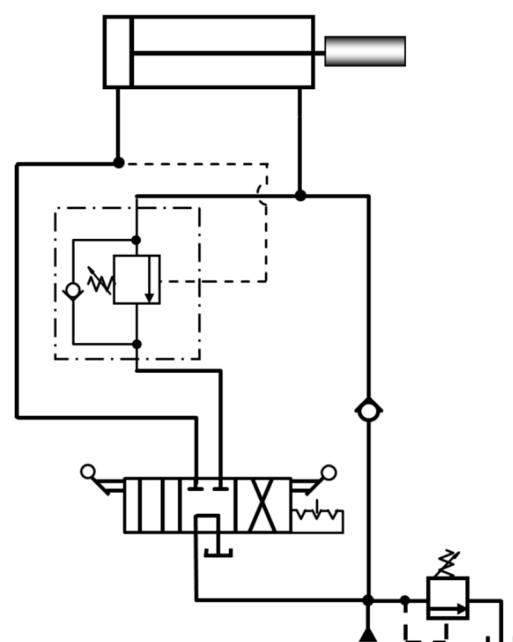
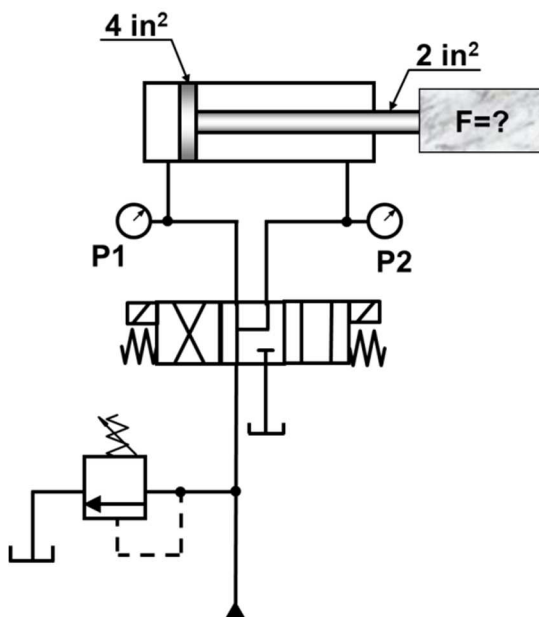
- Third, 7.5 GPM from the minor end is added to the pump flow, then the major end receives 17.5 GPM and 8.25 GPM will be leaving the minor end, which will be added to the pump's flow. The process keeps going until the amount of flow coming out the minor end will equal the pump flow, in other words the flow is regenerated.

Another way to explain the regenerative process is that the fluid which is coming out the minor end is used to fill half of the volume in the major end, the other half (which is equal to the rod cross-sectional area) is filled by the pump.

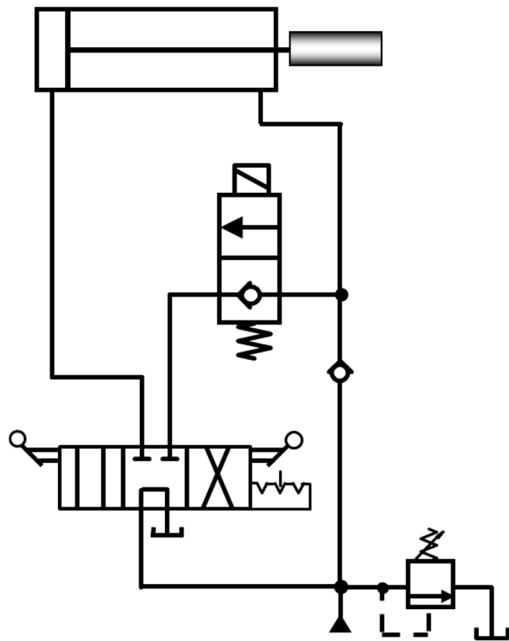
A disadvantage of regeneration is that output force is reduced. Since the fluid pressure is the same in both sides, the effective area on which the force is generated is the cross-sectional rod area.

To retract the cylinder rod, the directional control valve is shifted. The cap end of the cylinder is drained to tank; all pump flow and pressure is directed to the rod end. Since the pump is filling the same volume as at the cap end, the rod retracts at the same speed.

Since the force is reduced, regeneration is frequently employed only to extend the rod to the load, then when the load is going to be moved the rod end is drained/redirected so that full force in the major end could be exerted. The following circuits are examples of regenerative circuits which can be converted to normal circuits when full force is needed.



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Conclusions:

- Regeneration is used when the same rod speed is required on extension and retraction.
- Regeneration only occurs during rod extension with a 2:1 cylinder, when discharge flow from the rod end is diverted to the major end.
- In regenerative circuits, force on extension is reduced because of the counteracting force from the minor side of the piston.
- In calculations for extension speed and force during regeneration, the pump flow and the cross-sectional area of the rod is used, not the major piston area.

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Pressure Control Valves

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Introduction to Pressure Control Valves



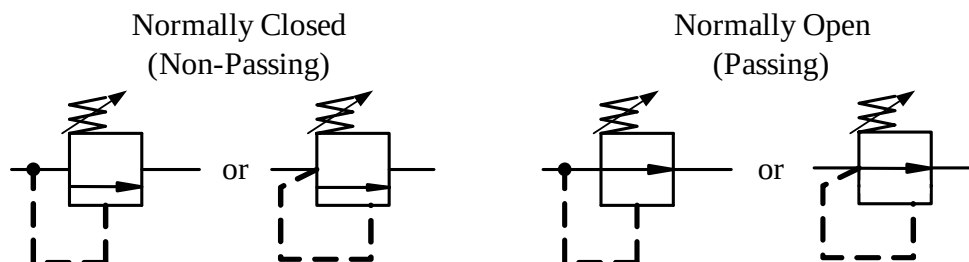
Pressure control valves have several different purposes in a hydraulic circuit. Some of the more common, but not limited to, purposes for pressure control valves are;

1. Limit pressure at the pump outlet or in any other part of the circuit.
2. Reduce pressure in an isolated part of the circuit.
3. Provide controlled deceleration for actuators.
4. Increase resistance for control of assisting or overrunning type loads.
5. Provide “Logic” functions in a circuit.

Usually the primary concern in hydraulic circuits is to either control flow rate and/or control the pressure. One common misconception in a fluid power circuit is that pressure can be controlled with a flow control valve. This is never the proper way to control pressure; a flow control valve will have an effect on the pressure in the circuit, but is not the right way to control pressure. Flow control valves control rate of flow; pressure control valves control pressure.

To control pressure, which also controls force; six types of pressure control valves have been designed. Since some of the symbols for the six pressure control valves are similar to each other, often it is only their purpose and location in the circuit that will indicate what type of pressure control valve is being used. Pressure control valves are usually named for their primary function in the system.

Since their primary purpose is to control pressure to perform a function in a hydraulic circuit, the issue of controlling pressure is a two-fold design issue. Pressure control valves may use pressure to start or stop flow. Thus the two different designs for the flow path in a pressure control valve; Normally Closed (Non-Passing) or Normally Open (Passing).



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Pressure control valves come in two different types, with several different names;

1. Simple or Direct Acting



A simple (direct acting) type of pressure control valve will use some type of closure, a ball or poppet is common, which is held in place by a spring force in the body. When system pressure at the inlet (primary) port is less than the spring force on the closure it will stay seated in the body. When system pressure exceeds the spring force being applied to the closure it will unseat and allow flow to divert through to the outlet (secondary) port, which is usually returning fluid to the tank or reservoir, as long as the pressure continues to exceed the spring value.

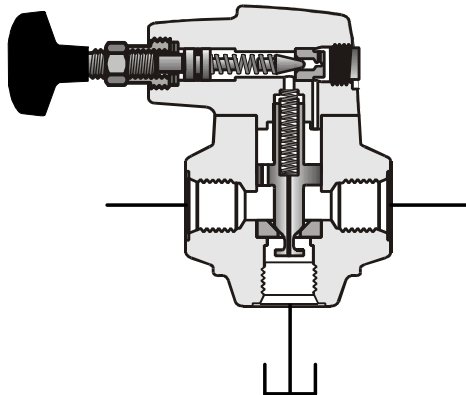
The main advantage a simple (direct acting) type valve has over a compound (pilot operated) type is there is only one moving part, the closure. The closure being in direct contact with the fluid pressure allows it to be more responsive to pressure changes in the system.

The main disadvantage for a simple (direct acting) type is they will partially open (cracking pressure) at approximately 100 to 200 psi (7 - 14 bar) below the spring setting on the valve. If you have a system that needs to operate at 1000 psi and the spring setting is adjusted to 1000 psi, the valve will begin to open at approximately 800 to 900 psi. This cracking pressure could reduce the efficiency or hinder the system from performing properly. Usually the spring setting on a simple (direct acting) type of pressure control valve is set at 100 to 200 psi (7 - 14 bar) **above** the maximum operating pressure. Some in the hydraulic industry use, instead of the hard value of 100 to 200 psi, a percentage value of approximately 10 to 20 percent as a value for cracking pressure.

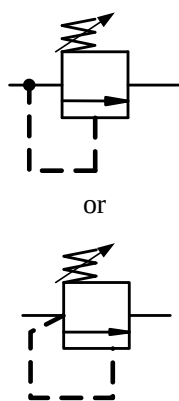
Simple (direct acting) type of valves can be noisy, if the fluid has a high velocity when bypassing to tank or reservoir. Because of their design they are typically used in low flow systems, usually around 5 gpm or less.

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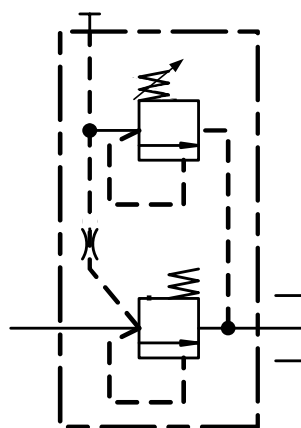
2. Compound or Balanced Spool or Pilot Operated



Simplified symbol



Detailed symbol



In a compound (pilot operated) type of valve there are many different designs, depending on the manufacturer, but the symbol and overall function or operation is the same.

Where the simple (direct acting) type of valve has to be piped into the system with a separate line, the compound (pilot operated) type actually is piped into the main system supply without a separate line. It is in the main system supply line handling main system flow at all times.

Main system flow enters the inlet (primary) port and exits the outlet (primary) port, as long as the main spool is seated, and continues to flow downstream to the system. The main spool will stay seated as long as the system pressure does not exceed the main spring setting on the valve. That same main system pressure flowing thru the valve is allowed to enter the upper chamber on the main spool, thru a small orifice (hole). This small orifice (hole) is the valve's internal pilot. The main spool has the same amount of surface area on both sides and with system pressure being the same, the same amount of force is created, which balances the main spool.

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This fluid pressure now acting on the bottom and top of the main spool is considered to be a balanced spool effect. A small, light duty spring (usually around 25 psi, and is called a bias spring) is situated on top of the main spool. This additional force on the top of the main spool is the only reason the spool stays seated and does not allow fluid to bypass the system and goes to reservoir, as long as everything else is working properly in the valve. That portion is the part of the valve that handles main system flow.

When main system pressure is allowed to enter the upper chamber on the main spool, thru the small orifice (hole), it also continues upward to the upper portion of the valve. Some will call this portion the pilot section or second stage. This portion handles main system pressure, not main system flow, and is where the main spring setting is set for the valves' main system pressure value. This part of the compound (pilot operated) valve is the same as simple (direct acting) type. As the fluid continues upward and begins to act on the closure, it is being held in place with the main spring. As long as the main system pressure is below the value created by the main spring the closure will stay seated and the valve will not open and bypass fluid to reservoir. If the fluid pressure is higher than the main spring setting, the closure will unseat and allow fluid to pass thru the valve and go straight to reservoir. When the closure is open allowing fluid to pass, the pressure created on the top of the main spool is reduced. When the pressure on the top cannot equal or balance the pressure on the bottom, the main spool will move up and allow main system flow to bypass to reservoir, essentially unbalancing the spool.

When the main system pressure decreases and allows the closure, in the top portion, to seat, pressure on the top of the main spool will begin to increase and balance the spool. When the main spool is in a balanced state the additional 25 psi bias spring force will make sure the main spool seats and stops main system flow from going to reservoir and allow it into the system.

The main advantage a compound (pilot operated) type has over a simple (direct acting) type is the balanced spool design is not as sensitive to pressure fluctuations. The allowance for a cracking pressure will be less, approximately 50 to 100 psi below the spring setting on the valve. Even though the spring setting should be 50 to 100 psi **above** the maximum operating pressure, most actually use 100 to 200 psi for both types. Again, some in the hydraulic industry use, instead of the hard value of 100 to 200 psi, a percentage value of approximately 10 to 20 percent as a value for cracking pressure. Compound (pilot operated) types of pressure control valves are designed to handle higher pressure values and higher flow rates than a simple (direct acting) type. This valve also has far greater flexibility in a hydraulic circuit and will be discussed later in this chapter.

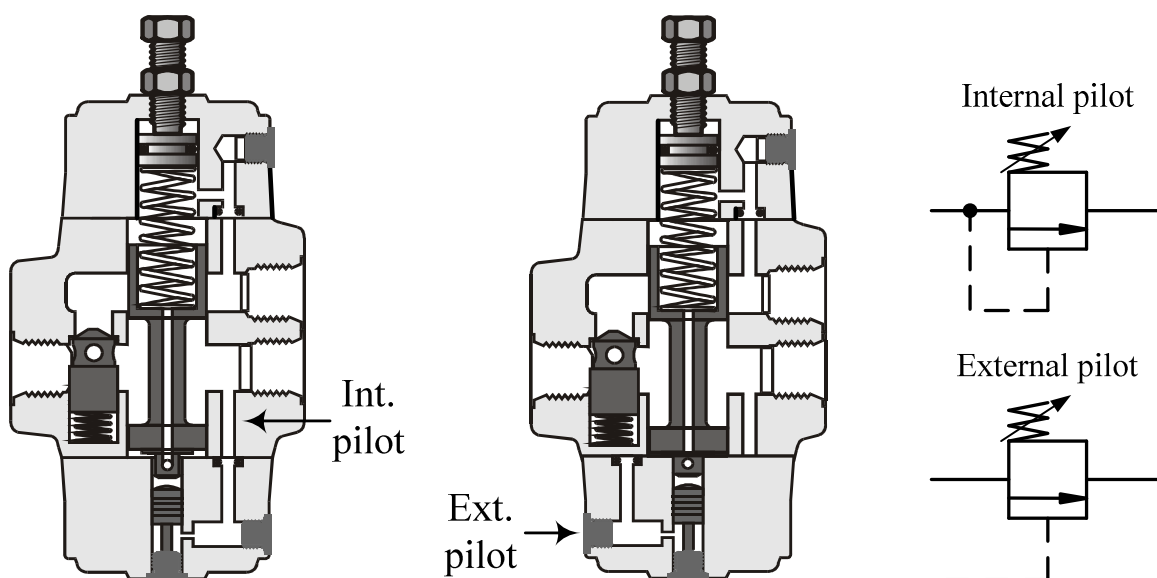
The main disadvantage a compound (pilot operated) type has over a simple (direct acting) type is with two moving devices and two springs is more parts for possible failure. Also the small orifice (hole) is prone to being partially blocked or completely plugged with contamination and interfering with the valves ability to function.

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Pressure Control Valve Pilots

Pressure control valves use pressure to control the moving device, usually a spool or poppet, inside the valve body. That pressure signal may be sensed inside the valve from the primary (inlet) port or from the secondary (outlet) port, depending on the valve type or purpose. This type of control would be called an Internal Pilot or Direct Operation.

If the pressure control valve senses a signal from a source outside of the valve body, it is called, External Pilot or Remote Operation. That pressure signal may be sensed from anywhere outside of the valve body and come from anywhere else in the hydraulic circuit.



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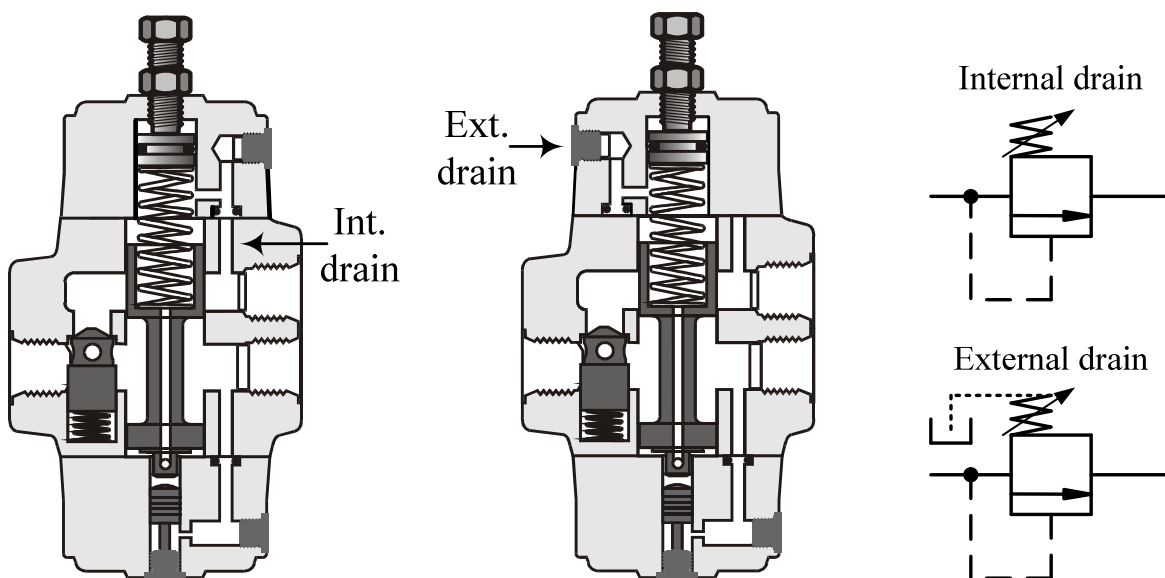
Pressure Control Valve Drains

The moving device, usually a spool or poppet, has to move inside the valve body to perform its function of controlling pressure. For the spool to move there has to be clearance between it and the valve body. If there is clearance, there is an opportunity for fluid to slip past and build up in the spring chamber. This fluid has to be drained out or eventually the spool will be prevented from moving. Some slippage or leakage is necessary to lubricate the spool, which is normal. However, the fluid cannot build up in the spring chamber to a point of stopping the spool. Also, if the valve has to handle reverse (bidirectional) flow, then there is a chance that fluid will fill up the spring chamber and, again, prevent the spool from moving.

In order for a pressure control valve to operate properly, the spring chamber (area above the spool) must be drained so the fluid does not prevent the spool from moving. This may be accomplished two different ways, Internal Drain or External Drain.

An internal drain is typically used when the secondary (outlet) port is connected to the reservoir, or is non-pressurized during the systems operation to prevent fluid from backing up in to spring chamber. Pressure Relief Valve and Counterbalance Valve applications are good examples of an internal drain.

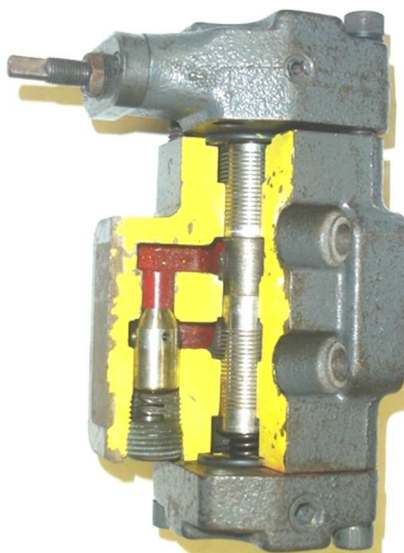
An external drain is typically used when the secondary (outlet) port does work, or you could say, has reverse (bidirectional) flow or is pressurized during its normal operation of the circuit. In either case fluid would back up into spring chamber. Sequence Valve and Pressure Reducing Valve applications are good examples of an external drain.



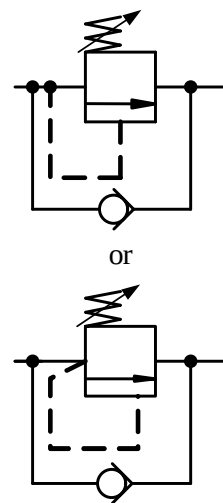
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Reverse Flow in a Pressure Control Valve

Depending on the circuit design and position of the pressure control valve in the circuit, the pressure control valve may have to handle reverse (bidirectional) flow. In some cases pressure control valves may be able to handle reverse (bidirectional) flow thru the valve. Typically if the system pressure is less than the spring setting the valve may be able to handle reverse flow. If the system pressure is higher, a by-pass check valve will be required to go around the valve. The by-pass check valve may be independent of the pressure control valve or built-in as an integral part of the valve.



Simplified symbol

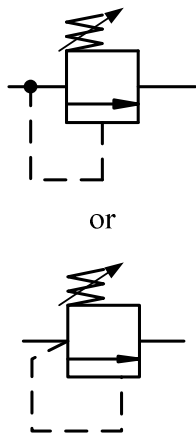


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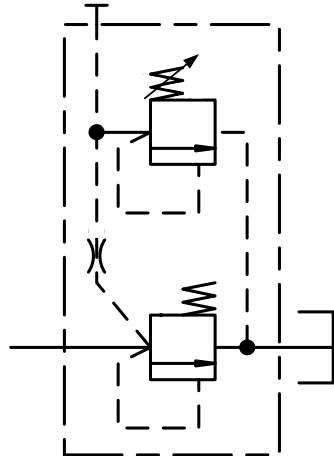
Types of Pressure Control Valves

Pressure Relief Valve

Simplified symbol

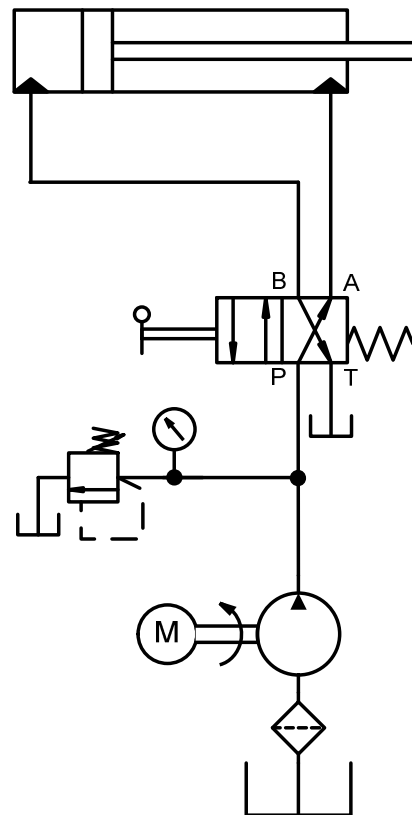
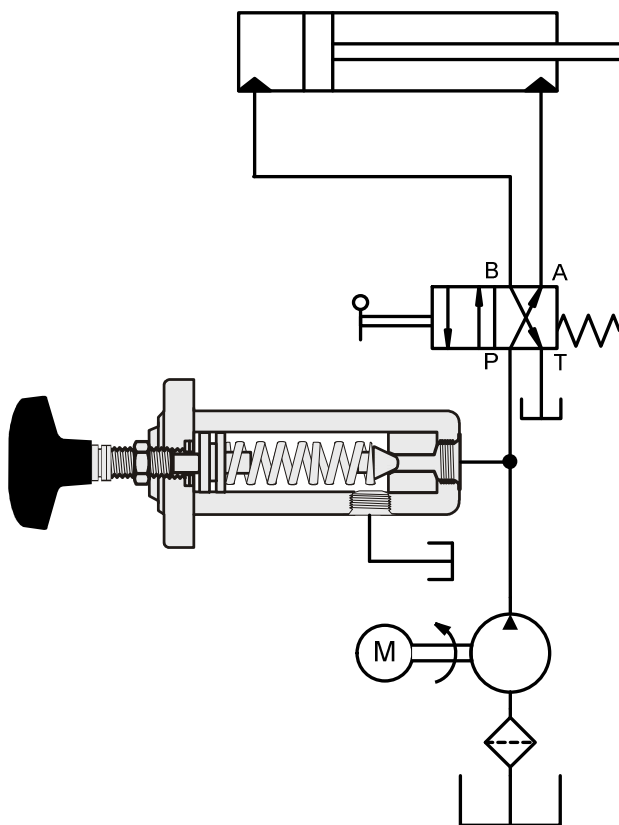


Detailed symbol
(Compound type only)

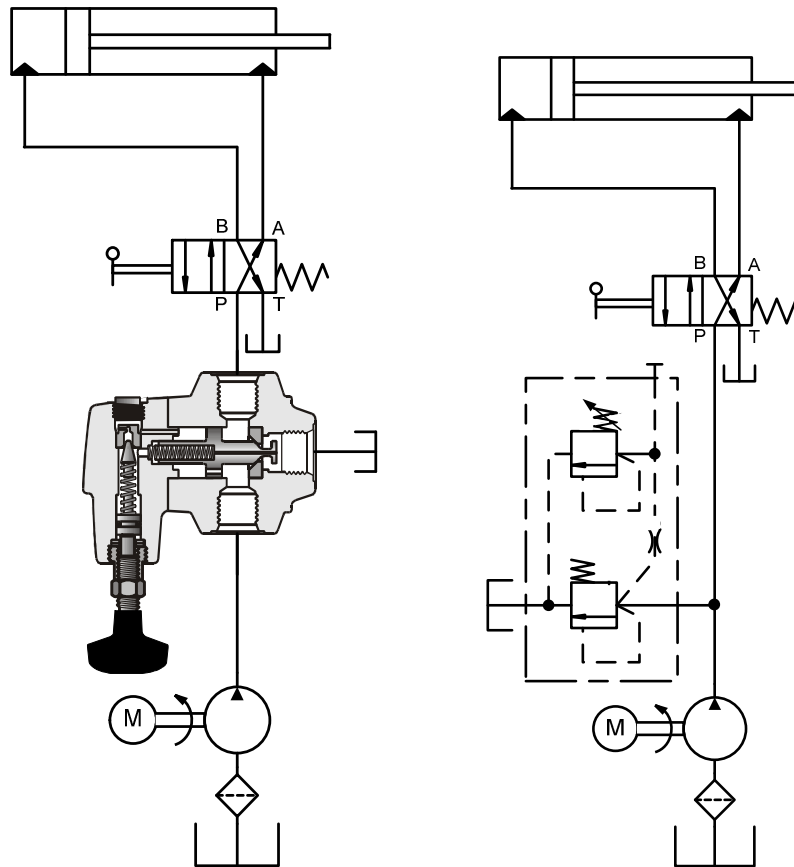


Purpose: Protect the pump and system piping.

Description: Normally closed
Spring loaded
Adjustable
Normally internal pilot
Normally internal drain



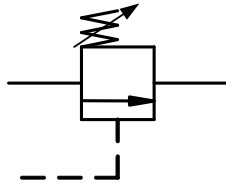
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Set the spring setting on the pressure relief valve at 100 to 200 psi above the maximum working pressure value in the circuit.

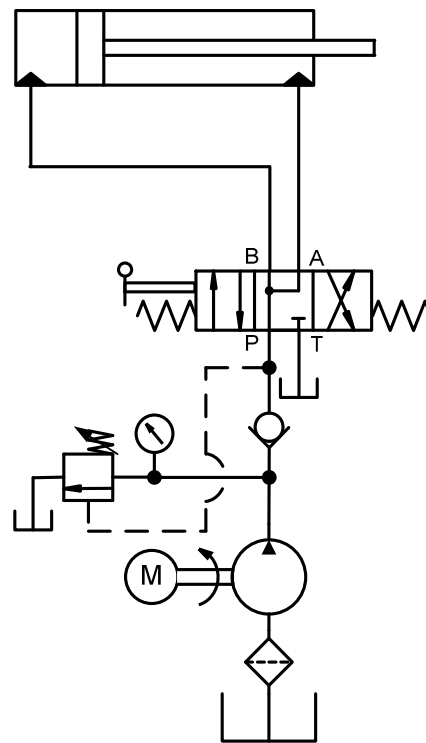
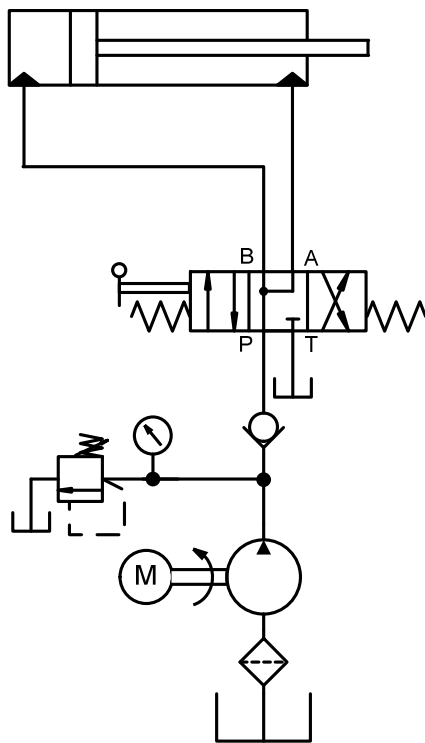
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Unloading Valve

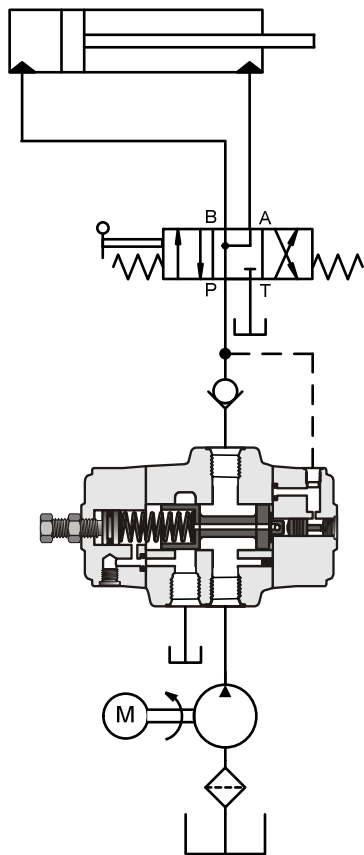


Purpose: Unload the pump or a portion of the circuit.

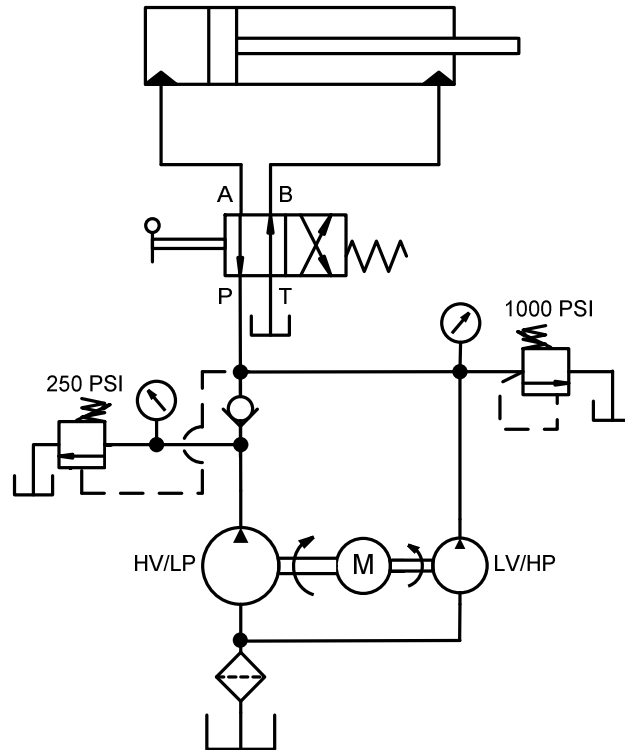
Description: Normally closed
Spring loaded
Adjustable
Normally external pilot
Normally internal drain



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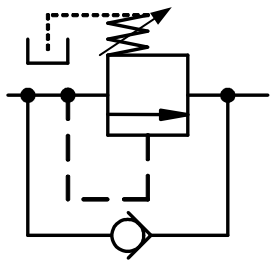
Common example of an Unloading Valve used in a HI/LO system



Set the spring setting on the unloading valve at 100 to 200 psi above the pressure value sensed by the external pilot.

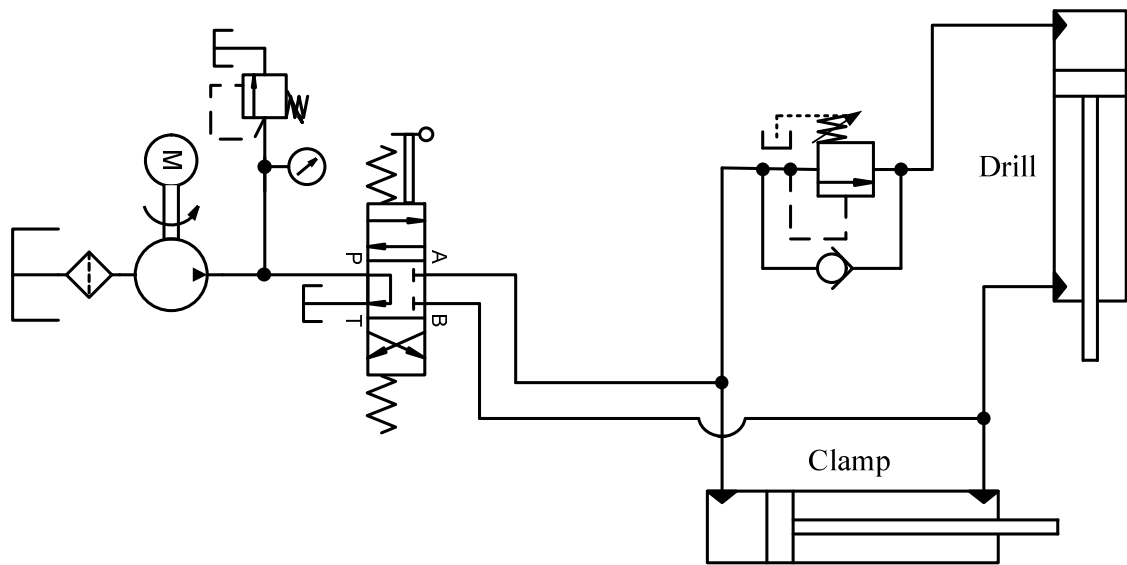
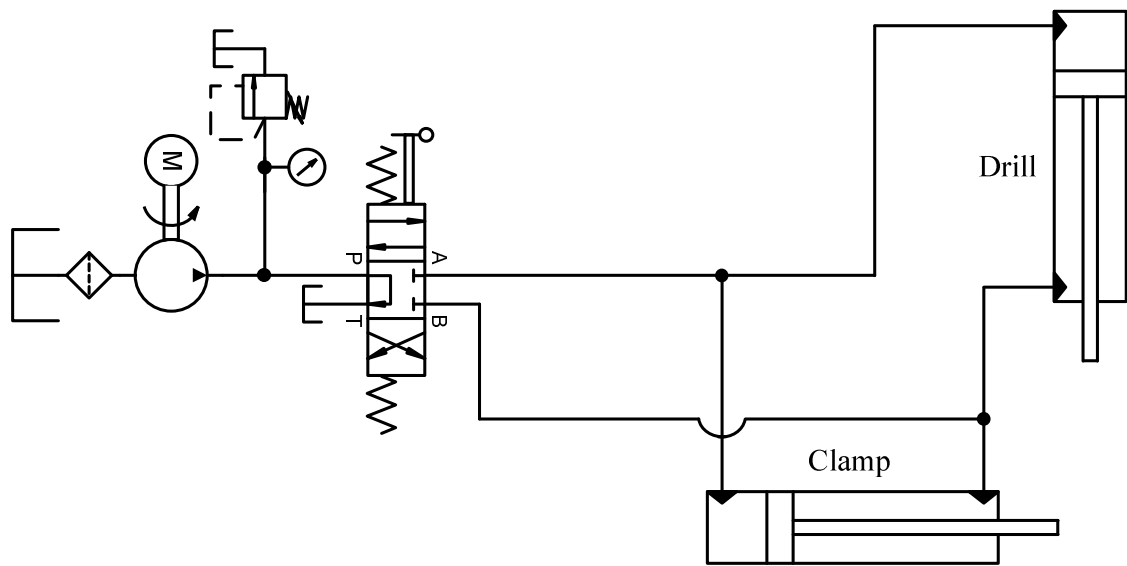
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Sequence Valve

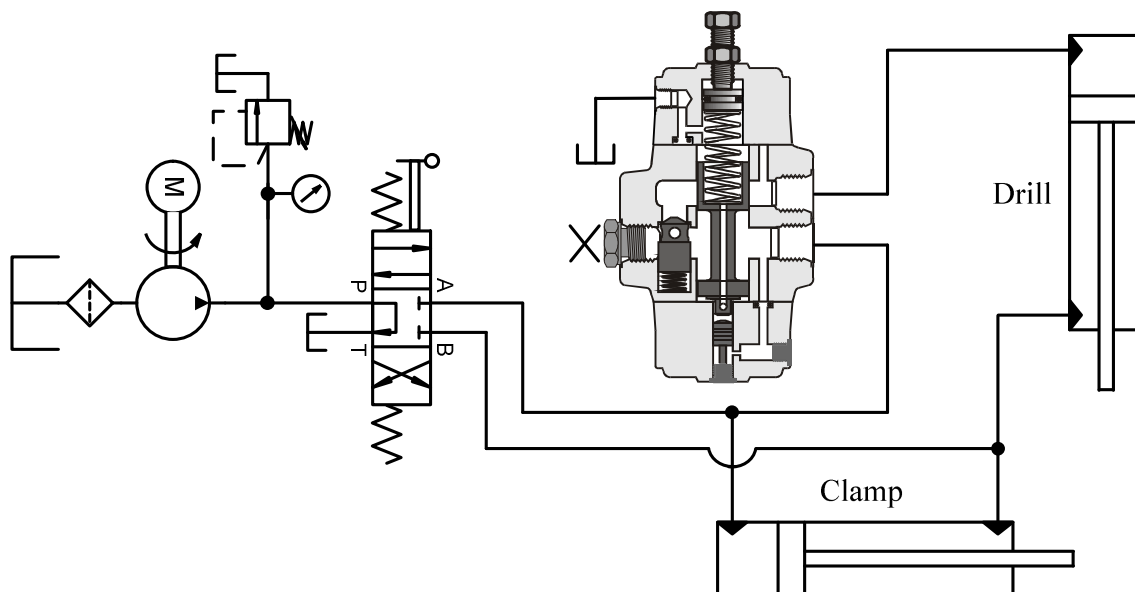


Purpose: Ensures that two operations happen in a pre-determined order.

Description: Normally closed
Spring loaded
Adjustable
Internal or external pilot
External drain
Bidirectional flow requires by-pass check valve



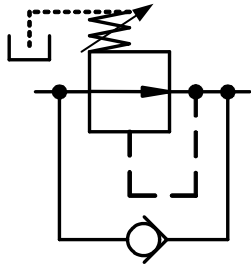
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Set the spring setting on the sequence valve at 100 to 200 psi above the pre-sequenced working pressure value. Set the spring setting on the pressure relief valve at 100 to 200 psi above the maximum pressure in the circuit.

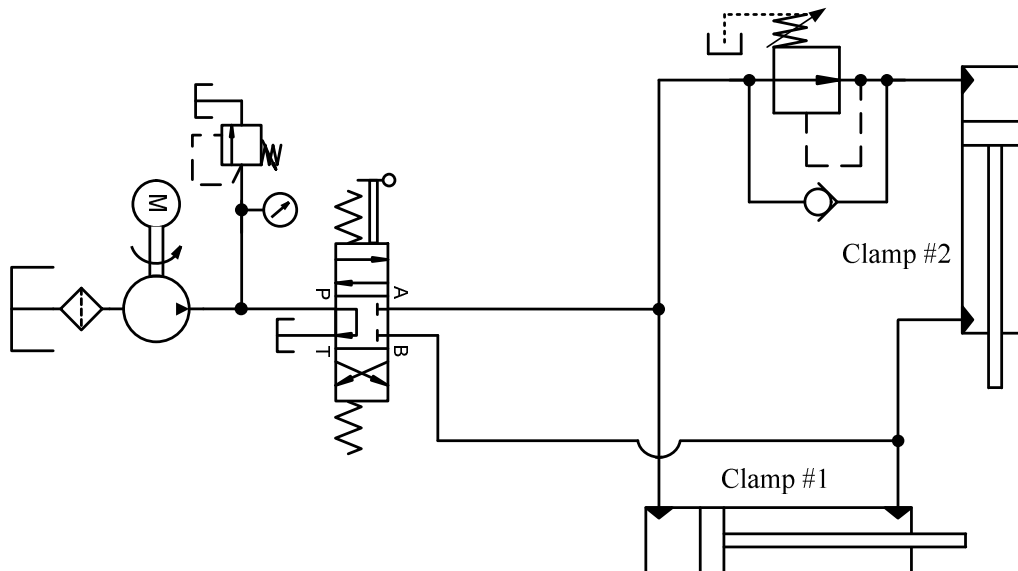
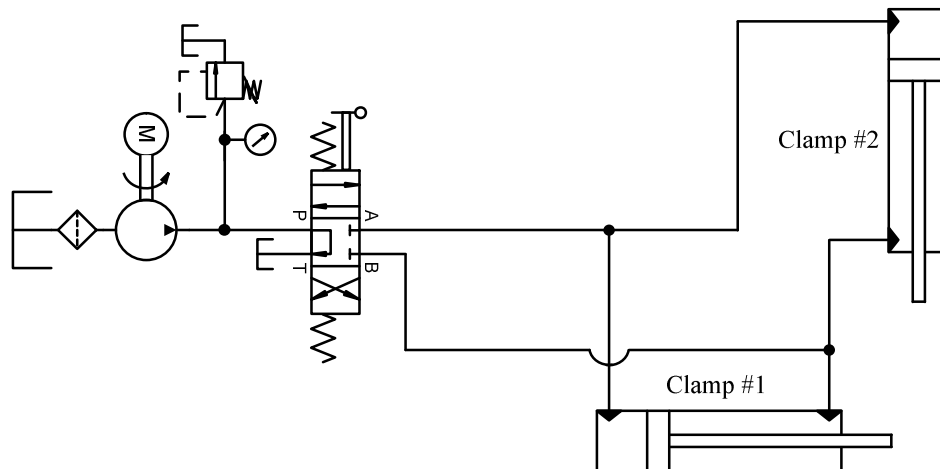
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Pressure Reducing Valve



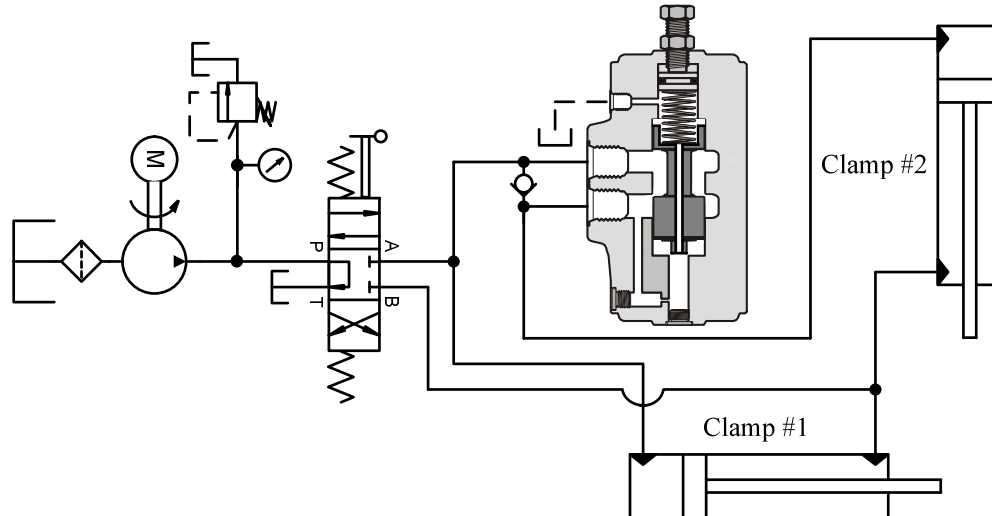
Purpose: Reduce main system pressure to a lower, more steady value.

Description: Normally open
Spring loaded
Adjustable
Internal pilot on the secondary (outlet) port
External drain
Bidirectional flow requires by-pass check valve

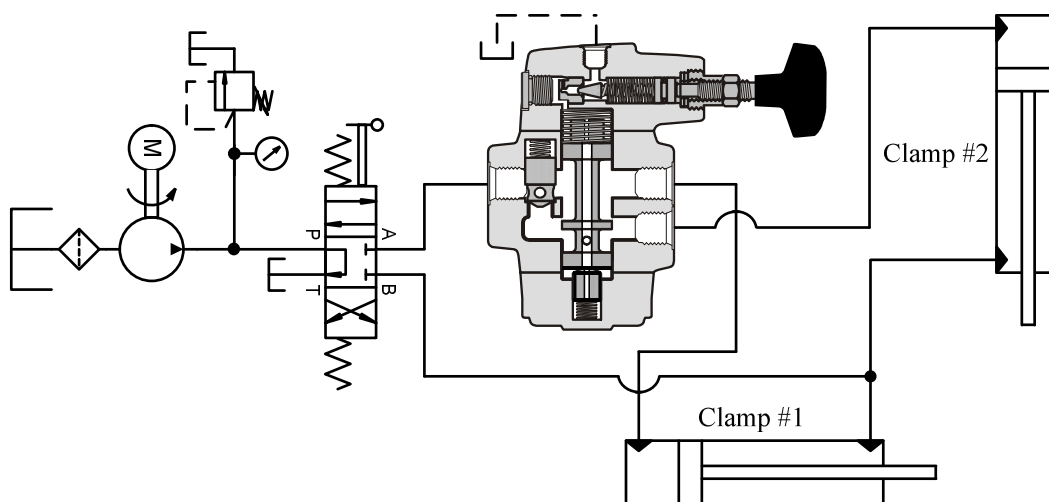


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Example of a Direct Acting type of Pressure Reducing Valve



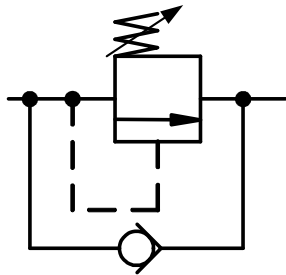
Example of a Pilot Operated type of Pressure Reducing Valve



The spring setting on a pressure reducing valve **does not** require any extra value due to cracking pressure. Set the spring setting at the working pressure required downstream.

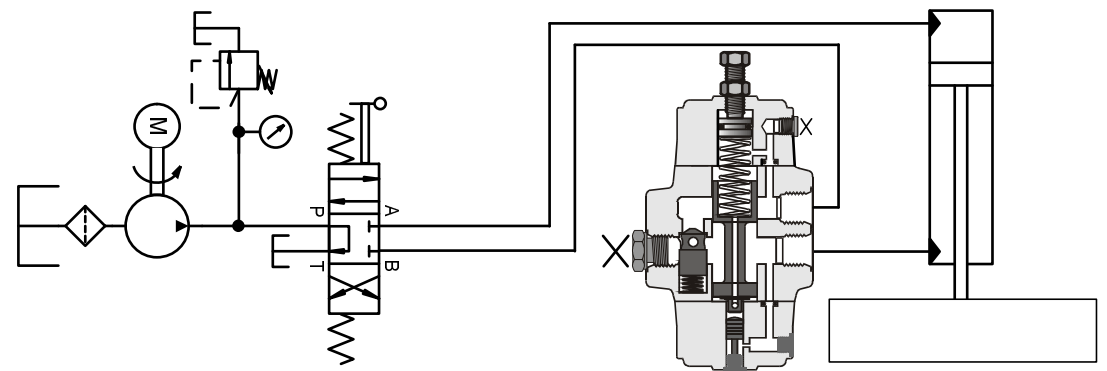
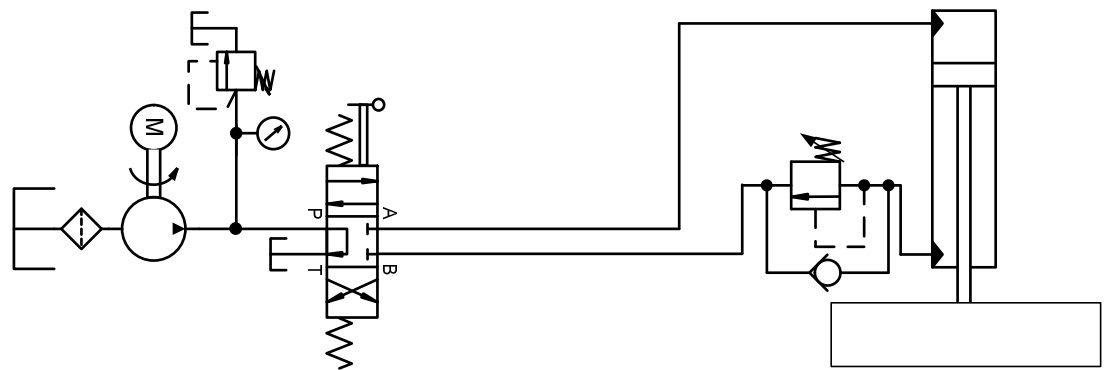
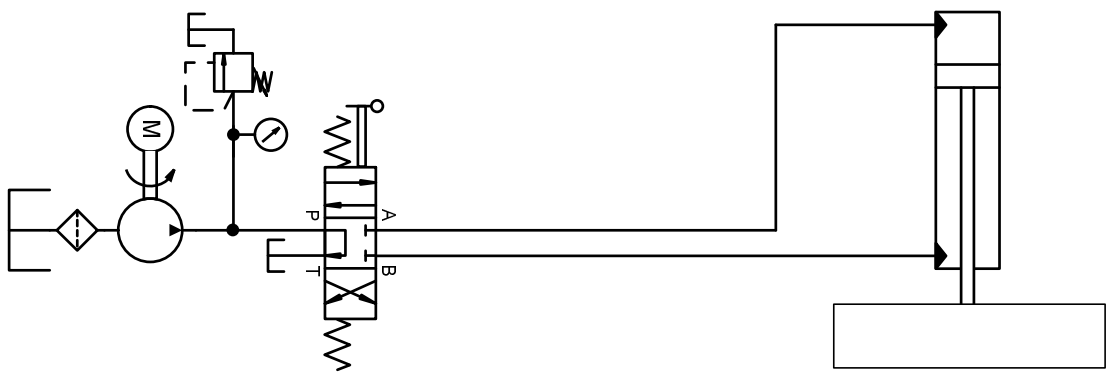
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Counter-Balance Valve

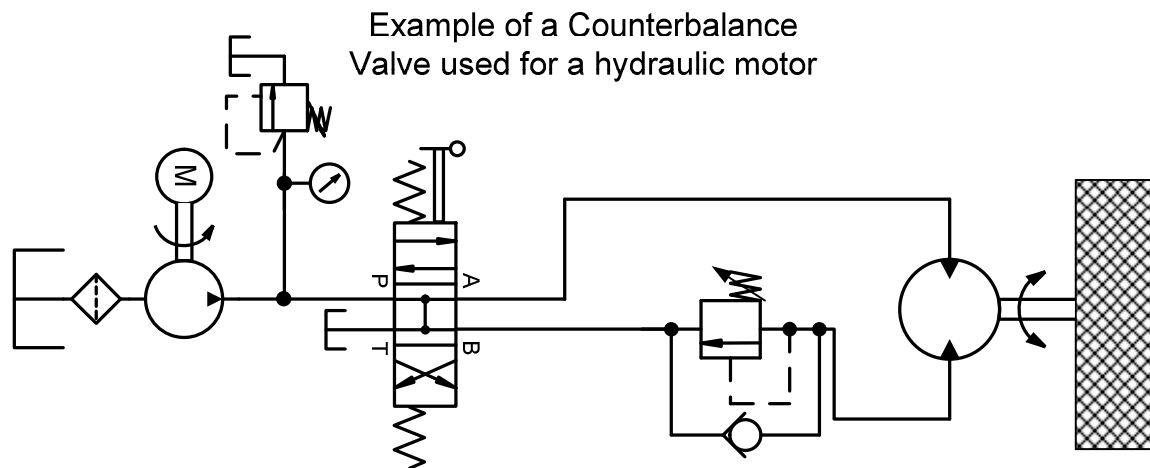


Purpose: Balance an over-running (assisting) load.

Description: Normally closed
 Spring loaded
 Adjustable
 Internal or external pilot
 Internal or external drain
 Bidirectional flow requires by-pass check valve



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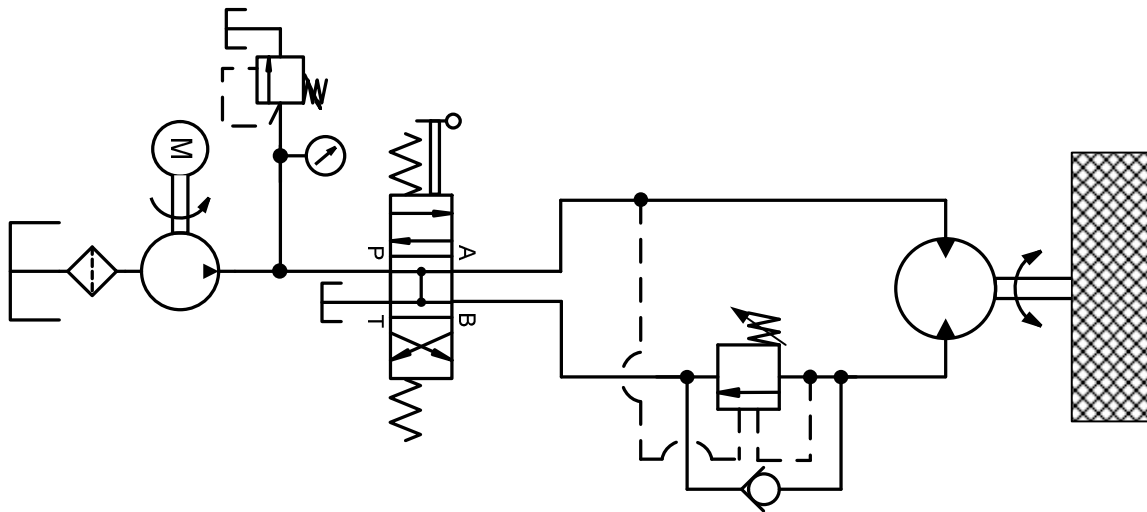
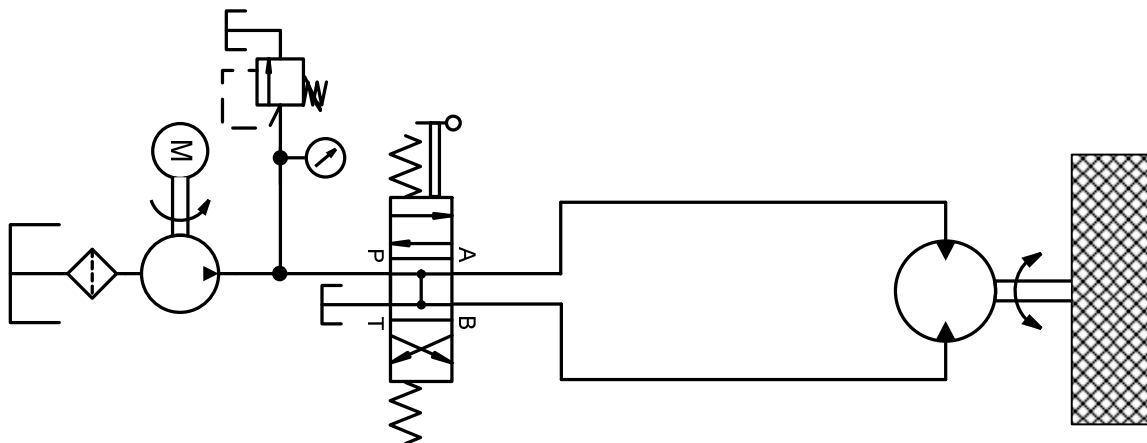
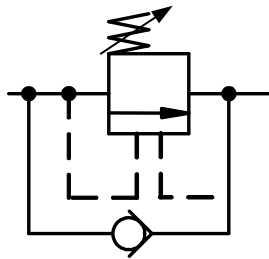
Usually the spring setting on the counter balance valve is set at 100 to 200 psi above the pressure created by the assisting load. Depending on circuit operation, response time and sensitivity of the circuit the spring setting may have to be adjusted to allow the circuit to operate properly.

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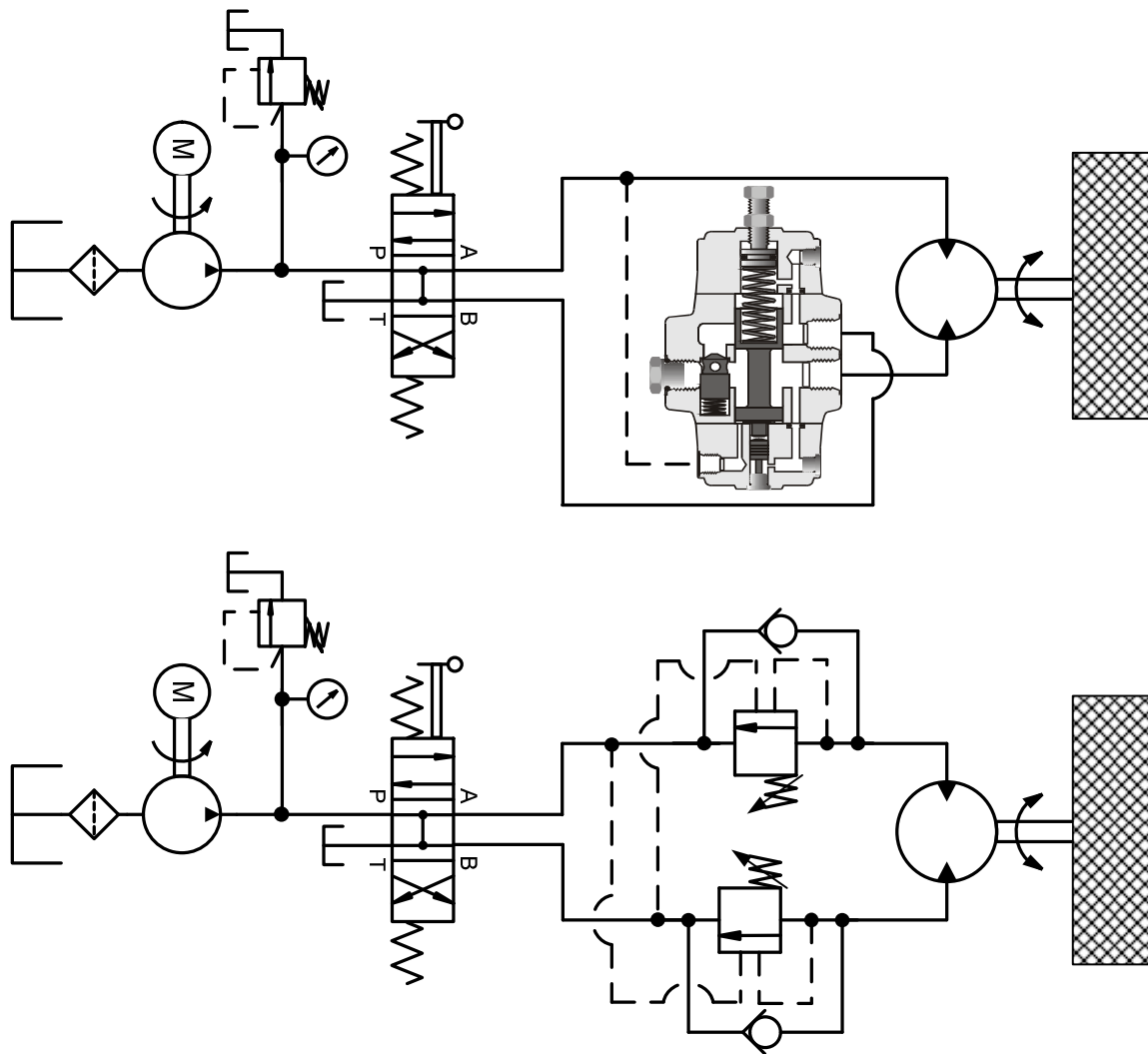
Brake Valve

Purpose: Brake or slow down a hydraulic motor.

Description: Normally closed
Spring loaded
Adjustable
Uses both internal and external pilot
Internal or external drain
Bidirectional flow requires by-pass check valve



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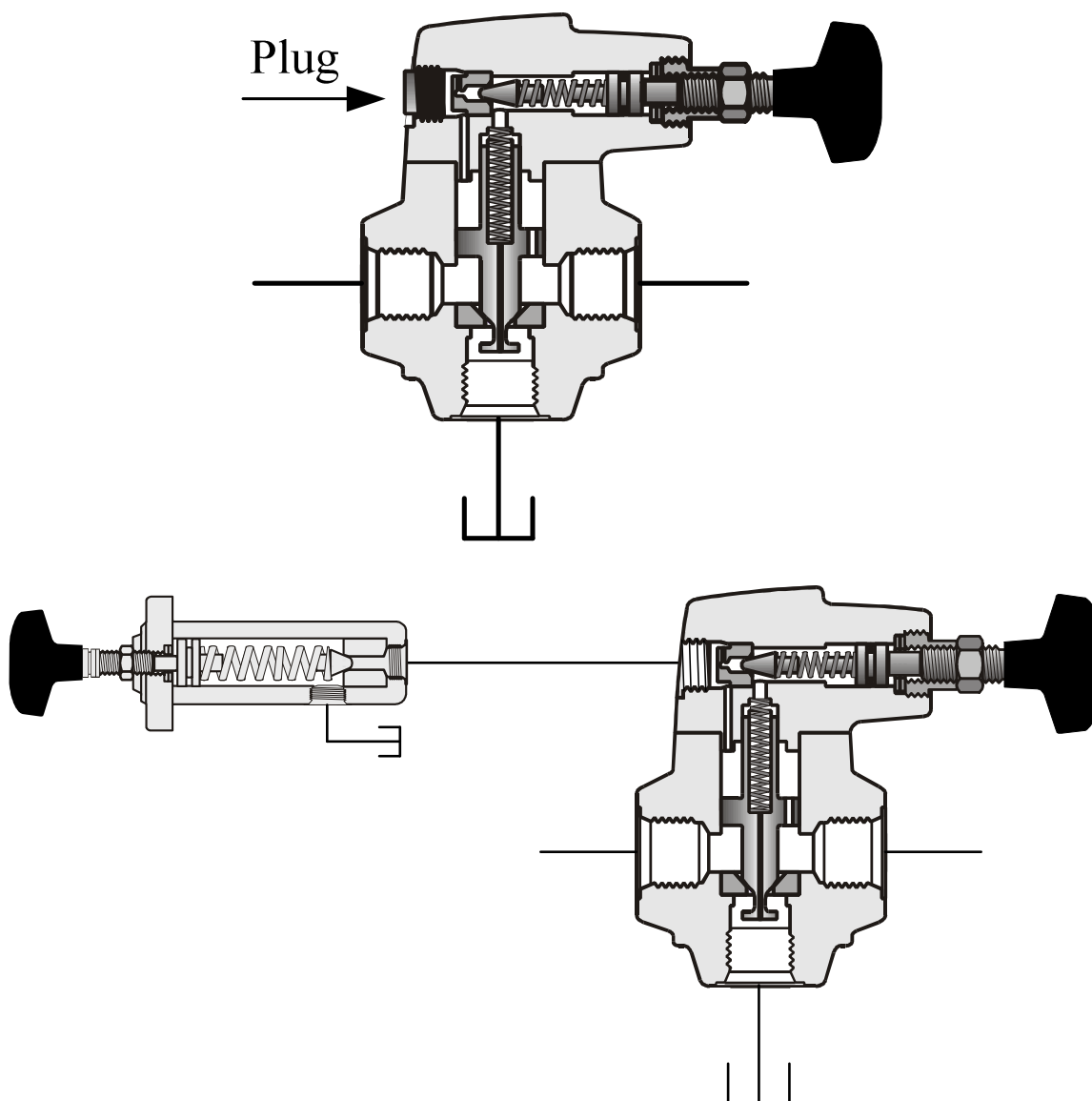
Set the spring setting on the brake valve at 100 to 200 psi above the required pressure to brake or slow down the motor, per the circuit requirements.

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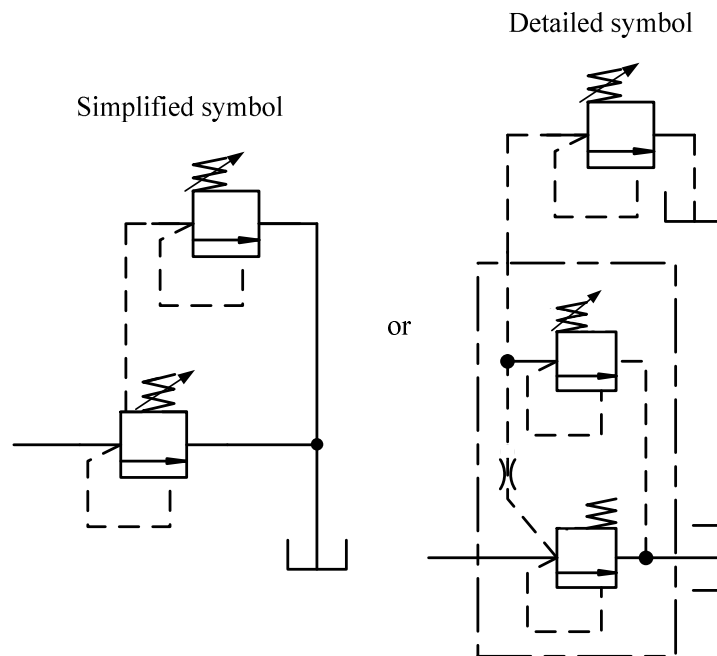
Remote Operation or Pilot operation of a Pilot Operated Pressure Control Valve

As was mentioned earlier, the compound (balanced spool or pilot operated) valve has far greater flexibility in a hydraulic circuit than the simple (Direct acting) type. Since the simple (Direct acting) type has only one closure and one spring the simple design doesn't allow it to be used in any other capacity than a pressure control valve. The two stage design of the compound (balanced spool or pilot operated) type is very useful in having the capacity to use it in variety of ways.

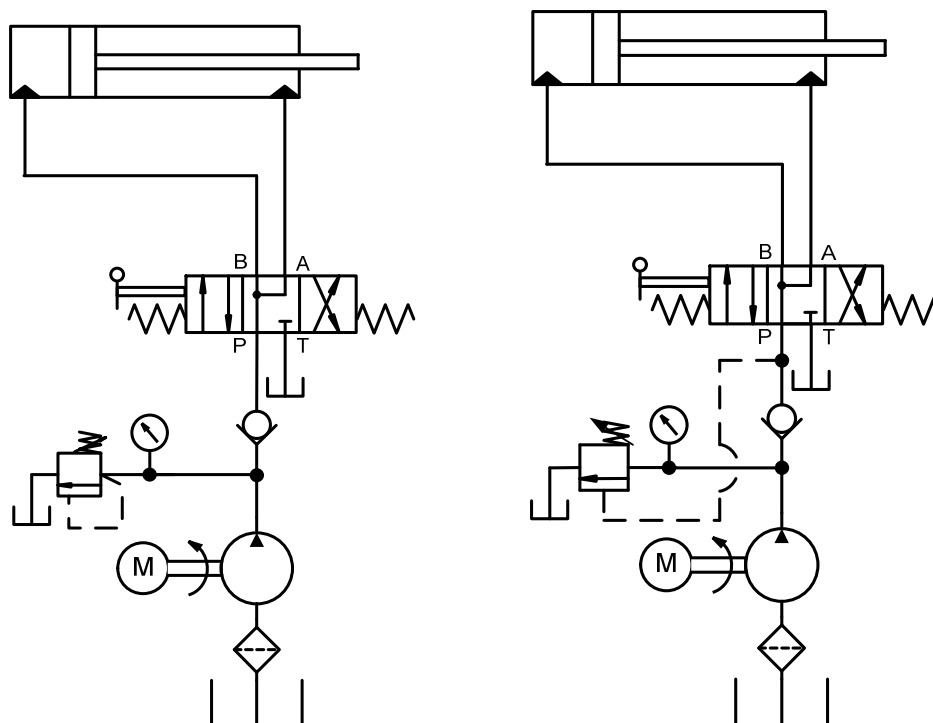
Usually when the compound (balanced spool or pilot operated) valve is used in this manner is when these names, remote operated or pilot operated, take precedence over the compound or balanced spool designations.



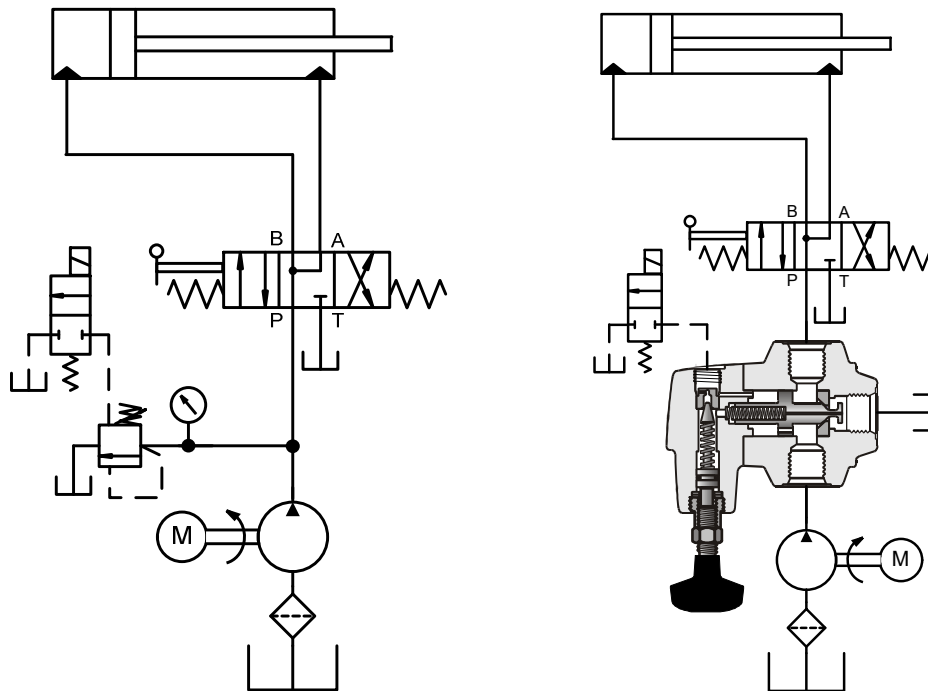
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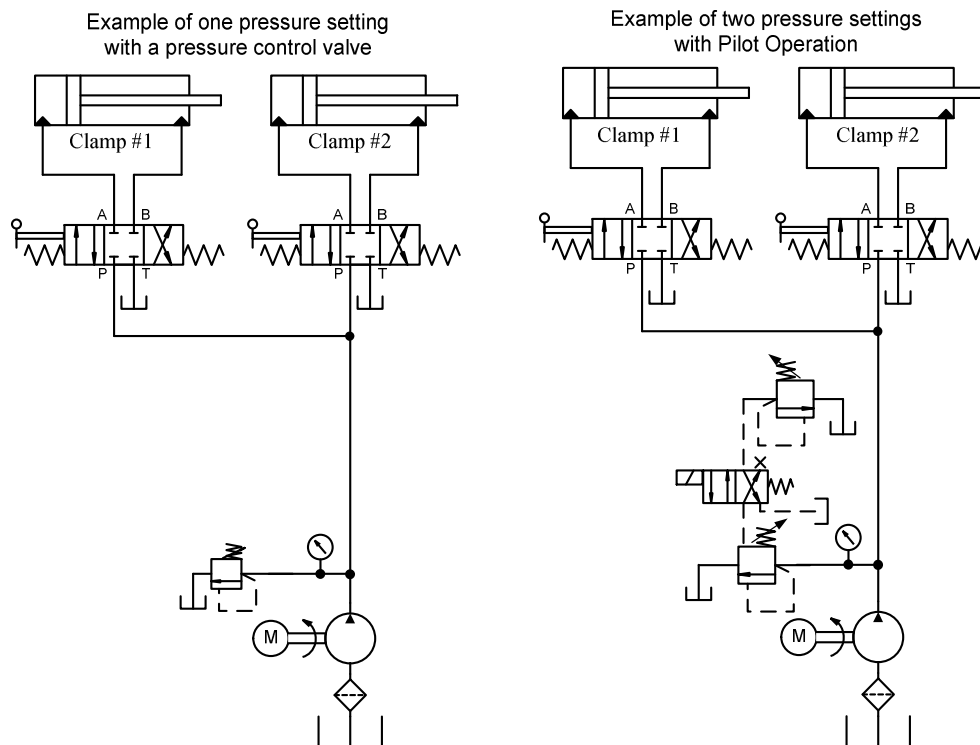
During idle time in a circuit a common way of reducing power generation and making the system more efficient is to unload the system or pump. Besides using a conventional unloading valve, it can be done other ways as illustrated below.



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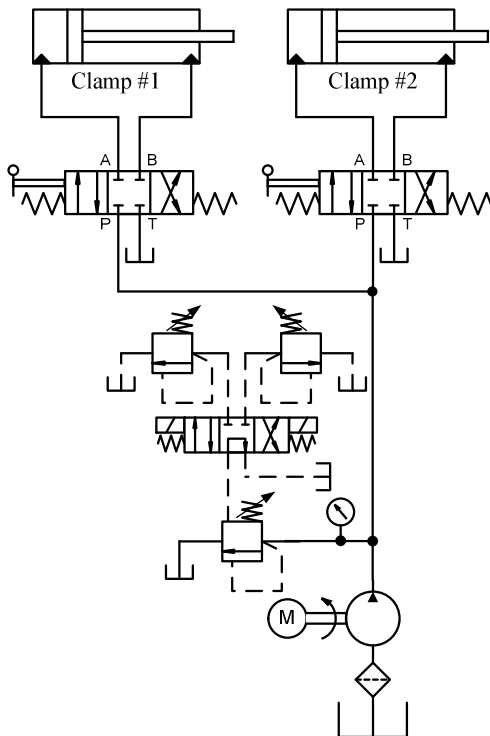


Remote / pilot operation allows the use of various other directional control and pressure control valve combinations to satisfy just about any circuit requirement. Some common examples are illustrated below.

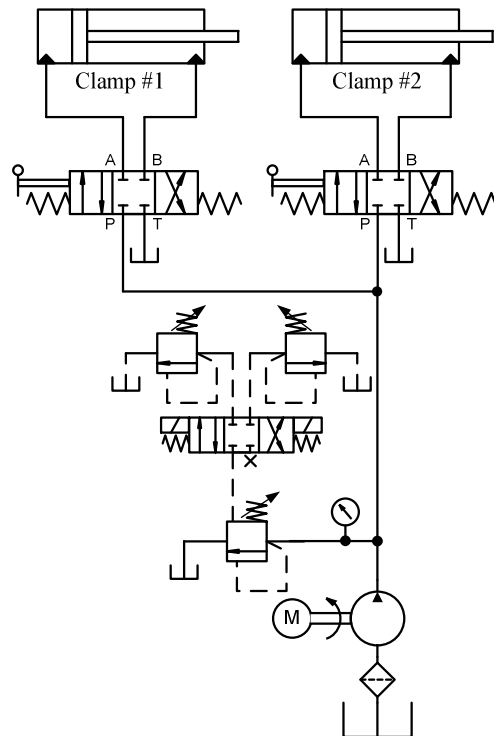


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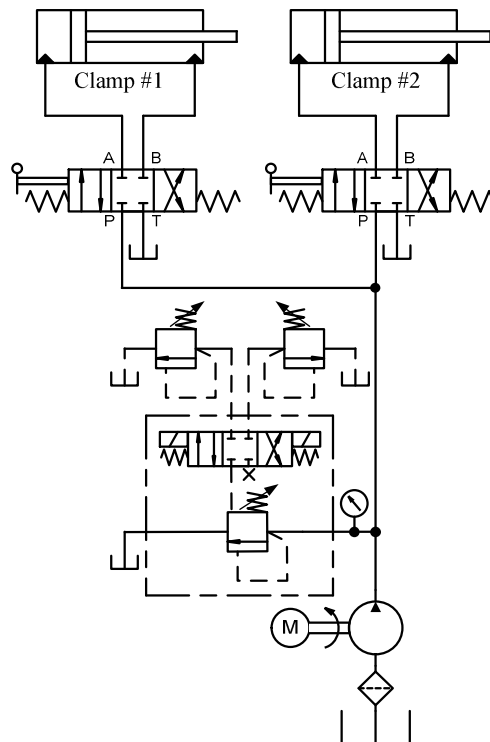
Example of two pressure settings with Pilot Operation & unloading the pump



Example of three pressure settings with Pilot Operation



Example of three pressure settings with built-in Solenoid/Pilot Operation

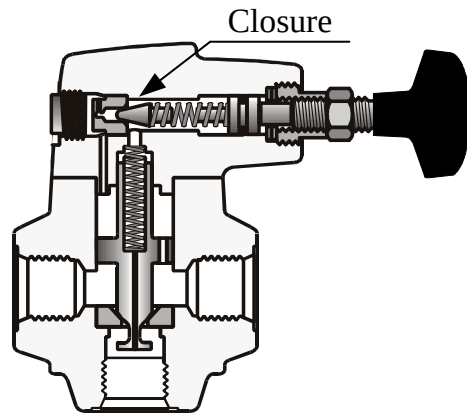


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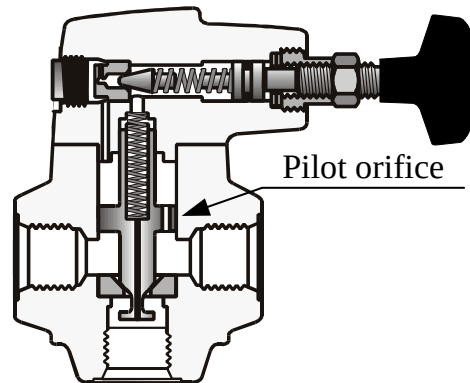
Operating Problems with Pilot Operated Pressure Control Valves

Specific problems may occur when using pilot operated types of pressure control valves. Most problems can be attributed to three main characteristics; closure sealing, clogging or plugging of the pilot orifice and line lengths that are excessive between the remote control valve and the main valve.

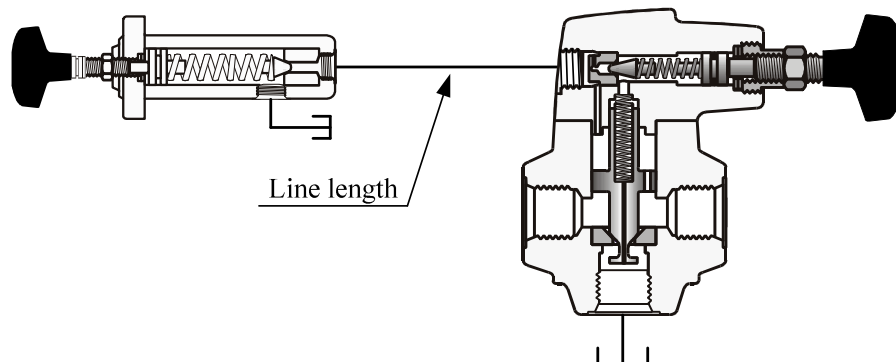
When a pilot operated valve operates, the closure moves in and out, seating and unseating. Wear and tear on the closure and seat, also if any contamination does not allow the closure to seat, may cause problems. If the closure cannot seat or seal properly, fluid will leak around the closure. If fluid is allowed to leak around the closure the top of the main spool cannot build pressure and balance the spool, which will not allow the valve to operate consistently. If this happens the pressure setting on the valve and system pressure will become erratic.



The pilot orifice is a very small hole or passage way. If the hole becomes partially clogged from contamination it will act erratic, since fluid cannot enter the top part of the spool and balance it out. If the hole is completely plugged, fluid cannot enter the top part of the spool at all. The spool may move or actuate one time but will stay open under low pressure, usually around 25 psi. The valve will dump full pump flow under low pressure.



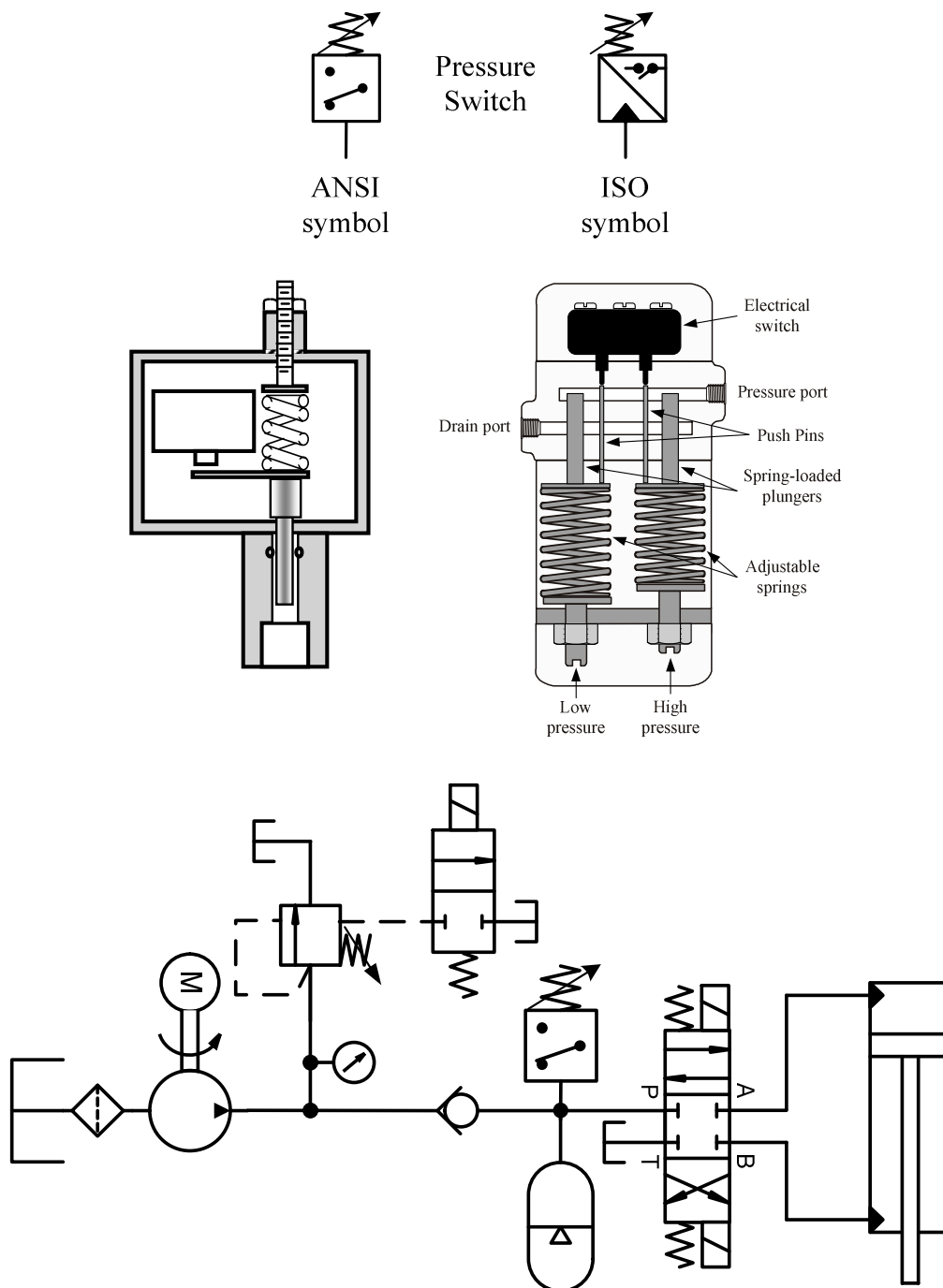
The farther a remote/pilot valve is positioned from the main valve may possibly create problems. Typically excessive line lengths allow a fluids' compressibility to be a problem, along with a chance of pressure pulsations and valve springs which interact and cause excessive noise and vibration. Line lengths should be kept to minimum or no greater than 20 feet, depending on the circuit.



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Pressure Switch

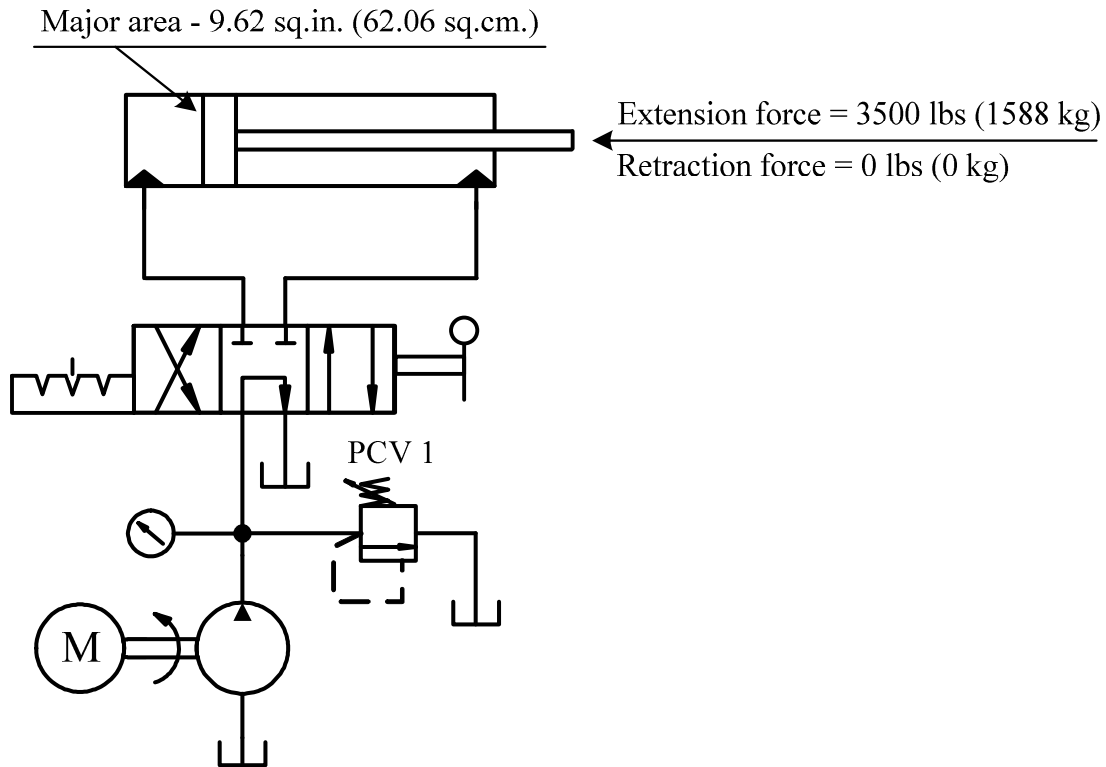
Pressure switches are used to open or close (make or break) electrical circuits at a desired pressure to control other devices. Pressure switches may be designed to use one pressure value or two pressure values.



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Pressure Control Valve Exercises

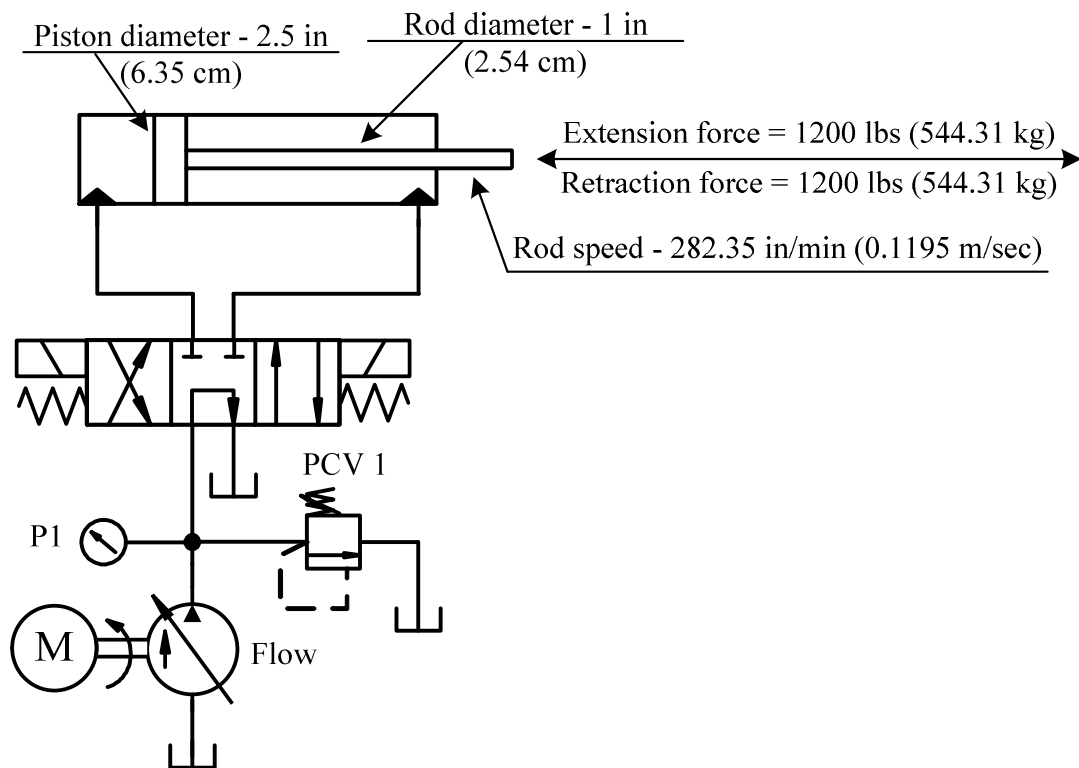
PCV Exercise #1



- 1) In the circuit illustrated, the horizontal mounted cylinder has a load during extension, but has no load during retraction.
(Disregard all other internal resistance. Use 150 psi (10.55 bar) for the cracking pressure)
 - a) What would be the spring setting for the pressure relief valve?

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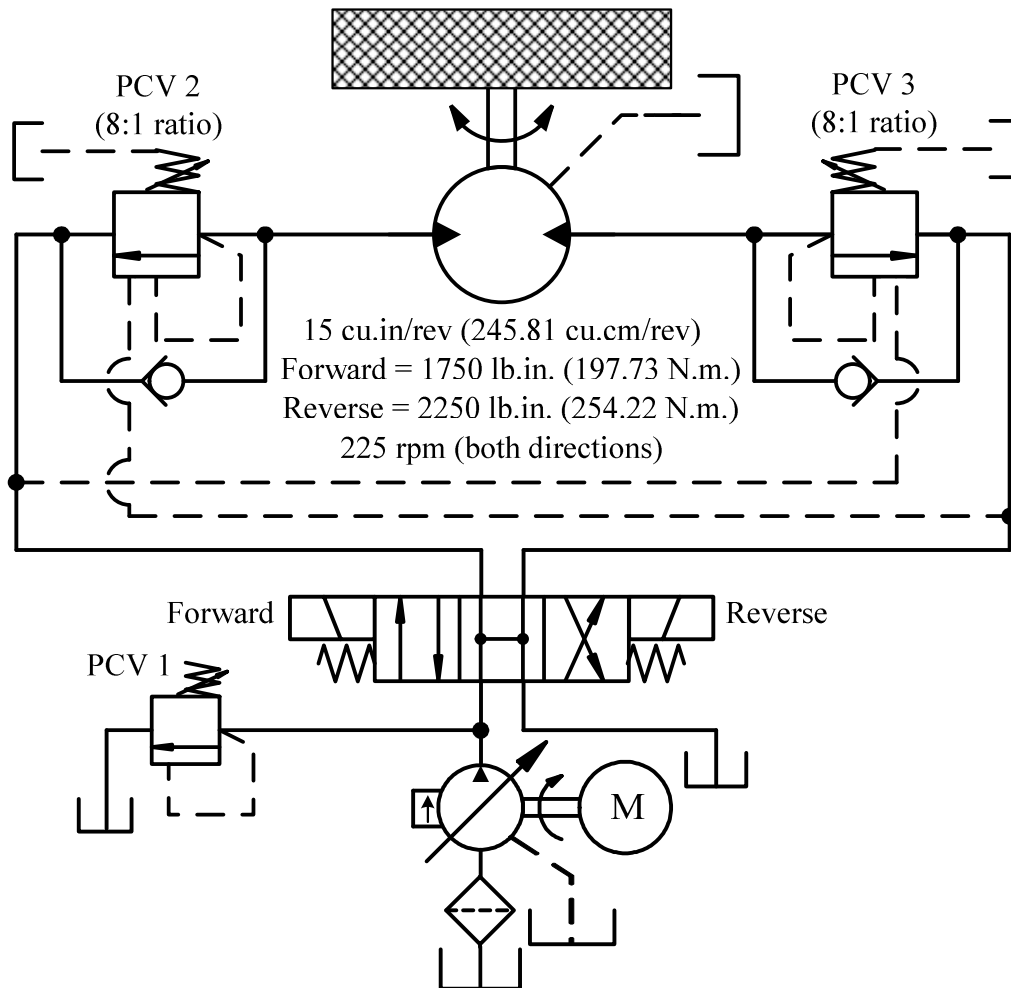
PCV Exercise #2



- 2) In the circuit illustrated, the horizontal mounted cylinder has the same load during extension and retraction.
(Disregard all other internal resistance. Use 150 psi (10.55 bar) for the cracking pressure)
 - a) What would be the pressure setting on PCV 1?

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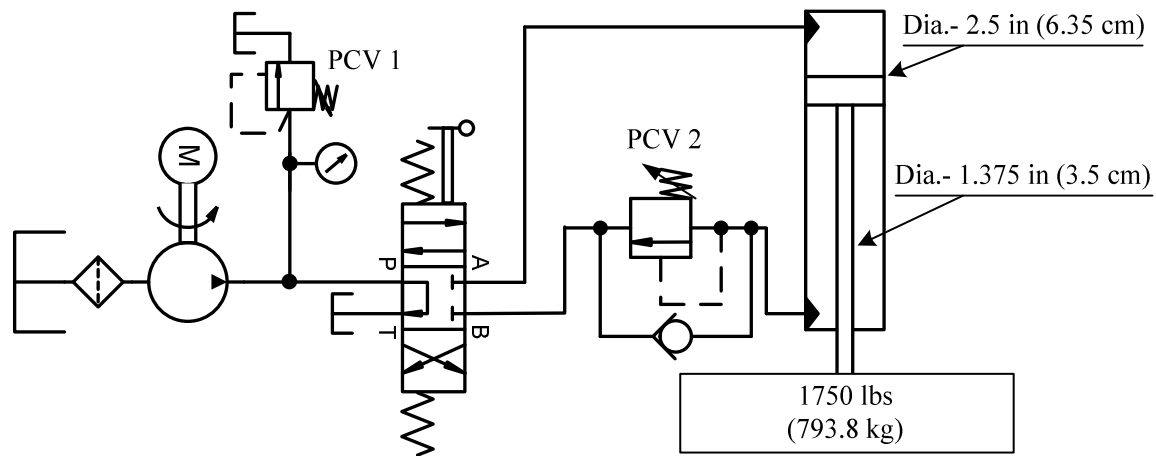
PCV Exercise #3



- 3) In the circuit illustrated:
(Disregard all other internal resistance. Use 150 psi (10.55 bar) for the cracking pressure.)
- What would be the spring setting of PCV 1?
 - What would be the spring setting of PCV 2?
 - What would be the spring setting of PCV 3?
 - What type of PCV is PCV 2?

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PCV Exercise #4



- 4) In the circuit illustrated, the vertical mounted cylinder has the same load during extension and retraction.
(Disregard all other internal resistance. Use 150 psi (10.55 bar) for the cracking pressure.)

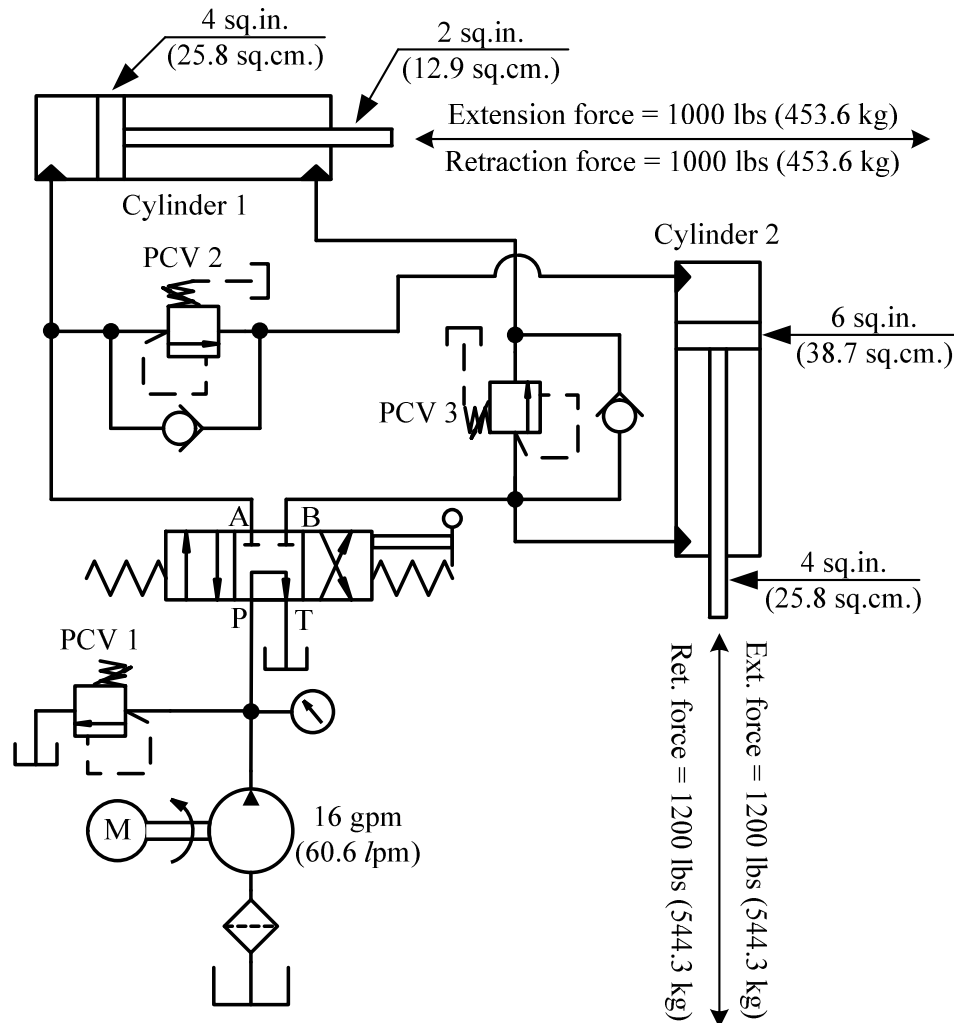
a) What would be the spring setting of PCV 1?

b) What would be the spring setting of PCV 2?

c) What type of PCV is PCV 2?

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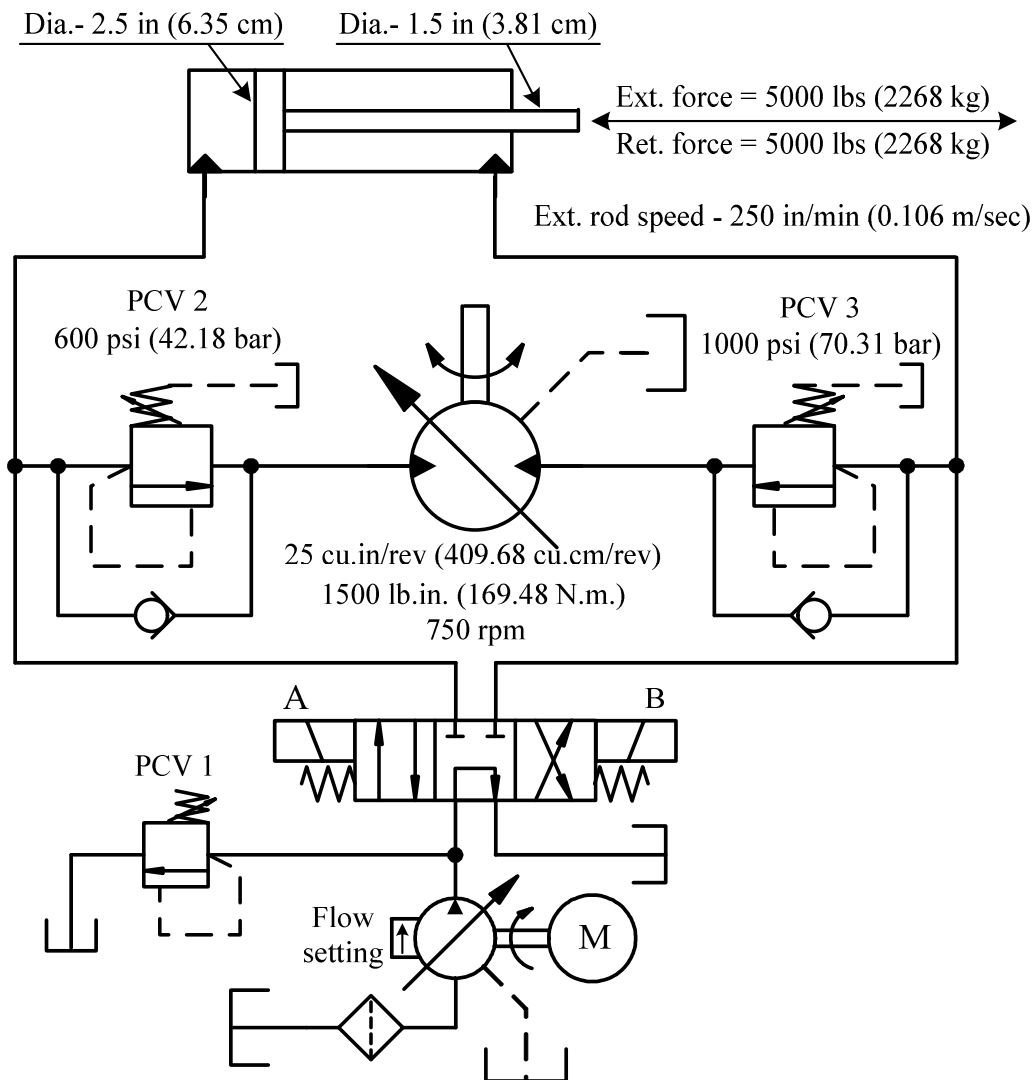
PCV Exercise #5



- 5) In the circuit illustrated, both cylinders are horizontal mounted and have the same load during extension and retraction.
(Disregard all other internal resistance. Use 150 psi (10.55 bar) for the cracking pressure.)
 - a) What would be the spring setting of PCV 1?
 - b) What would be the spring setting of PCV 2?
 - c) What would be the spring setting of PCV 3?
 - d) What type of PCV is PCV 3?

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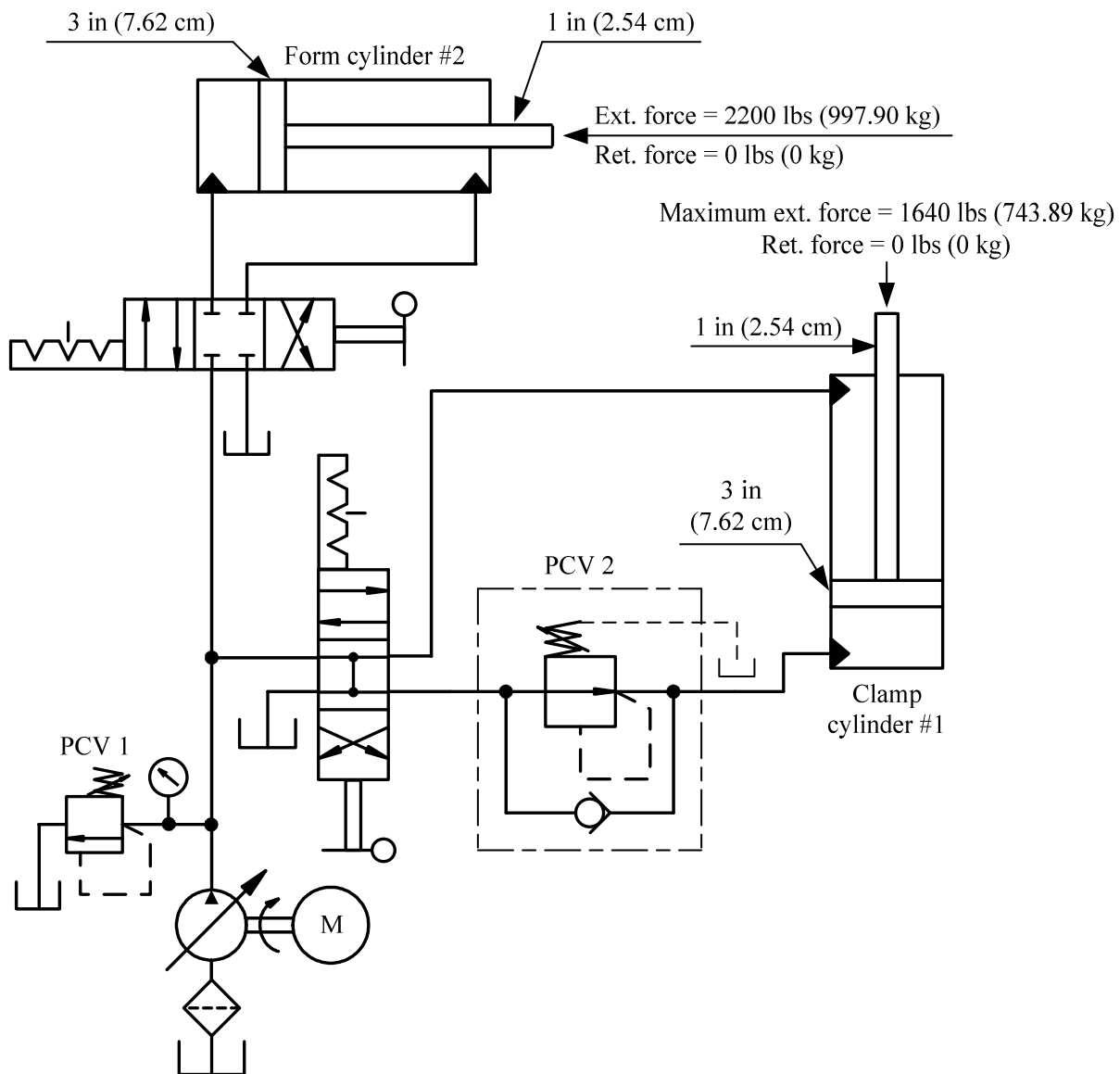
PCV Exercise #7



- 7) In the circuit illustrated, the horizontal mounted cylinder has the same load during extension and retraction.
(Disregard all other internal resistance. Use 150 psi (10.55 bar) for the cracking pressure.)
- Are the pressure settings for PCV 2 and PCV 3 high enough? If not, what should they be?
 - What would be the spring setting for PCV 1?

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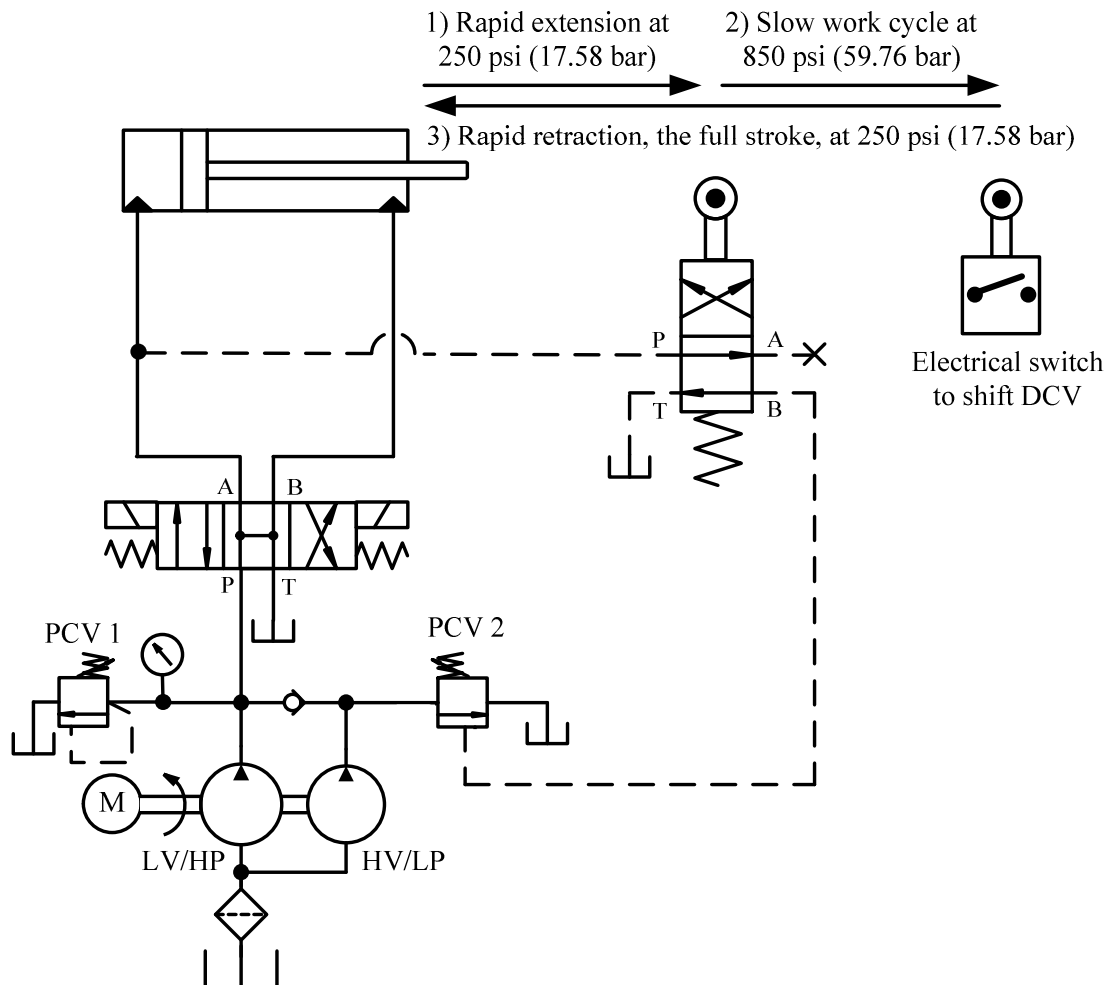
PCV Exercise #8



8. In the circuit illustrated, both horizontal mounted cylinders have a load during extension but have no load during retraction.
(Disregard all other internal resistance. Use 150 psi (10.55 bar) for the cracking pressure.)
 - a. What would be the spring setting for PCV 2? Clamp cylinder #1 cannot exceed 1640 lbs (743.89 kg).
 - b. What would be the spring setting for PCV 1?
 - c. What type of PCV is PCV 2?

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PCV Exercise #9



9. In the circuit illustrated, the cylinder is horizontal mounted and the pressures are listed above. (Disregard all other internal resistance. Use 150 psi (10.55 bar) for the cracking pressure.)
 - a. What would be the spring setting for PCV #1?
 - b. What would be the spring setting for PCV #2?
 - c. What type of PCV is PCV #2?

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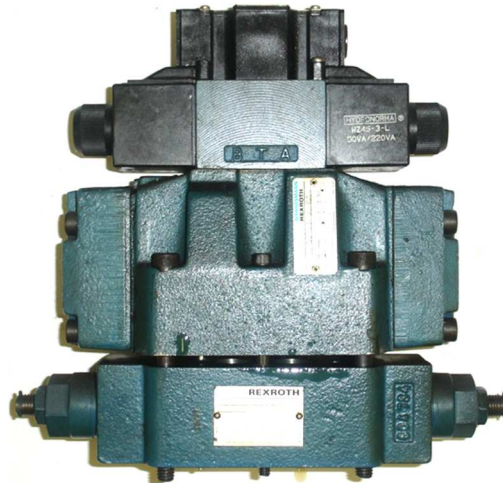
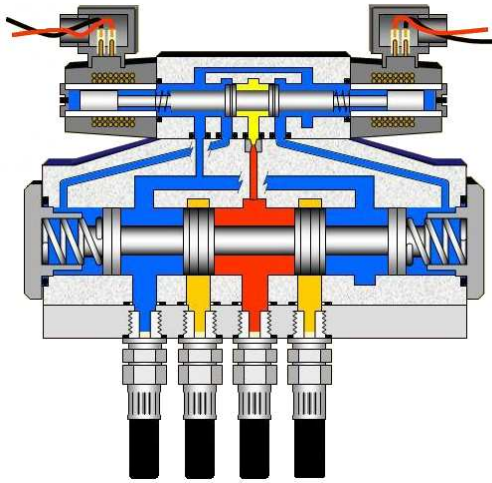
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Directional Control Valves

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Introduction to Directional Control Valves



Directional control valves provide three important functions in a hydraulic circuit;

1. It can stop flow.
2. It can allow flow.
3. It can change direction of flow.

To perform these functions a directional control valve uses a body that houses a moving device which connects and disconnects internal passages. The moving device may be of several different designs;

1. Ball
2. Poppet
3. Dart
4. Sliding spool
5. Rotary spool
6. Linear sliding plate
7. Rotary sliding plate

Directional control valves are defined/classified by their principal characteristics;

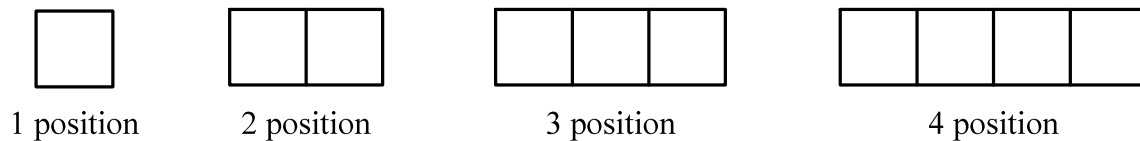
1. Number of positions.
2. Number of ports.
3. Number of ways (flow paths).
4. Condition of the valve (If necessary).
5. Type of center. (If necessary)
6. Methods of control/operation (actuators).
7. Size or Maximum ratings.
8. Type of connections.

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Directional Control Valve Characteristics

1. Number of positions.

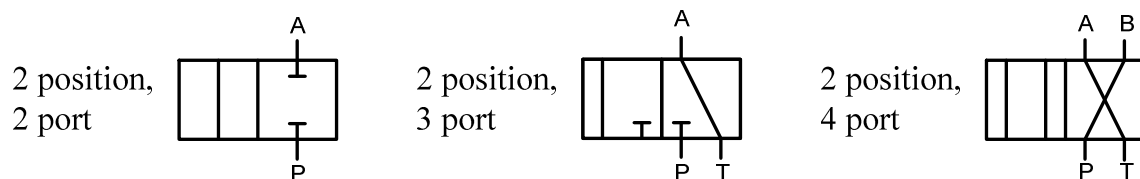
The number of positions is defined as how many boxes (envelopes) are included in the valve symbol. Each box (envelope) contains the symbolic representation of the condition or flow pattern for that particular position.



2. Number of ports.

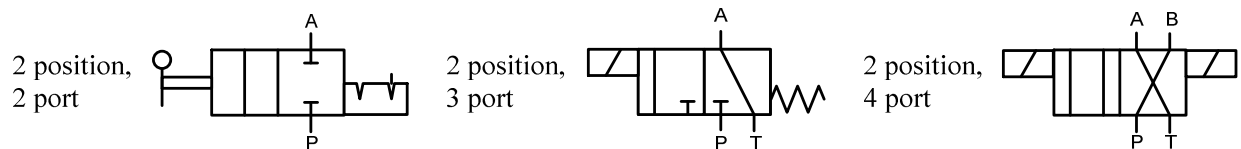
The number of ports is defined as how many fluid lines (physical connections) are included in one position of the valve symbol. **Do not** count the number of ports across the entire valve symbol, the valve can only utilize one position at any time. Only count the number of ports in the at home (at rest or de-energized) position.

Several factors will determine the at home (at rest or de-energized) position for a valve. **First**, hydraulic schematics/blueprints are drawn in the at home (at rest or de-energized) position. **Secondly**, that means the valves contained in the circuit will probably be drawn in the at home (at rest or de-energized) position, unless the circuit dictates otherwise. Sometimes the overall purpose or a specific operation in the circuit means that one or more valves in the circuit may be drawn in the energized or made position. Other times a valve may be drawn in the energized or made position for safety reasons. Those are a few of the reasons why a valve representation may deviate from the at home (at rest or de-energized) guideline. Only the circuits' purpose or operation can dictate whether a valve may be drawn in the energized or de-energized position. More than likely, most valves are drawn in the at home (at rest or de-energized) position.

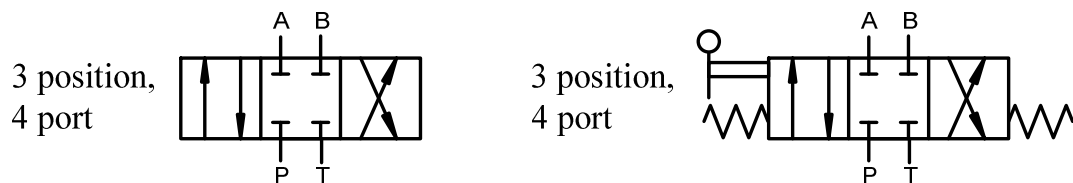


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Also, certain symbols associated with the valve will be a determining factor in the valves representation in the circuit. For example, a 2 position valve can use an actuator to shift the valve and a spring to return the valve when that energy source is removed. The spring is used to return the valve to the home position. The valve will probably stay in this position more than it will in the opposing position and usually it is not practical to leave an energy source engaged for long periods of time. Also, valves are typically drawn in their home position because that particular flow condition that exists is what the valve needs to satisfy the circuits' purpose.



For a 3 position valve, the theory is slightly different. In a two position valve, the valve is always working, it does not have an opportunity to rest or idle the circuit. Since a 3 position valve has an extra position, a common purpose for that position is to rest or idle the circuit when work is not taking place. This is commonly referred to as a home or neutral position. In a 3 position valve the center position is referred to as the home position. However since this position may not be related to any work taking place it is not included in the description of a valve, at least as far as the ports or ways are concerned. When describing the number of ports, do not use the home position on a 3 position valve. Move left or right to a working position and count them from there.

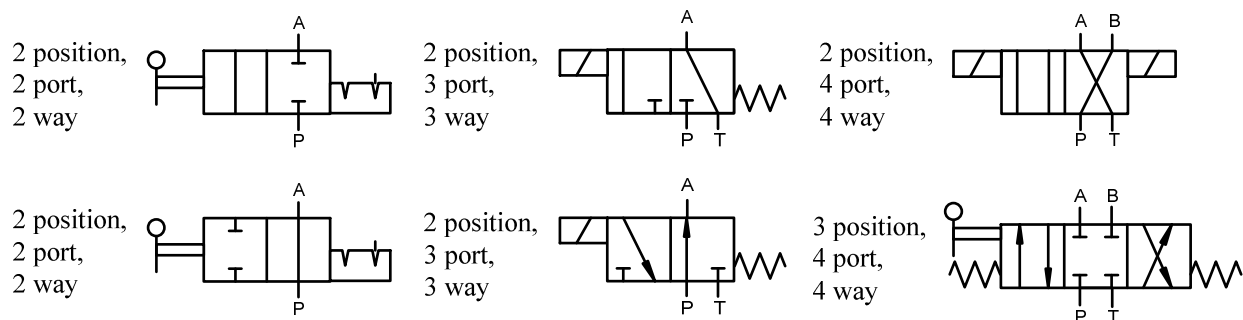


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3. Number of ways (flow paths).

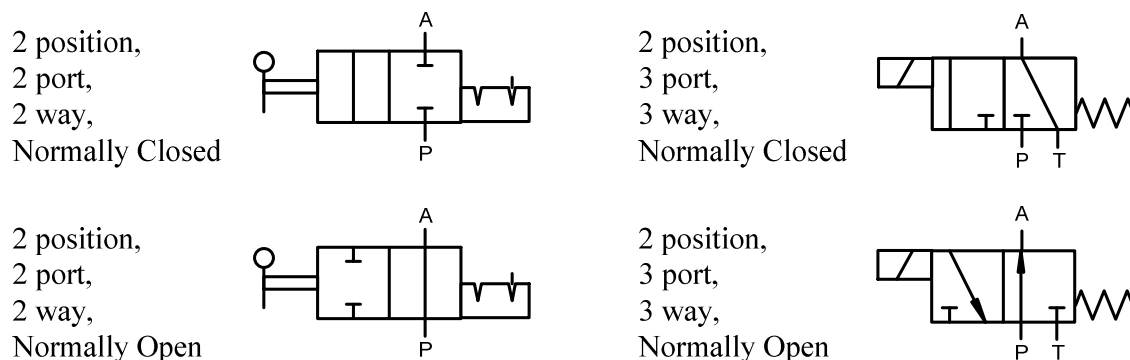
The number of ways (flow paths) is defined as how many flow paths are established in one position of the valve symbol. Again, **do not** count the number of ways across the entire valve symbol, the valve can only utilize one position at any time. Only count the number of ways in the at home (at rest or de-energized) position.

The same basic guidelines, for describing a valve, that were discussed in the previous topic of number of ports also apply with the number of ways. Establish the at home (at rest or de-energized) position for a 2 position valve and count the number of ways from there. The center position of a 3 position valve is considered a neutral, at home, position, but do not count the number of ways from the center position. Move left or right to a working position and count them from there.



4. Condition of the Valve (If necessary).

The condition of the valve refers to the flow path from the pressure port and its condition. If the flow path is allowing flow to enter the circuit then the position has a Normally Passing (Normally Open) condition. If the flow path is blocked then the position has a Normally Non-Passing (Normally Closed) condition. These terms are typically only used on 2 position, 2 way and 3 way valves.

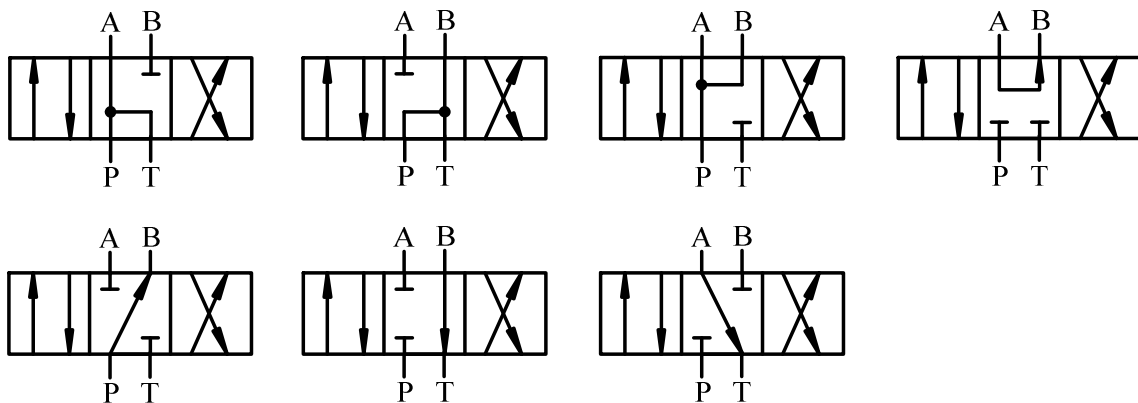


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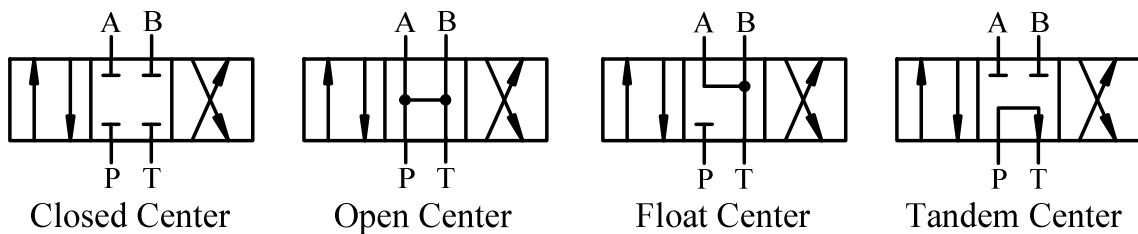
5. Type of center (If necessary).

If the directional control valve is a 3 position type, the two extreme (outside) positions are the power positions and are directly related to the actuator motion. They control the movement of the actuator in one direction and then in the other direction. The third position, or better known as the center condition, typically isn't related to the working aspect of the circuit. It is designed to perform logic, or to satisfy a specific need in the circuit.

There are a variety of flow patterns and conditions related to the center position of a 3 position valve. Some are illustrated below;



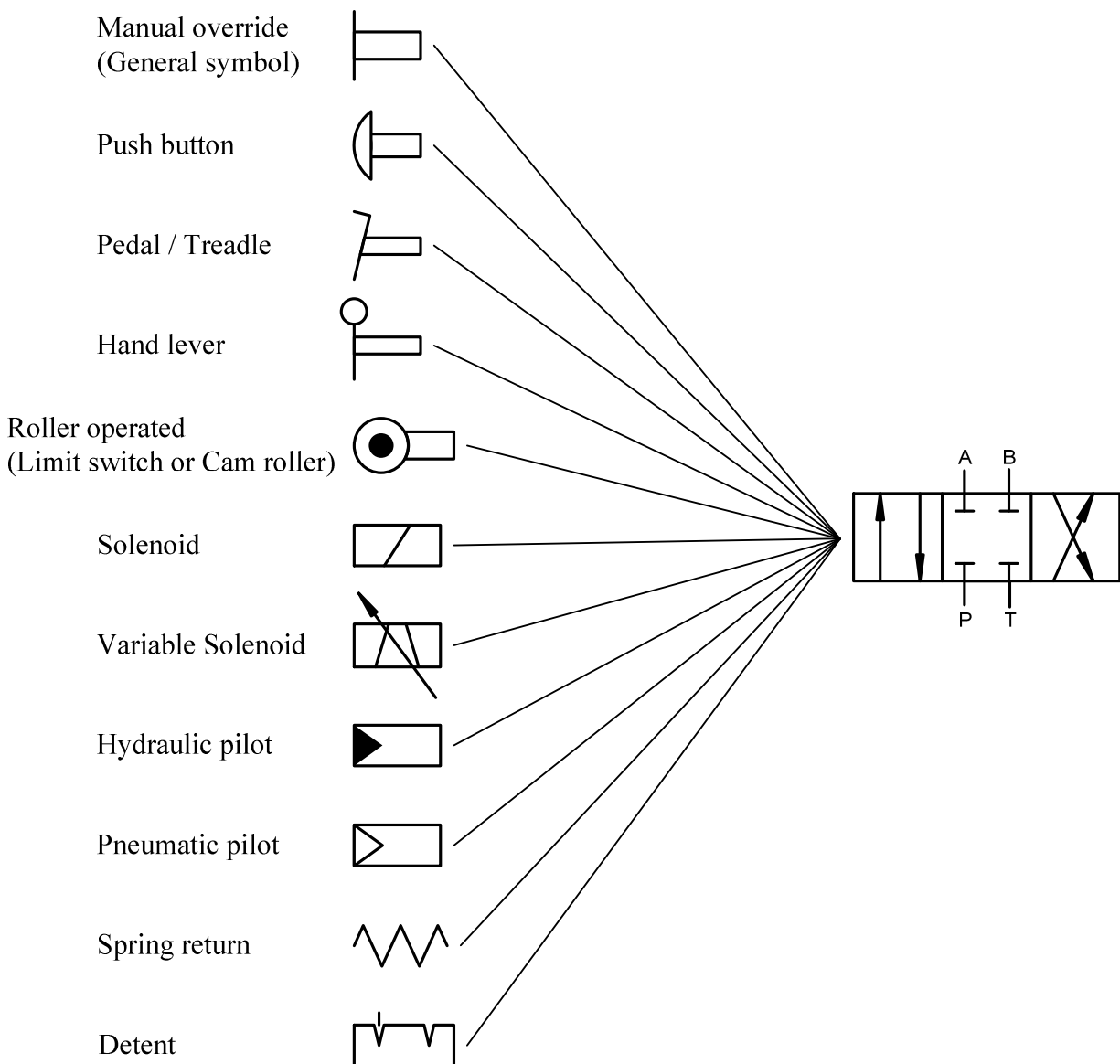
The four most common center conditions for a 3 position valve are illustrated below;



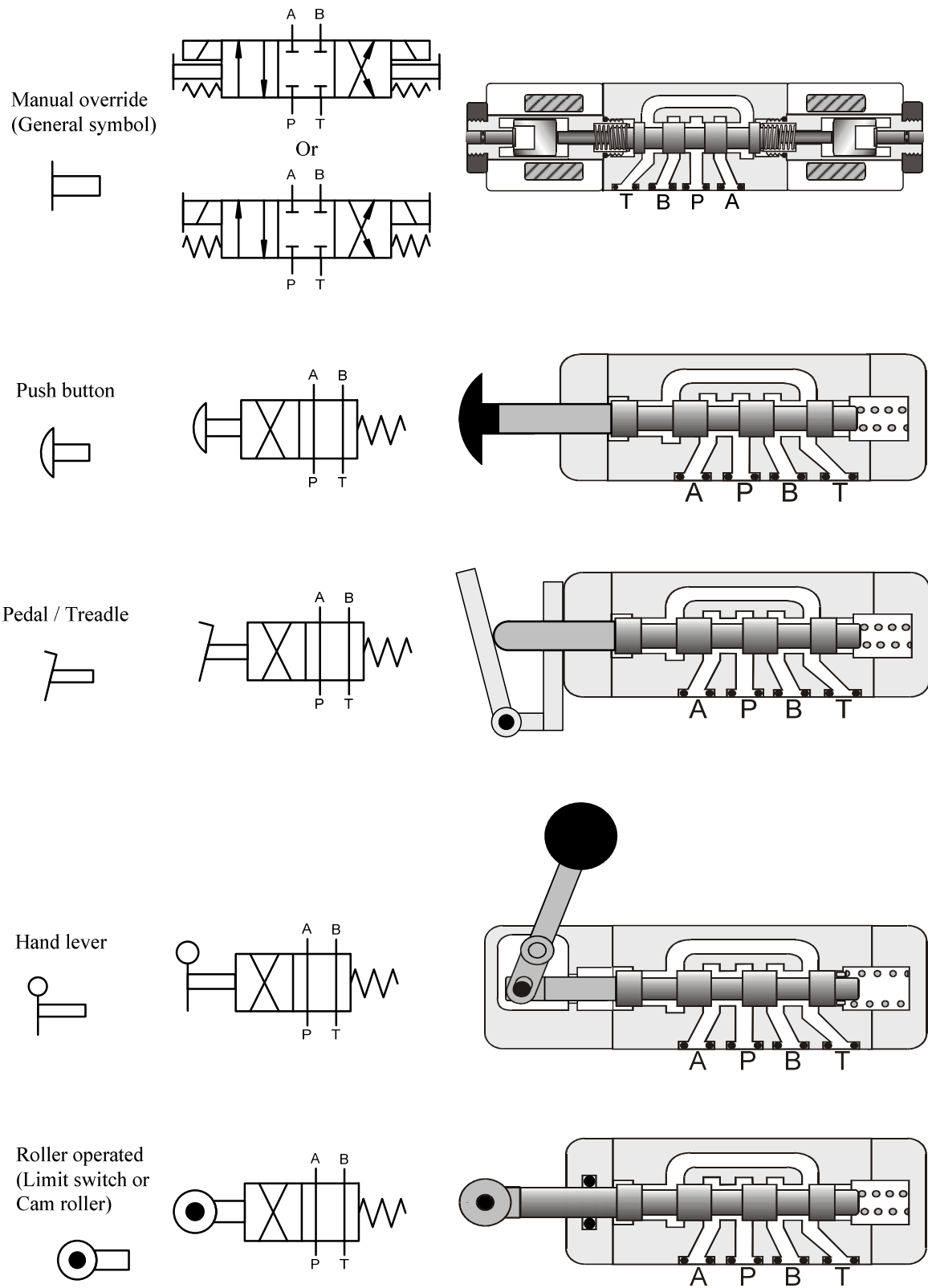
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6. Methods of control/operation (actuators).

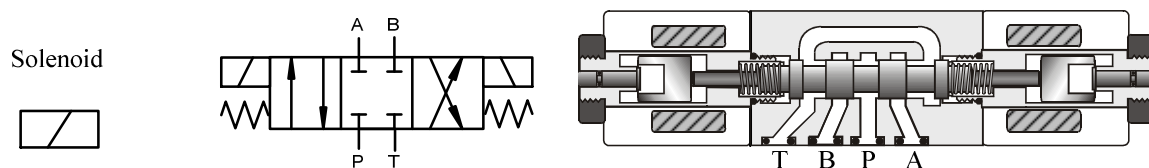
Any directional control valve will require some method of control/operation (or as some call them actuators) to move, hold in place or return to home the moving device. What dictates the method is the type of control or operation required of the circuit. Several different types of methods, even a combination, of controls may be used.



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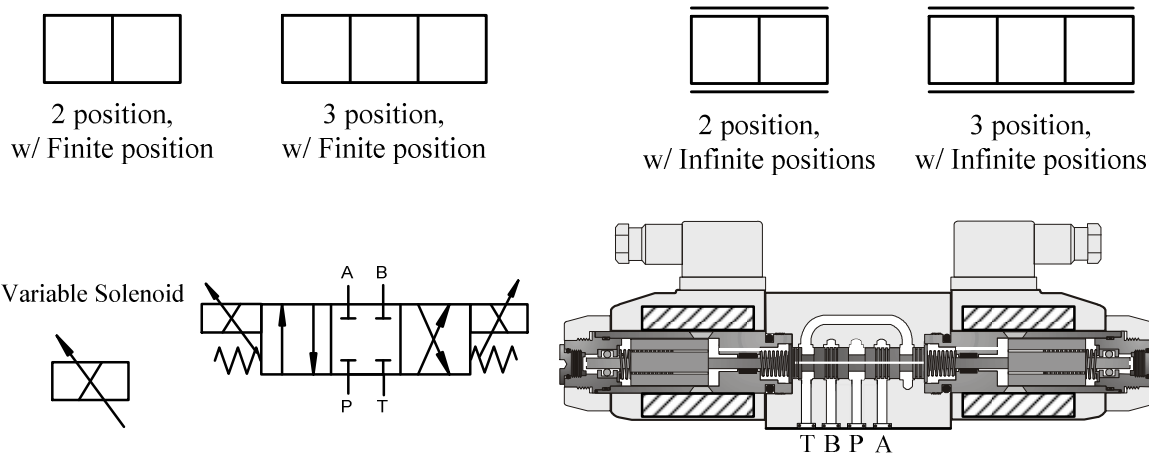
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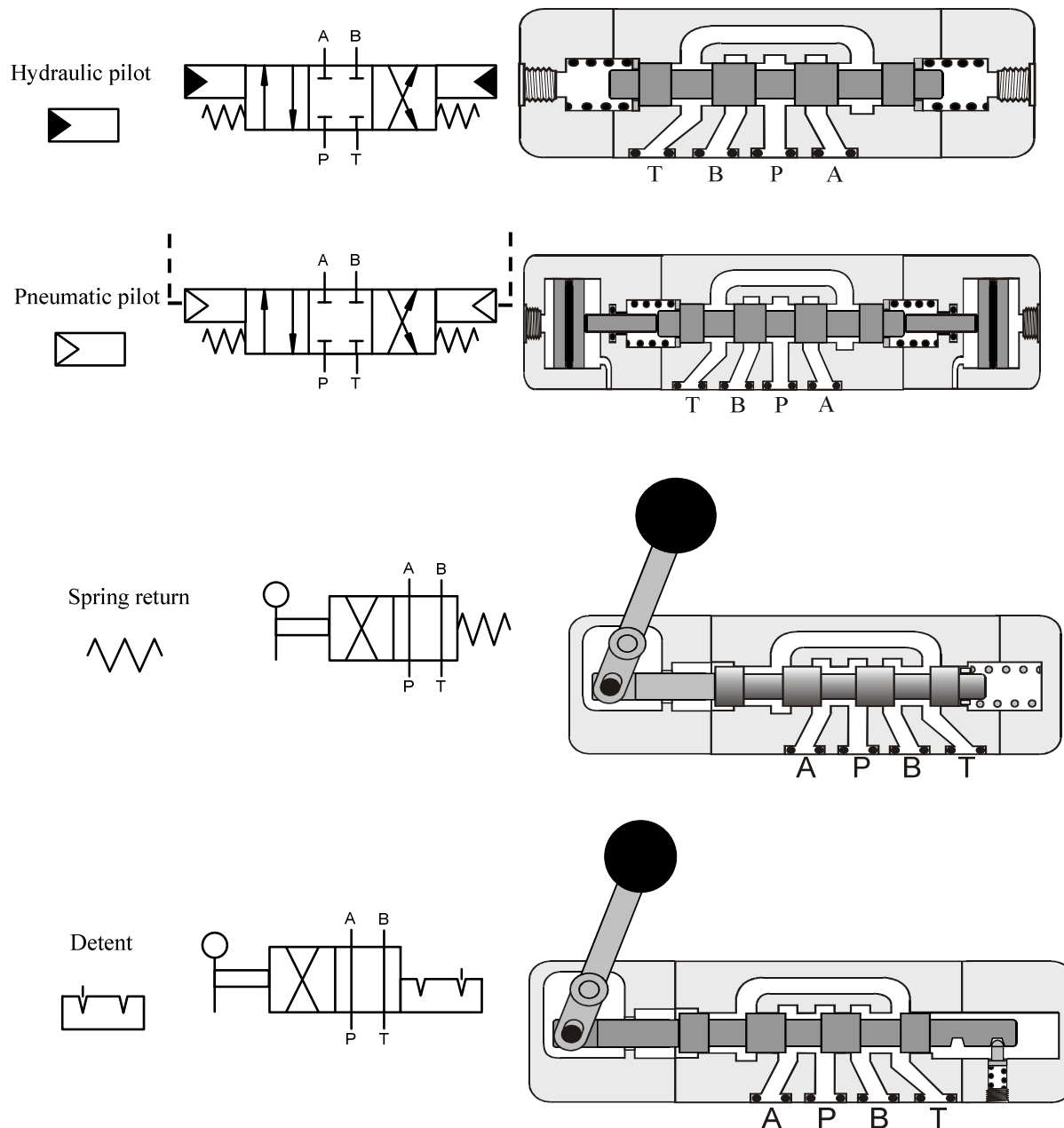
a. Finite or Infinite Positioning.

When a directional control valve is shifted it may be using a finite positioning or infinite positioning operation. Finite positioning means the valve will shift and achieve a defined or finite position. And depending on the flow pattern represented on that position it may or may not be allowing flow to happen. It will be one position or another, nothing variable in the movement. All of the methods of control/operation/actuators are considered finite positioning however some have the capability to be both, finite or infinite.

Infinite positioning means the method of control/operation/actuator is variable. When a directional control valve is shifted by a variable method it provides an opportunity for the moving device, probably a spool, to move back and forth to control the flow passage through the valve. Controlling the flow controls the position, speed or force for an actuator. Basically the valve has an infinite number of positions within the valves full movement. Proportional valves and Servo valves use this type of operation.



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7. Size or Maximum ratings.

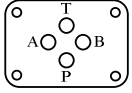
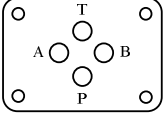
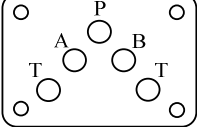
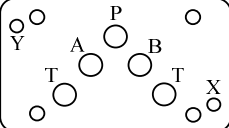
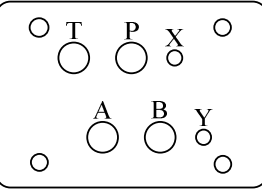
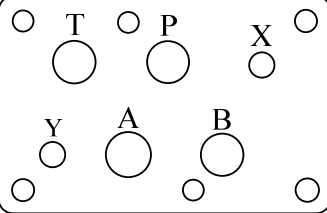
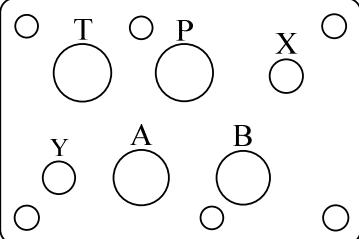
The size or maximum rating indications can include, but not limited to, everything from nominal size of the pipe connections, size of the mounting plate, maximum rated gpm to the maximum pressure rating.

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8. Type of connections.

The type of connection the dcv has to the system could be by the use of pipe threads, straight threads, flanged pipe connections, or manifold (sub-plate) mounted. Usually it is more practical to use a manifold or sub-plate mounting for ease of installation and replacement of valves because the existing piping does not have to be disturbed.

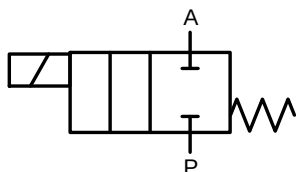
NFPA and ISO Manifold (Sub-Base) information & porting for hydraulic DCV

<u>NFPA</u>	<u>ISO</u>	<u>CETOP</u>	<u>NG</u>	<u>Port Diameter</u>	<u>Flow Rating</u>	<u>Port Configuration</u>
D02	02	2	4	0.178"	5 gpm	
D03	03	3	6	0.295"	10 gpm	
D05	05	5	10	0.440"	20 gpm	
D05H	05	NA	NA	0.440"	25 gpm	
D07	07	7	16	0.690"	30 gpm	
D08	08	8	25	0.984"	60 gpm	
D10	10	10	32	1.250"	100 gpm	

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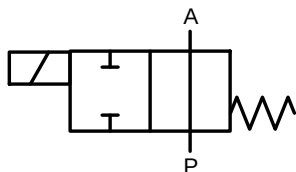
Directional Control Valve Circuit Applications

2 position,
2 port,
2 way,
Normally Closed



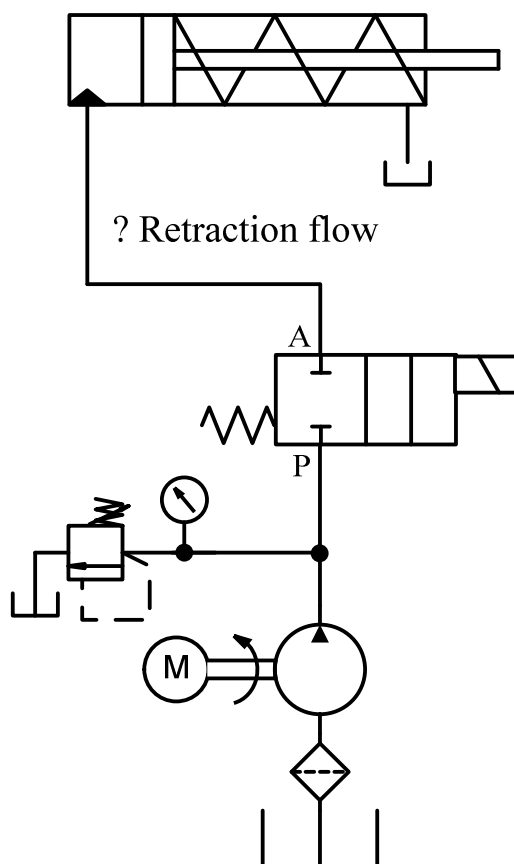
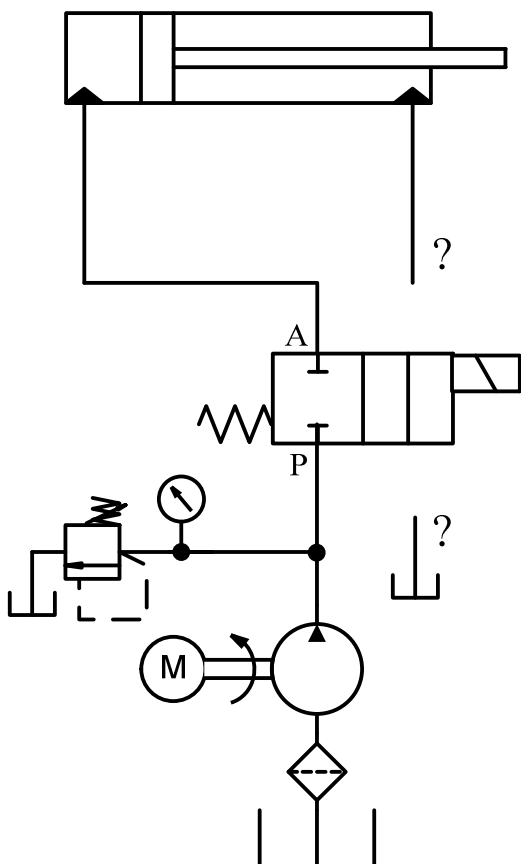
A 2 position – 2 port – 2 way dcv has two passage ways which are connected or disconnected by the moving device.

2 position,
2 port,
2 way,
Normally Open



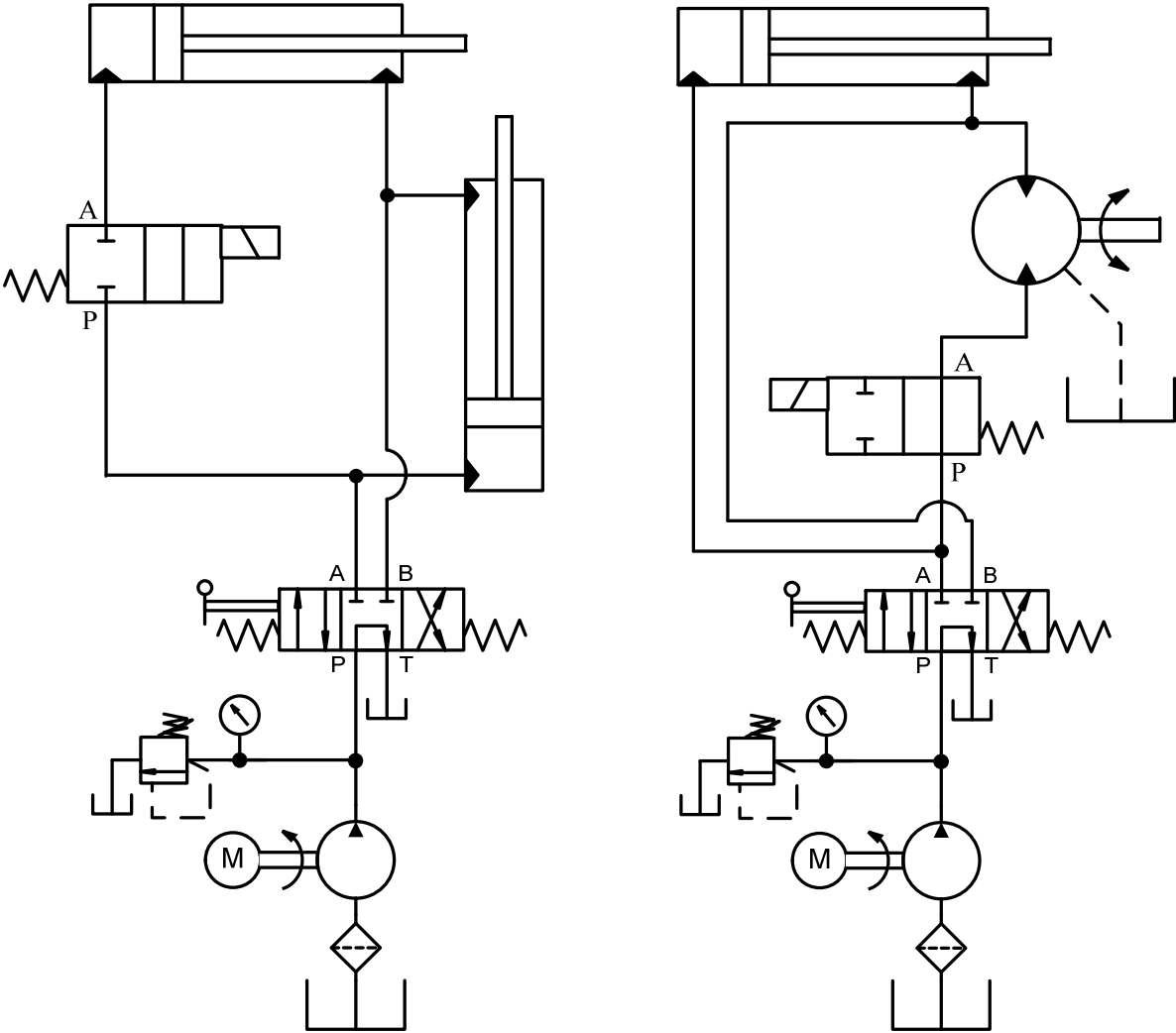
This type of valve provides an on / off function and is not practical to power an actuator. For example it can be used as a safety lockout, to isolate the circuit or a certain component or to connect or disconnect various parts of the system.

Examples of why a 2 position, 2 port, 2 way will not work in a conventional application.



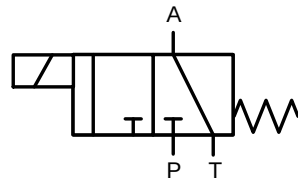
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Examples of how a 2 position, 2 port, 2 way can work in a conventional application.

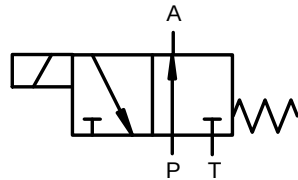


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2 position,
3 port,
3 way,
Normally Closed



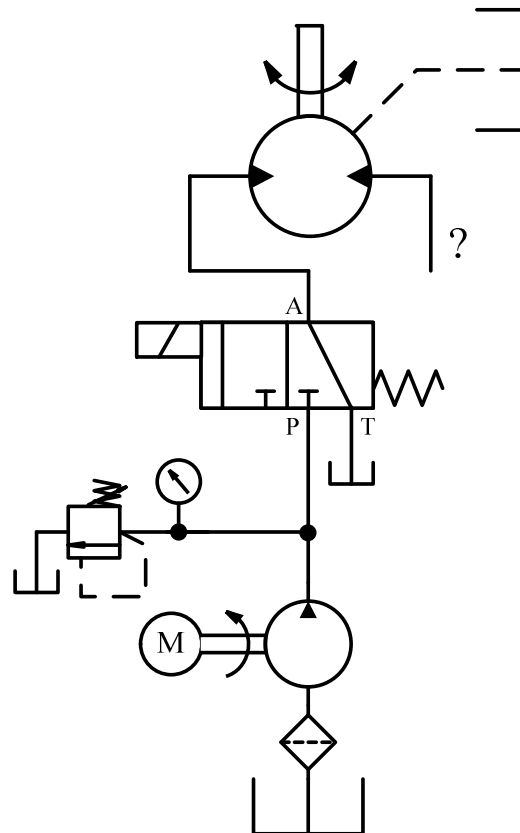
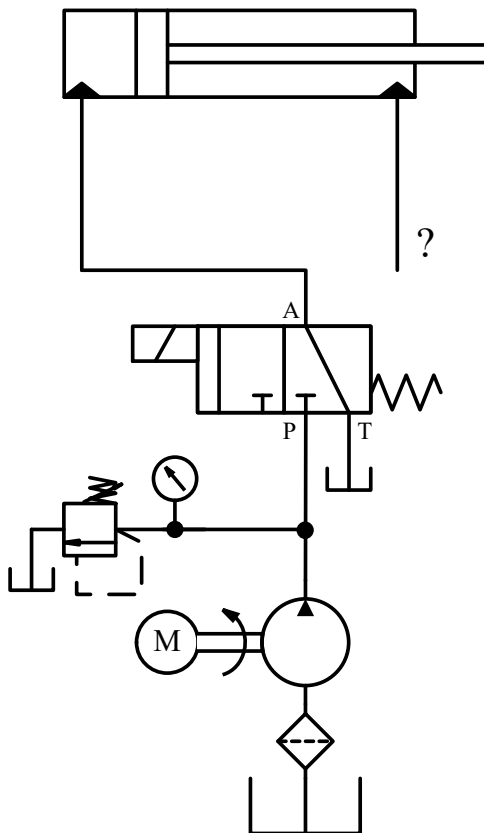
2 position,
3 port,
3 way,
Normally Open



A 2 position – 3 port – 3 way dcv has three passage ways; one pressure, one actuator (utilization) and one tank (reservoir). The extra passage way provides more flexibility in a circuit application.

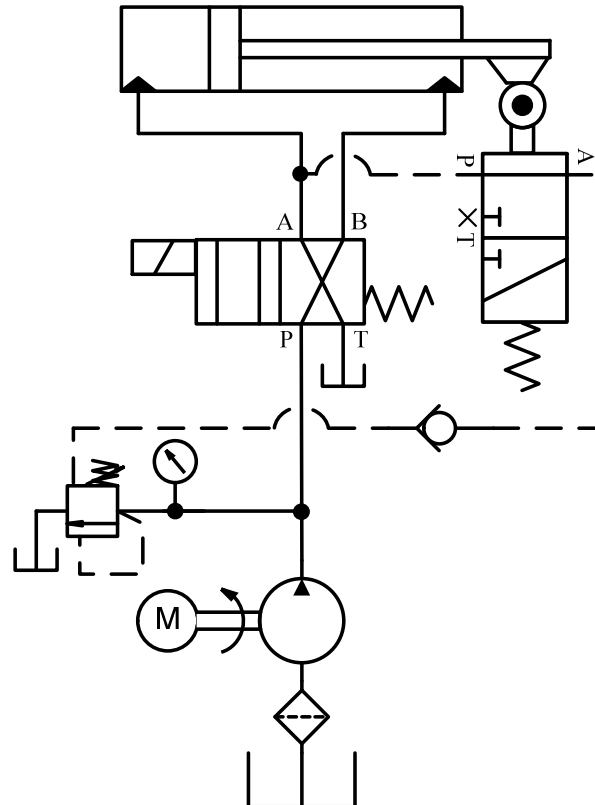
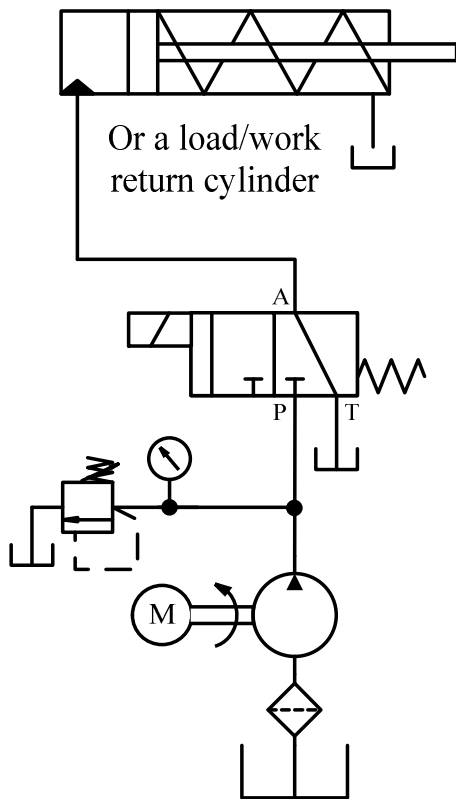
The function of this valve is to pressurize an actuator port in one position and exhaust (depressurize) the actuator in the other position. This valve has the capability of being used as a control valve or a power valve. However, the disadvantage is not having a fourth port, which means it cannot be used in a double acting actuator application.

Examples of why a 2 position, 3 port, 3 way will not work in a conventional application.

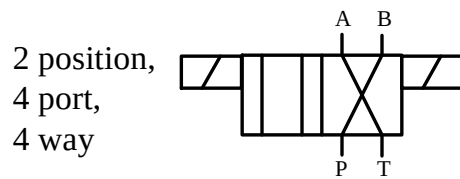


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Examples of how a 2 position, 3 port, 3 way can work in a conventional or typical application.



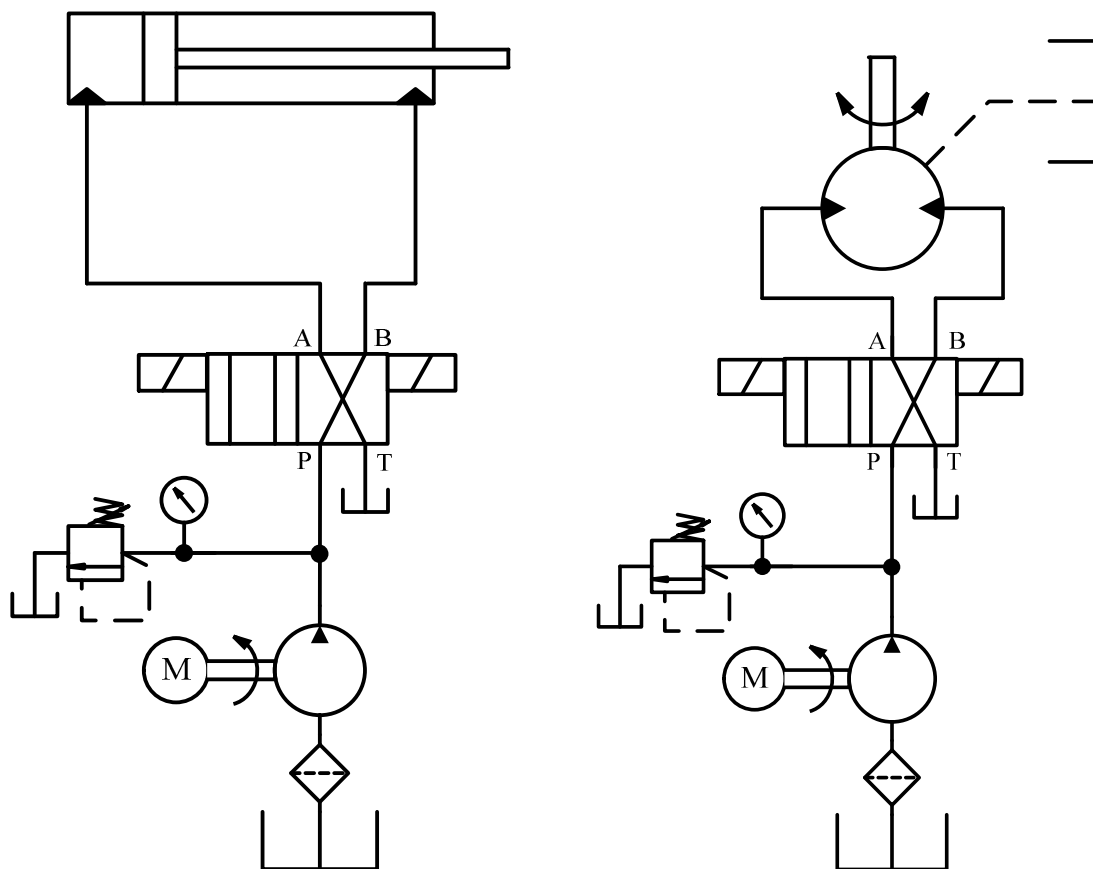
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A 2 position – 4 port – 4 way dcv is one of the more popular 2 position valves. With four passage ways; one pressure, two actuator (utilization) and one tank (reservoir) provides more flexibility than any other two position in a circuit application.

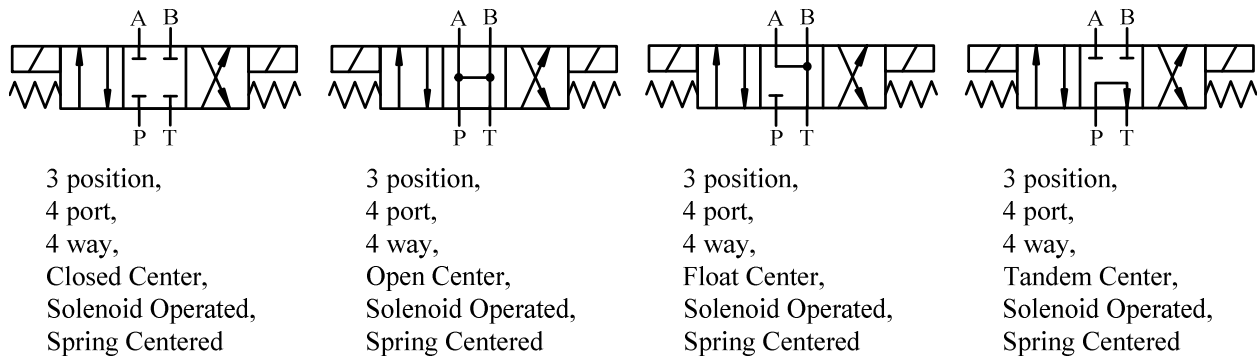
The function of this valve is to operate a double acting (bidirectional) actuators. This valve has the capability of being used as a control valve or a power valve.

Some common examples of a 2 position, 4 port, 4 way in a conventional application.



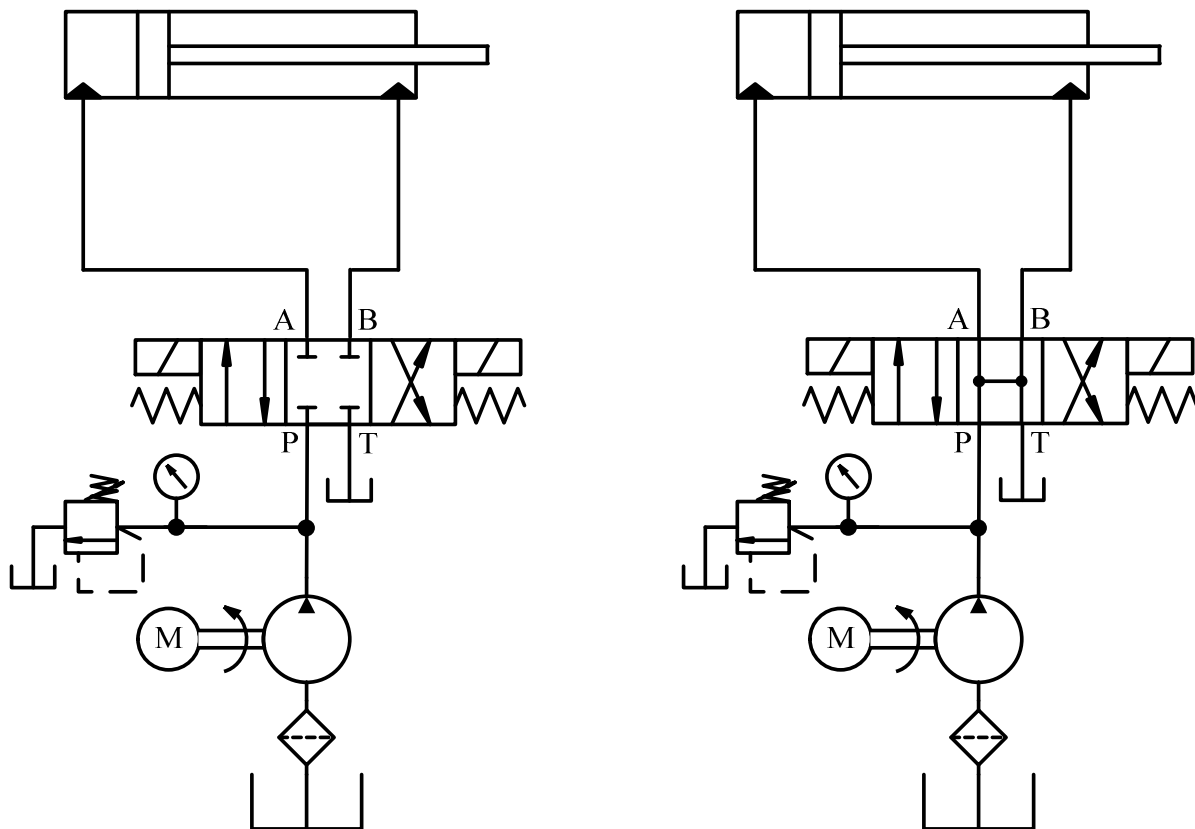
The only disadvantage of using a 2 position, 4 port, 4 way valve is with only 2 positions, the circuit is always in a working position. If the circuit needs to idle or in a neutral (non-working) condition for any considerable amount of time other design considerations would have to be utilized.

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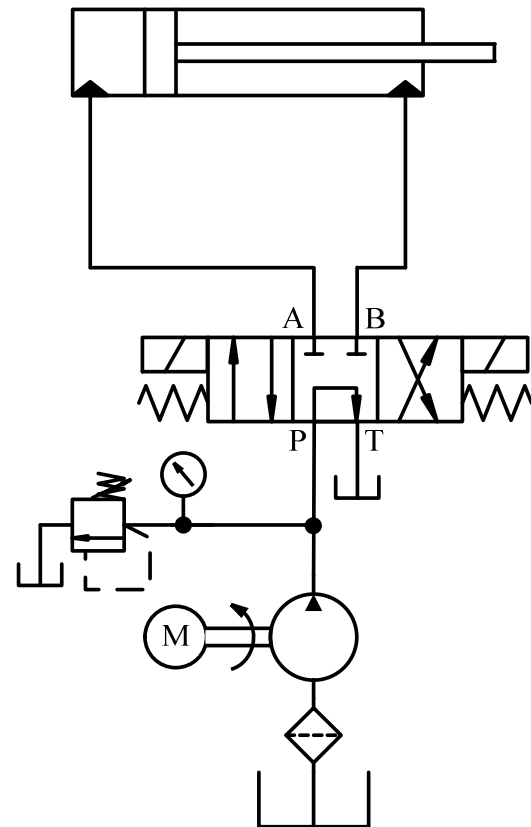
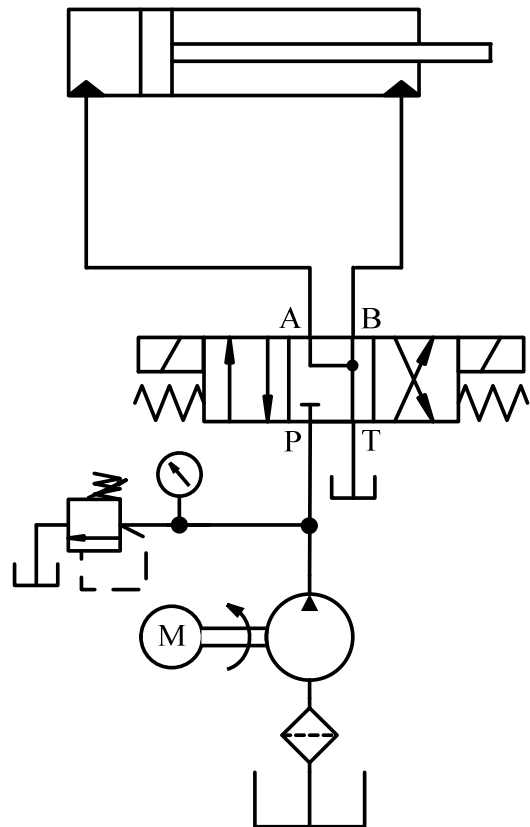


3 position – 4 port – 4 way directional control valves are the most popular dcv used in hydraulic applications. With four passage ways; one pressure, two actuator (utilization) and one tank (reservoir) provide greater flexibility and with the addition of a center condition, three position valves offer more circuit control.

Some common examples of a 3 position, 4 port, 4 way in a conventional application.



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Directional Control Valve Terminology

Direct-Acting (Directly Operated) type of Directional Control Valve

A Direct-Acting or Directly Operated type of directional control valve indicates how the force from the methods of control/operation (actuators), discussed earlier, act on/with the moving device inside the directional control valve. If the energy acts directly against the moving device to control or move it, then it is considered to be direct-acting or directly operated. All of the, previously mentioned, methods of control/operation (actuators) can be direct acting or directly operate the valve.

Depending on the size, pressure rating or flow rating of the valve, some methods become impractical; solenoids are excellent examples of this limitation. When that is the case certain methods are chosen that are more capable of handling the circuit or a combination of methods may be used.

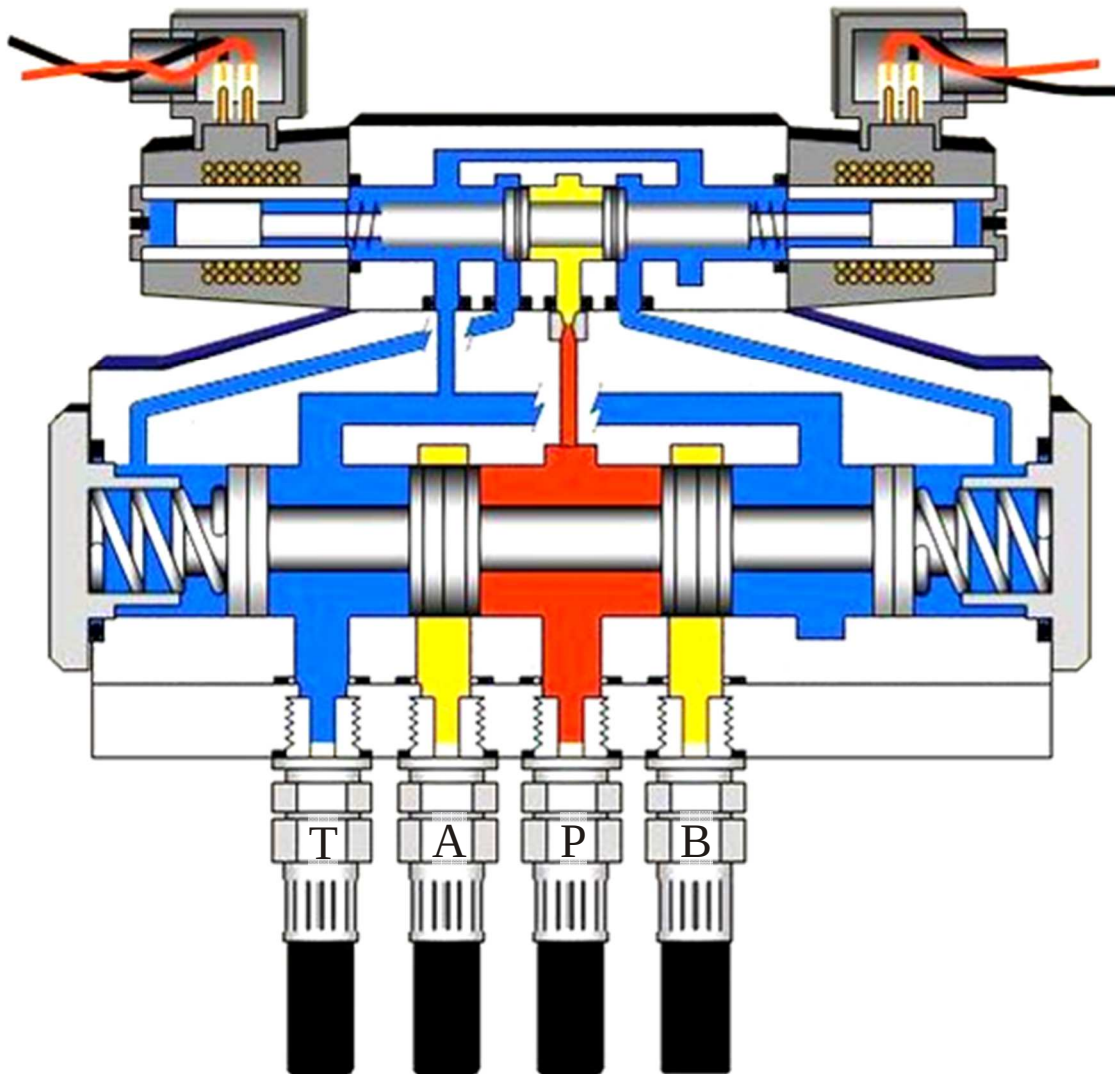
Controlled type of Directional Control Valve

Several factors may affect the decision to use a controlled type of setup versus a direct-acting or directly operated setup for a directional control valve. One factor may be the more desirable electrical/electronic controls used on modern equipment today and the advantages it brings. The largest factor that dictates switching to this type of valve setup is usually the shifting forces required to shift the moving device. On larger valves (high flow rate) or with valves that handle very high pressure values, the force needed to shift the moving device is considerable. More than would be practical to use on most of the direct acting type of operators.

If this is the case a unique/special type of valve setup is used. When the force required to shift the valve is too large for most of the direct acting operators an energy source (or operator) is chosen that can handle the increased shifting force, typically pilot operation is used. Pilot operation uses the already existing system pressure needed for the work and used it to shift the valve. That means a way then has to be installed to control that source of energy, which is usually another directional control valve. In this setup two directional control valves are used to control the circuit. Now the terminology can become confusing; controlled or operated, two valves with different controllers/operators (or actuators).

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The more common names for this type of valve setup are; **Solenoid Controlled-Pilot Operated** or **2 Stage DCV** or **Piggy-Back DCV**. As illustrated below, the larger valve on the bottom handles main system flow and main system pressure, it is shifted (pilot operated) by main system pressure from the smaller valve on top. The smaller valve on top is not handling main system flow, but it is experiencing main system pressure. The smaller valve is used to control (solenoid controlled) the pilot pressure used to shift the larger main valve on the bottom. Pilot pressure moves the main valve, but the solenoid controls the pilot pressure.



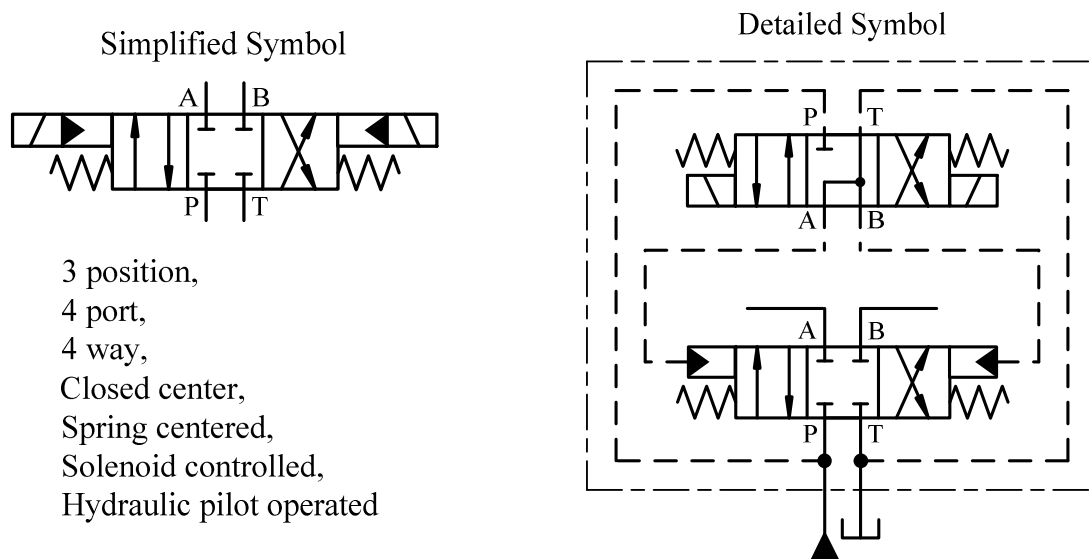
This type of setup may be used with two position and three position valves, with virtually any port or flow pattern. The main valve is operated by pilot pressure, but several different combinations of controllers/operators (actuators) may be used on the smaller pilot valve.

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Solenoid Controlled, Pilot Operated (2 Stage DCV or Piggy-Back DCV) setup

Some things have to be taken into consideration when using this type of setup. Mainly it is the pilot operation pressure/fluid and how it is being handled within the valve itself.

Solenoid controlled, pilot operated directional control valve symbols.



Solenoid controlled, pilot operated directional control valve drain

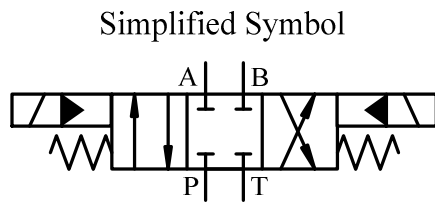
The pilot valve tank (reservoir) passage for a solenoid controlled – pilot operated directional control valve is called the pilot drain. As the main valve shifts pilot pressure provides the force to shift, however two conditions have to exist around the moving device or it will not budge. **One**, the valve has to have an energy source to create a force to shift the moving device. **Two**, the moving device cannot have a resisting force on the back side in the form of trapped fluid or pressurized fluid. The pilot valve is responsible for pressurizing one side of the valve and de-pressurizing the opposite side and providing an opportunity for the de-pressurized fluid to escape. Typically it is directed back to tank (or reservoir).

Two methods are used to drain fluid back to tank (or reservoir); Internal Drain or External Drain.

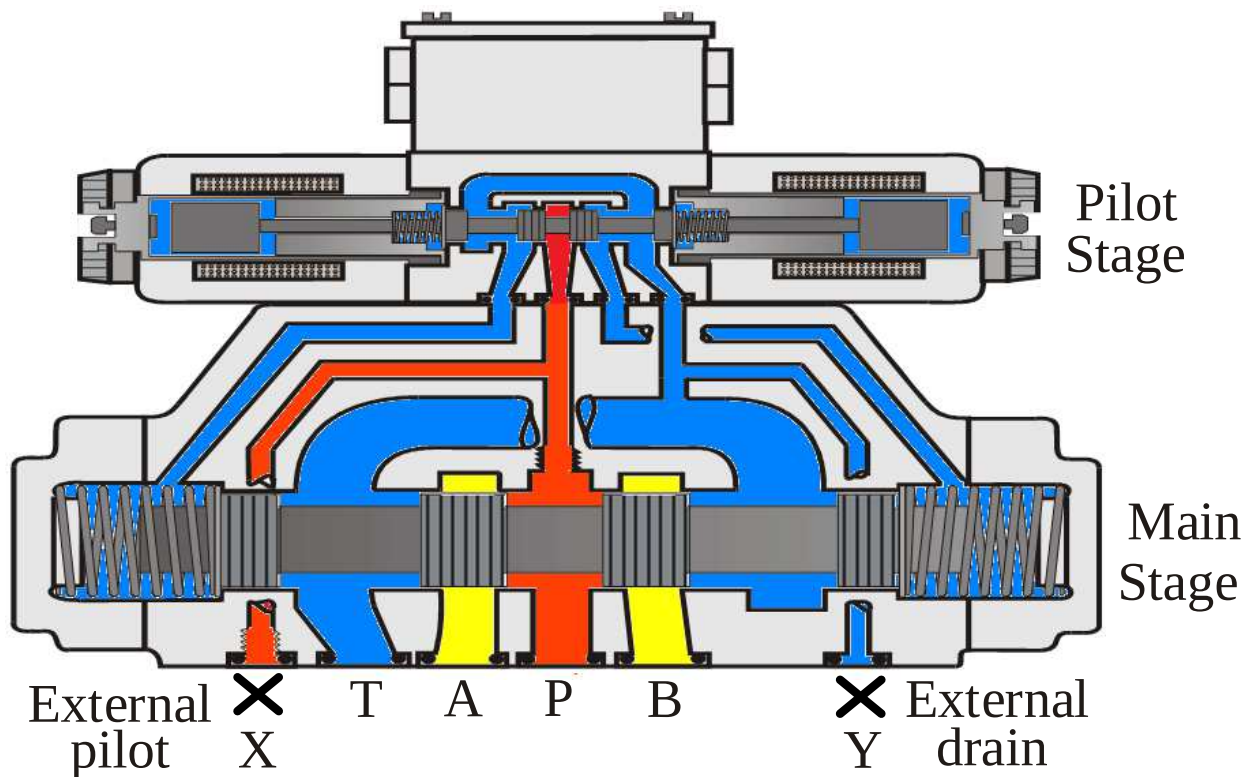
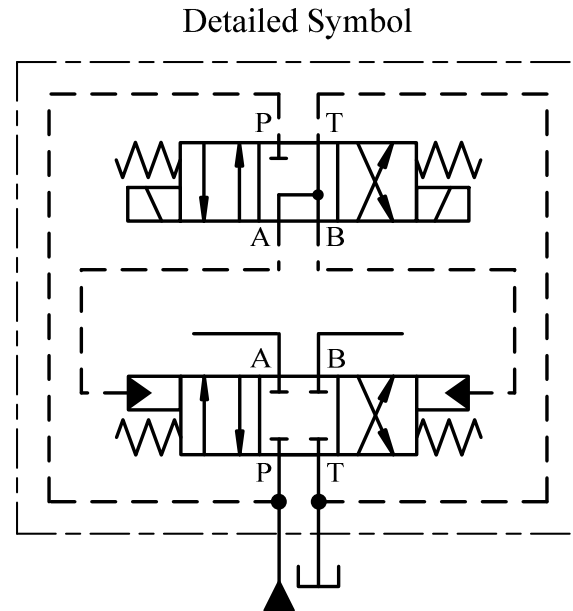
Internal drain means the pilot valve will drain the fluid through the main valves tank (or reservoir) port. External drain means the pilot valve will drain the fluid through the external drain port on the mounting plate (or sub-base) with external piping.

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Internal pilot drain example;



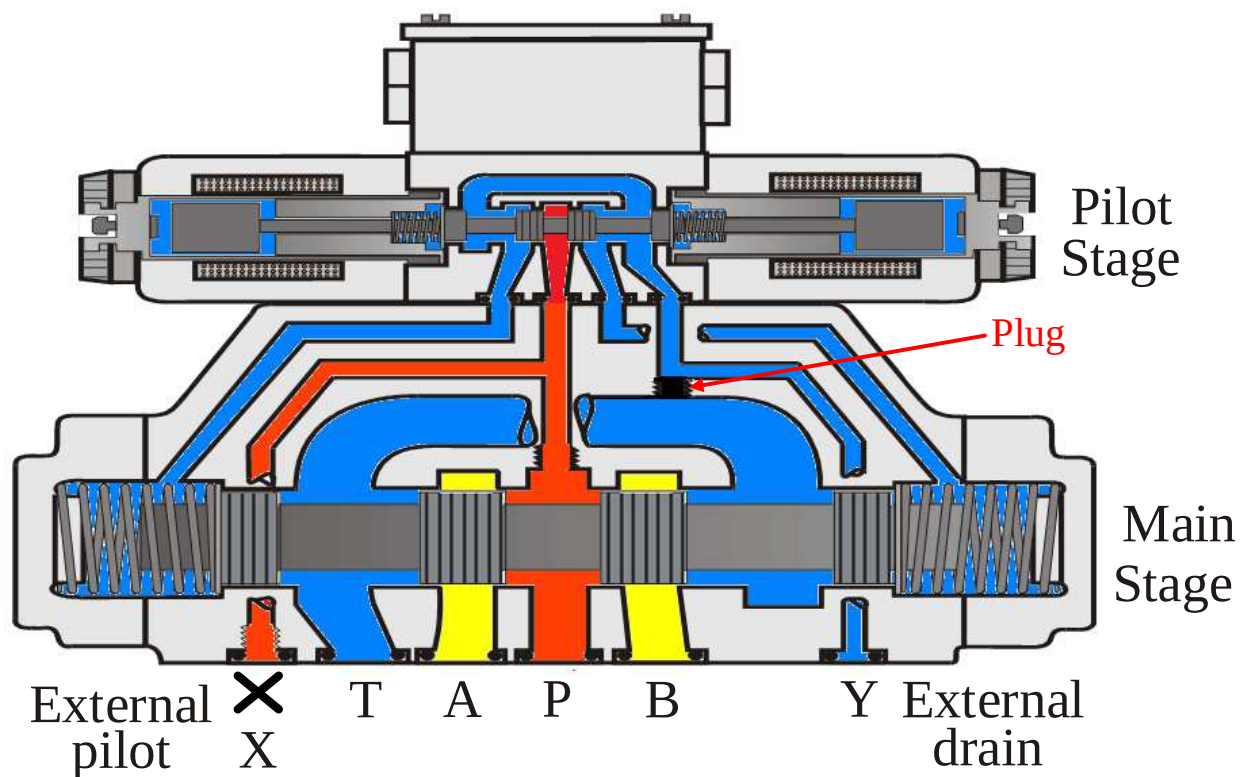
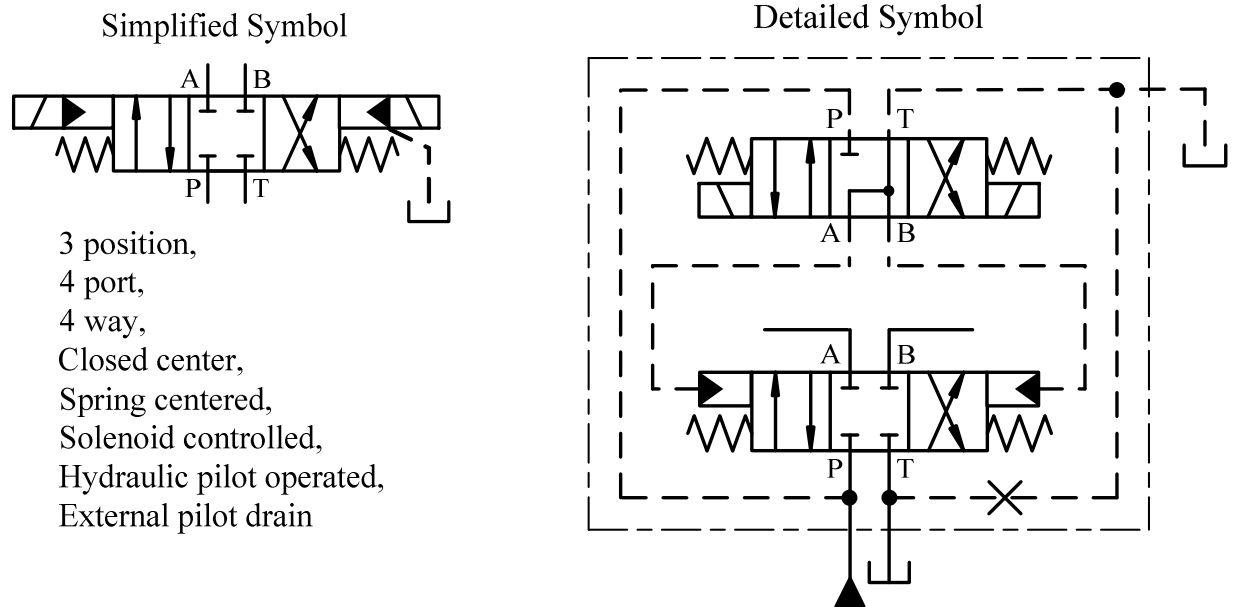
3 position,
4 port,
4 way,
Closed center,
Spring centered,
Solenoid controlled,
Hydraulic pilot operated,
Internal pilot drain



**** Safety note:** Before removing or installing any plugs to change from one drain setup to another, always refer to the valves documents or consult with the manufacturer.**

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External pilot drain example;



**** Safety note:** Before removing or installing any plugs to change from one drain setup to another, always refer to the valves documents or consult with the manufacturer.**

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Solenoid controlled, pilot operated directional control valve pilot pressure

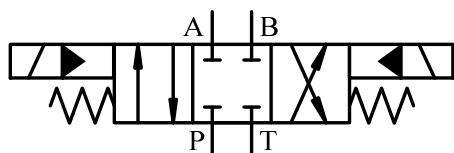
The pilot valve pressure (pump) passage for a solenoid controlled – pilot operated directional control valve is the passage way that takes main system pressure, not main system flow, and uses the energy to shift the main valve. The pilot valve is responsible for directing the pilot pressure to one side of the valve or the other to control the main valve.

Two methods are used for supplying pilot pressure to the pilot valves pressure (pump) port; Internal Pilot or External Pilot.

Internal pilot means the pilot valve will use the same pressure supplied to the main valves pressure (pump) port for system operation. External pilot means the pilot valve will use another source of pressure outside of the main valves pressure (pump) port. Usually from some other source through the external pilot port on the mounting plate (or sub-base) with external piping.

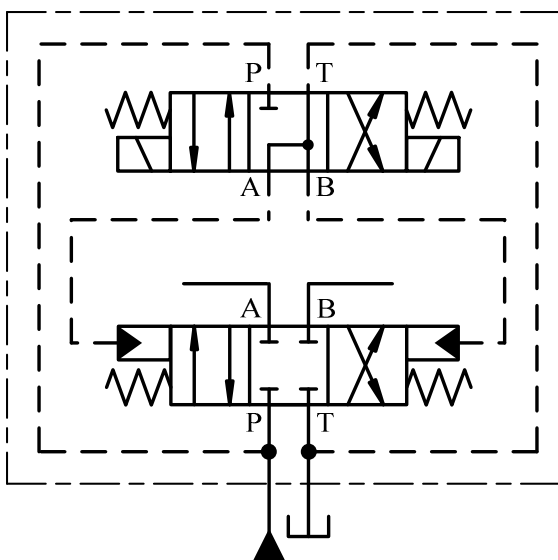
Internal pilot pressure example;

Simplified Symbol



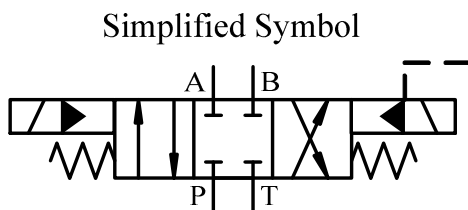
3 position,
4 port,
4 way,
Closed center,
Spring centered,
Solenoid controlled,
Hydraulic pilot,
Internal pilot operated,
Internal pilot drain

Detailed Symbol

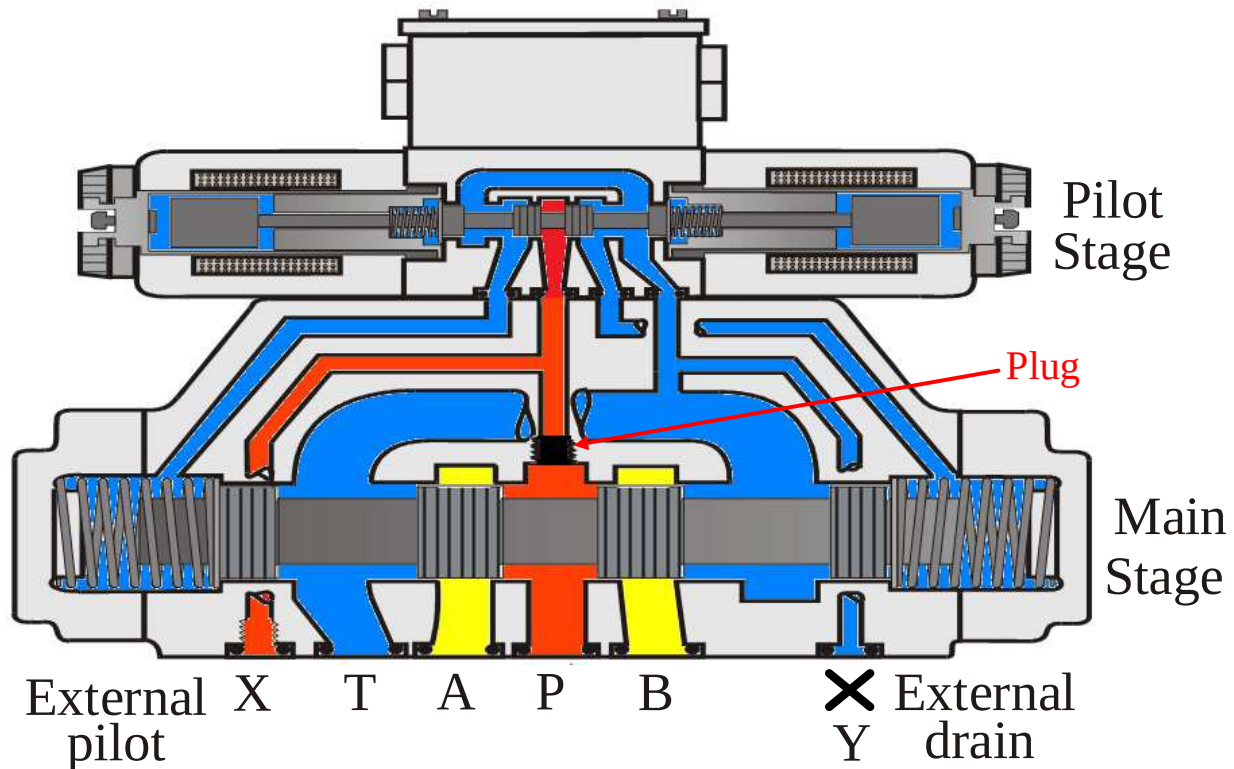
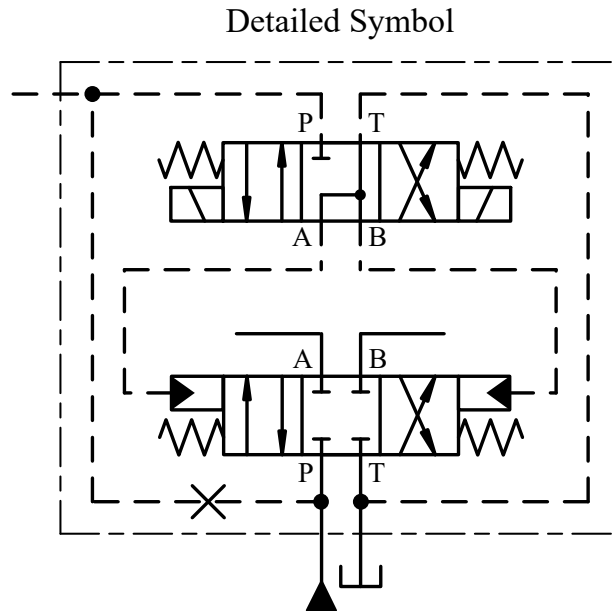


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External pilot pressure example ;



3 position,
4 port,
4 way,
Closed center,
Spring centered,
Solenoid controlled,
Hydraulic pilot,
External pilot operated,
Internal pilot drain



**** Safety note:** Before removing or installing any plugs to change from one pilot setup to another, always refer to the valves documents or consult with the manufacturer.**

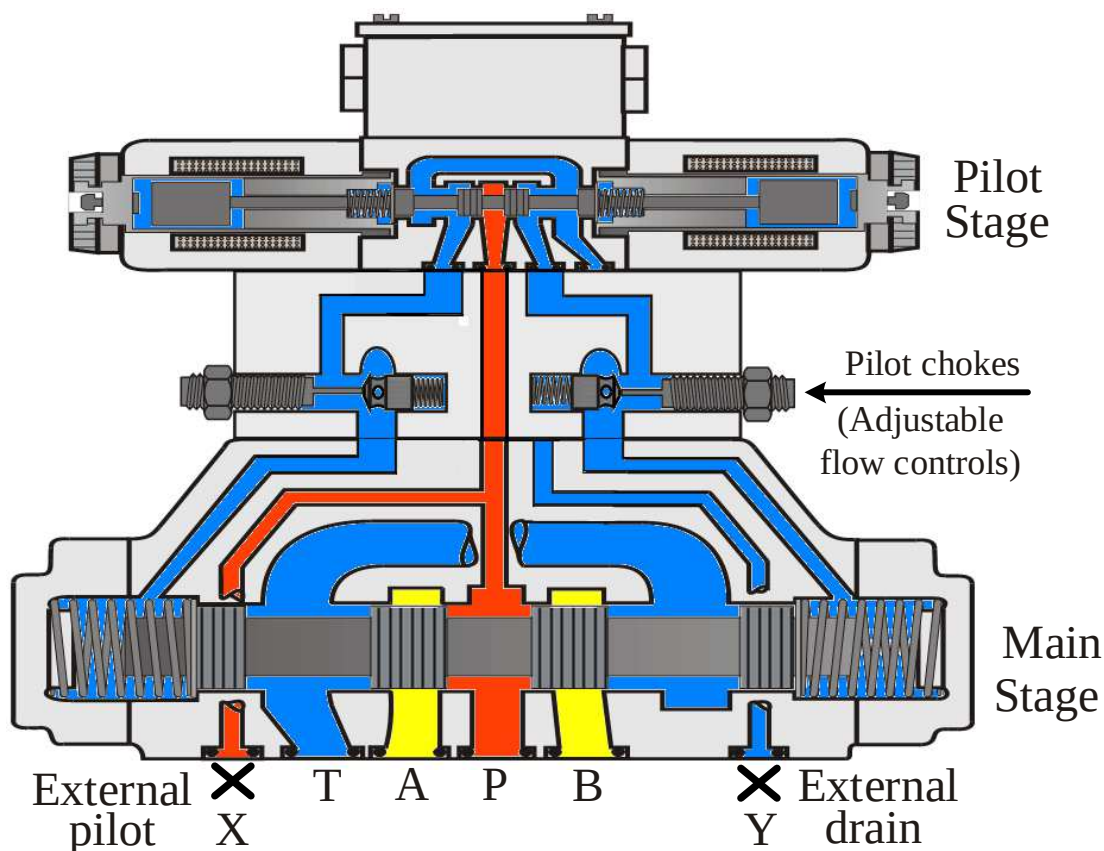
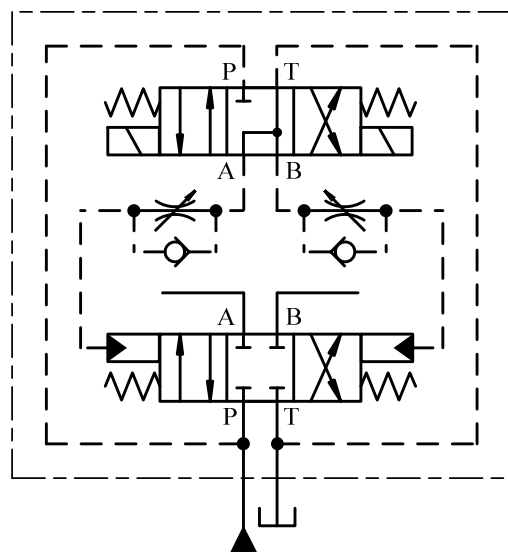
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Solenoid controlled, pilot operated directional control valve pilot choke control

When the spool of a pilot operated directional control valve shifts, the pilot fluid can experience a shock load as the large amount of fluid is forced out of the pilot area quickly. A choke control slows the shifting speed of the spool of the main valve (2nd stage) which reduces the fluid shock.

A choke control is essentially a flow control for the pilot flow leaving the main valve and entering the solenoid controlled valve. A choke control does not eliminate the shock, it reduces the shock loading applied to the fluid which reduces the shock felt by the solenoid controlled valve (1st stage) also.

Some advantages to choke controls are reducing fluid shock, smooth acceleration and deceleration. One disadvantage is the main valve (2nd stage) spool speed is reduced which could affect the response time or speed of the circuit.



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Check Valves

A check valve is a one-way directional control valve, it allows free flow in one direction and blocks flow in the opposite direction. Two basic types of check valves exist with several different designs or variations of each available, as listed below;

1. Unpiloted Check Valves (In-Line Versions)
 - a. Straight In-Line Check Valve
 - b. Right Angle (90°) Check Valve
 - c. Restrictor Check Valve
 - d. Spring-Loaded Check Valve
2. Piloted Check Valves (Pilot Operated Check Valves)
 - a. Pilot to Open
 - b. Pilot to Close

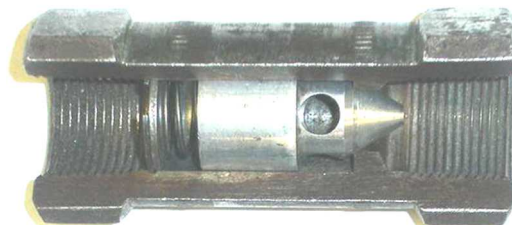
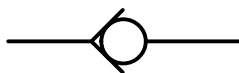
Unpiloted Check Valves (In-Line Versions)

Straight Check Valve (Standard Check Valve)

Straight check valves are simple devices with a valve body containing a closure and a non-adjustable bias spring. The closure may be of a ball or poppet design. The spring is a light duty spring, usually in the range of 5 to 25 psi, although higher spring ratings are available for special requirements.

A straight check valve is a one-way directional control valve, it allows free flow in one direction and blocks flow in the opposite direction. The light duty spring holds the closure seated blocking flow in the no-flow direction. In the free flow direction, the fluid will only have to generate enough pressure to overcome the spring rating and unseat the closure allowing flow in the opposite direction.

**Straight (Standard)
check valve**



Use a check valve, in general, in any line where reversed flow (or back flow) cannot be allowed.

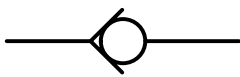
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Right Angle Check Valve (90° Check Valve)

Right angle check valves operate the same as a straight (standard) check valve, except the inlet and outlet ports are at an angle of ninety degrees in relation to each other. Right angle check valves are heavier duty units, have a higher flow rating and have a lower pressure drop across the valve. The symbol for a right angle check valve is the same as the straight check valve.

Because of the heavy duty nature of a right angle check valve, the closure is typically a poppet design and the closure and seat are hardened for durability. The spring is usually in the range of 5 to 25 psi, although higher spring ratings are available for special requirements.

Right angle (90 deg.)
check valve

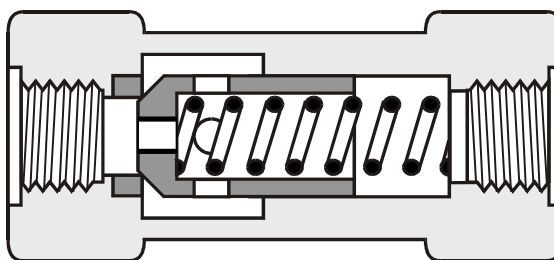
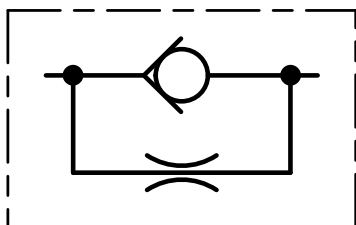


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Restrictor Check Valve

Restrictor check valves operate the same as a straight or right angle check valve, however there is one critical difference. The restrictor check valve does allow free flow in one direction and instead of blocked flow in the opposite direction it has a fixed orifice in the closure, which allows restricted flow in the opposite direction. The restricted flow direction acts as a flow control device.

**Restrictor
check valve**



Spring Loaded Check Valve

Spring loaded check valves operate the same as a straight or right angle check valve, except for substantial difference. The spring loaded check valve has a larger spring rating biasing the closure. All of the previously mentioned check valves had a light duty spring biasing the closure, even-though none of the symbols represented a spring, it was assumed or understood the spring was there for the check valve to function properly.

With a higher spring rating, a spring loaded check valve can be used in situations where a higher back pressure is required within the circuit, versus the lighter spring rating of 5 to 25 psi. Thus another common name for them would be a Back Pressure Check Valve. Spring loaded check valves can have a spring rating as high as 150 psi and be adjustable.

**Spring loaded
check valve**

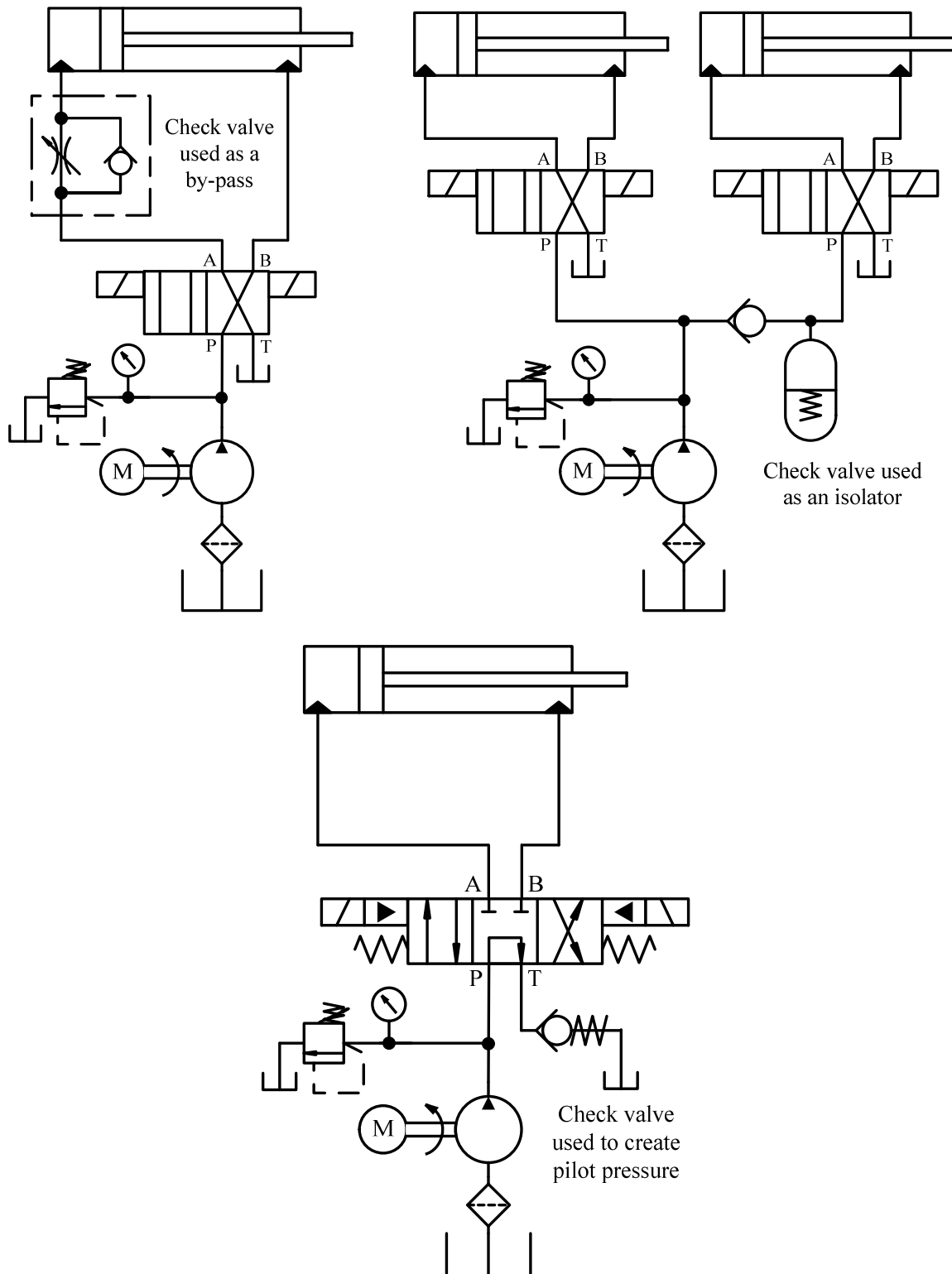


**Adjustable spring
loaded check valve**



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Examples of unpiloted check valve applications;



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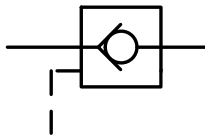
Piloted Check Valves (Pilot Operated Check Valves)

Piloted check valves operate in the same manner as an unpiloted (or conventional) check valve, allowing free flow in one direction and blocked flow in the opposite direction. The difference is this valve uses a separate pilot (signal) pressure to open or close the closure when the pilot reaches a high enough value. Two types of pilot operated check valves exist; a **Pilot to Open** and **Pilot to Close** version.

Pilot to Open Check Valve

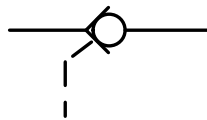
A pilot to open check valve is designed to allow free flow in one direction and blocked flow in the opposite direction. The difference with a pilot to open version is free flow is permitted, in the blocked flow direction, when a pilot (signal) pressure is applied to the pilot port/piston.

Detailed symbol

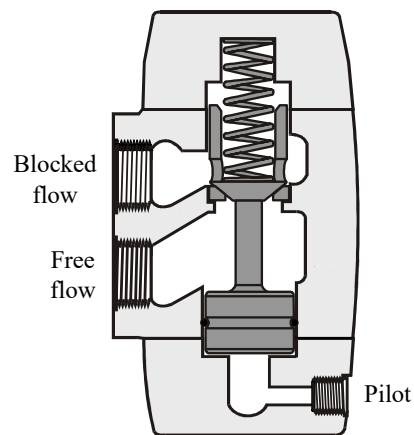


Piloted check valve,
pilot to open

Simplified symbol



Piloted check valve,
pilot to open

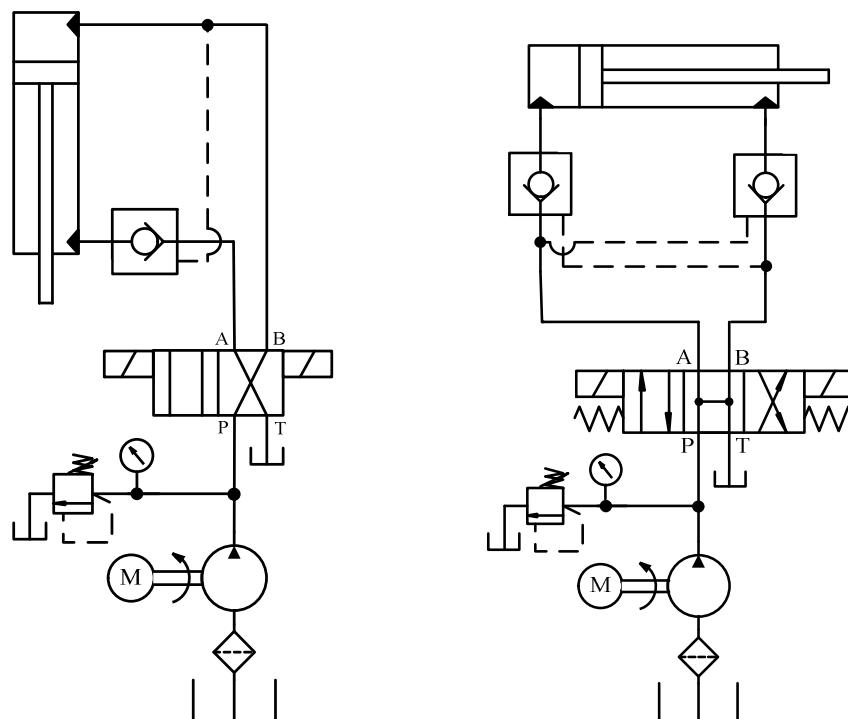


Since the closure has system pressure acting against it, in the blocked flow direction, when the valve has a pilot (signal) pressure acting on the pilot port/piston there could be a balancing of forces across the valve. For the pilot (signal) pressure to be able to unseat the closure the pressure/force/area would need to be greater than the resistant force on the back of the closure. To ensure the pilot can overcome the closure pilot to close check valve are designed with the pilot piston having more surface area, which will provide more force, to provide a dominate signal and ensure the closure can be overcome.

Manufacturers of pilot to open check valves will design them around a ratio to ensure the proper size valve is used in a given application.

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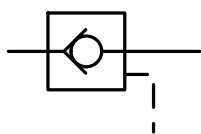
Common examples of Pilot to Open Check Valve applications;



Pilot to Close Check Valve

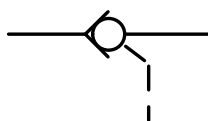
A seldom used valve is the pilot to close check valve. It is designed to allow free flow in one direction and blocked flow in the opposite direction, until a pilot (signal) pressure is applied to close the valve. It is usually not necessary to have a valve that always blocks flow in one direction and also capable of stopping it in the opposite direction.

Detailed symbol

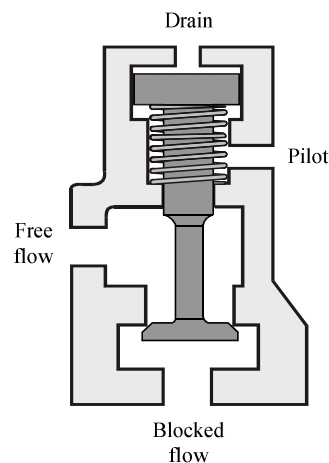


Piloted check valve,
pilot to close

Simplified symbol



Piloted check valve,
pilot to close



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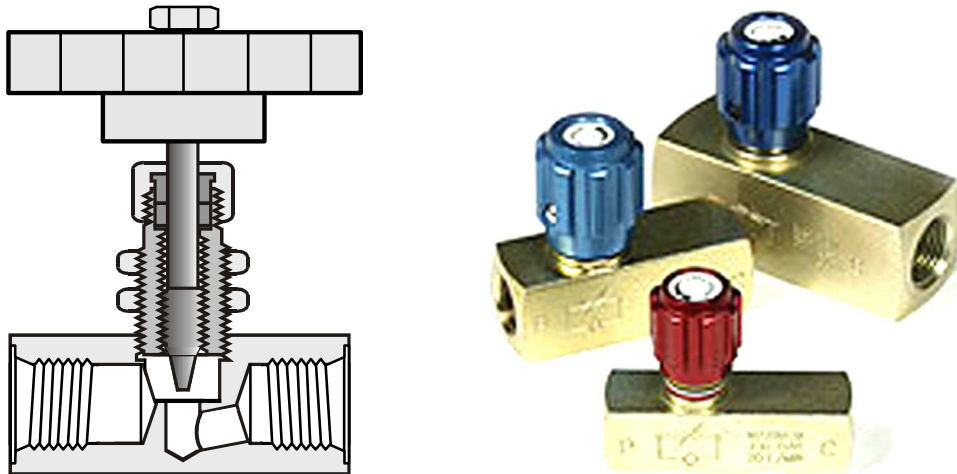
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Flow Control Valves

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Introduction to Flow Control Valves

The speed of an actuator is determined by the volume of fluid entering or exiting the actuator. If the speed of the actuator is of no great concern, then the volume of fluid is of no concern. However, if the speed of the actuator is important then some way must be used to control the volume of fluid in that part of the circuit.



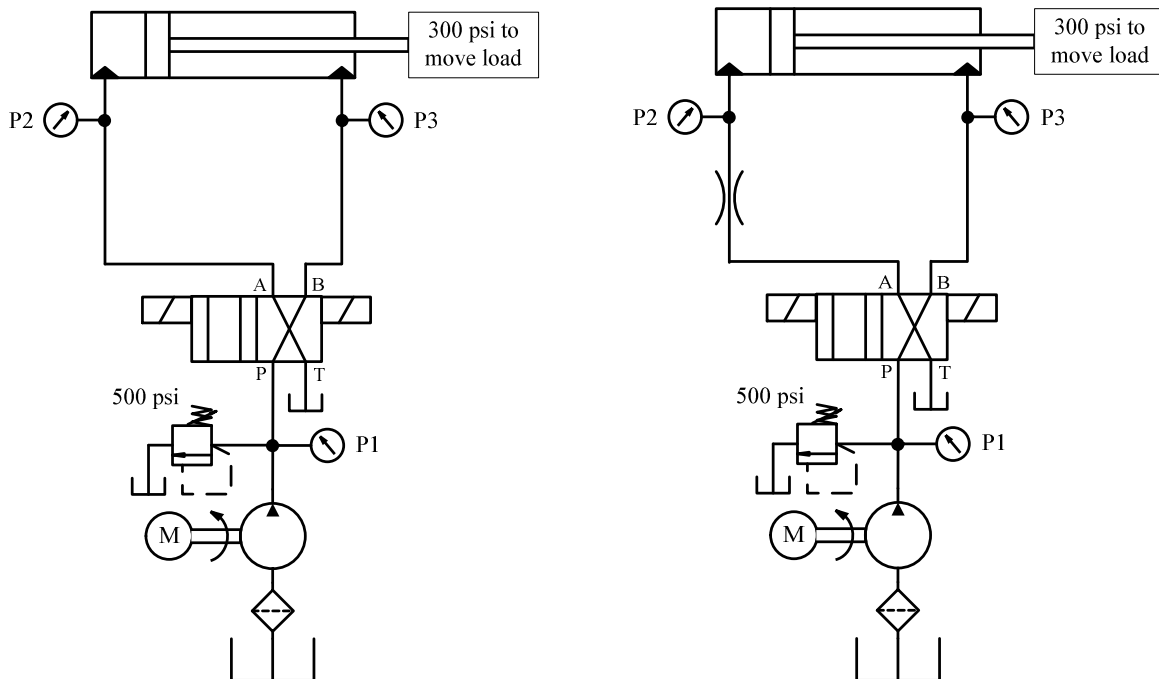
The function of a flow control is to reduce/restrict the volume of fluid in a part or leg of a circuit. Reducing/restricting the rate of flow will directly result in a speed reduction in that part of the circuit, which will result in a speed reduction of the actuator.

Flow control valve



The flow control valve performs this function by being a higher than normal resistance for the circuit. To overcome the extra resistance, the positive displacement pump will apply a greater pressure to the fluid, which will force the circuit to utilize other components in the circuit to control the extra pressure generated in the circuit. When the extra resistance forces the pressure to rise to the next least possible resistance, which is usually the systems pressure relief valve. When the pressure relief valve opens, the extra fluid will take another pathway, over the relief valve. So, typically the flow control valve and pressure relief valve work together to handle the flow and pressure in a flow control type circuit.

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Fluid Flow through a Flow Control Valve

A flow control valve (restrictor) is a smaller than normal opening (hole) in the fluid's flow path. Flow through a flow control valve can be affected by several factors. The three most important factors are;

1. Size of the opening (hole)
2. Pressure differential across the valve
3. Temperature of the fluid

Size of the opening (hole)

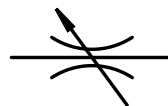
The size of the opening (hole) relates directly to the flow rate in that part of the circuit. The larger the opening (hole), the more fluid can pass through; the smaller the opening (hole) the less fluid can pass through.

To control the size of the opening (hole) either a fixed or adjustable flow control may be used. The most widely used type is the adjustable flow control.

Fixed flow control

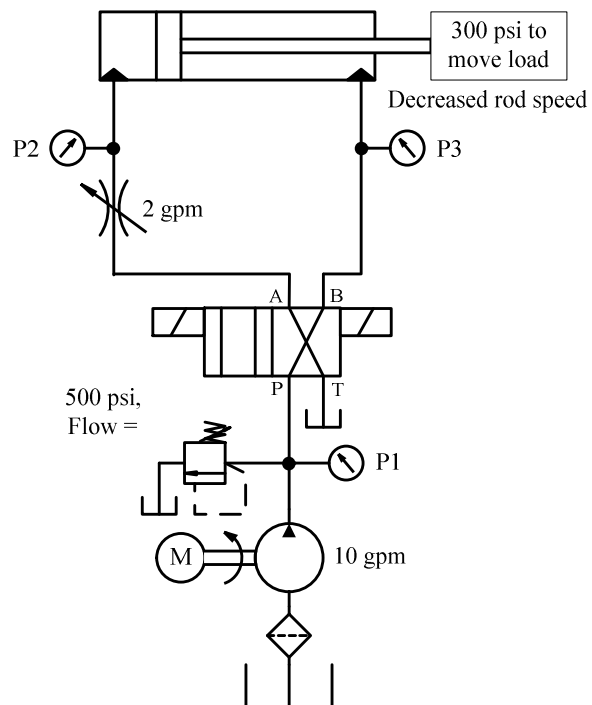
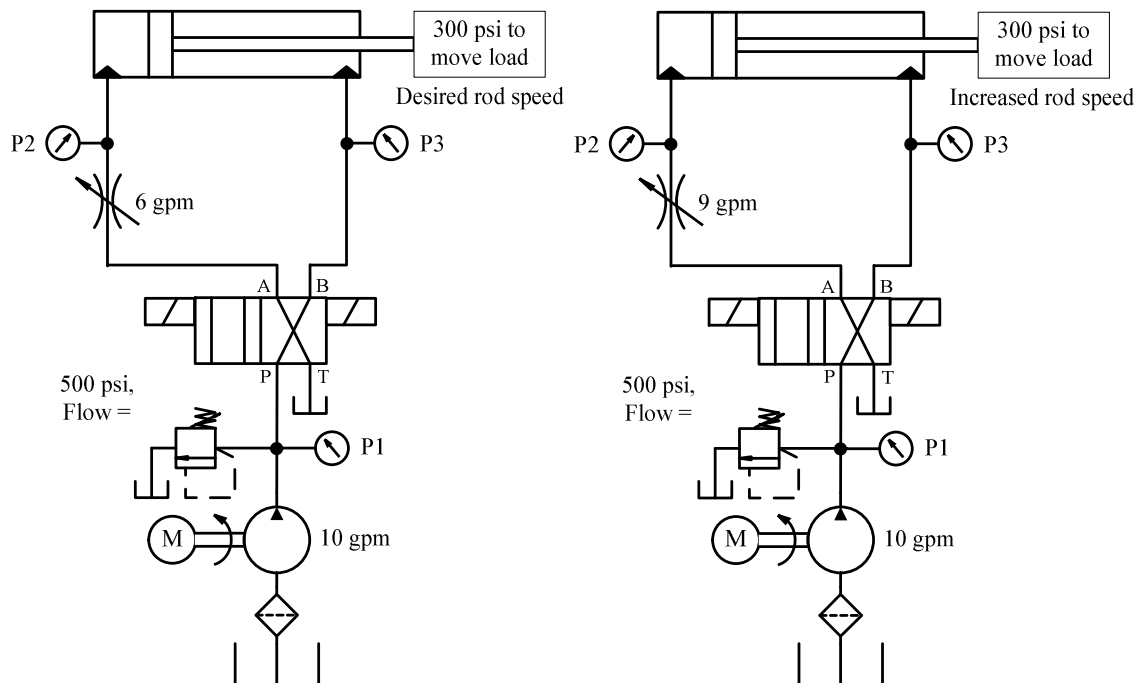


Adjustable flow control



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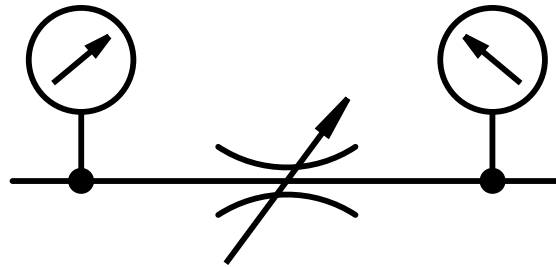
The Effects of Adjusting a Flow Control Circuit



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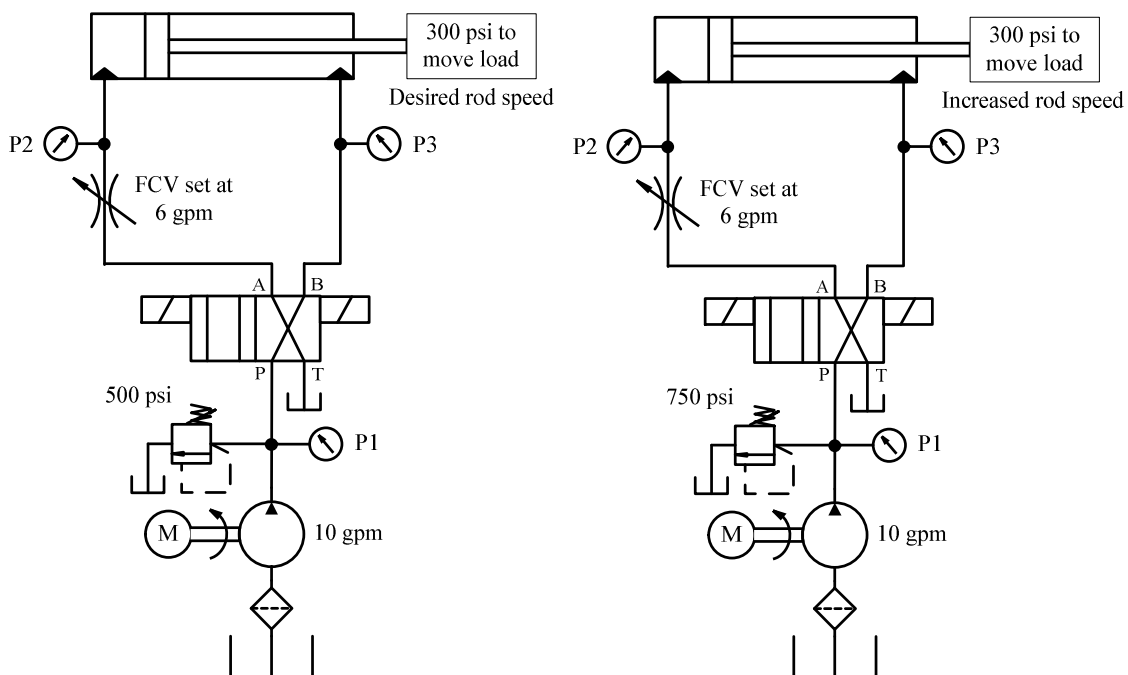
Pressure differential across the valve

Flow through a flow control valve is also affected by the pressure differential (difference in pressure from one side of the valve to the other side of the valve). Since pressure in a hydraulic system is energy, an increase in the pressure differential across the valve will mean more flow downstream. The reverse is true also, a decrease in the pressure differential across the valve will mean less flow downstream.

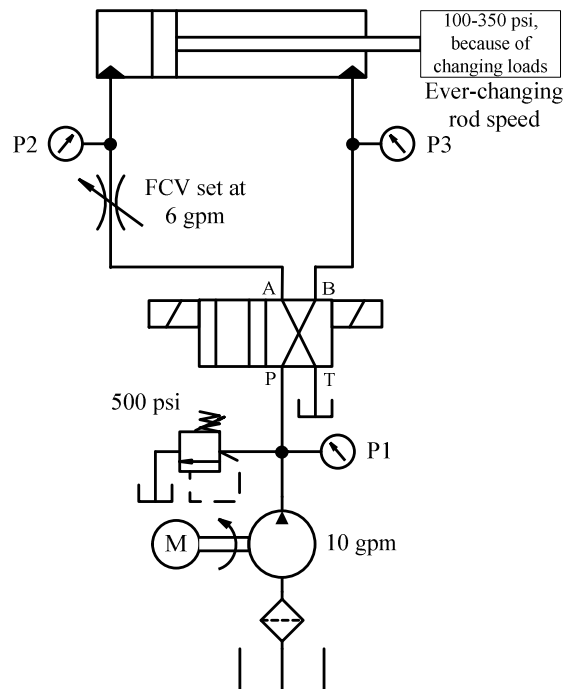


The Effects of Pressure Differential on a Flow Control Circuit

Once the flow control valve is set to achieve the desired actuator speed, if the system pressures through-out the circuit stay consistent then the flow will be consistent through the flow control valve. If the pressures do not stay consistent, for example; from employee intervention, changing loads on the actuator, a complex circuit with multiple directional control valves or multiple actuators could result in inconsistent pressures.



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Types of Flow Control Valves

As was illustrated previously, any change in the pressure before or after the flow control valve will affect the flow through the valve, which will affect actuator speed. These pressure changes must be compensated for before the flow control valve can control flow or control actuator speed.

All the flow control valves mentioned up to this point are **Non-Pressure Compensated Flow Control Valves**; fixed flow control and the adjustable flow control. They are good flow control devices as long as the pressure differential across the valve stays constant and the valve is not adjusted they will provide good speed control for the actuator.

If pressure fluctuations are inevitable, then a more sophisticated type of flow control valve must be chosen, one that will compensate for any pressure fluctuations in the circuit. This type of valve is called a **Pressure Compensated Flow Control Valve**.

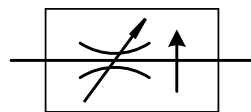
Pressure compensated flow control valves fall into two basic categories; **Restrictor Type and By-Pass Type**.

1. **Restrictor Type**; A restrictor type **does** have a direct working relationship with the pressure relief valve.
2. **By-Pass Type**; A by-pass type **does not** have a direct working relationship with the pressure relief valve.

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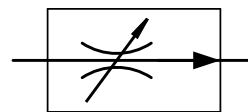
Adjustable Flow Control Valve, Pressure Compensated, Restrictor Type

Adjustable flow control,
Pressure Compensated,
Restrictor type



(ANSI symbol)

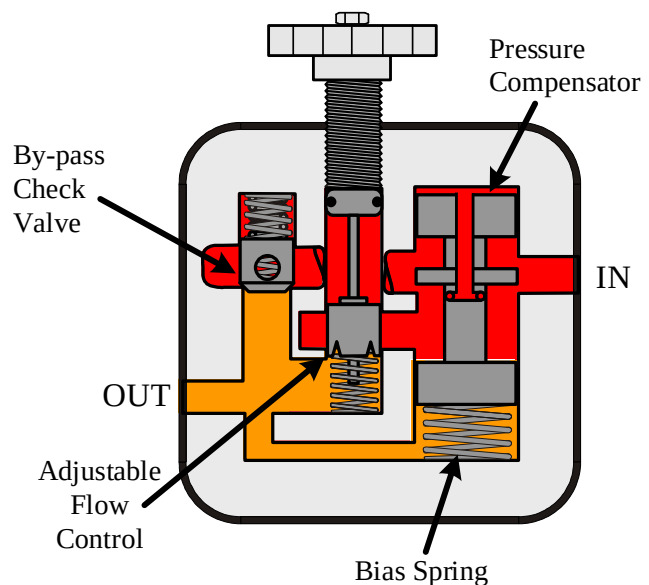
Adjustable flow control,
Pressure Compensated,
Restrictor type



(ISO symbol)

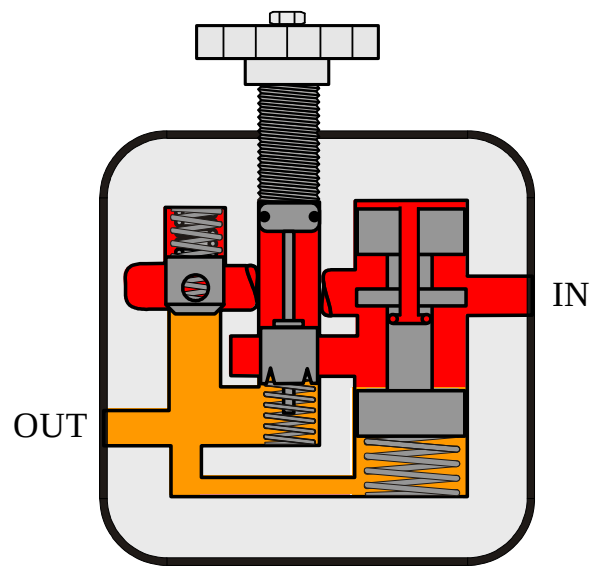
This type of valve has a compensator spool, built-in, to eliminate pressure changes through the valve. As fluid enters the valve it immediately encounters the compensator spool, pressure compensation happens before the flow controlling aspect. Since the compensator spool feels incoming pressure and outgoing pressure values, to fully understand the compensator spools operation we will by-pass the spool for now and readdress it later. Incoming fluid flows past the compensator spool and goes through the flow control. At this point remember, we are reducing/restricting fluid flow which will inherently be a pressure differential across the flow control. After passing through the flow control the fluid then will exit the valve.

The reduced flow rate leaves the flow control and will usually go downstream and interact with the actuator. The only resistance to flow from the flow control to the actuator is the work/load on the actuator. That is the pressure being felt in the outgoing side of the flow control valve. That downstream pressure also is diverted through a passageway to one side of the compensator spool. Downstream pressure plus a bias spring (usually around 100 – 200 psi) is the pressure/force being applied to that side of the compensator spool. Remember the incoming pressure, in a restrictor type of flow control valve it is the pressure relief valve setting, it is entering a pilot port in the spool and is diverted to the opposite side of the compensator spool. The incoming pressure opposes the outgoing pressure, plus the bias spring and this is where pressure compensation takes place.

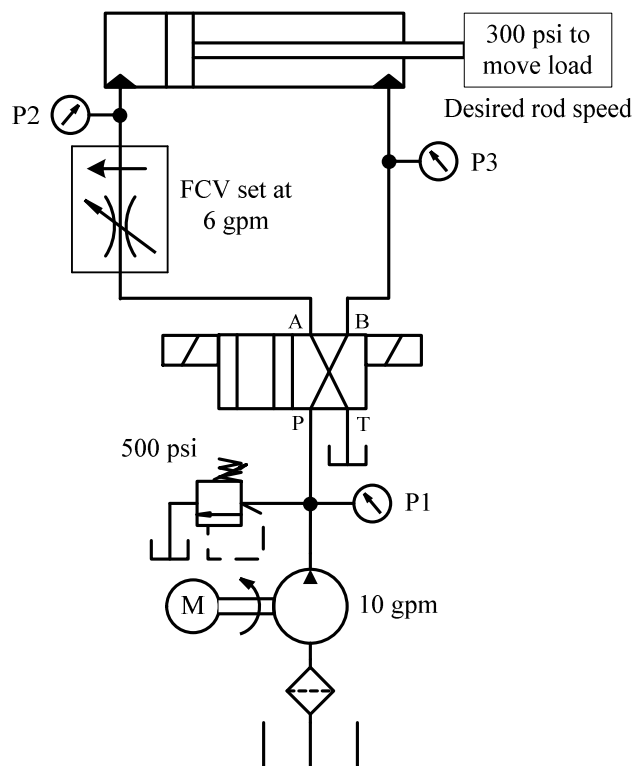


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The compensator spool senses/feels both pressure values and shuttles back and forth as pressure builds on one side or the other. As it shuttles back and forth the spool acts as a restriction to keep a steady flow/pressure differential across the flow control. When the compensator spool reacts to pressure it becomes a flow control or restrictor itself. It creates a constant pressure differential across the flow control in the form of the bias spring on the compensator spool.



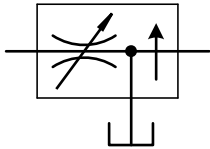
In effect, the spool regulates the difference in pressure across the flow control to a value equal to the bias spring. If the bias spring is 100 psi, the pressure differential will be 100 psi. While the pressure differential is limited to the value of the spring, the pressure differential across the whole is actually controlled by the pressure relief valve and downstream actuator pressure. There is actually three different pressures interacting with the valve; incoming pressure, an intermediate pressure (inside the valve) set by the compensator spool and downstream pressure.



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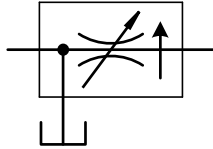
Adjustable Flow Control Valve, Pressure Compensated, By-Pass Type

Adjustable flow control,
Pressure Compensated,
By-Pass type



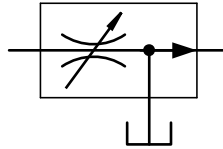
(ANSI symbol)

Adjustable flow control,
Pressure Compensated,
By-Pass type



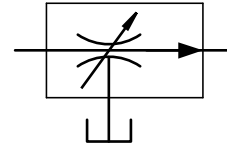
(Alternate ANSI symbol)

Adjustable flow control,
Pressure Compensated,
By-Pass type



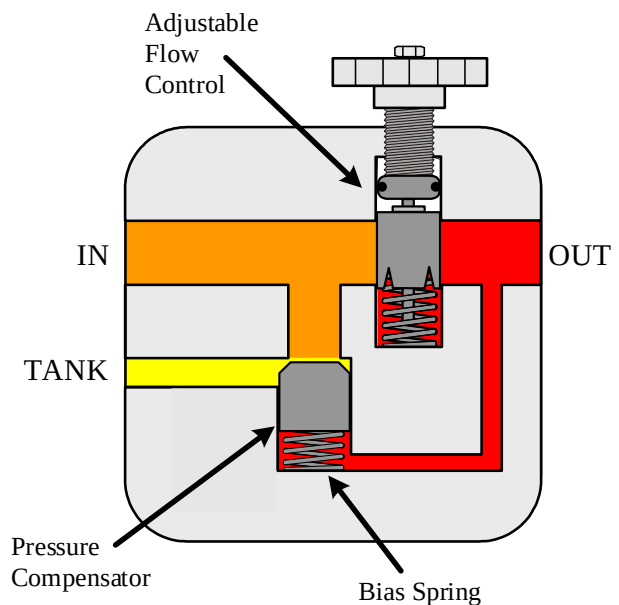
(ISO symbol)

Adjustable flow control,
Pressure Compensated,
By-Pass type



(Alternate ISO symbol)

Similar to the restrictor type, this type of valve has a compensator spool also to eliminate pressure changes through the valve. As fluid enters the valve it simultaneously encounters the compensator spool and flow control, the fluid has basically two options. Some fluid will go through the flow control, based on its setting, and the remainder will go through the compensator spool. Since the compensator spool feels incoming pressure and outgoing pressure values, to fully understand the compensator spools operation we will by-pass the spool for now and readdress it later. Incoming flow has two options; some will go through the flow control and interact with the actuator, some will by-pass the flow control and is diverted to tank/reservoir.



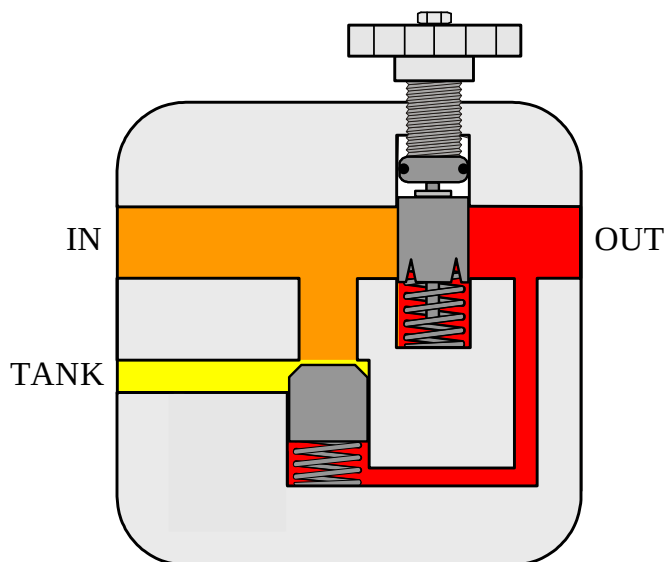
The reduced flow rate leaves the flow control and will usually go downstream and interact with the actuator. The only resistance to flow from the flow control to the actuator is the work/load on the actuator. That is the pressure being felt in the outgoing side of the flow control valve. That downstream pressure also is diverted through a passageway to one side of the compensator spool. Downstream pressure plus a bias spring (usually around 100 – 200 psi) is the pressure/force being applied to that side of the compensator spool.

Remember, this is a by-pass type of flow control valve. It does not create an extra resistance to flow in the circuit, thus it does not have to work with the pressure relief valve to handle the extra flow, it simply by-passes it to tank/reservoir. The incoming pressure is a combination of downstream pressure, at the actuator, to do the work and the bias spring behind the compensator spool. That is all the incoming pressure has to overcome.

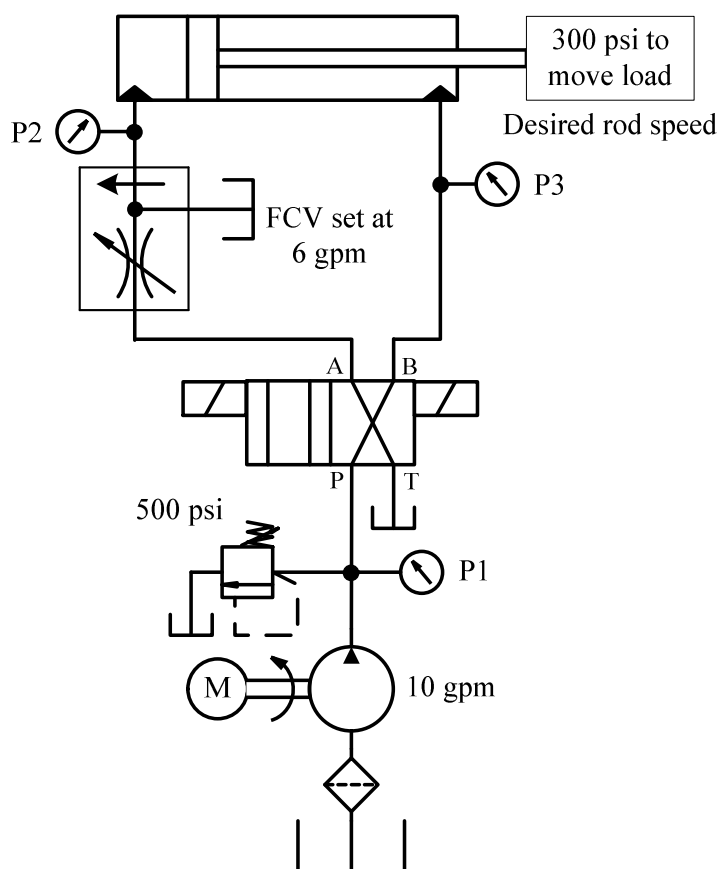
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The incoming pressure opposes the outgoing pressure, plus the bias spring and this is where pressure compensation takes place.

The compensator spool senses/feels both pressure values and shuttles back and forth as pressure builds on one side or the other. As it shuttles back and forth the spool acts as a restriction to keep a steady flow/pressure differential across the flow control. When the compensator spool reacts to pressure it becomes a flow control or restrictor itself. It creates a constant pressure differential across the flow control in the form of the bias spring on the compensator spool.



In effect, the spool regulates the difference in pressure across the flow control to a value equal to the bias spring, by by-passing more or less fluid to tank/reservoir. If the bias spring is 100 psi, the pressure differential will be 100 psi.



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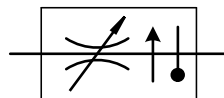
Temperature of the fluid

Flow through a flow control valve can also be affected by the temperature of the fluid. When the temperature of the fluid rises or falls it will have a direct impact on the viscosity of the hydraulic fluid. If the fluid is hotter, it will become thinner and more fluid will flow through a flow control valve, thusly if the fluid is colder, it will become thicker and less fluid will flow through a flow control.

Usually the temperature of a hydraulic fluid will stay fairly stable or consistent. However, when the temperature is not stable or consistent and it has an effect on the systems. A flow control with temperature compensation would be the choice.

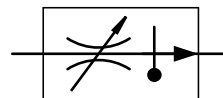
Adjustable Flow Control Valve, Pressure Compensated, Restrictor Type, Temperature Compensated

Adjustable flow control,
Pressure Compensated,
Restrictor type,
Temperature Compensated



(ANSI symbol)

Adjustable flow control,
Pressure Compensated,
Restrictor type,
Temperature Compensated



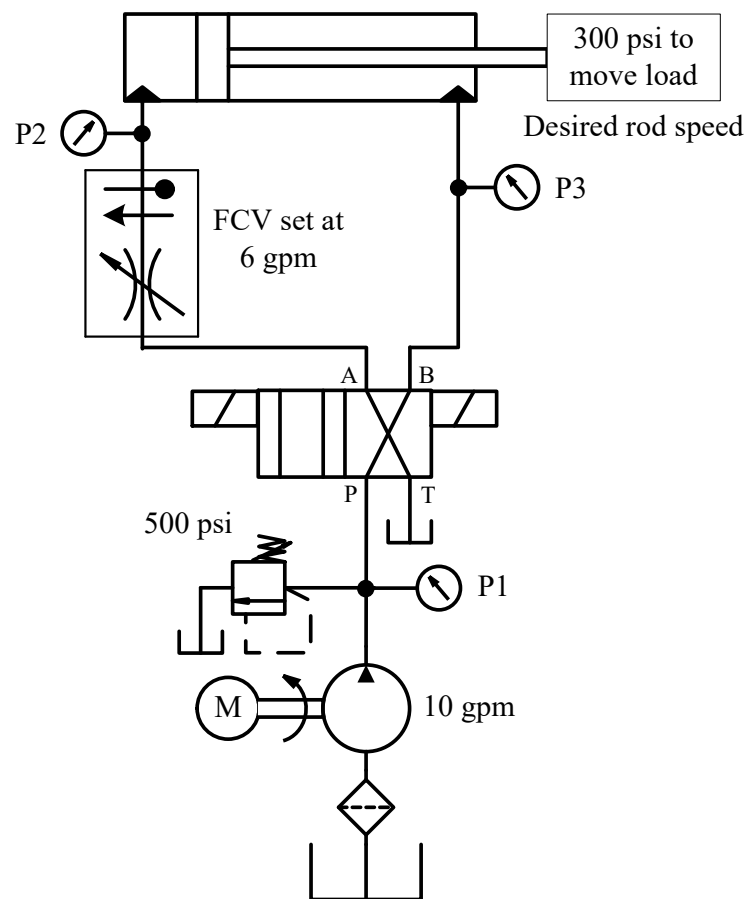
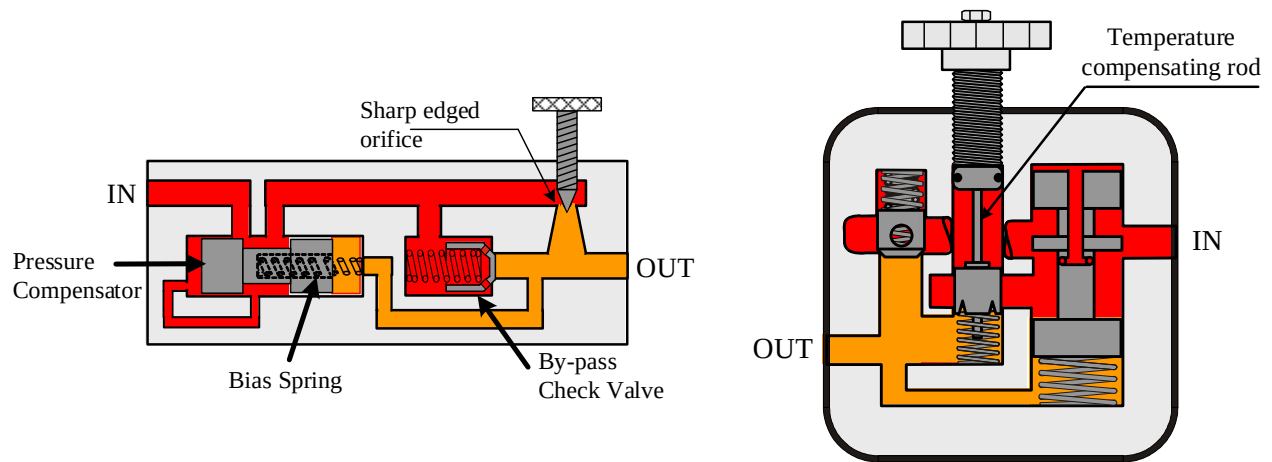
(ISO symbol)

Two methods are used to compensate for temperature fluctuations in a pressure compensated flow control valve; sharp edge orifice and temperature compensating rod.

Research has shown that when a liquid flows through a properly shaped opening (hole or orifice), the rate of flow is not affected by temperature. When liquid is sheared while flowing through a sharp edge it actually neutralizes the effect of a fluid's changing viscosity. The problem with a sharp edge orifice is the changing variables in a hydraulic circuit, the complicated mathematical calculations involved with the size/design, precision tolerances and machining required to manufacture one.

Another method of temperature compensation is the use of an aluminum or bi-metallic rod. The rod is attached to the movable section of the adjustable flow control and controls the size of the opening (hole or orifice) when the temperature changes. The material the rod is made of is very sensitive to temperature. When the temperature increases it will expand through thermal expansion, and when the temperature decreases it will contract.

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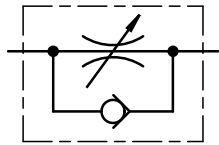
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Adjustable Flow Control Valve with By-Pass Check Valve

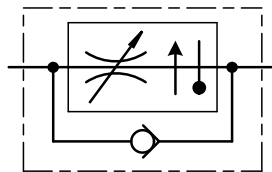
As was mentioned at the beginning of this chapter, the function of a flow control is to reduce/restrict the volume of fluid in a leg of a circuit, which will result in a speed reduction of the actuator. Since most transmission lines have bi-directional flow and most actuators are expected to move in two directions, the flow would actually be restricted in both directions. If controlling flow in both directions is not the desired effect, then some way has to be provided to go around/by-pass the flow control.

Usually a check valve is used, as a separate component or built in, with the flow control to have controlled speed in one direction and free flow in the opposite direction.

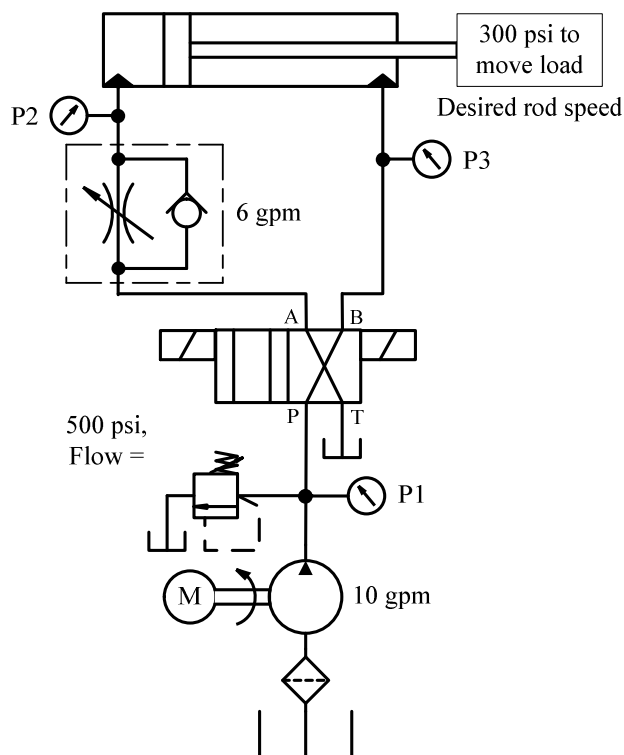
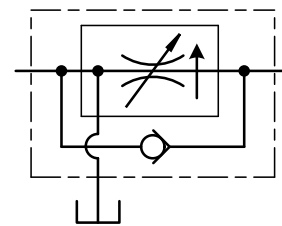
Adjustable flow control,
Restrictor type,
w/ By-Pass Check Valve



Adjustable flow control,
Restrictor type,
Pressure Compensated,
Temperature Compensated,
w/ By-Pass Check Valve



Adjustable flow control,
By-Pass type,
Pressure Compensated,
w/ By-Pass Check Valve



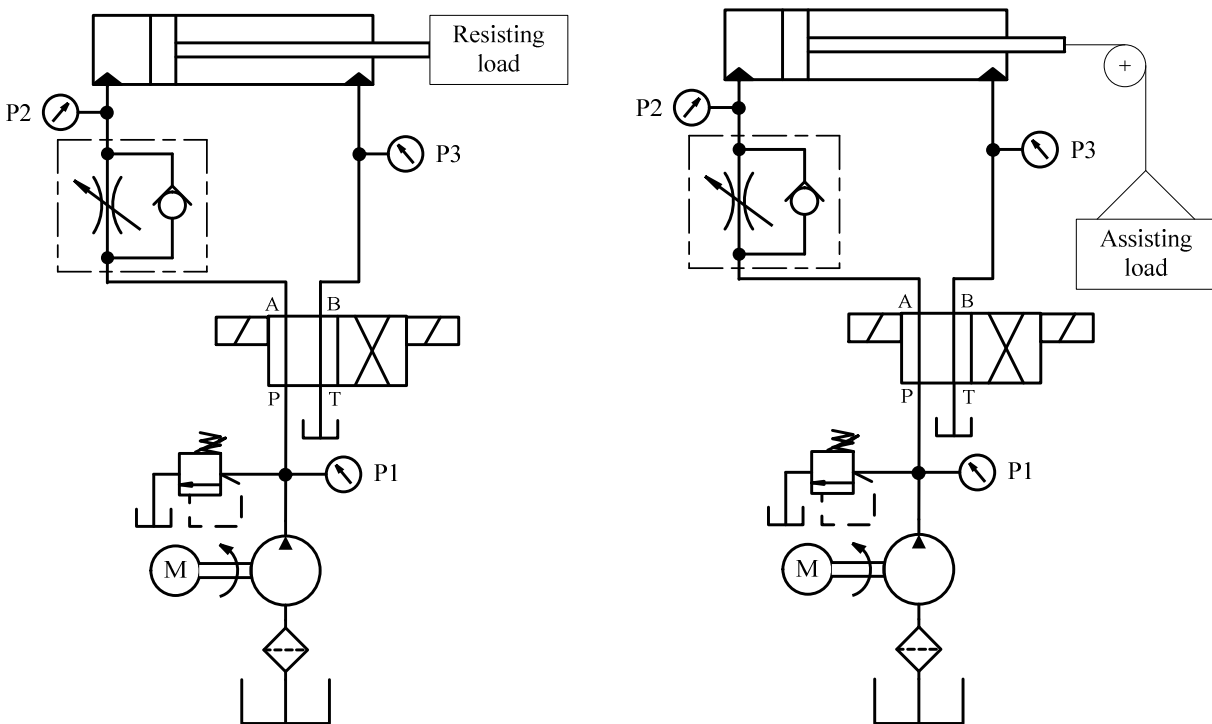
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Flow Control Methods

There are three basic methods of controlling flow in a circuit; **Meter-In**, **Meter-Out**, and **Bleed-Off**.

Meter-In Method

Meter-In is the method of placing the flow control in the circuit so as the fluid is metered (or controlled) before it enters the actuator.



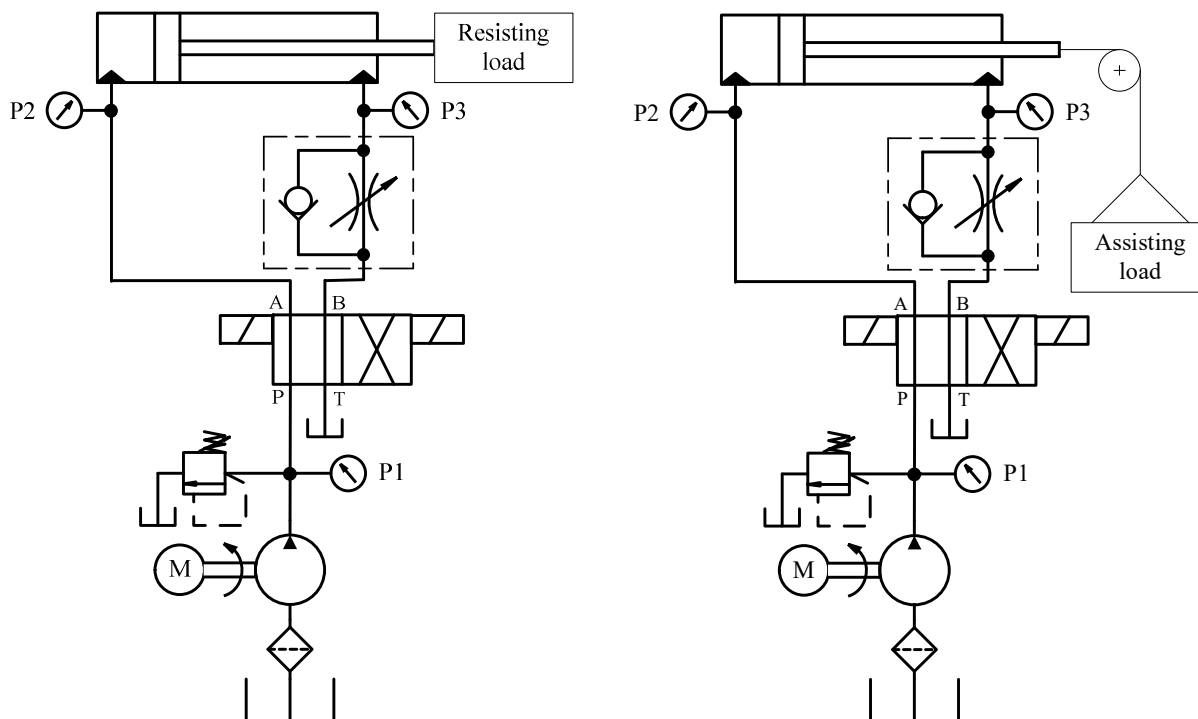
Some advantages of a meter-in method would be; it is an extremely accurate method of controlling flow in resisting load applications, it can dampen certain flow and pressure fluctuations (within reason), it is the only type of circuit application in which a by-pass type of pressure compensated flow control can be used.

A disadvantage of a meter-in method would be; it cannot be used in an assisting load application.

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Meter-Out Method

Meter-Out is the method of placing the flow control in the circuit so as the fluid is metered (or controlled) as it exits or leaves the actuator.



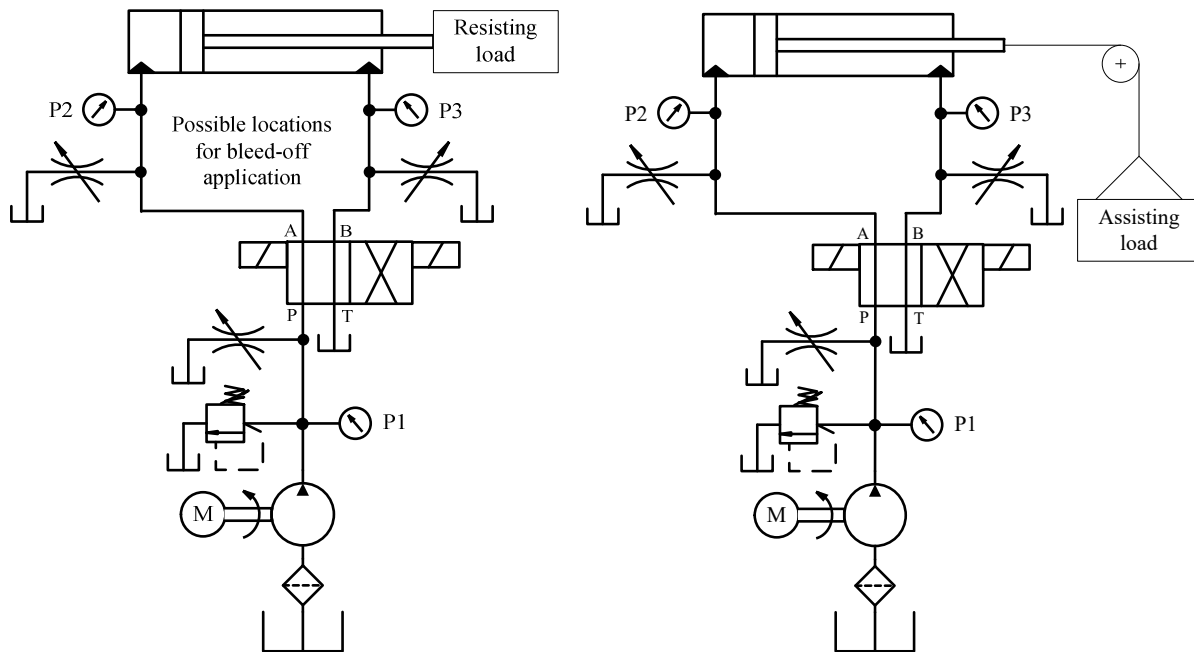
Some advantages of a meter-out method would be; it is an extremely accurate method of controlling flow, it can handle all load applications (resisting and assisting), it is the most popular method of flow controlling, it can use all types of flow controls except the by-pass type.

A disadvantage of a meter-out method would be; depending on the circuit dynamics - it can generate a large, even dangerous, pressure intensification situation at times.

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Bleed-Off Method

Bleed-Off is a method of placing the flow control in a way that the flow control is teed off of the supply line to the actuator and diverts the unwanted flow back to tank/reservoir. It acts as a controlled leak.



An advantage of a bleed-off method would be; the pump operates at the pressure required to do the work – since excess flow returns to tank/reservoir instead of through the relief valve and it generates less heat.

A disadvantage of a bleed-off method would be; there is considerable loss of accuracy because the measured flow is actually going to tank and not the actuator. Making the accuracy of the actuator subject to any variations in flow and pressure in the circuit, which means the actuator would become undependable. Also, a bleed-off method should never be used in assisting load applications, because it cannot control the load. Even though a bleed-off has certain advantages, its disadvantages far outweigh the advantages. For this reason the bleed-off is the least popular of the three methods.

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In summary, the meter-in and meter-out methods are very accurate, but not very efficient. The bleed-off method is very efficient, but not very accurate. ****When in doubt; meter-out.****

The placement/position of the flow control valve in the circuit is irrelevant, it is the effect the placement/position has on the actuator that is of primary concern. Proper placement/position of a flow control can be decided by several factors, however these three factors tend to be the most important when trying to satisfy an actuator speed application. Always ask these questions when deciding where to place/position a flow control valve.

With this place/position of the flow control valve;

1. can I control the speed of the actuator?
2. can I control the load/work?
3. is the location too close, or too far away from the actuator?

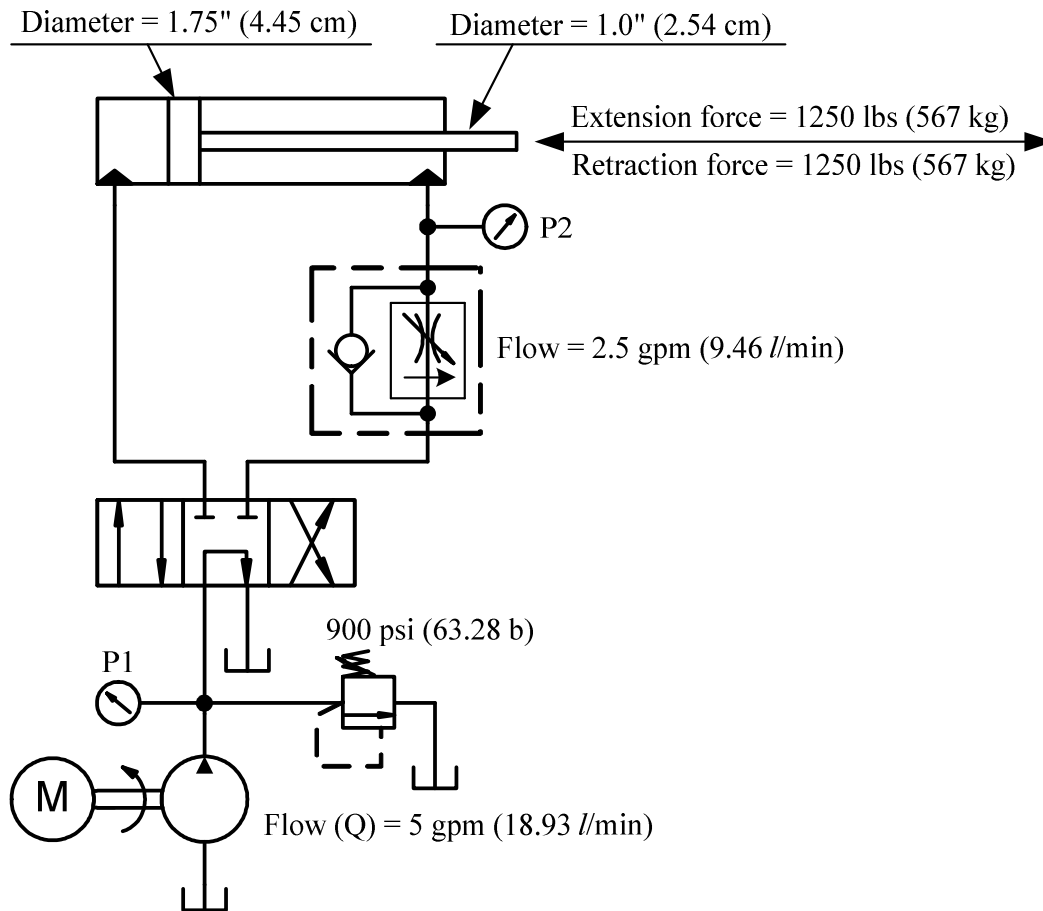
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Flow Control Valve Exercises

FCV Exercise #1



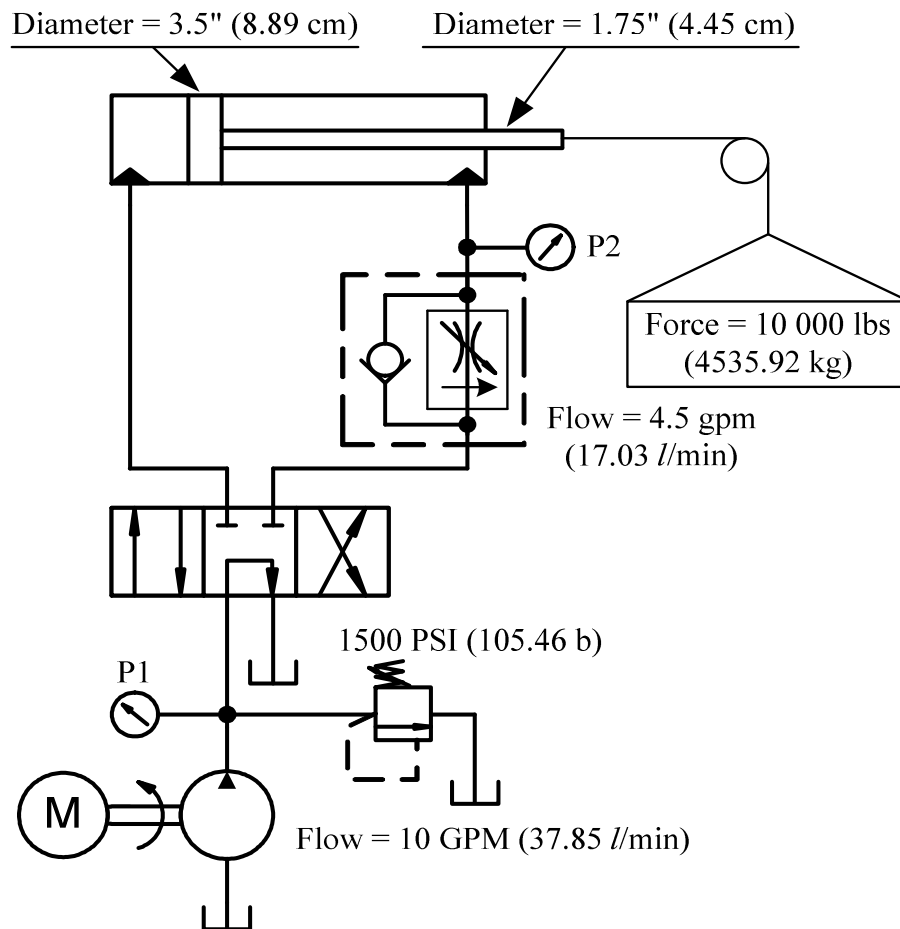
- 1) In the circuit illustrated, the horizontal mounted cylinder has the same load during extension and retraction: (Disregard all other internal resistance.)
 - a) Calculate P1 during extension.
 - b) Calculate P2 during extension.
 - c) Calculate rod speed during extension.

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FCV Exercise #2



2) In the circuit illustrated, the horizontal mounted cylinder has the same load during extension and retraction. (Disregard all other internal resistance.)

a) Calculate P1 during extension.

b) Calculate P2 during extension.

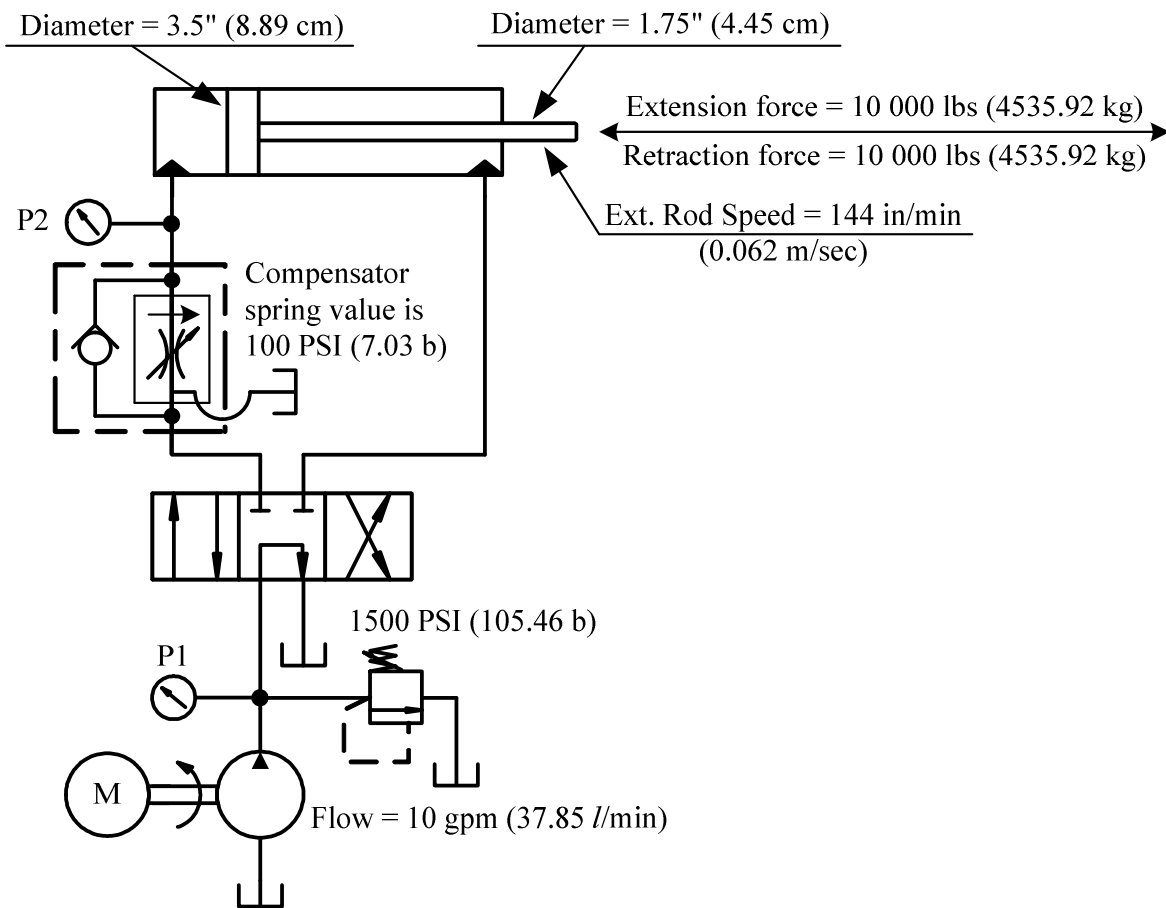
c) Calculate rod speed during extension.

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FCV Exercise #3



3) In the circuit illustrated, the horizontal mounted cylinder has the same load during extension and retraction. (Disregard all other internal resistance.)

a) Calculate P1 during extension.

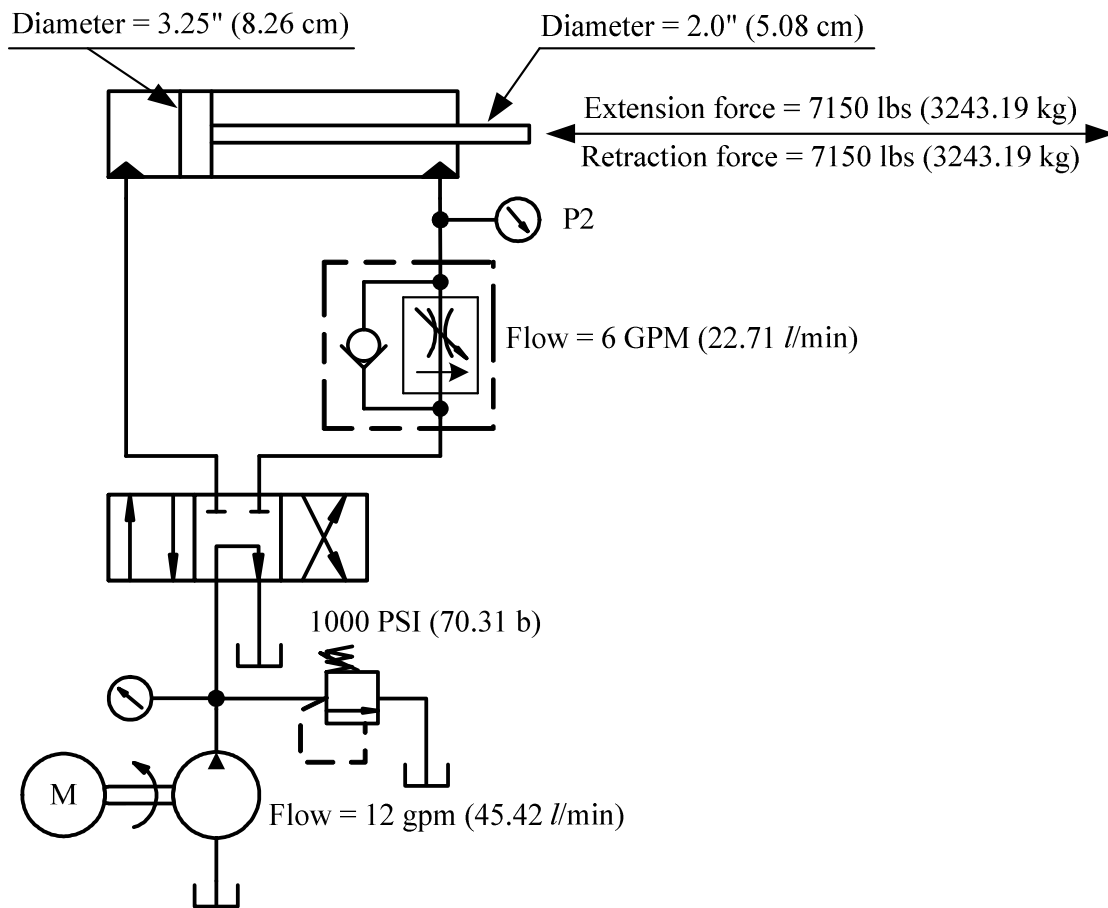
b) Calculate P2 during extension.

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FCV Exercise #4



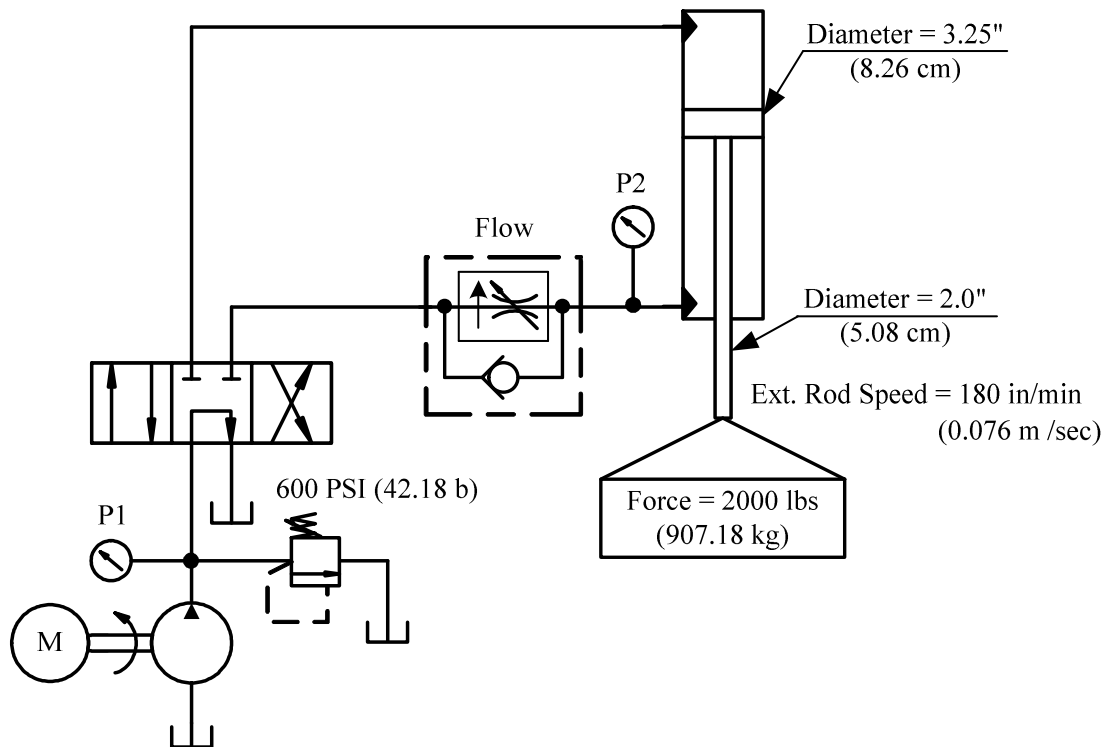
- 4) In the circuit illustrated, the horizontal mounted cylinder has the same load during extension and retraction. (Disregard all other internal resistance.)
 - a) Calculate P2 during extension.
 - b) Calculate rod speed during extension.
 - c) Calculate P2 during retraction.
 - d) Calculate retraction rod speed using the current cylinder and relief valve settings.

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FCV Exercise #5



5) In the circuit illustrated, the vertical mounted cylinder has the same load during extension and retraction. (Disregard all other internal resistance.)

a) Calculate P2 during extension.

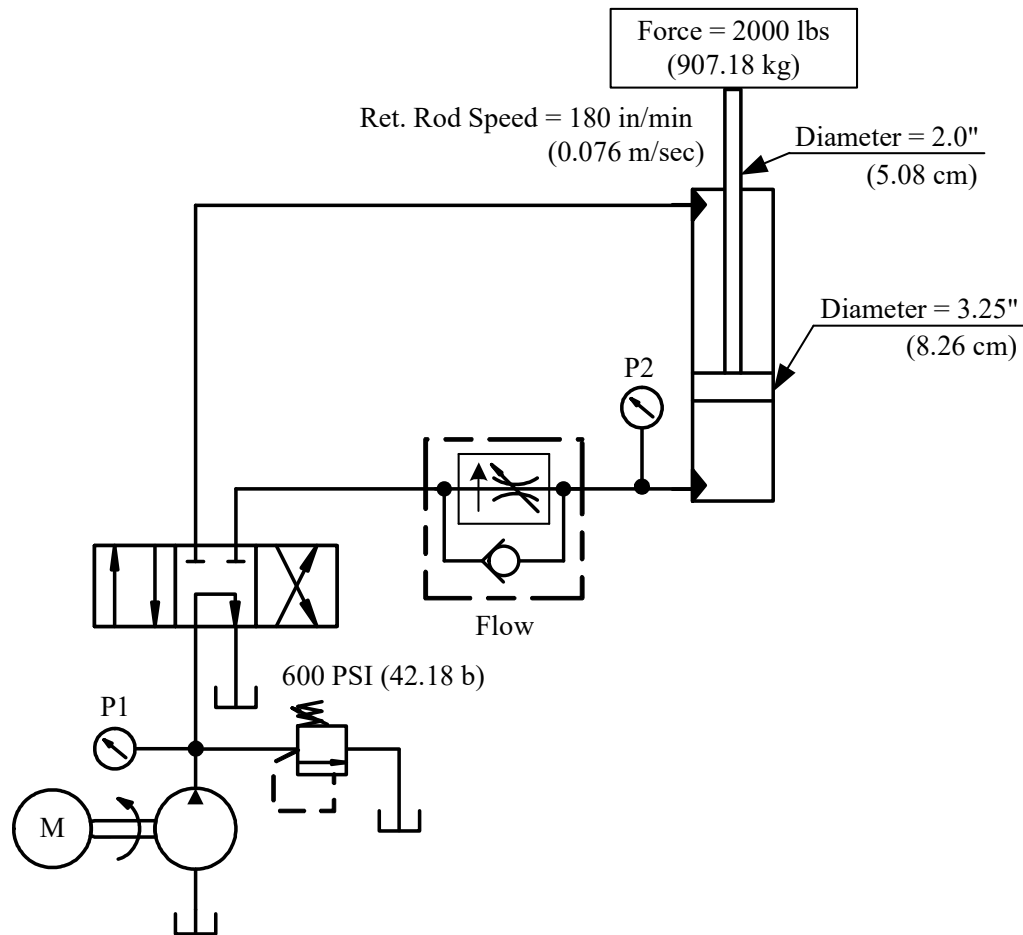
b) Calculate flow through the flow control during extension.

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FCV Exercise #6



6) In the circuit illustrated, the vertical mounted cylinder has the same load during extension and retraction. (Disregard all other internal resistance.)

a) Calculate P2 during retraction.

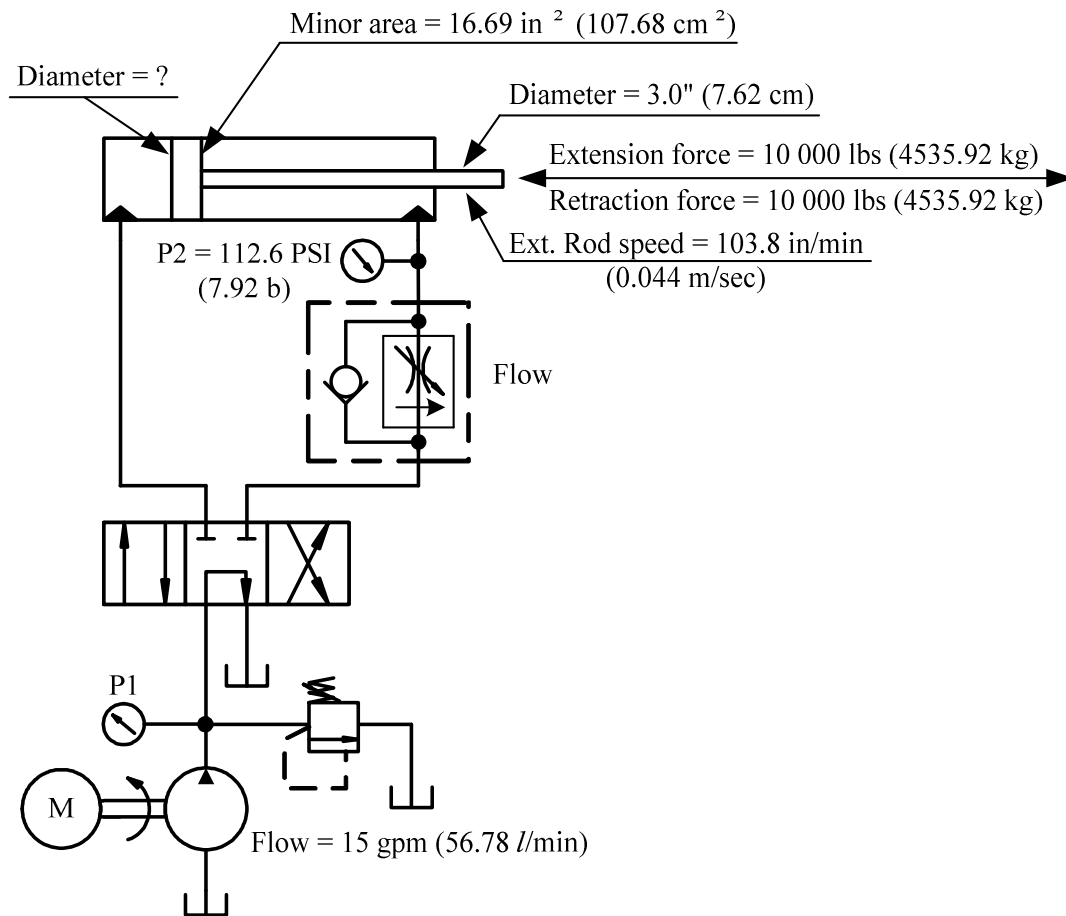
b) Calculate flow through the flow control during retraction.

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FCV Exercise #7



7) In the circuit illustrated, the horizontal mounted cylinder has the same load during extension and retraction. (Disregard all other internal resistance.)

a) Calculate P1 during extension.

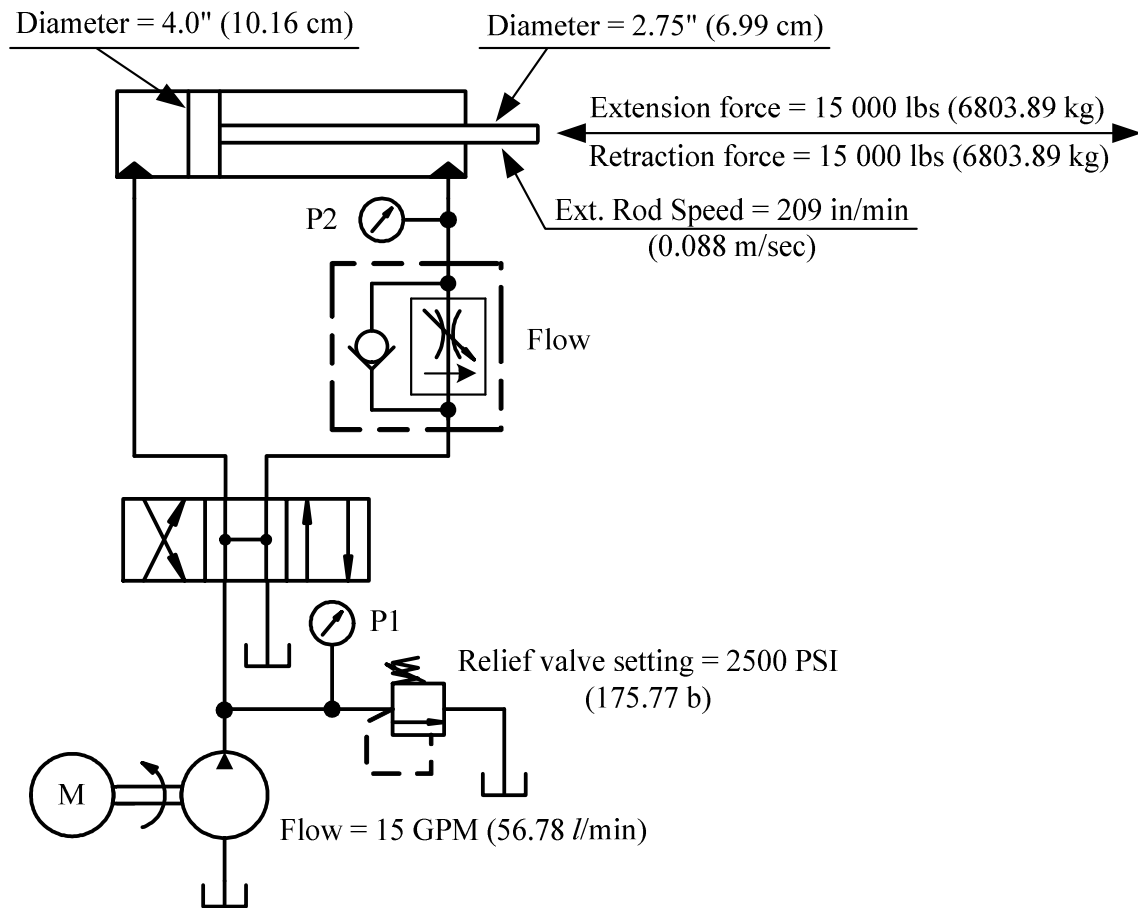
b) Calculate flow through restrictor during extension.

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FCV Exercise #8



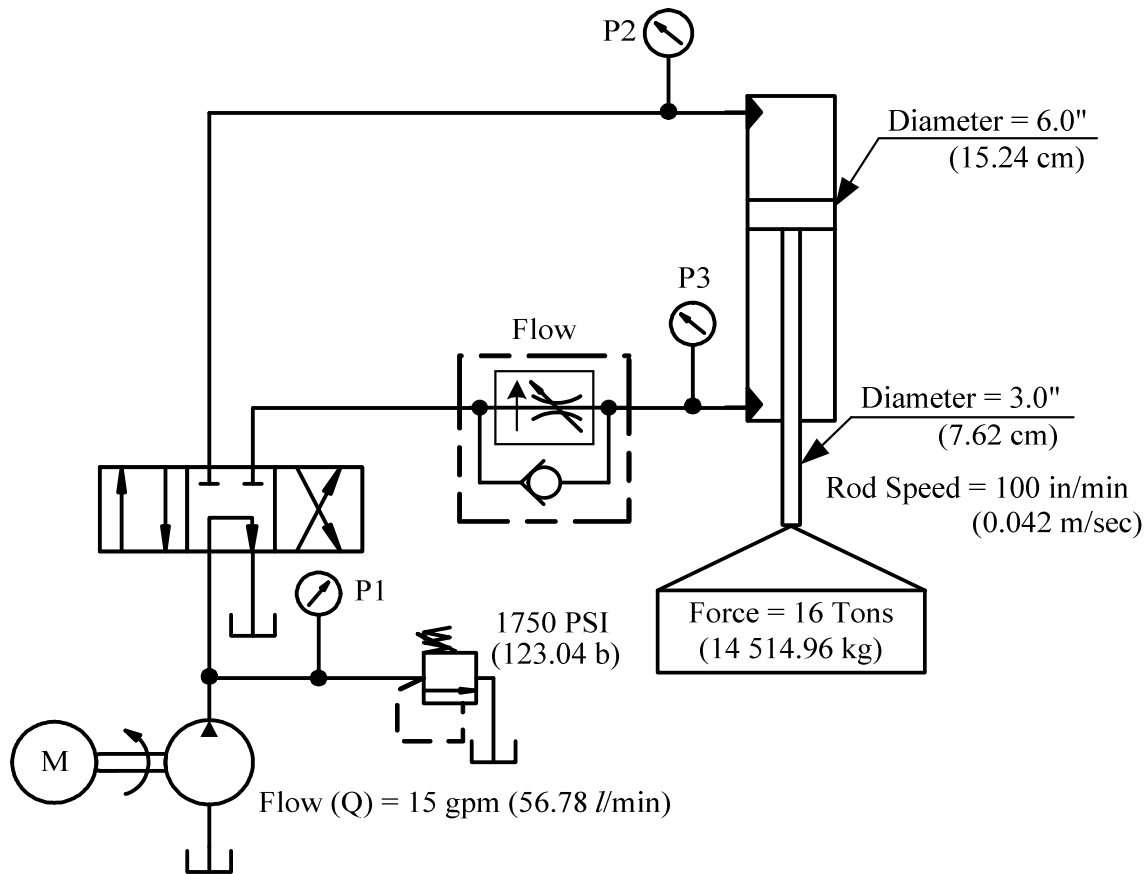
- 8) In the circuit illustrated, the horizontal mounted cylinder has the same load during extension and retraction. (Disregard all other internal resistance.)
- Calculate P1 during extension.
 - Calculate P2 during extension.
 - Calculate flow through flow control during extension.
 - Calculate P1 during retraction.
 - Calculate P2 during retraction.
 - Calculate rod speed during retraction.

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FCV Exercise #9



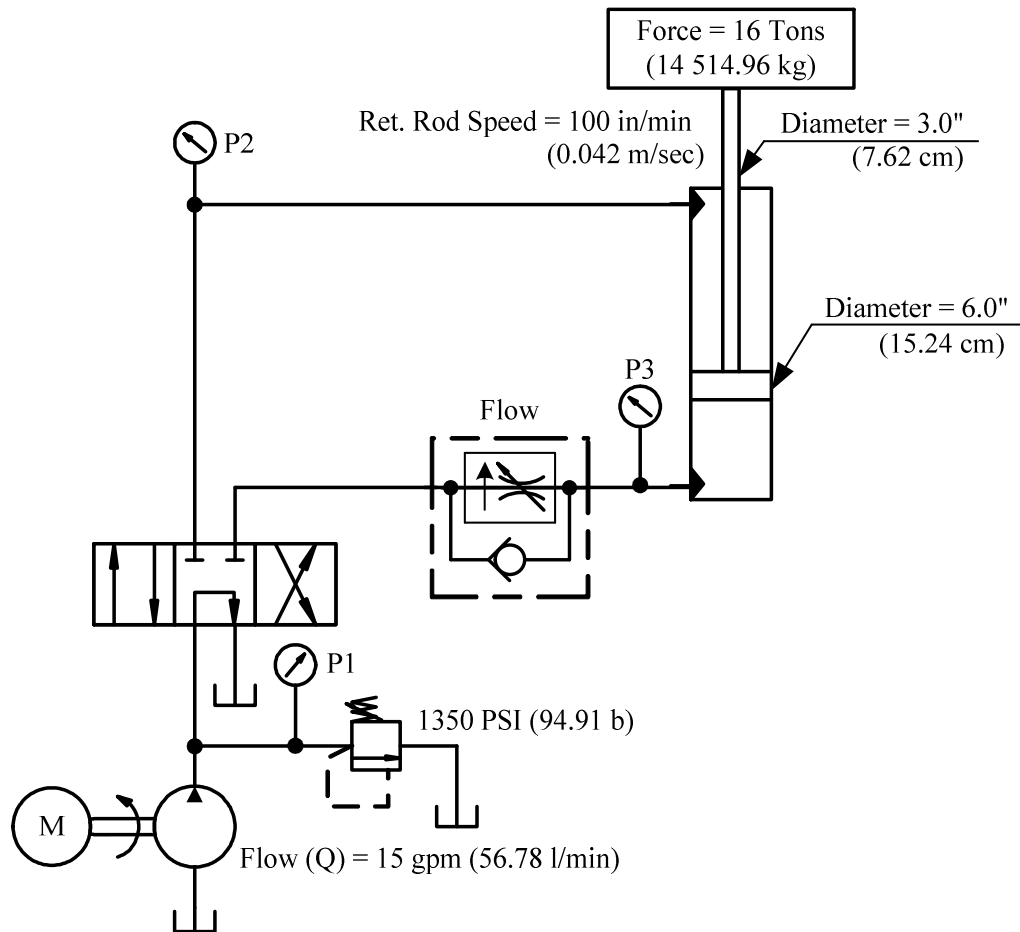
- 9) In the circuit illustrated, the vertical mounted cylinder has the same load during extension and retraction. (Disregard all other internal resistance.)
 - a) Calculate P1 during extension.
 - b) Calculate P2 during extension.
 - c) Calculate P3 during extension.
 - d) Calculate flow through flow control during extension.
 - e) Calculate P1 during retraction.
 - f) Calculate P2 during retraction.
 - g) Calculate rod speed during retraction.

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FCV Exercise #10



10) In the circuit illustrated, the vertical mounted cylinder has the same load during extension and retraction. (Disregard all other internal resistance.)

- Calculate P1 during retraction.
- Calculate P2 during retraction.
- Calculate P3 during retraction.
- Calculate flow through flow control during retraction.
- Calculate P1 during extension.
- Calculate P2 during extension.
- Calculate rod speed during extension.

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Accumulators

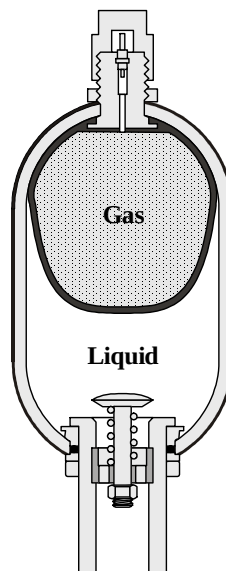
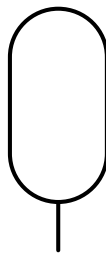
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Introduction to Accumulators

Unlike a gas in a pneumatic system, the liquid used in a hydraulic system cannot be compressed and stored for use at a different time or place. Where it can be advantageous to store an incompressible liquid under pressure is with a hydraulic accumulator.

Accumulators are components that store energy in the form of a pressurized fluid by using a compressible gas as the means to create a force on the liquid to pressurize it. Because of their ability to store extra energy and release it when needed, accumulators are very useful in improving hydraulic efficiency and as you will see later have several other purposes in a hydraulic system.

**Accumulator
(Basic symbol)**



Types of Accumulators

The different types of accumulators include;

1. Weight Loaded
2. Spring Loaded
3. Hydro-Pneumatic (Gas Charged) Types
 - a. Piston Type
 - b. Diaphragm Type
 - c. Bladder Type

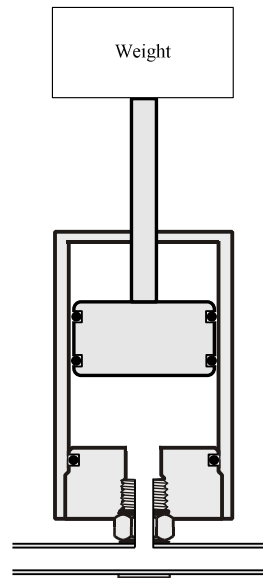
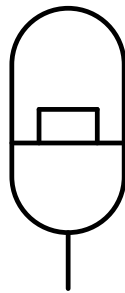
Each type identifies by what means an accumulator maintains a force on the liquid while it is stored.

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Weight Loaded Accumulator

A weight loaded accumulator maintains a force on the stored liquid by some heavy weight acting on a piston or ram. The weight may be any heavy material such as iron block, concrete block, or even a water bag or container.

Weight loaded accumulator



The advantages of a weight loaded accumulator would be; this is the only type that creates a constant pressure/flow whether it is full or nearly empty, can supply large volumes of fluid, can serve several hydraulic circuits or systems at a time.

The disadvantages of a weight loaded accumulator would be; it is the oldest of all the basic types, are usually large, heavy and bulky, difficult to size for smaller applications, generally have to be mounted vertically, pressure intensification (or shock loading) of the fluid during rapid discharge and sudden stop, changing the load may be impractical.

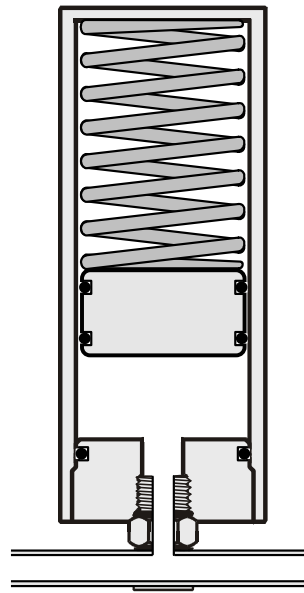
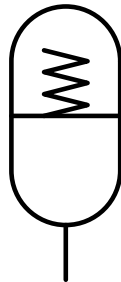
Typical applications would be central hydraulic systems, heavy presses or mills where a constant pressure is required or in applications where a large volume of fluid is necessary.

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Spring Loaded Accumulator

A spring loaded accumulator applies a force on the stored liquid by a spring acting on a piston or ram.

Spring loaded accumulator



The advantages of a spring loaded accumulator would be; can be sized to fit an application better than the weight loaded– the only limitation to sizing would be spring size or rate, can be mounted in any position, absorbs sudden pressure surges (shock loading) very well (acts like a shock absorber on a car).

The disadvantages of a spring loaded accumulator would be; it does not create a constant pressure/flow during discharge, when large volumes of fluid are needed the forces required would make the spring sizes impractical, typically operate at lower pressure values, changing the spring load/rate may be impractical also.

Typical applications would be individual central hydraulic systems, heavy presses or mills where a large volume of fluid is necessary.

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Hydro-Pneumatic Accumulators

A hydro-pneumatic (gas charged) type is the most common type of accumulator used in industrial hydraulic systems. This type applies a force on the stored liquid by using a compressed gas.

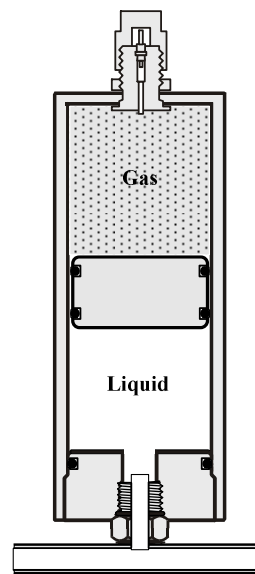
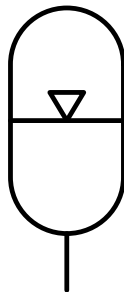
Hydro-pneumatic accumulators are divided into three basic types; Piston Type, Diaphragm Type, and Bladder Type. The name of each type indicates the type of device that separates the gas from liquid.

In all types of hydro-pneumatic accumulators **oxygen or compressed air should never be used**, because of the danger of exploding an air/oil vapor mixture. Always use an inert gas for the gas charge. The most commonly used gas is dry nitrogen.

Piston Type (Hydro-Pneumatic) Accumulator

A piston type (hydro-pneumatic) accumulator applies a force on the stored liquid by compressed gas acting on a piston.

**Piston type
accumulator**



The advantages of a piston type accumulator would be; can be sized to fit virtually any application, can be mounted in any position, absorbs sudden pressure surges (shock loading) very well, can change the gas charge to match an application.

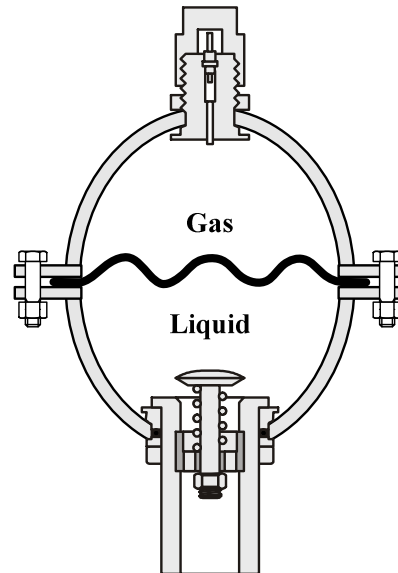
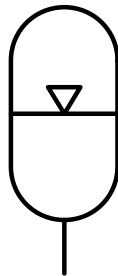
The disadvantages of a piston type accumulator would be; it does not create a constant pressure/flow during discharge.

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Diaphragm Type (Hydro-Pneumatic) Accumulator

A diaphragm type (hydro-pneumatic) accumulator applies a force on the stored liquid by compressed gas acting on a diaphragm which separates two halves of the housing.

**Diaphragm type
accumulator**



The advantages of a diaphragm type accumulator would be; can be sized to fit virtually any application – typically used in smaller volume applications, can be mounted in any position, absorbs sudden pressure surges (shock loading) very well, can change the gas charge to match an application.

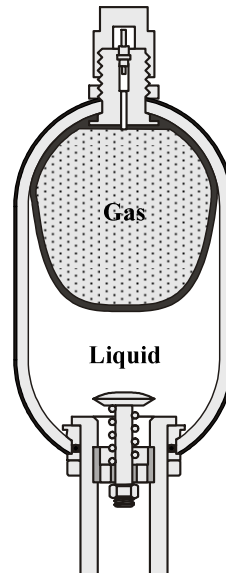
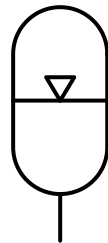
The disadvantages of a diaphragm type accumulator would be; typically used in small volume applications, it does not create a constant pressure/flow during discharge.

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Bladder Type (Hydro-Pneumatic) Accumulator

A bladder type (hydro-pneumatic) accumulator applies a force on the stored liquid by compressed gas acting in a bladder (bag).

Bladder type accumulator



The advantages of a bladder type accumulator would be; can be sized to fit virtually any application, absorbs sudden pressure surges (shock loading) very well, can change the gas charge to match an application.

The disadvantages of a bladder type accumulator would be; it should be mounted in a vertical position, it does not create a constant pressure/flow during discharge.

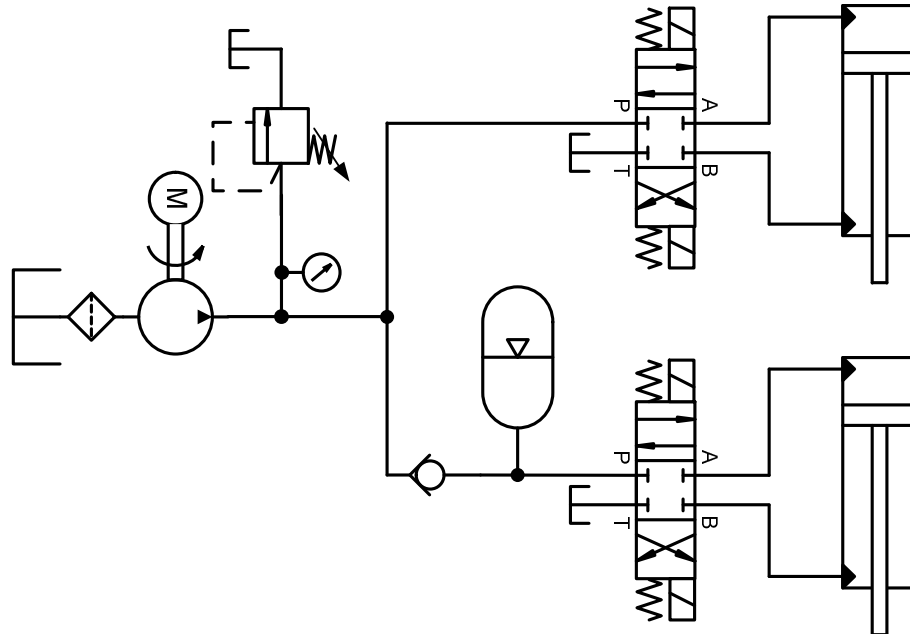
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Accumulator applications in a Circuit

Since accumulators are stored sources of energy they can have a variety of purposes in a hydraulic system. Some examples would be;

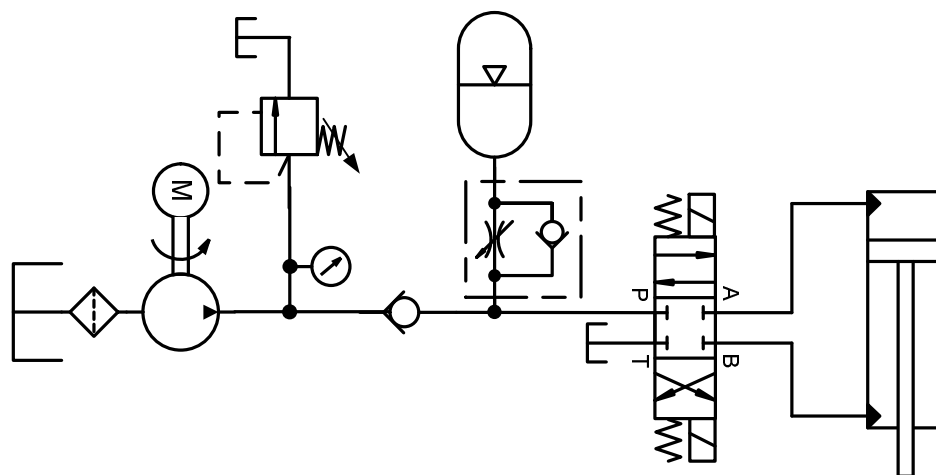
1. Maintain system pressure

Maintain system pressure



2. Increase system flow/speed

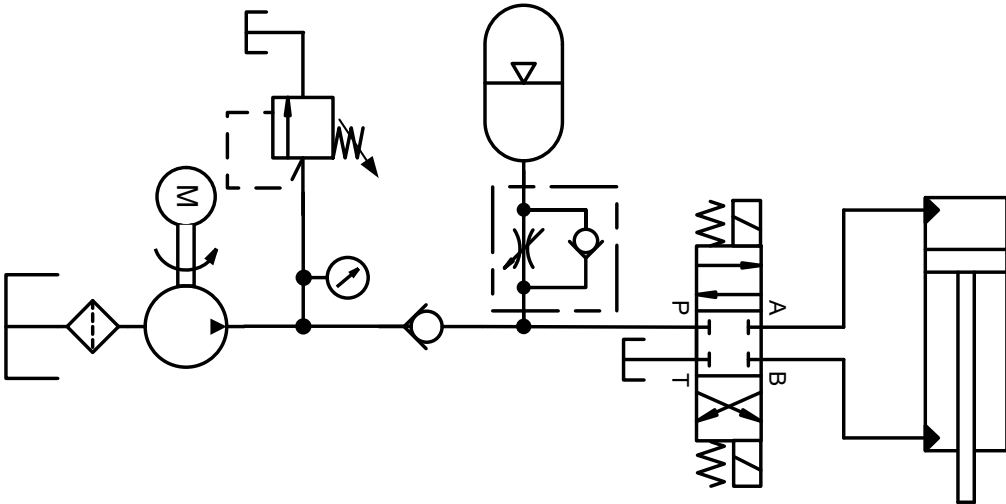
Increase system flow/speed



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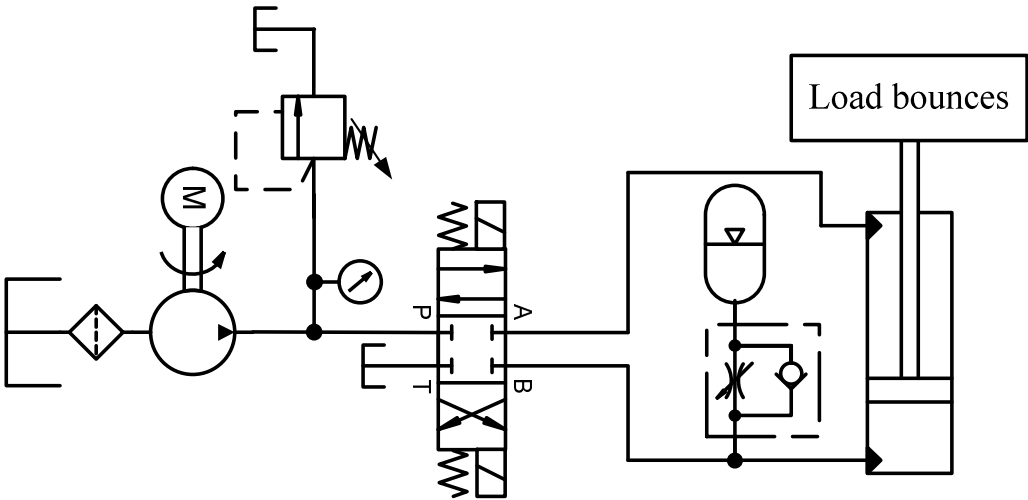
3. Absorb pressure surge/shock

Absorb pressure surge/shock



Or

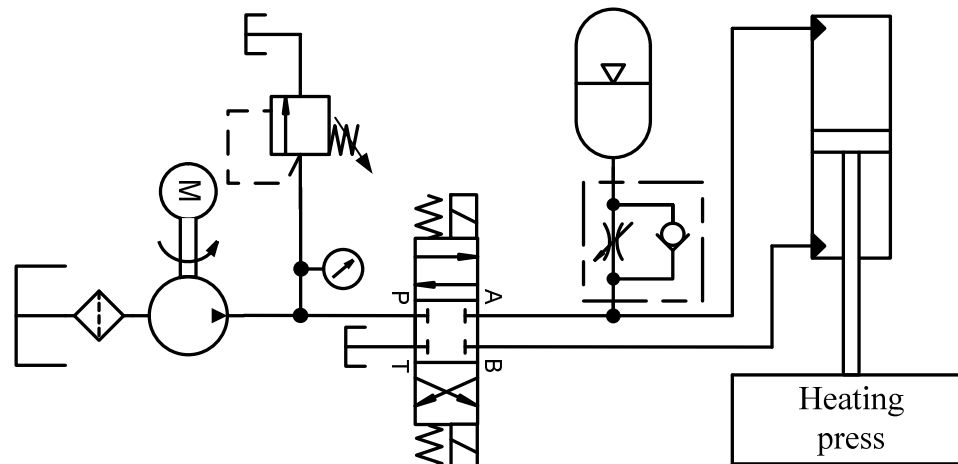
Absorb pressure surge/shock



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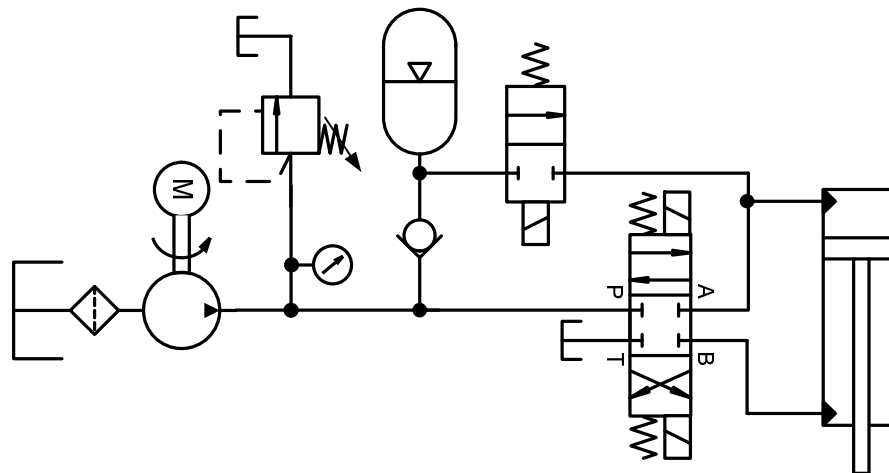
4. Absorb extra volume (thermal expansion)

Absorb extra volume
(Thermal expansion)



5. As a standby (emergency) power source

As a standby (emergency)
power source



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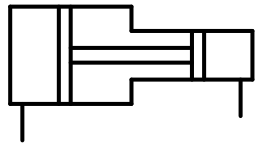
Accumulator Precautions

1. **Never work on a circuit with an accumulator until checking for the absence of pressure.**
2. In any accumulator application, always use a method (manual or automatic) to de-energize the accumulator when the machine is shut down or requires maintenance.
3. Always isolate the pump from the accumulator with a check valve so the accumulator flow cannot back up into the pump.
4. To avoid the danger of explosion, air or oxygen should not be used or allowed to enter the system.
5. A repair or connection of a pressure gauge should be done after de-energizing the accumulator.
6. For bladder maintenance;
 - a. Bladders should never be stored folded.
 - b. Bladders should be stored by hanging from the gas valve.
 - c. Bladders should be stored inflated to their full length.
7. The gas pre-charge pressure should always be below the minimum operating pressure.
8. If you are not familiar with the proper procedures for charging/pre-charging an accumulator, **do not take it upon yourself.** To try and service an accumulator without the proper training or knowledge is dangerous. An accumulator has the potential to become a bomb when improperly handled.

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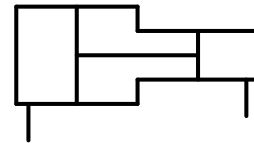
Intensifier

An intensifier is a device used to increase/multiply fluid pressure. An intensifier can develop pressures several times higher than developed by the pump.

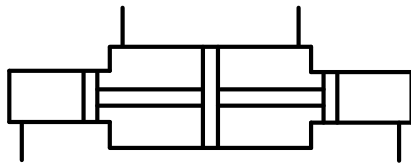


(Detailed symbol)

Intensifier,
single acting

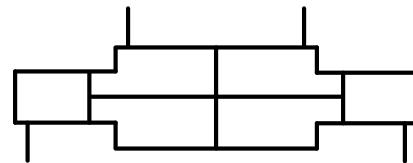


(Simplified symbol)



(Detailed symbol)

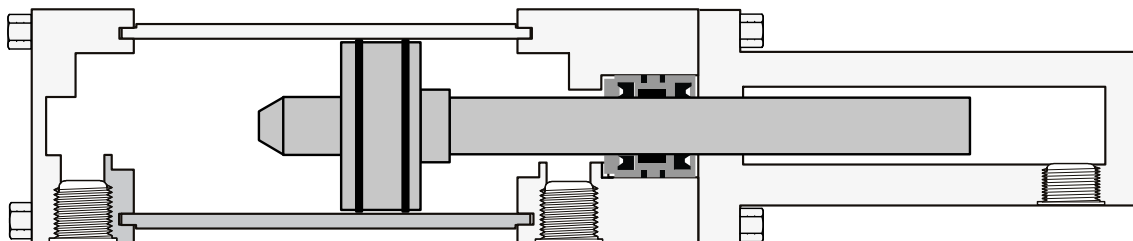
Intensifier,
double acting



(Simplified symbol)

System pressure enters and acts on the large area side of the intensifier, which creates a force that is delivered down the connecting rod to the small area side. That force against the small area piston creates a higher discharge pressure.

Each side of the intensifier will have the same force, since both sides are connected thru the connecting rod. The low pressure against a large area will always equal the high pressure against the small area, thus the pressure and area values in an intensifier are always inversely proportional to each other.



The disadvantage to using an intensifier is the volume of fluid being discharged. Since the advantage is a higher pressure is created by the smaller area, the smaller area also creates a smaller volume. What is gained in pressure is lost in volume.

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Priming

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Definition of Priming

The definition of priming indicates that a component or circuit has been purged of all air and is filled with liquid, or in this case, hydraulic fluid. Without properly removing the air, the system could be seriously damaged from lack of lubrication.

After a modification or repair has been made several factors need to be taken into consideration before operating a component or system. Any time a repair is made to a hydraulic system the opportunity to lose fluid and gain air in your system exists. It could be as simple as removing fluid lines and replacing a component. Since so many different manufacturers and components exist, an exact procedure for each circumstance is not practical, but a general procedure can be used to help in most cases.

General Component / System Priming

1. If the fluid lines are flexible, ensure they are secured as high as possible to prevent fluid loss during repair.
2. **Never perform a dry installation!** When installing a repaired or new component always ensure it contains hydraulic fluid for the initial start-up. Operating a system or component dry at initial start-up would result in costly damage from the lack of lubrication from the hydraulic fluid.
3. Once the component has been repaired or replaced, if the component has a priming (purge or vent) port, it should then be filled with fluid. If the component does not have a priming port, then it may be possible to remove a line and fill it from the open port.
4. Disconnect any inlet and outlet lines from the component and fill those lines with fluid and reconnect.
5. When the installation and initial priming has been performed, it is likely the component or system still has a small amount of air. Hydraulic components and systems have a hard time purging trapped air. To help the air escape, we have to reduce the resistance, which means; running the fluid line directly back to tank/reservoir if possible, reduce the spring pressures/settings on all components.
6. Check the pump for proper rotation, then slowly rotate or jog the pump several times (for example; on two seconds - off three seconds) until the pressure begins to rise, and run the pump at the lowest rpm possible until the system is free of air and is fully primed with fluid.
7. After the system has been primed, run it for several minutes at the lower pressure settings, this will purge any remaining trapped air.
8. When the system is primed, the system can now be closed and all the spring/pressure settings and adjustments made for proper system operation.

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Adjustment and Priming Procedures Exercise

Organize the following adjustment and priming procedures in the proper order by numbering 1 thru 8.

- ____. After the system has been primed, run it for several minutes at the lower pressure settings, this will purge any remaining trapped air.
- ____. Once the component has been repaired or replaced, if the component has a priming (purge or vent) port, it should then be filled with fluid. If the component does not have a priming port, then it may be possible to remove a line and fill it from the open port.
- ____. When the system is primed, the system can now be closed and all the spring/pressure settings and adjustments made for proper system operation
- ____. When the installation and initial priming has been performed, it is likely the component or system still has a small amount of air. Hydraulic components and systems have a hard time purging trapped air. To help the air escape, we have to reduce the resistance, which means; running the fluid line directly back to tank/reservoir if possible, reduce the spring pressures/settings on all components.
- ____. **Never perform a dry installation!** When installing a repaired or new component always ensure it contains hydraulic fluid for the initial start-up. Operating a system or component dry at initial start-up would result in costly damage from the lack of lubrication from the hydraulic fluid.
- ____. Check the pump for proper rotation, then slowly rotate or jog the pump several times (for example; on two seconds - off three seconds) until the pressure begins to rise, and run the pump at the lowest rpm possible until the system is free of air and is fully primed with fluid.
- ____. If the fluid lines are flexible, ensure they are secured as high as possible to prevent fluid loss during repair.
- ____. Disconnect any inlet and outlet lines from the component and fill those lines with fluid and reconnect.

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Nameplate Data

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Introduction to Nameplate Data

The data supplied by the manufacturer on the nameplate or ID tag of a hydraulic component can vary widely. This material will emphasize the most common data supplied on the nameplate and use basic components found in a hydraulic system.

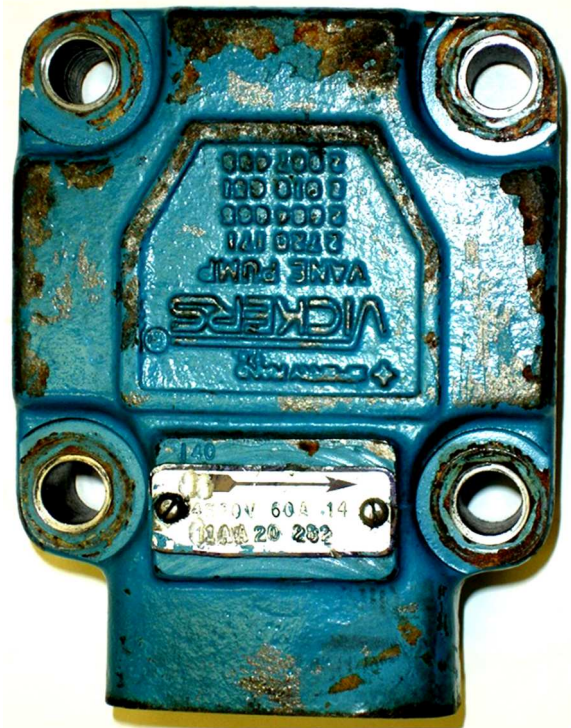
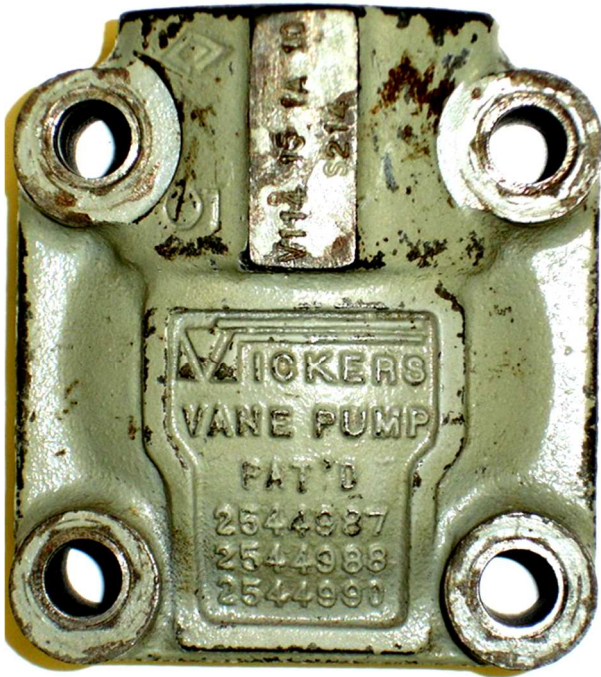
Pump Nameplate Data

Common data that may be included on a pump nameplate (but not limited to);

1. Manufacturer
2. Name (Type) of component (not required)
3. Symbol for component (not required)
4. Model, Serial, Code or Manufacturers number (any combination)
5. Flow rating
6. Pressure rating
7. RPM rating
8. Viscosity rating
9. Direction of rotation (if required)
10. Direction of fluid flow (if required)



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Type			
Model			
Q	GPM	H	Feet
P	HP	N	RPM
PSI max		°F max	⌚

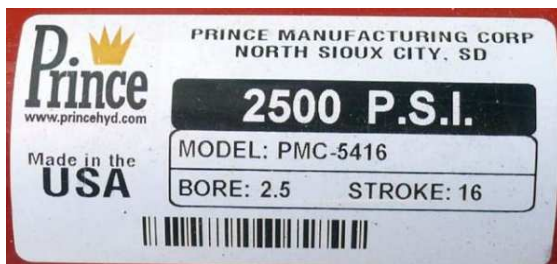
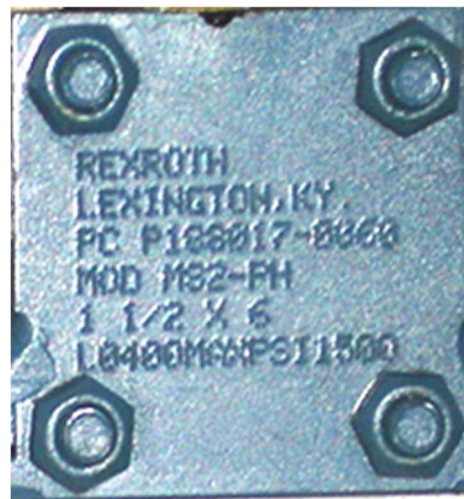
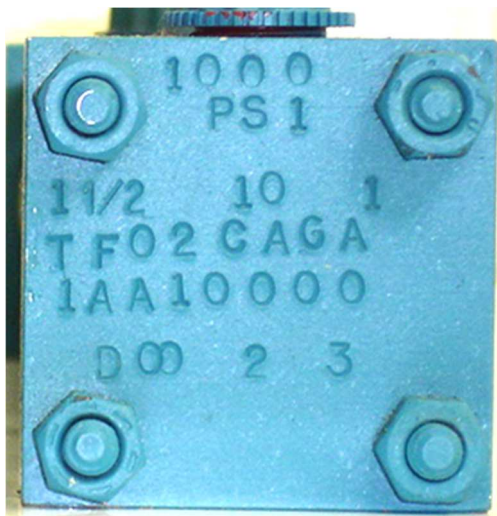
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Actuator Nameplate Data

Common data that may be included on an actuator nameplate (but not limited to);

1. Manufacturer
2. Name (Type) of component (not required)
3. Symbol for component (not required)
4. Model, Serial, Code or Manufacturers number (any combination)
5. Basic size (example; piston diameter and stroke length)
6. Flow rating (if applicable)
7. Pressure rating (if applicable)
8. RPM rating (if applicable)
9. Direction of rotation (if required)

Hydraulic cylinder (linear actuator) examples



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Hydraulic motor (rotary actuator) examples

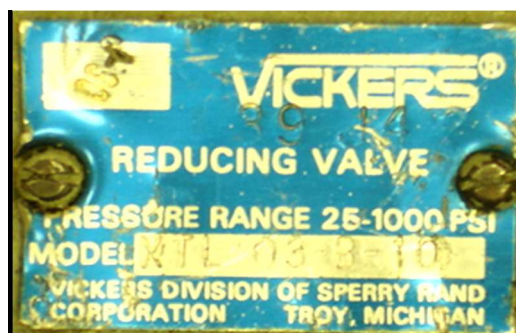


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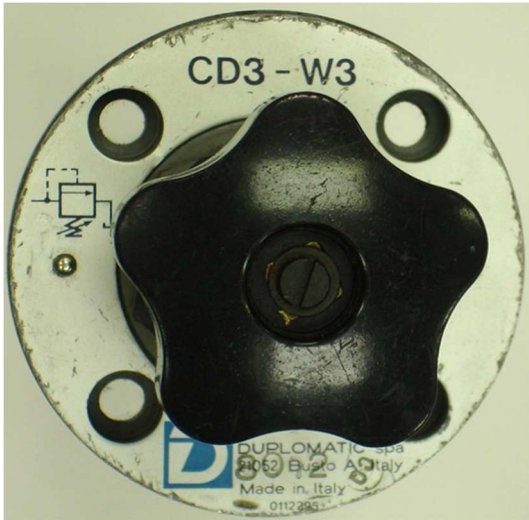
Pressure Control Valve Nameplate Data

Common data that may be included on a PCV nameplate (but not limited to);

1. Manufacturer
2. Name (Type) of component (not required)
3. Symbol for component (not required)
4. Model, Serial, Code or Manufacturers number (any combination)
5. Pressure rating
6. Port identification



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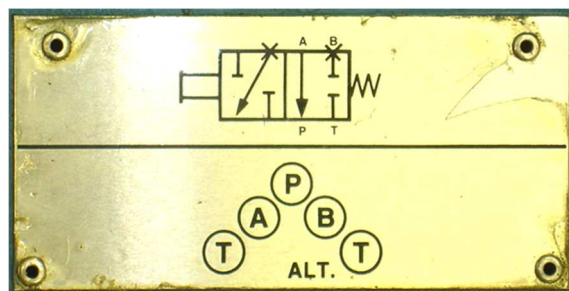
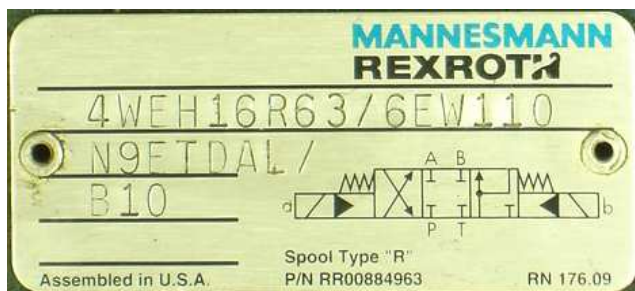


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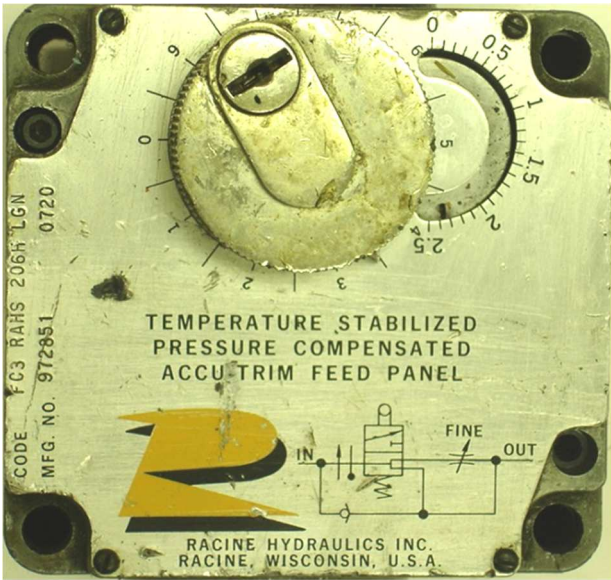
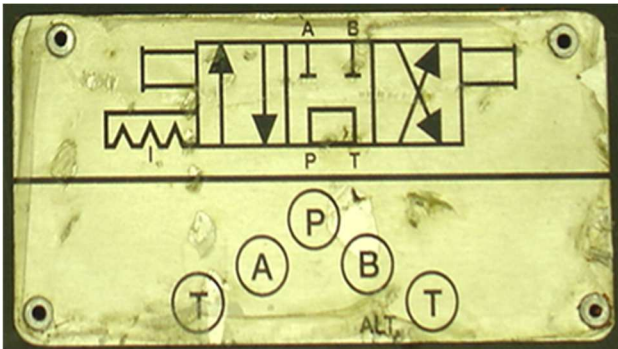
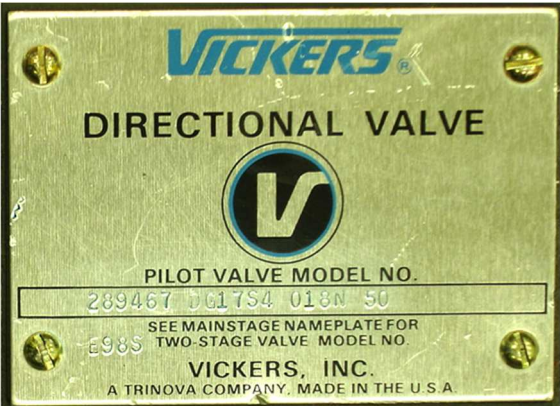
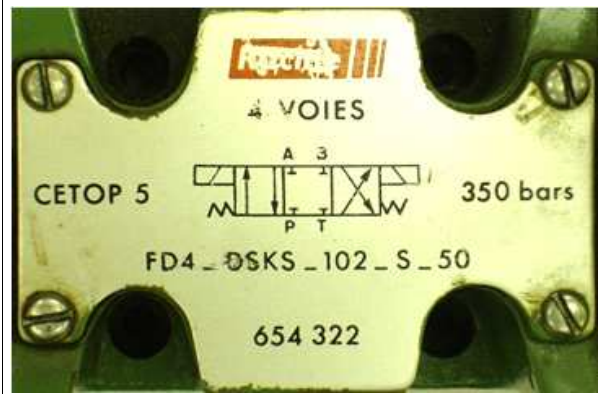
Directional Control Valve Nameplate Data

Common data that may be included on a DCV nameplate (but not limited to);

1. Manufacturer
2. Name (Type) of component (not required)
3. Symbol for component (not required)
4. Model, Serial, Code or Manufacturers number (any combination)
5. Pressure rating
6. Port identification (not required)



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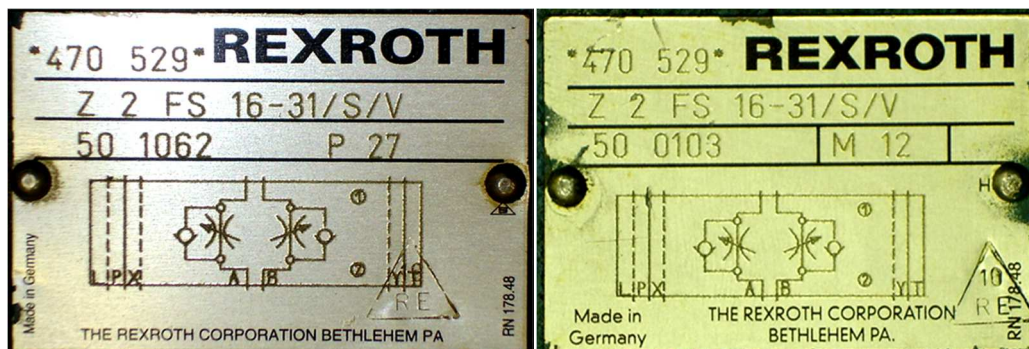


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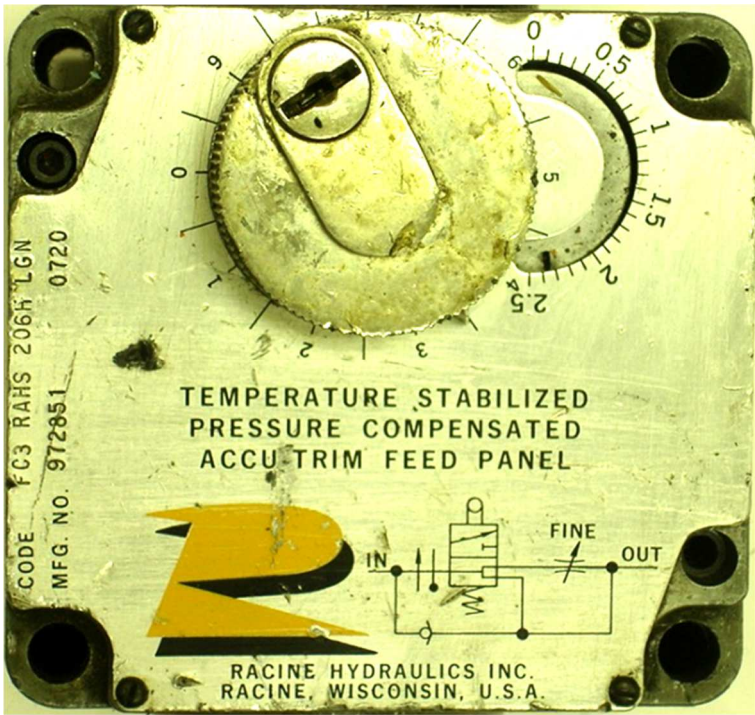
Flow Control Valve Nameplate Data

Common data that may be included on a FCV nameplate (but not limited to);

1. Manufacturer
2. Name (Type) of component (not required)
3. Symbol for component (not required)
4. Model, Serial, Code or Manufacturers number (any combination)
5. Flow rating
6. Pressure rating (if applicable)
7. Port identification



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Accumulator Nameplate Data

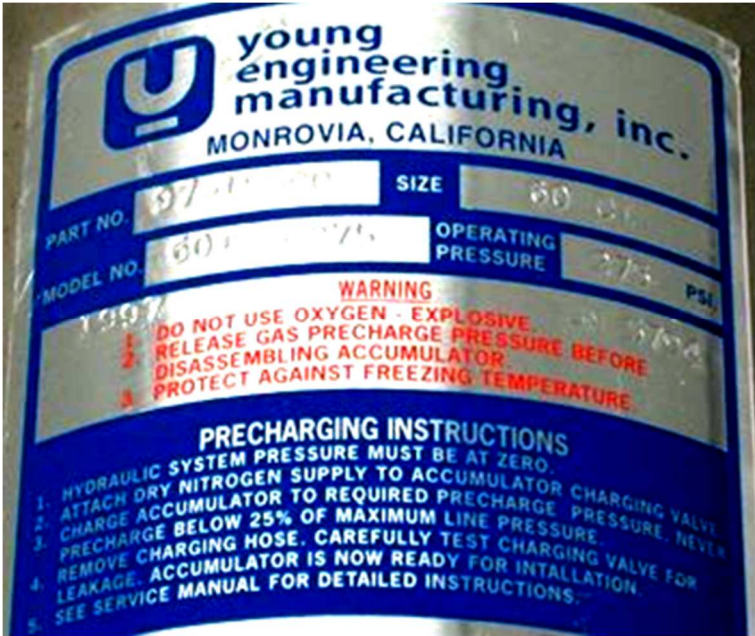
Common data that may be included on an accumulator nameplate (but not limited to);

1. Manufacturer
2. Name (Type) of component (not required)
3. Symbol for component (not required)
4. Model, Serial, Code or Manufacturers number (any combination)
5. Pressure rating
6. Temperature rating.
7. Size or capacity.
8. Precharge rating/value.



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GREER HYDRAULICS, INC
MAX. WORKING PRESS. 3000 PSI
MAX. TEMP. 150°F
SERIAL NO. 2602 1972



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Nameplate Interpretation Exercises

Interpretation Exercises Master List

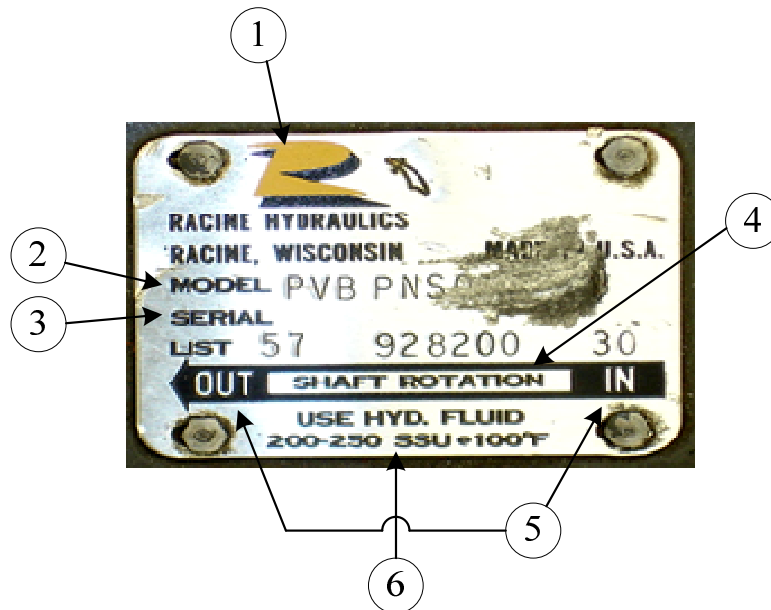
- | | |
|-----------------------------|---------------------------|
| A) Basic size | N) Port identification |
| B) Code # | O) Precharge rating/value |
| C) Configuration | P) Pressure rating |
| D) Direction of fluid flow | Q) Recommended position |
| E) Direction of rotation | R) RPM rating |
| F) Flow rating | S) Safety rating |
| G) Horsepower rating | T) Serial # |
| H) Manufacturer | U) Size or capacity |
| I) Manufacturers # | V) Style |
| J) Model # | W) Symbol for component |
| K) Name (Type) of component | X) Temperature rating |
| L) Orientation | Y) Viscosity rating |
| M) Part # | Z) Volts/Hz |

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Nameplate Interpretation Exercise #1



Choose from the master list to identify the pump nameplate data.

1. _____

4. _____

2. _____

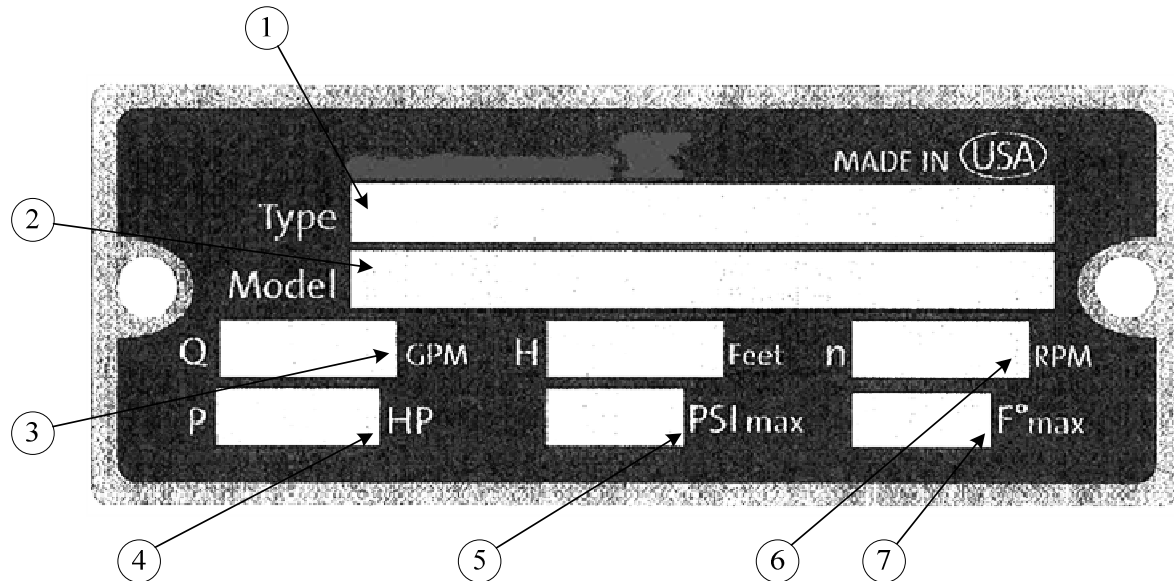
5. _____

3. _____

6. _____

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Nameplate Interpretation Exercise #2



Choose from the master list to identify the pump nameplate data.

1. _____

5. _____

2. _____

6. _____

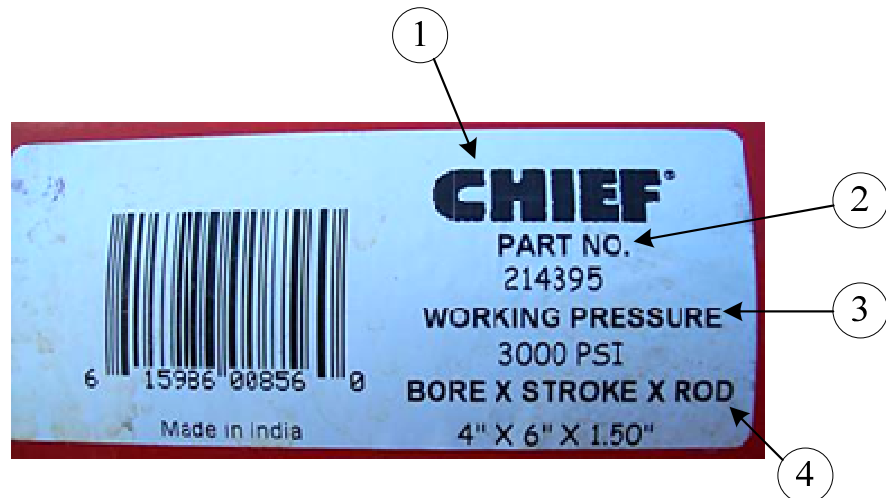
3. _____

7. _____

4. _____

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Nameplate Interpretation Exercise #3



Choose from the master list to identify the actuator nameplate data.

1. _____

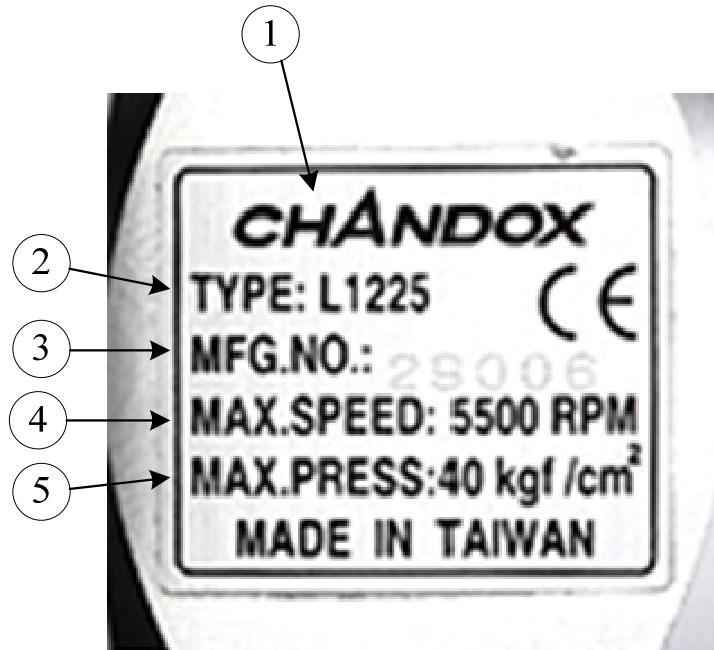
3. _____

2. _____

4. _____

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Nameplate Interpretation Exercise #4



Choose from the master list to identify the actuator nameplate data.

1. _____

4. _____

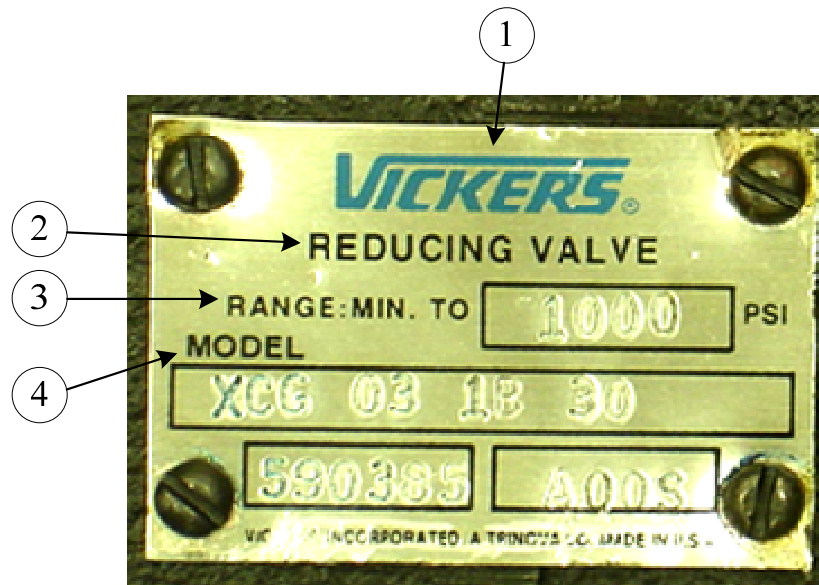
2. _____

5. _____

3. _____

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Nameplate Interpretation Exercise #5



Choose from the master list to identify the pressure control valve nameplate data.

1. _____

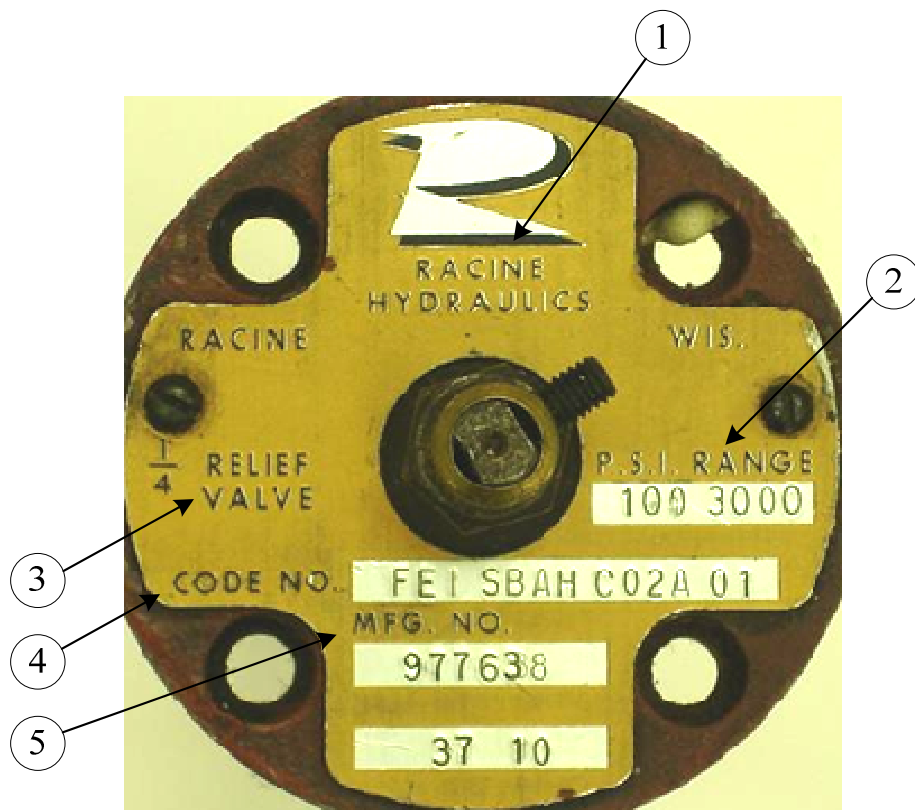
3. _____

2. _____

4. _____

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Nameplate Interpretation Exercise #6



Choose from the master list to identify the pressure control valve nameplate data.

1. _____

4. _____

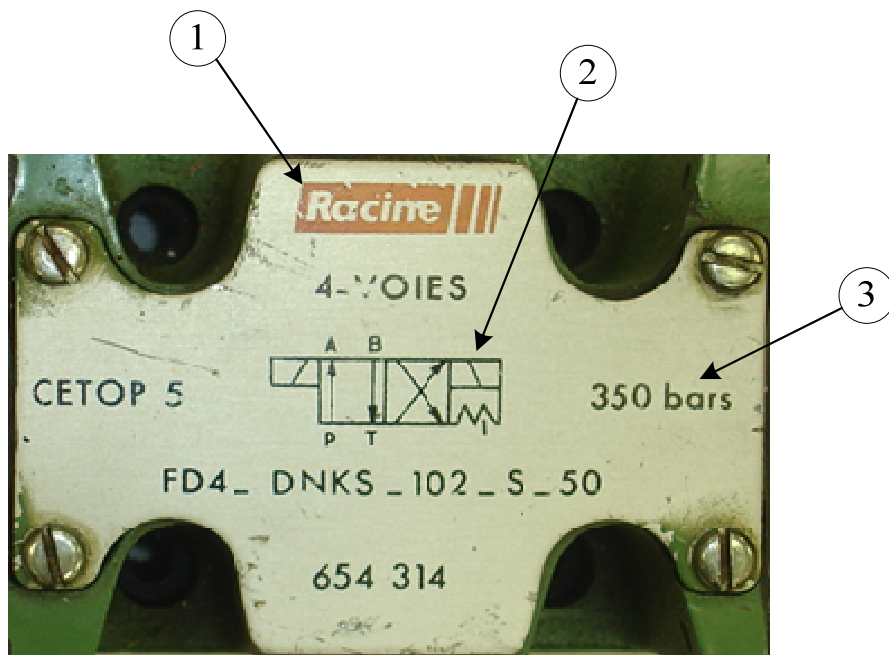
2. _____

5. _____

3. _____

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Nameplate Interpretation Exercise #7



Choose from the master list to identify the directional control valve nameplate data.

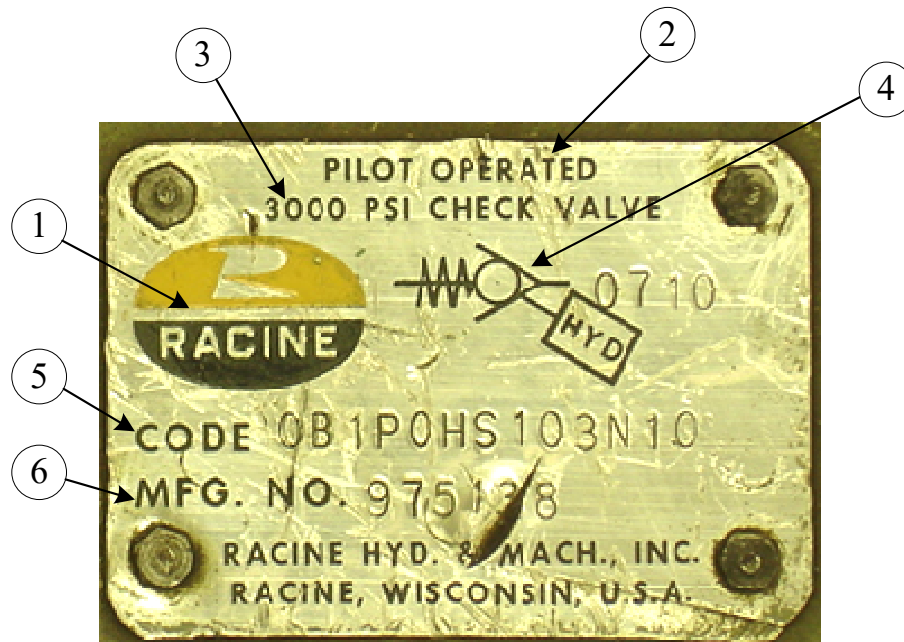
1. _____

3. _____

2. _____

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Nameplate Interpretation Exercise #8



Choose from the master list to identify the directional control valve nameplate data.

1. _____

4. _____

2. _____

5. _____

3. _____

6. _____

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Nameplate Interpretation Exercise #9



Choose from the master list to identify the flow control valve nameplate data.

1. _____

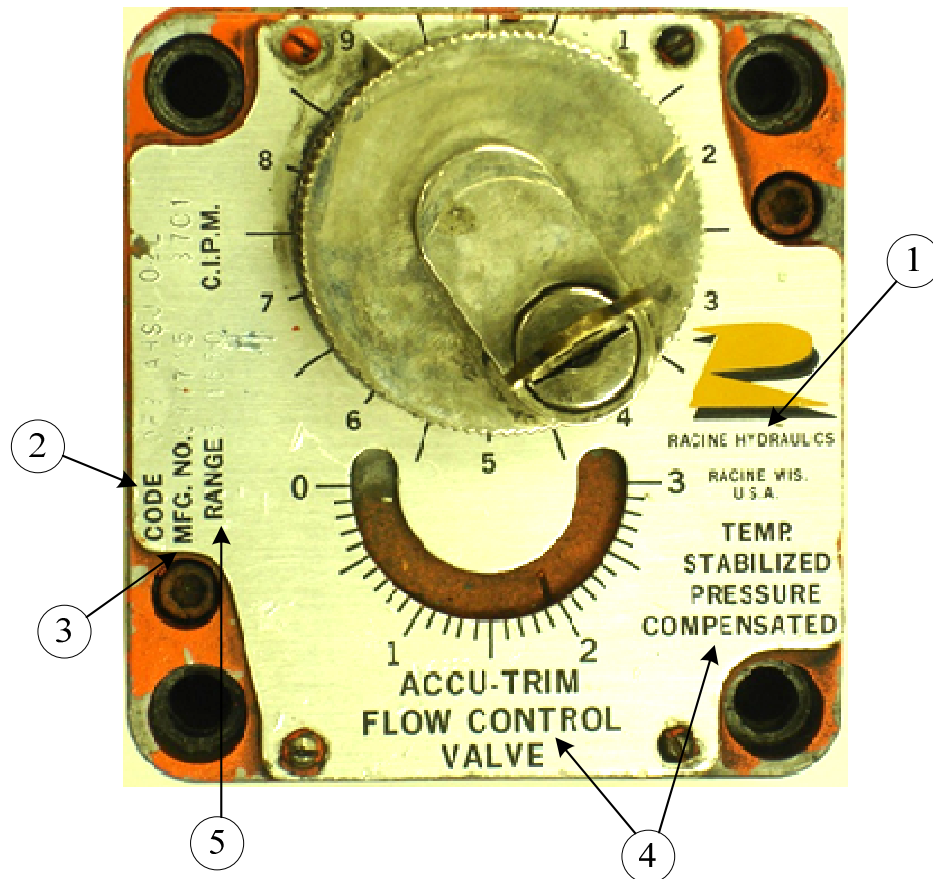
3. _____

2. _____

4. _____

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Nameplate Interpretation Exercise #10



Choose from the master list to identify the flow control valve nameplate data.

1. _____

4. _____

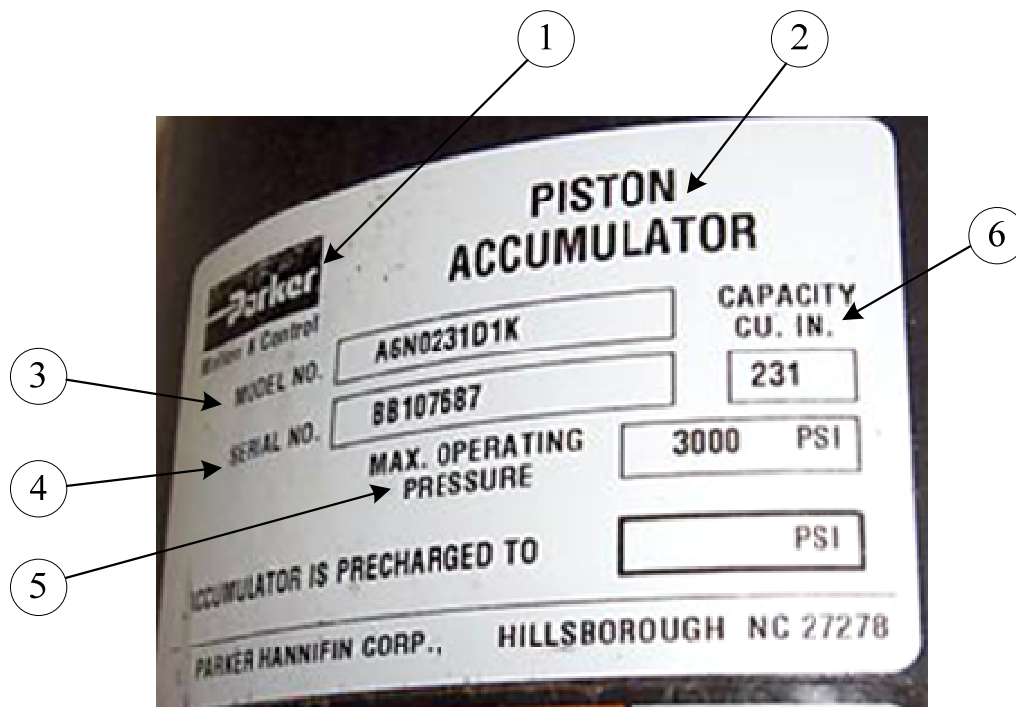
2. _____

5. _____

3. _____

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Nameplate Interpretation Exercise #11



Choose from the master list to identify the accumulator nameplate data.

1. _____

4. _____

2. _____

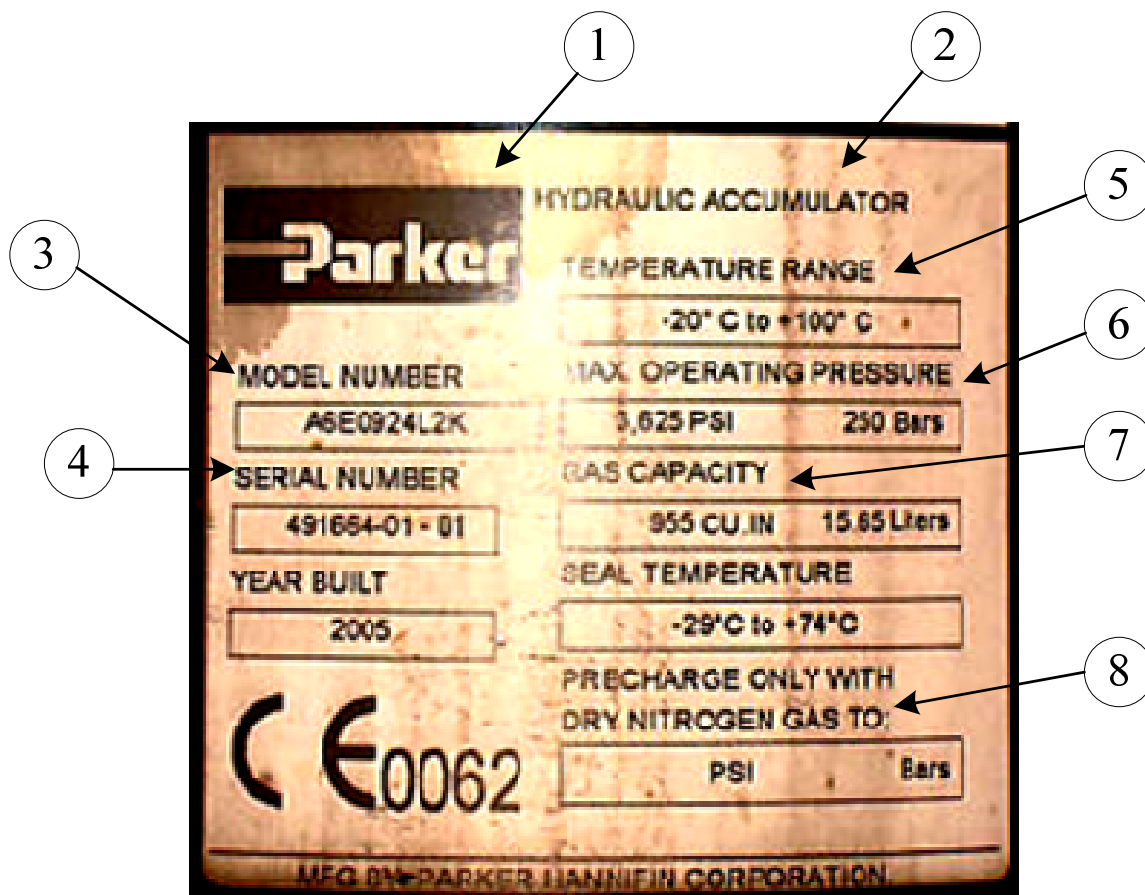
5. _____

3. _____

6. _____

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Nameplate Interpretation Exercise #12



Choose from the master list to identify the accumulator nameplate data.

1. _____

5. _____

2. _____

6. _____

3. _____

7. _____

4. _____

8. _____

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Schematics

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Schematic Interpretation

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Introduction to Schematics

The application and possible combinations of principles and components are countless. It would be impossible to illustrate every hydraulic circuit/system in this section. For that reason, those described in this material are a sampling of typical hydraulic systems.

Schematic Symbols

Most of the component symbols were illustrated as each topic was covered in the previous chapter. The following is a summary of the most common ISO symbols.

INSTRUMENTS & ACCESSORIES (MISCELLANEOUS EQUIPMENT)

Pressure gauge



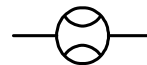
Differential pressure gauge



Temperature gauge



Flow meter



Volumetric flow meter



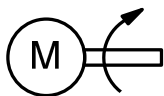
Pressure switch



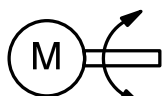
Indicator



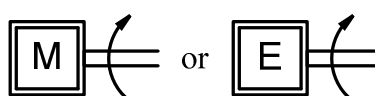
PRIME MOVERS



Electric motor
(Uni-directional)



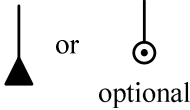
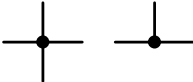
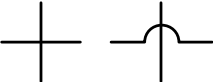
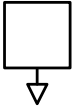
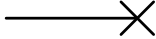
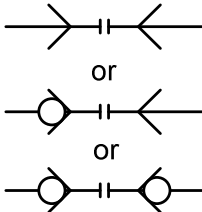
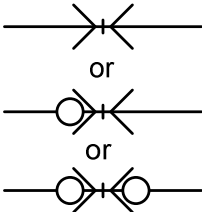
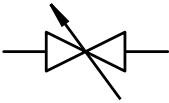
Electric motor
(Bi-directional)



Internal combustion
engine

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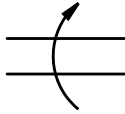
ENERGY TRANSMISSION (Fluid Conductors)

Pressure (Energy) source		Working line	
Control line (Pilot line)		Flexible pipeline	
Exhaust line or drain line		Containment line (Enclosure line)	
Line connection (Fixed)		Lines crossing	
Electrical line		Outlet with pipe connection (Threaded port)	
Pressure connection point closed		Pressure connection point with connecting line	
Quick disconnect coupling, disconnected (Without checks, with one check or two checks)		Quick disconnect coupling, connected (Without checks, with one check or two checks)	
Shut-off valve (adjustable)		Rotary coupling	
Shut-off valve (non-adjustable)			

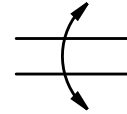
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MECHANICAL COMPONENTS

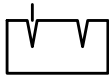
Shaft, rotary movement,
1 direction



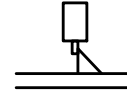
Shaft, rotary movement,
2 directions



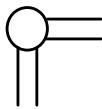
Detents for a DCV



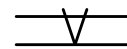
Latch (Locking device)



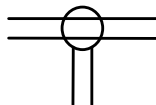
Hinge joint, simple



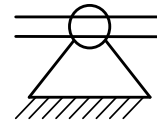
Over-center device



Hinge joint with
extended lever



Hinge joint with
fixed pivot

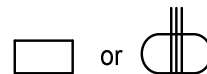


ENERGY & FLUID STORAGE

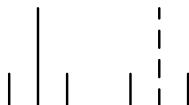
Reservoir, vented



Reservoir, pressurized



Return or Drain line
below fluid level



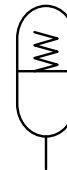
Return or Drain line
above fluid level



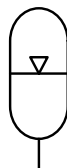
Accumulator



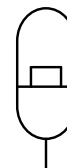
Accumulator,
Spring Loaded



Accumulator,
Gas Charged



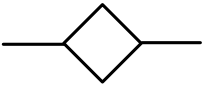
Accumulator,
Weight loaded



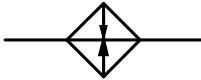
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FLUID CONDITIONERS

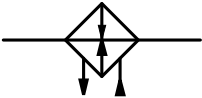
Conditioner



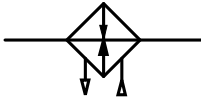
Heater



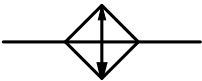
Heater, heating medium is liquid



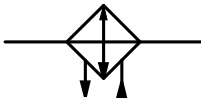
Heater, heating medium is gaseous



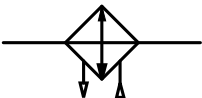
Cooler



Cooler, cooling medium is liquid



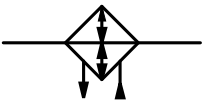
Cooler, cooling medium is gaseous



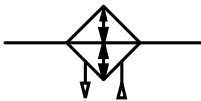
Temperature controller



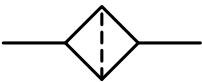
Temperature Controller, medium is liquid



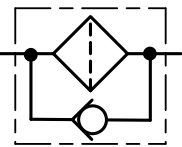
Temperature Controller, medium is gaseous



Filter



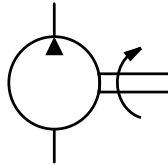
Filter, w/By-pass check valve



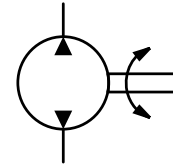
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HYDRAULIC PUMPS

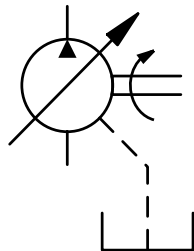
Hydraulic pump,
fixed displacement,
1 direction of flow,
1 direction of rotation



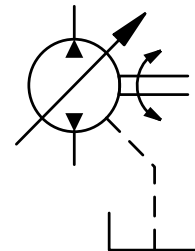
Hydraulic pump,
fixed displacement,
2 directions of flow,
2 directions of rotation



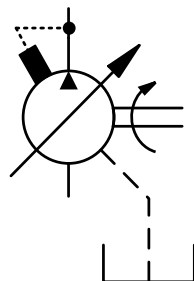
Hydraulic pump,
variable displacement,
1 direction of flow,
1 direction of rotation,
with case drain



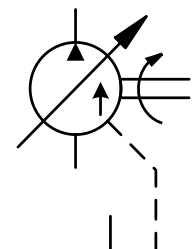
Hydraulic pump,
variable displacement,
2 directions of flow,
2 directions of rotation,
with case drain



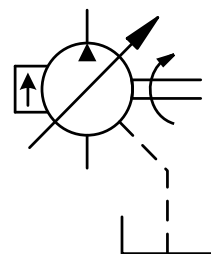
Hydraulic pump,
variable displacement,
pressure compensated,
1 direction of flow,
1 direction of rotation,
with case drain



Hydraulic pump,
variable displacement,
pressure compensated,
1 direction of flow,
1 direction of rotation,
with case drain



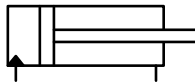
Hydraulic pump,
variable displacement,
pressure compensated,
1 direction of flow,
1 direction of rotation,
with case drain



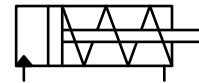
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ACTUATORS

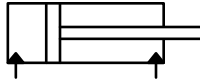
Single acting cylinder,
single rod,
load return



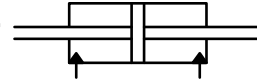
Single acting cylinder,
single rod,
spring return



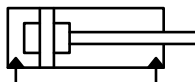
Double acting cylinder,
single rod



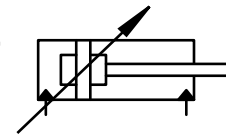
Double acting cylinder,
double rod



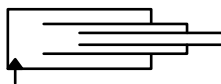
Double acting cylinder,
single rod, with fixed
cushioning in both
directions



Double acting cylinder,
single rod, with
adjustable cushioning
in both directions



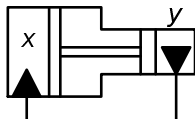
Single acting,
telescopic cylinder,
load return



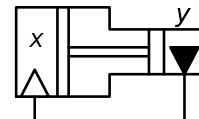
Double acting,
telescopic cylinder



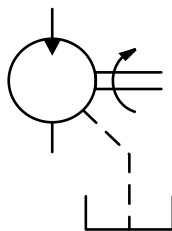
Pressure intensifier
(Same fluid)



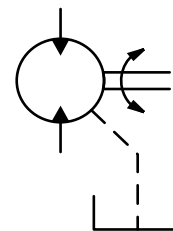
Pressure intensifier
(Different fluids)



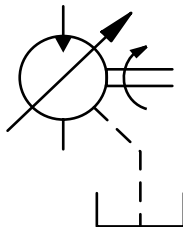
Hydraulic motor,
fixed displacement,
1 direction of flow,
1 direction of rotation,
with case drain



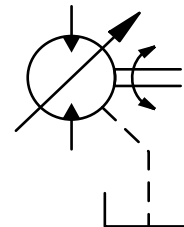
Hydraulic motor,
fixed displacement,
2 directions of flow,
2 directions of rotation,
with case drain



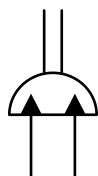
Hydraulic motor,
variable displacement,
1 direction of flow,
1 direction of rotation,
with case drain



Hydraulic motor,
variable displacement,
2 directions of flow,
2 directions of rotation,
with case drain



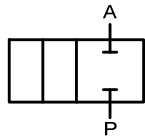
Rotary actuator,
Single vane or
Double Vane



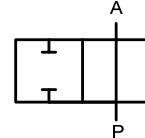
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DIRECTIONAL CONTROL VALVES

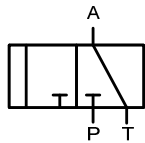
2 position,
2 port, 2 way,
normally closed



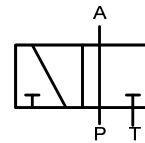
2 position,
2 port, 2 way,
normally open



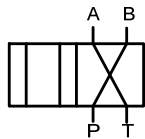
2 position,
3 port, 3 way,
normally closed



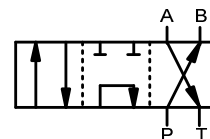
2 position,
3 port, 3 way,
normally open



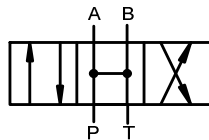
2 position,
4 port, 4 way



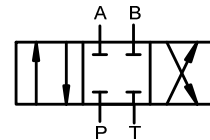
3 position,
Transitional center,
4 port, 4 way



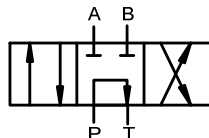
3 position,
4 port, 4 way,
open center



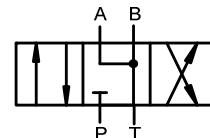
3 position,
4 port, 4 way,
closed center



3 position,
4 port, 4 way,
tandem center



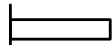
3 position,
4 port, 4 way,
floating center



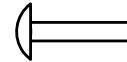
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VALVE OPERATORS

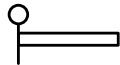
Manual actuator
(General symbol)



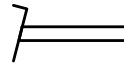
Push button



Hand lever



Pedal / Treadle



Detent



Spring return



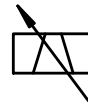
Roller operated
(Limit switch or Cam roller)



Solenoid



Variable Solenoid



Pneumatic pilot

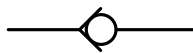


Hydraulic pilot

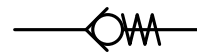


CHECK VALVES

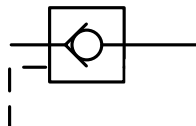
Standard Check
valve



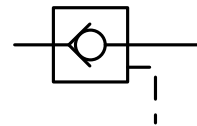
Back-Pressure
Check valve



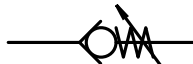
Piloted check valve,
pilot to open



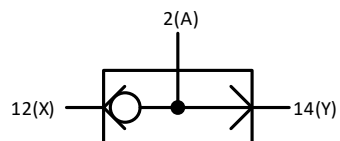
Piloted check valve,
pilot to close



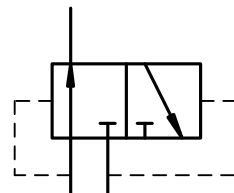
Adjustable Check
valve



Shuttle valve
("or" gate), old
simplified symbol



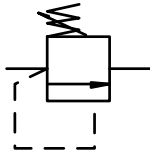
Shuttle valve
("or" gate), new
detailed symbol



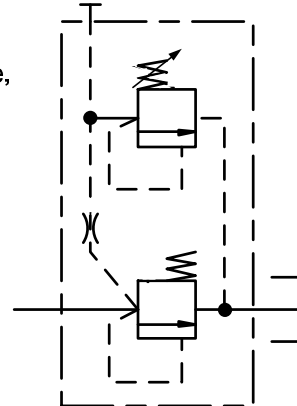
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PRESSURE CONTROLS

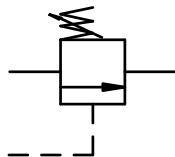
Pressure relief valve,
direct acting



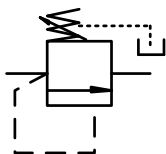
Pressure relief valve,
pilot operated



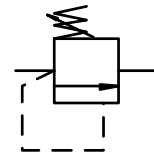
Unloading valve



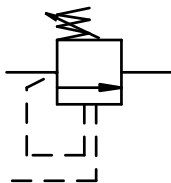
Sequence valve, with
external spring drain



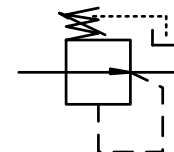
Counterbalance valve



Brake valve



Pressure reducing
valve, adjustable, with
external spring drain

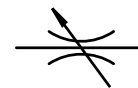


FLOW CONTROLS

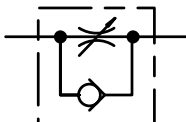
Fixed flow control
(Restrictor)



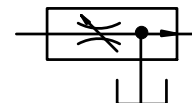
Adjustable flow control
(Restrictor type)



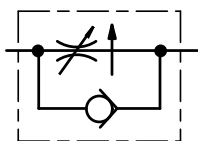
Adjustable flow control
(Restrictor type), with
by-pass check valve



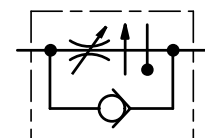
Adjustable flow control
(By-Pass type)



Adjustable flow control
(Restrictor type), with
pressure compensation



Adjustable flow control
(Restrictor type), with
pressure & temperature
compensation



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Schematic Line Technology Explanation

Solid line 

(Working/Main line - It handles main system pressure and main system flow)

Long dash line 

(Pilot line - It handles main system pressure, **not** main system flow)

Short dash line 

(Drain line - It handles drainage/leakage fluid back to reservoir, **not** main system pressure/flow)

Mixed dash line 

(Enclosure line - It is used when two different components are built into one valve body)

Double solid line 

(Mechanical connection - It is used for shafts, couplings, levers, piston rods (optional))

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Schematic Interpretation Exercises Component Master List

Conditioning

1. Inlet strainer filter
2. Suction line filter
3. Pressure line filter
4. Return line filter
5. Vented reservoir w/ drain line below fluid level
6. Vented reservoir w/drain line above fluid level
7. Vented reservoir, w/main line below fluid level
8. Vented reservoir, w/main line above fluid level
9. Air cooler
10. Water cooler

Pumps

11. Unidirectional, fixed displacement
12. Bi-Directional, fixed displacement
13. Unidirectional, variable displacement
14. Bi-Directional, variable displacement
15. Unidirectional, fixed displacement, pressure compensated, internal pilot
16. Unidirectional, variable displacement, pressure compensated, w/ case drain
17. Bi-directional, variable displacement, pressure compensated, w/ case drain
18. HV/LP, unidirectional, fixed displacement
19. LV/HP, unidirectional, fixed displacement

Actuators

20. Single rod, single acting, load return hydraulic cylinder
21. Single rod, single acting, spring return, hydraulic cylinder
22. Single rod, double acting, hydraulic cylinder
23. Single rod, double acting, w/ fixed cushions - both ends, hydraulic cylinder
24. Single rod, double acting, w/ adjustable cushions – both ends, hydraulic cylinder
25. Single rod, double acting, w/ adjustable cushion - piston end, hydraulic cylinder
26. Single rod, double acting, w/ adjustable cushion - rod end, hydraulic cylinder
27. Single rod, double acting hydraulic cylinder, w/ stop tube
28. Single rod, single acting, hydraulic cylinder, w/ spring, stroke adjuster - rod end
29. Double rod, double acting, hydraulic cylinder
30. Rotary actuator
31. Fixed displacement, unidirectional, hydraulic motor
32. Fixed displacement, bidirectional, hydraulic motor
33. Variable displacement, bidirectional, external pilot, hydraulic motor
34. Variable displacement, bidirectional, external drain, hydraulic motor
35. Variable displacement, bidirectional, pressure compensated, hydraulic motor
36. Variable displacement, bidirectional, internal pilot, external drain, hydraulic motor

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Pressure Control Valves (PCV's)

37. Pressure relief valve, normally closed, spring loaded, adjustable, internal pilot, internal drain
38. Pressure relief valve, normally open, spring loaded, adjustable, internal pilot, internal drain
39. Pressure relief valve, normally closed, spring loaded, adjustable, internal pilot, internal drain, pilot/remote operated
40. Pressure relief valve, normally closed, spring loaded, adjustable, external pilot, internal drain, pilot/remote operated
41. Pressure relief valve, normally open, spring loaded, adjustable, external and internal pilot, external drain, pilot/remote operated
42. Pressure relief valve, normally closed, spring loaded, adjustable, internal pilot, internal spring drain until "c" is energized creating external/remote pilot operation
43. Unloading valve, normally closed, spring loaded, adjustable, external pilot, internal spring drain
44. Unloading valve, normally open, spring loaded, adjustable, external pilot, internal spring drain, w/ by-pass check valve
45. Unloading valve, normally closed, spring loaded, adjustable, external pilot, internal spring drain, w/ by-pass check valve
46. Sequence valve, normally closed, spring loaded, adjustable, external pilot, external spring drain
47. Sequence valve, normally closed, spring loaded, adjustable, internal pilot, internal spring drain
48. Sequence valve, normally closed, spring loaded, adjustable, internal pilot, external spring drain
49. Counter-Balance valve, normally closed, spring loaded, adjustable, external pilot, external spring drain
50. Counter-Balance valve, normally closed, spring loaded, adjustable, external pilot, internal spring drain
51. Pressure reducing valve, normally closed, spring loaded, adjustable, internal pilot on the outlet (secondary) port, external spring drain
52. Pressure reducing valve, normally open, spring loaded, adjustable, internal pilot on the outlet (secondary) port, external spring drain
53. Pressure reducing valve, normally open, spring loaded, adjustable, internal pilot on the inlet (primary) port, external spring drain
54. Brake valve, normally open, spring loaded, adjustable, internal and external pilot, external drain
55. Brake valve, normally closed, spring loaded, adjustable, internal and external pilot, internal drain
56. Brake valve, normally closed, spring loaded, adjustable, internal and external pilot, internal drain, w/ by-pass check valve
57. Electrical switch
58. Pressure switch

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Directional Control Valves (DCV's)

- 59. 2 Positions, 2 ports, 2 ways, normally open, spring return, solenoid operated
- 60. 2 Positions, 2 ports, 2 ways, normally closed, spring return solenoid, operated
- 61. 2 Positions, 3 ports, 3 ways, normally closed, spring return, roller operated
- 62. 2 Positions, 3 ports, 3 ways, normally open, spring return, roller operated
- 63. 2 Positions, 3 ports, 3 ways, normally closed, spring return, solenoid operated,
- 64. 2 Positions, 4 ports, 4 ways, spring return, hand lever operated
- 65. 2 Positions, 4 ports, 4 ways, spring return, roller operated
- 66. 2 Positions, 4 ports, 4 ways, solenoid operated
- 67. 2 Positions, 4 ports, 4 ways, solenoid controlled
- 68. 2 Positions, 5 ports, 5 ways, spring centered, w/ detents, roller operated
- 69. 3 Positions, 4 ports, 4 ways, float center, spring centered, hand lever operated
- 70. 3 Positions, 4 ports, 4 ways, tandem center, spring centered, hand lever operated
- 71. 3 Positions, 4 ports, 4 ways, closed center, spring centered, hand lever operated
- 72. 3 Positions, 4 ports, 4 ways, float center, spring centered, solenoid operated
- 73. 3 Positions, 4 ports, 4 ways, tandem center, spring centered, solenoid operated
- 74. 3 Positions, 4 ports, 4 ways, open center, spring centered, solenoid operated
- 75. 3 Positions, 4 ports, 4 ways, closed center, spring centered, solenoid operated
- 76. 3 Positions, 4 ports, 4 ways, float center, spring centered, solenoid controlled, hydraulic pilot, internal pilot operated, internal pilot drain
- 77. 3 Positions, 4 ports, 4 ways, tandem center, spring centered, solenoid controlled, hydraulic pilot, external pilot operated, external pilot drain
- 78. 3 Positions, 4 ports, 4 ways, tandem center, spring centered, solenoid controlled, hydraulic pilot, internal pilot operated, external pilot drain
- 79. 3 Positions, 4 ports, 4 ways, closed center, spring centered, solenoid controlled, hydraulic pilot, external pilot operated, external pilot drain
- 80. 3 Positions, 4 ports, 4 ways, open center, spring centered, solenoid controlled, hydraulic pilot, internal pilot operated, external pilot drain
- 81. Shut-off valve, non-adjustable

Check Valves

- 82. Check valve
- 83. Spring loaded check valve
- 84. Spring loaded, internal pilot, check valve
- 85. Piloted check valve, pilot to open
- 86. Piloted check valve, pilot to close
- 87. Piloted check valve, pilot to open, internal pilot
- 88. Piloted check valve, pilot to close, external drain

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Flow Control Valves (FCV's)

- 89. Fixed flow control valve, restrictor type
- 90. Fixed flow control valve, restrictor type, w/ by-pass check valve
- 91. Adjustable flow control valve, restrictor type
- 92. Adjustable flow control valve, restrictor type, w/ by-pass check valve
- 93. Adjustable flow control valve, by-pass type, w/ by-pass check valve
- 94. Adjustable flow control, pressure compensated, by-pass type
- 95. Adjustable flow control, pressure compensated, restrictor type
- 96. Adjustable flow control, pressure compensated, restrictor type, w/ by-pass check valve
- 97. Adjustable flow control, pressure compensated, by-pass type, temperature compensated
- 98. Adjustable flow control, pressure compensated, restrictor type, temperature compensated

Accumulators

- 99. Weight loaded accumulator
- 100. Spring loaded accumulator
- 101. Work loaded accumulator
- 102. Hydro-Pneumatic accumulator

Line Technology

- 103. Main line
- 104. Pilot line
- 105. Enclosure line
- 106. Mechanical connection
- 107. Flexible line

Accessories

- 108. Pressure Gage
- 109. Differential pressure gage
- 110. Temperature gage
- 111. Flow meter
- 112. Indicator

Power Source

- 113. Prime mover (Electrical motor)
- 114. Prime mover (Internal combustion engine)

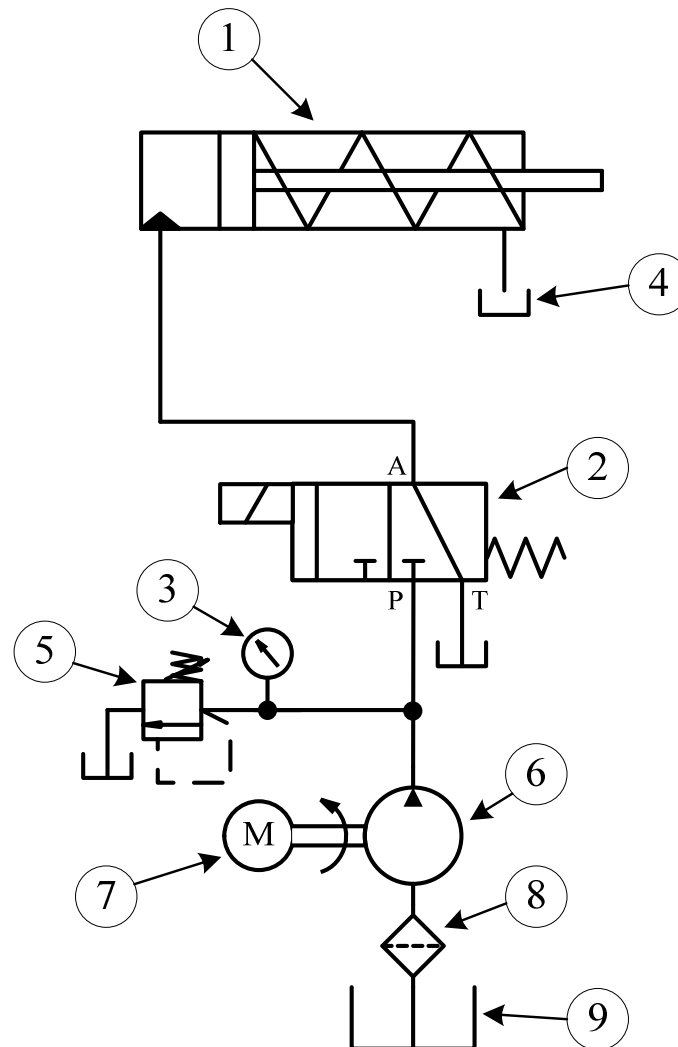
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Schematic Interpretation Exercises

Schematic Exercise #1



Choose from the master list to identify the components in Schematic Exercise #1

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

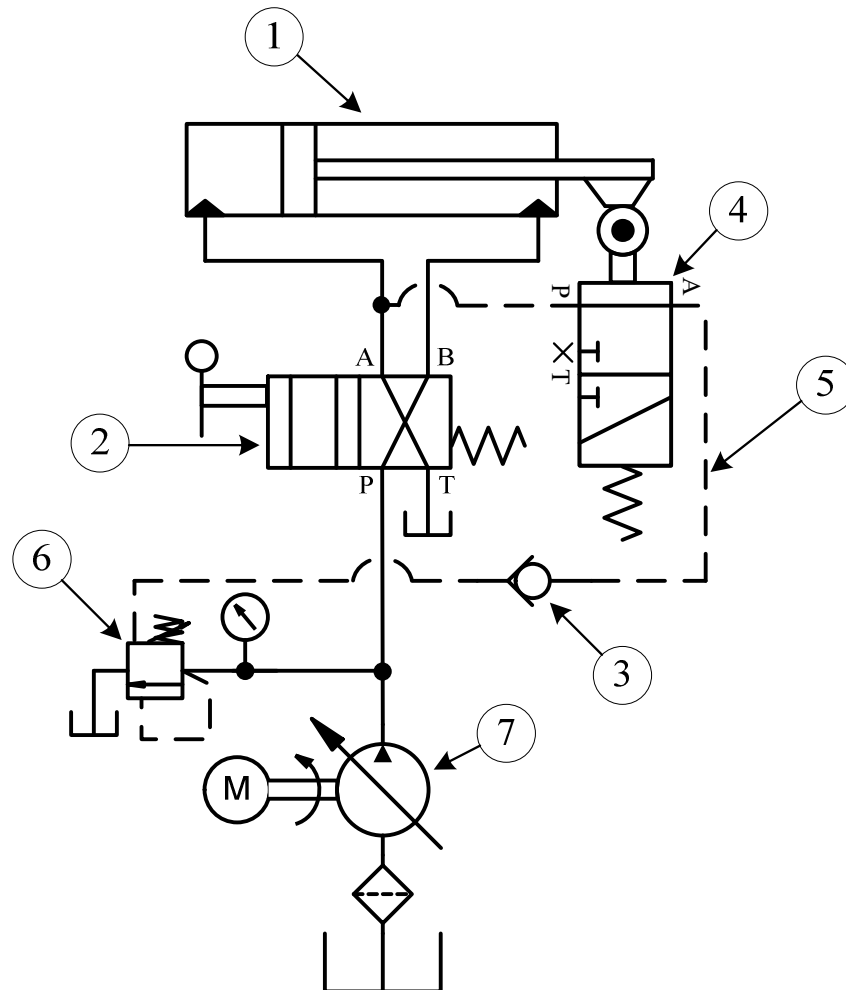
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Number the statements in sequential order to describe the proper operation of Schematic Exercise #1. (Note: all the statements will be used)

1. When the solenoid is energized
- _____ The DCV shifts and the flow path is open from P to A
- _____ The cylinder retracts
- _____ When the solenoid is de-energized, the spring will bring the DCV back to home position
- _____ The cylinder extends
- _____ At the end of the extension stroke the pump flow is sent to reservoir via PCV5
- _____ The flow path is open from A to T and pump flow is sent to the reservoir through PCV5

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Schematic Exercise #2



Choose from the master list to identify the components in Schematic Exercise #2

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

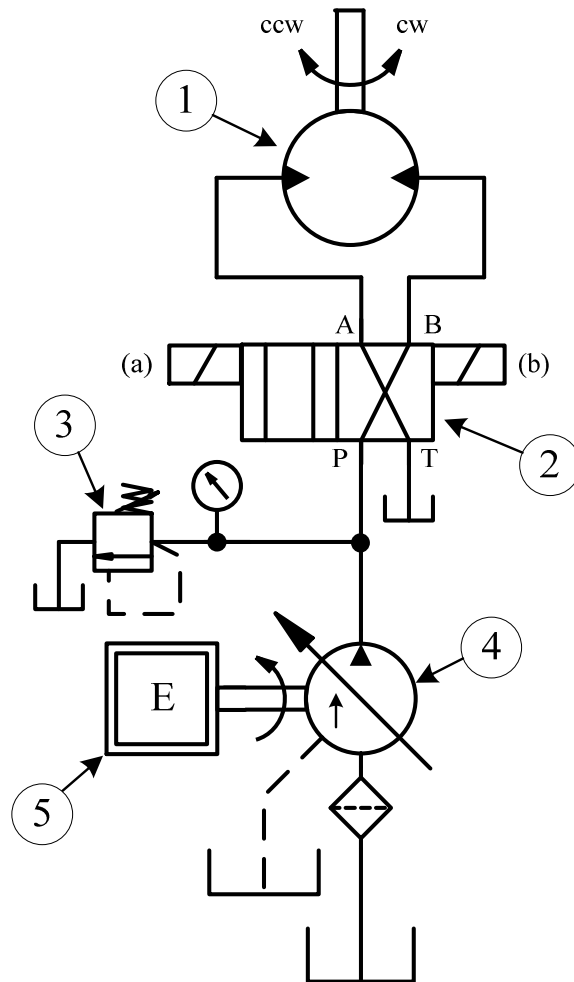
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Number the statements in sequential order to describe the proper operation of Schematic Exercise #2. (Note: all the statements will be used)

1. When the DCV2 is shifted
- _____ At the end of the retraction stroke, the DCV4 is activated opening its flow path from **P** to **A** unloading the pump at low pressure through the PCV6
- _____ When the DCV's operator is released, the spring will bring the DCV2 back to home position
- _____ The flow through DCV4 is blocked by the valve3 and the cylinder extends, the circuit is capable to use the PCV6 setting
- _____ The flow paths are connected from **P** to **B** and from **A** to **T**
- _____ Once the DCV4 is de-energized its flow path is block. The cylinder is capable to use the PCV6 setting
- _____ At the end of the extension stroke the pump's flow is sent to reservoir via the PCV6 and the pressure gage read the PCV6 setting
- _____ The flow paths are connected from **P** to **A** and from **B** to **T**
- _____ The cylinder retracts and the circuit is capable to use the PCV6 setting

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Schematic Exercise #3



Choose from the master list to identify the components in Schematic Exercise #3

1. _____

2. _____

3. _____

4. _____

5. _____

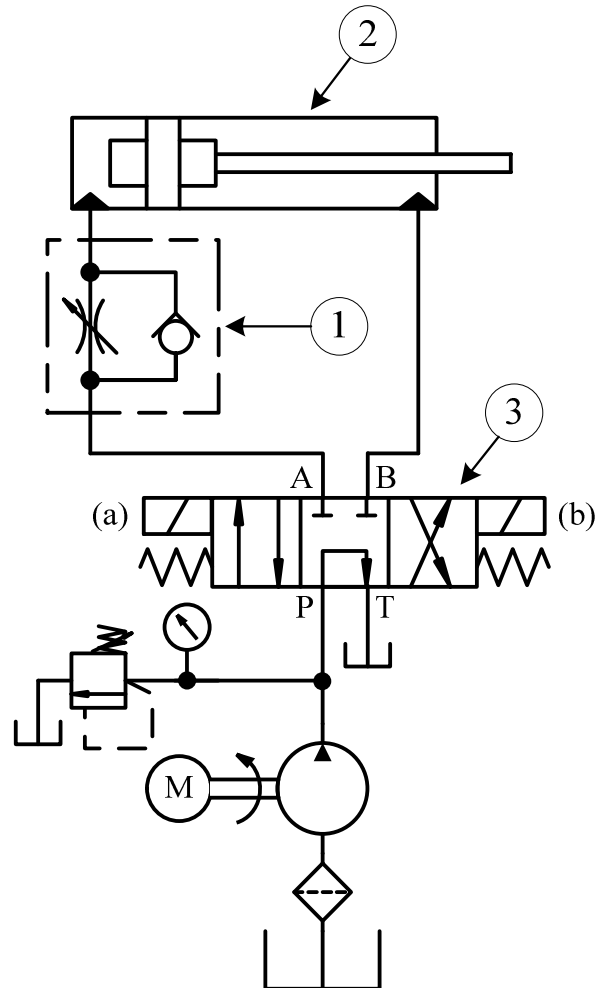
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Number the statements in sequential order to describe the proper operation of Schematic Exercise #3. (Note: all the statements will be used)

- _____ When operator **a** is de-energized, and operator **b** is energized the DCV will return to home position
- _____ The DCV shifts and the flow path is open from **P** to **A** and from **B** to **T**
- _____ The motor rotates CW. The pressure gage will read the pressure necessary to move the load
- _____ The flow paths are connected from **P** to **B** and from **A** to **T** and the motor rotates CCW, the pressure gage will read the pressure necessary to move the load
- 1.** When the operator **a** is energized

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Schematic Exercise #4



Choose from the master list to identify the components in Schematic Exercise #4

1.(Component) _____

2. _____

3. _____

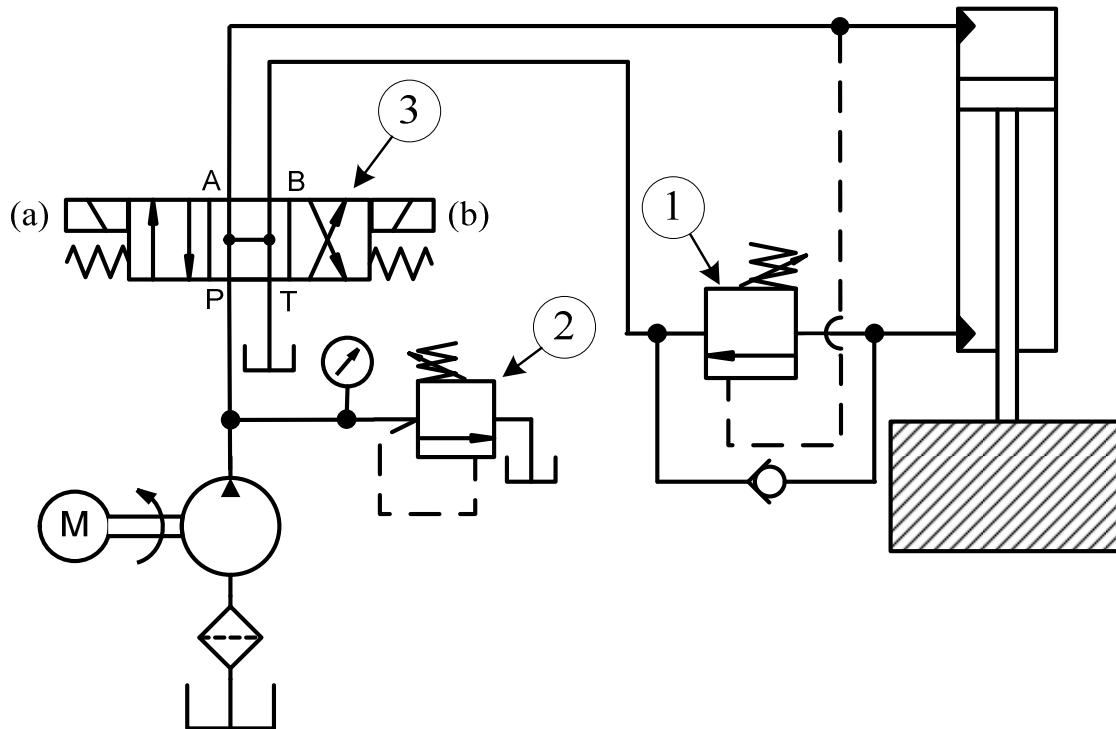
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Number the statements in sequential order to describe the proper operation of Schematic Exercise #4. (Note: all the statements will be used)

- _____ When solenoid **b** is de-energized, the springs shift the DCV back to home position and the flow path is open from **P** to **T** and the ports **A** and **B** will be blocked; the cylinder stop at any position, it can't be moved from the actual position, and the pump will be unloaded through the DCV
- _____ The DCV shifts and the flow path is open from **P** to **A** and from **B** to **T**; the FCV1 meter in the cylinder extension
- _____ When solenoid **a** is de-energized, the springs shift the DCV back to home position and the flow path is open from **P** to **T** and the ports **A** and **B** will be blocked; the cylinder stop at any position, it can't be moved from the actual position, and the pump will be unloaded through the DCV
- 1.** When the solenoid **a** is energized
- _____ When the solenoid **b** is energized, the DCV shifts and the flow path is open from **P** to **B** and from **A** to **T**
- _____ The FCV1 is by-passed and the cylinder retracts at maximum speed

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Schematic Exercise #5



Choose from the master list to identify the components in Schematic Exercise #5

1. _____

2. _____

3. _____

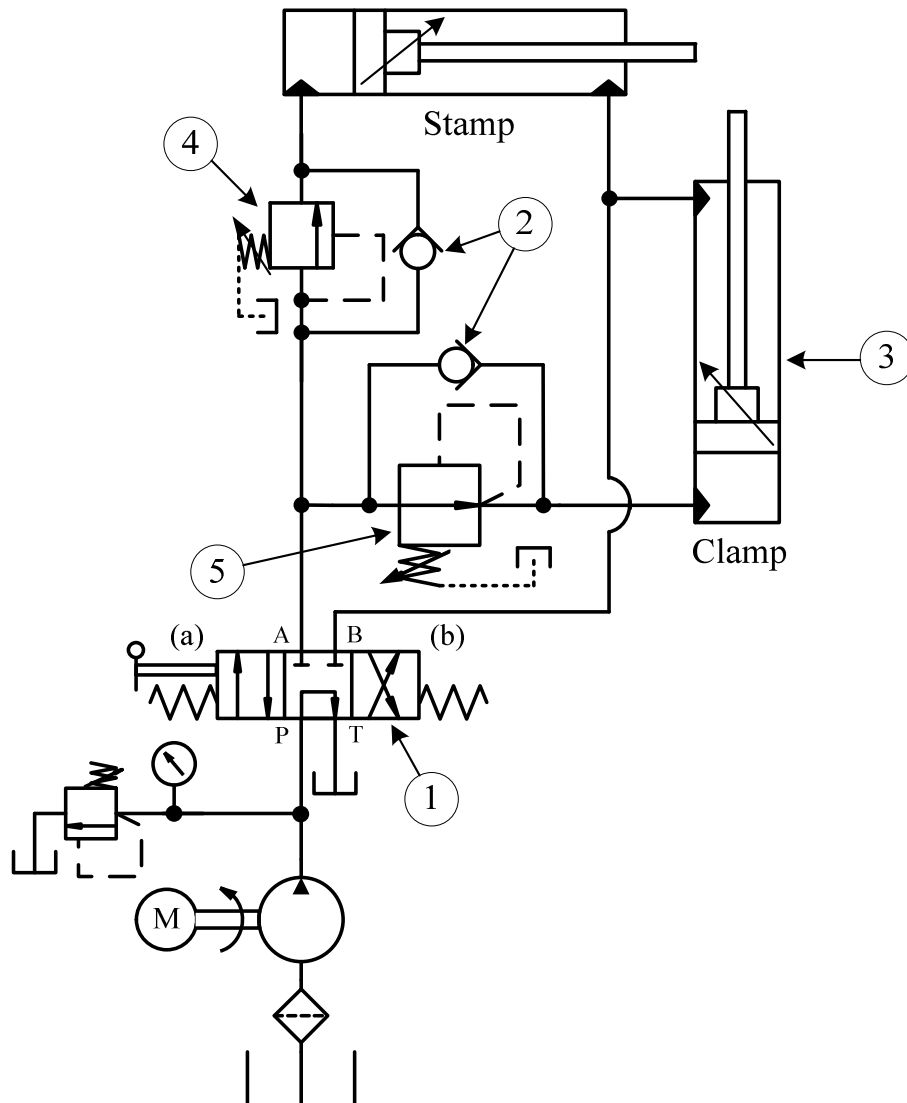
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Number the statements in sequential order to describe the proper operation of Schematic Exercise #5. (Note: all the statements will be used)

- _____ When the pressure at the cylinder cap end reaches the PCV1 setting the valve open and the load goes down under control
- _____ The DCV shifts and the flow path is open from **P** to **A** and from **B** to **T**
- _____ When solenoid **a** is de-energized, the springs shift the DCV back to home position and the ports **A**, **B**, and **P** will be connected to **T**; the pressure at the cylinder cap end goes close to zero, the PCV1 close, the cylinder stop at any position and the pump will be unloaded through the DCV
- _____ When the solenoid b is energized, the DCV shifts and the flow path is open from **P** to **B** and from **A** to **T**; the flow by-passes the PCV1 and the load goes up
- 1.** _____ When the solenoid **a** is energized
- _____ When solenoid **b** is de-energized, the springs shift the DCV back to home position and the ports **A**, **B**, and **P** will be connected to **T**; the pressure at the cylinder cap end goes close to zero, the PCV1 close, the cylinder stop at any position and the pump will be unloaded through the DCV

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Schematic Exercise #6



Choose from the master list to identify the components in Schematic Exercise #6

1. _____

2. _____

3. _____

4. _____

5. _____

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Number the statements in sequential order to describe the proper operation of Schematic Exercise #6. (Note: all the statements will be used)

_____ The flow path is open from **P** to **A** and from **B** to **T**

1. When the DCV1's operator is pushed to have **a**

_____ Once the cap end clamp cylinder reaches the PCV5 setting, the PCV5 closes, and the pressure at the PCV4 primary port goes up

_____ The flow is blocked at the PCV4, by-passes the PCV5, pressurizes the clamp cylinder cap end, and the clamp cylinder extends

_____ When the pressure at the PCV4 primary port reaches the PCV4 setting, the PCV4 open and the stamp cylinder extends

_____ The rod ends in both cylinders are pressurized and the flow from the cap ends of both cylinders by-passes the PCV4 and PCV5, allowing the retraction of both cylinders

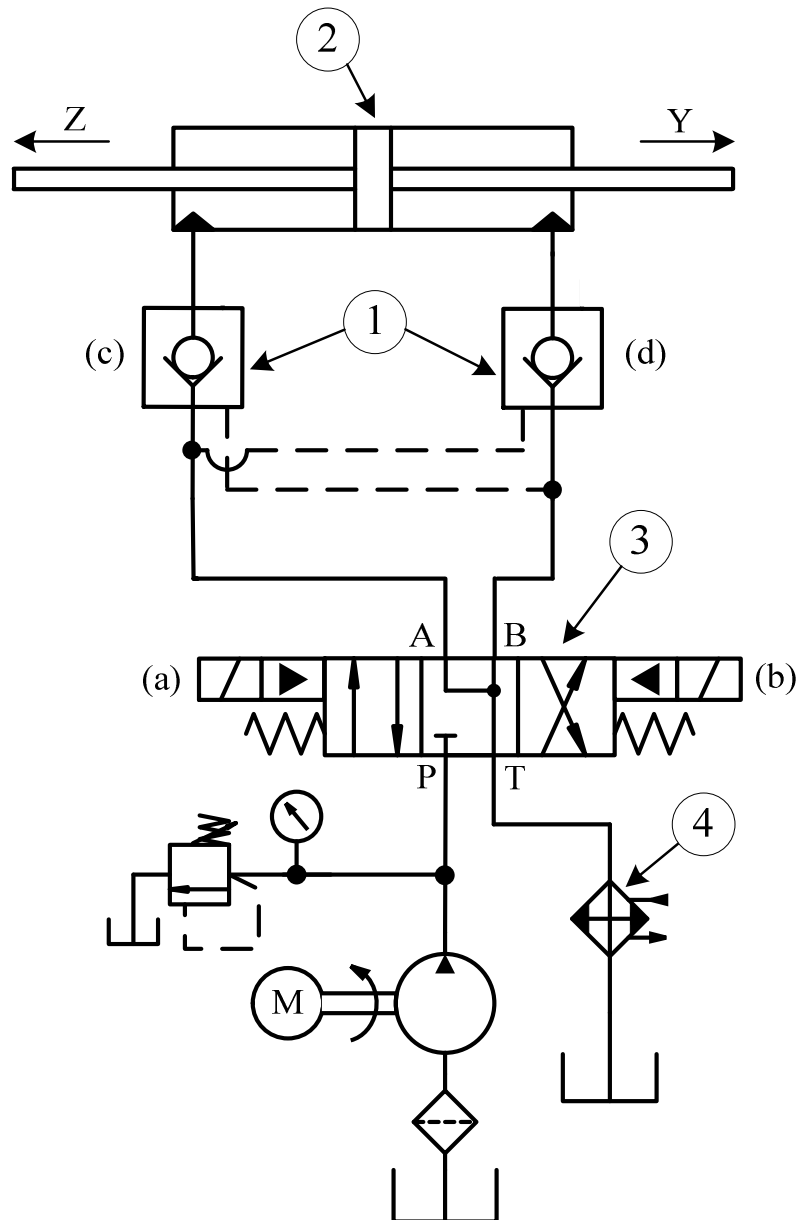
_____ When the DCV1's operator is released, the spring shifts the DCV back to home position and the flow path is open from **P** to **T** and the ports **A** and **B** will be blocked, the cylinders extension stop at any position and the pump will be unloaded through the DCV

_____ When the operator is pull to get **b** , the flow path is open from **P** to **B** and from **A** to **T**

_____ When the operator is released, the spring shifts the DCV back to home position and the flow path is open from **P** to **T** and the ports **A** and **B** will be blocked; the cylinders retraction stop at any position and the pump will be unloaded through the DCV

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Schematic Exercise #7



Choose from the master list to identify the components in Schematic Exercise #7

1. _____

2. _____

3. _____

4. _____

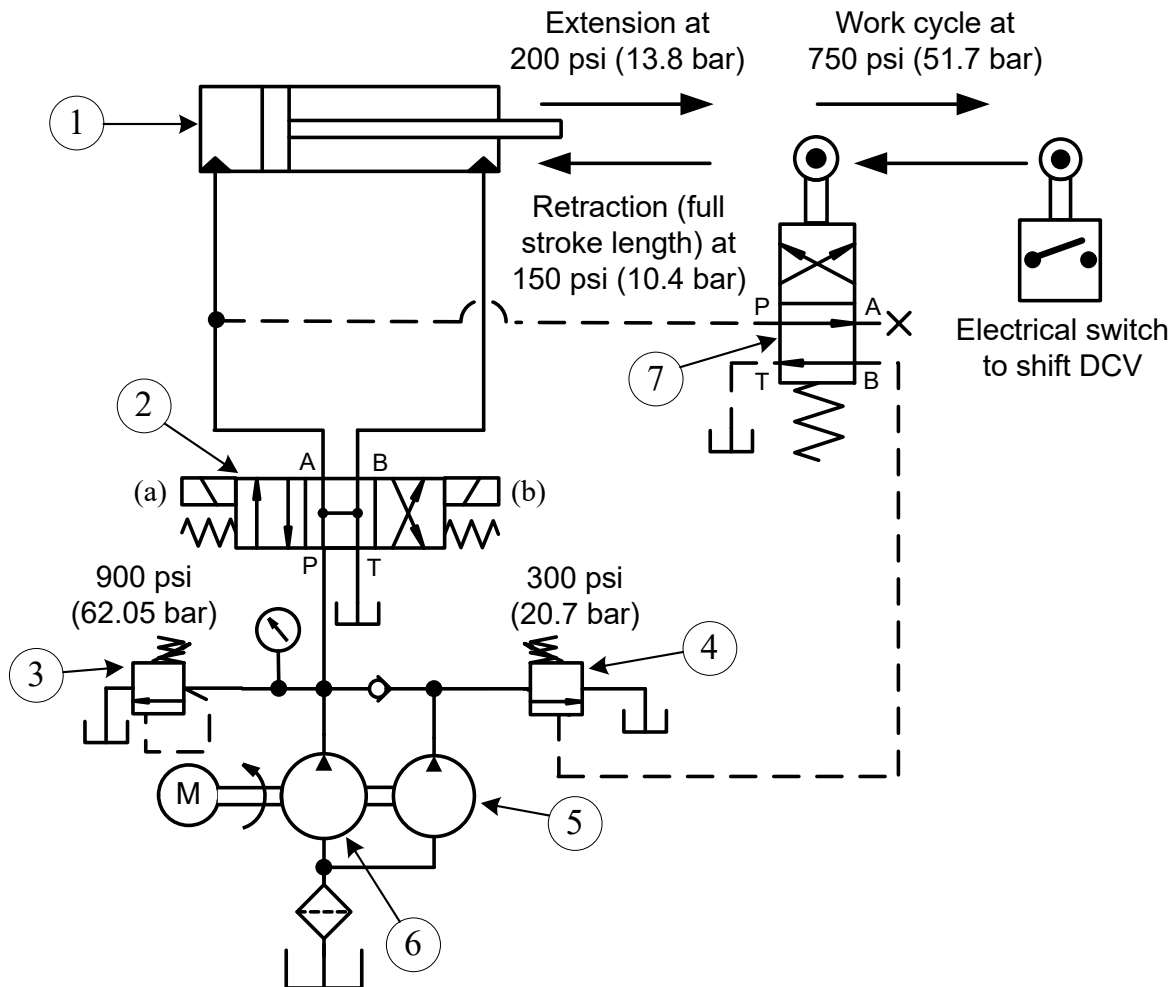
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Number the statements in sequential order to describe the proper operation of Schematic Exercise #7. (Note: all the statements will be used)

- _____ The DCV shifts and the flow path is open from **P** to **A** and **B** to **T**; the valve **c** open
- _____ The pressure at the valve **d** inlet port goes up opening the valve **c** allowing the cylinder to move in the **Z** direction
- _____ The pressure at the valve **c** inlet port goes up opening the valve **d** allowing the cylinder to move in the **Y** direction
- 1.** When the solenoid **a** is energized
- _____ When solenoid **b** is de-energized, the springs shift the DCV back to home position blocking the port **P** and connecting ports **A** and **B** to **T**; the cylinder stop at any position and it can't be moved from the actual position. The pressure gage reads the pressure relief valve setting
- _____ When solenoid **a** is de-energized, the springs shift the DCV back to home position blocking the port **P** and connecting ports **A** and **B** to **T**; the cylinder stop at any position and it can't be moved from the actual position. The pressure gage reads the PCV setting
- _____ When the solenoid **b** is energized, the DCV shifts and the flow path is open from **P** to **B** and **A** to **T**; the valve **d** open

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Schematic Exercise #8



Choose from the master list to identify the components in Schematic Exercise #8

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

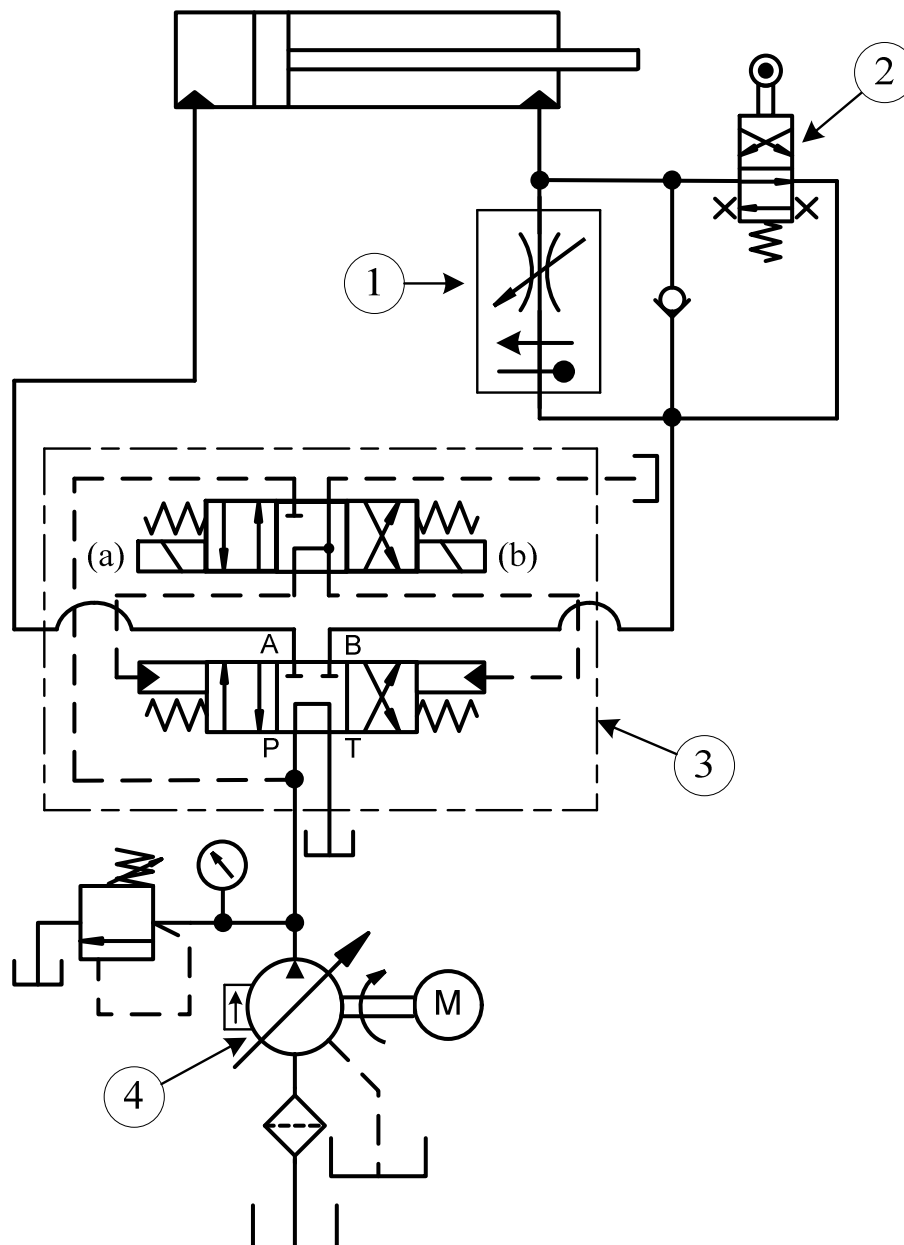
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Number the statements in sequential order to describe the proper operation of Schematic Exercise #8. (Note: all the statements will be used)

1. When the solenoid **a** is energized
- _____ When the DCV7 is de-activated its flow path from P to B is close and the cylinder retraction continues at its maximum speed, pressure gage reads 150PSI
- _____ Both pumps send their flow to the cylinder rod end and the cylinder retracts at its maximum speed, pressure gage reads 150PSI
- _____ The DCV2 shifts and the flow path is open from **P** to **A** and from **B** to **T**
- _____ Both pumps sent their flow to the cylinder cap end and the cylinder has a fast Extension, pressure gage reads 200PSI
- _____ When solenoid **b** is de-energized, the springs shift the DCV2 back to home position and the ports **A**, **B**, and **P** will be connected to **T**; the pumps will be unloaded through the DCV2 and the cylinder can be moved of its position
- _____ When the DCV7 is activated its flow path from P to B is open
- _____ When the pressure gage read 300PSI, the PCV4 open unloading the pump **5** and the cylinder cap end receives only the pump **6** flow slowing down the extension, pressure gage reads 750PSI
- _____ At the end of the extension stroke, the solenoid **a** is de-energized, the springs shift the DCV2 back to home position and the ports **A**, **B**, and **P** will be connected to **T**; the PCV4 close, and the pumps will be unloaded through the DCV2
- _____ At the end of the extension stroke, the solenoid **b** shifts the DCV2, the flow paths are open from **P** to **B** and from **A** to **T**

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Schematic Exercise #9



Choose from the master list to identify the components in Schematic Exercise #9

1. _____ 2. _____

3.(Line Technology _____ 4. _____

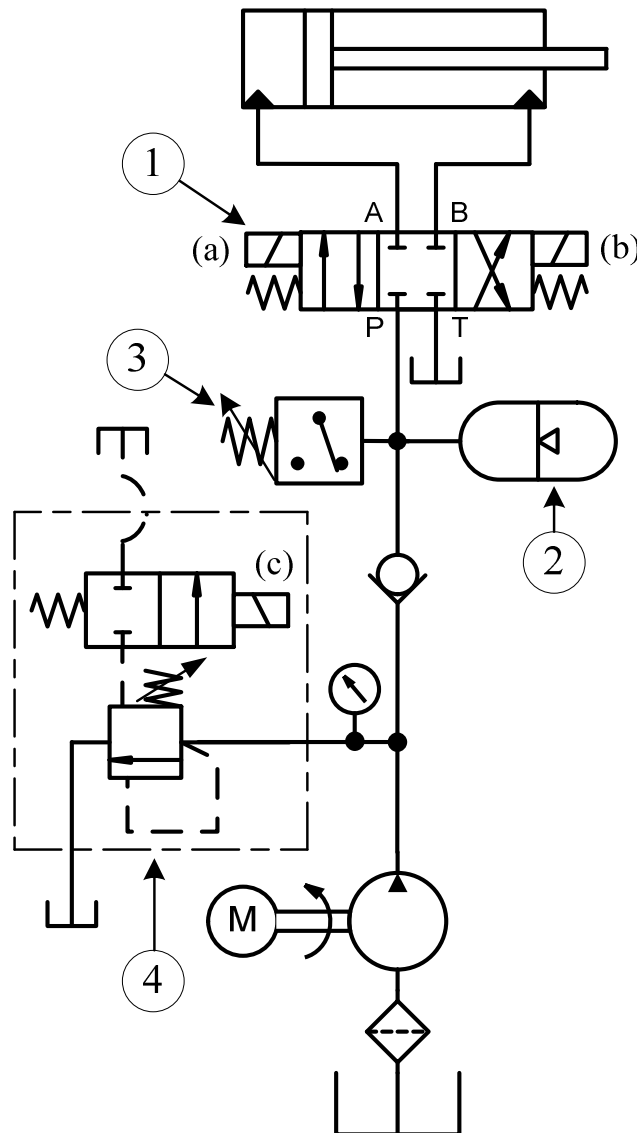
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Number the statements in sequential order to describe the proper operation of Schematic Exercise #9. (Note: all the statements will be used)

- _____ When solenoid **b** is de-energized, the springs shift the DCV back to home position the flow path is open from **P** to **T** and the ports **A** and **B** will be blocked; the cylinders retraction stop at any position, it can't be moved from the actual position, and the pump will be unloaded through the DCV3
- _____ When the DCV2 is activated its flow path is closed
- _____ When solenoid **a** is de-energized, the springs shift the DCV back to home position. The flow path is open from **P** to **T** and the ports **A** and **B** will be blocked, the cylinders extension stop at any position, it can't be moved from the actual position, and the pump will be unloaded through the DCV3
- _____ The FCV is by-passed through the check valve and the cylinder retracts at maximum speed
- _____ The cylinder rod end flow is meter out through the FCV and the cylinder extension is slowing down
- _____ When the solenoid **b** is energized, the DCV3 shifts and the flow path is open from **P** to **B** and from **A** to **T**
- _____ When the DCV2 is de-activated its flow path is open and the cylinder retraction continues at its maximum speed
- _____ The DCV3 shifts and the flow path is open from **P** to **A** and from **B** to **T**
- _____ The cylinder rod end flow by-passes the FCV through the DCV2 and the cylinder has a fast extension
- 1.** When the solenoid **a** is energized

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Schematic Exercise #10



Choose from the master list to identify the components in Schematic Exercise #10

1. _____

2. _____

3. _____

4.(Component) _____

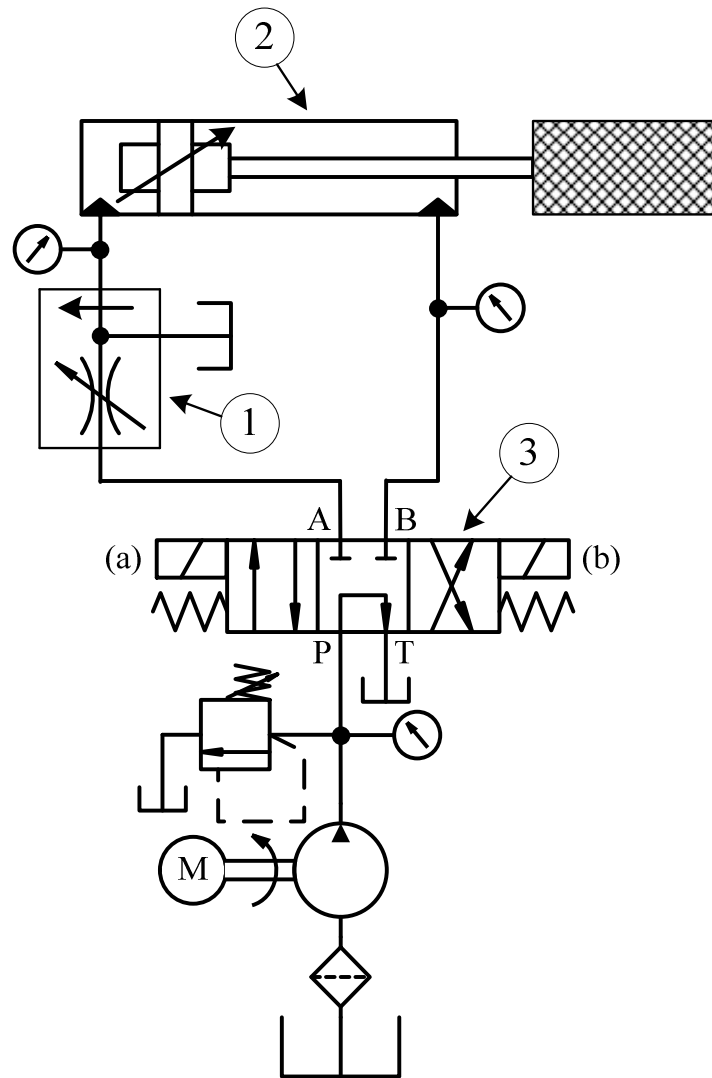
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Number the statements in sequential order to describe the proper operation of Schematic Exercise #10. (Note: all the statements will be used)

- _____ When solenoid **b** is de-energized, the springs shift the DCV1 back to home position and all the flow paths will be blocked; the cylinder stop at any position and it can't be moved; the pump will be charging the device 2. When the device 2 reaches the pressure device 3 hi setting, the pump will be unloaded through the PCV4
- _____ The check valve open, and the device 2 and/or the pump flow extend the cylinder
- _____ The pilot DCV shifts and opens the PCV4 pilot port to tank
- _____ When solenoid **a** is de-energized, the springs shift the DCV1 back to home position and all the flow paths will be blocked
- _____ When the device 2 reaches the device 3 hi setting, the check valve will close and the solenoid **c** will be activated
- _____ The PCV4 open at low pressure and the pump is unloaded through the PCV4
- _____ The DCV1 shifts and the flow path is open from **P** to **A** and from **B** to **T**
- _____ When the solenoid **b** is energized, the DCV1 shifts and the flow path is open from **P** to **B** and from **A** to **T**
- _____ The device 2 flow retracts the cylinder till its pressure drops below of the device 3 low setting
- _____ The cylinder stop at any position and it can't be moved; the pump will be charging the device 2
- _____ When the pressure is bellow of the device 3 low setting the solenoid **c** is de-activated
- _____ The pilot DCV goes back to home position blocking the PCV4 pilot port
- 1.** _____ When the solenoid **a** is energized
- _____ The PCV4 close, the check valve open and the pump flow finishes the retraction

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Schematic Exercise #11



Choose from the master list to identify the components in Schematic Exercise #11

1. _____

2. _____

3. _____

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Number the statements in sequential order to describe the proper operation of Schematic Exercise #11. (Note: all the statements will be used)

_____ The DCV shifts and the flow path is open from **P** to **A** and from **B** to **T**

1. When the solenoid **a** is energized

_____ When solenoid **a** is de-energized, the springs shift the DCV back to home position and the flow path is open from **P** to **T** and the ports **A** and **B** will be blocked; the cylinder stop at any position, the cylinder can't be moved from the actual position, and the pump will be unloaded through the DCV

_____ The FCV meter out the cylinder retraction without compensation and the pump extra flow is sent to tank though the PCV

_____ When solenoid **b** is de-energized, the springs shift the DCV back to home position and the flow path is open from **P** to **T** and the ports **A** and **B** will be blocked; the cylinder stop at any position, the cylinder can't be moved from the actual position, and the pump will be unloaded through the DCV

_____ The FCV meter in the cylinder extension, its pressure is compensated and the pump extra flow is sent to tank though the FCV

_____ When the solenoid **b** is energized, the DCV shifts and the flow path is open from **P** to **B** and from **A** to **T**

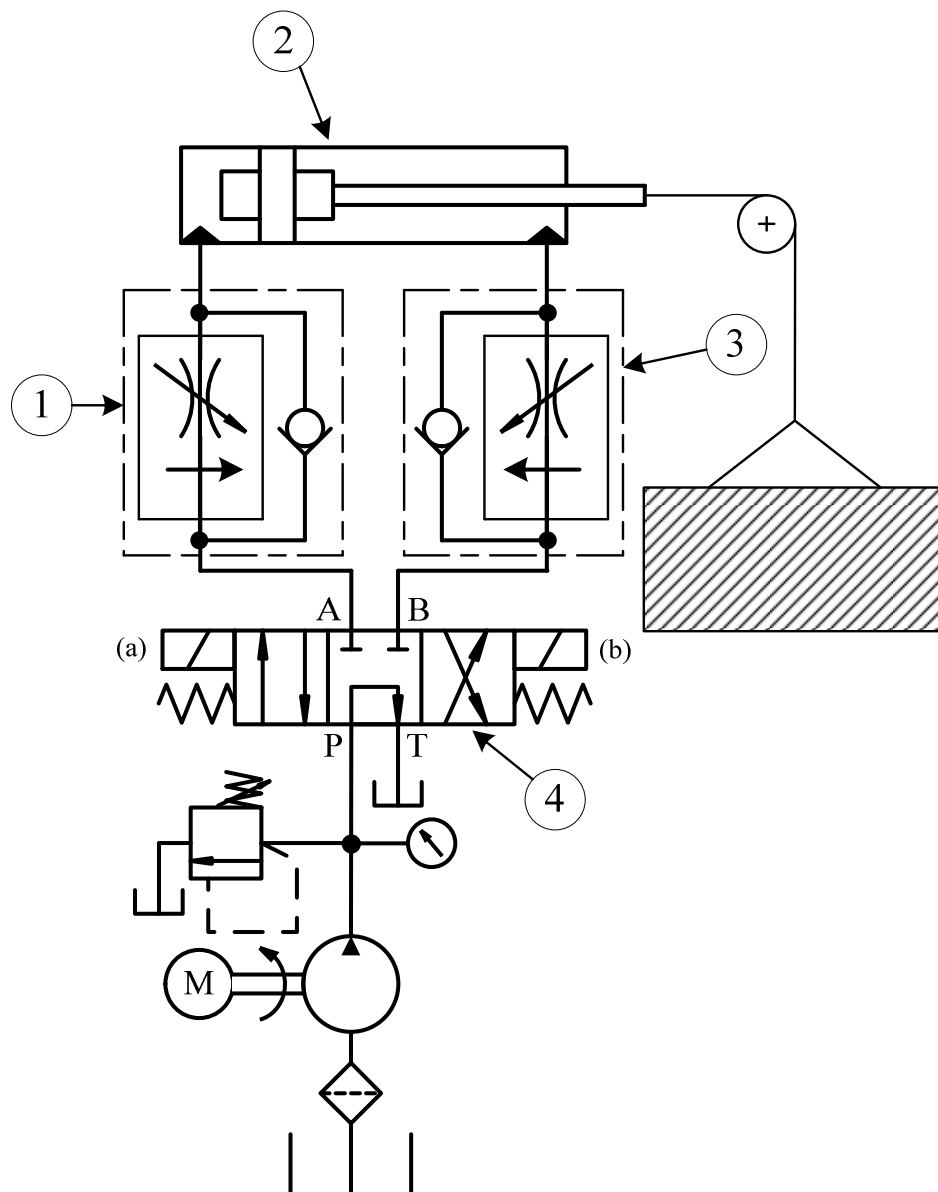
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Number the statements in sequential order to describe the proper operation of Schematic Exercise #12. (Note: all the statements will be used)

1. When the solenoid **a** is energized
- _____ The PCV2 is by-passed
- _____ When solenoid **b** is de-energized, the springs shift the DCV back to home position and the ports **A**, **B**, and **P** will be connected to **T**; the actuator stop at any position and the pump will be unloaded through the DCV
- _____ The DCV shifts and the flow path is open from **P** to **A** and from **B** to **T**
- _____ When solenoid **a** is de-energized, the springs shift the DCV back to home position and the ports **A**, **B**, and **P** will be connected to **T**; the actuator stop at any position and the pump will be unloaded through the DCV
- _____ When the pressure in the PCV3 inlet port reaches its spring setting, the PCV3 open and the actuator rotates **CCW**
- _____ When the solenoid **b** is energized, the DCV shifts and the flow path is open from **P** to **B** and from **A** to **T**
- _____ When the pressure in the PCV2 inlet port reaches its spring setting, the PCV2 open and the actuator rotates **CW**
- _____ The PCV3 is by-passed

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Schematic Exercise #13



Choose from the master list to identify the components in Schematic Exercise #13

1.(Component) _____

2. _____

3.(Component) _____

4. _____

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Number the statements in sequential order to describe the proper operation of Schematic Exercise #13. (Note: all the statements will be used)

_____ When the solenoid **b** is energized, the DCV shifts and the flow path is open from **P** to **B** and from **A** to **T**

_____ The load goes down under control

_____ When solenoid **b** is de-energized, the springs shift the DCV back to home position and the flow path is open from **P** to **T** and the ports **A** and **B** will be blocked; the cylinder stop at any position, the cylinder can't be moved from the actual position, and the pump will be unloaded through the DCV

_____ The FCV3 is bypassed, the cylinder cap end flow is meter out, and its pressure is compensated

_____ The DCV shifts and the flow path is open from **P** to **A** and from **B** to **T**

_____ The FCV1 is bypassed, the cylinder rod end flow is meter out and its pressure is compensated

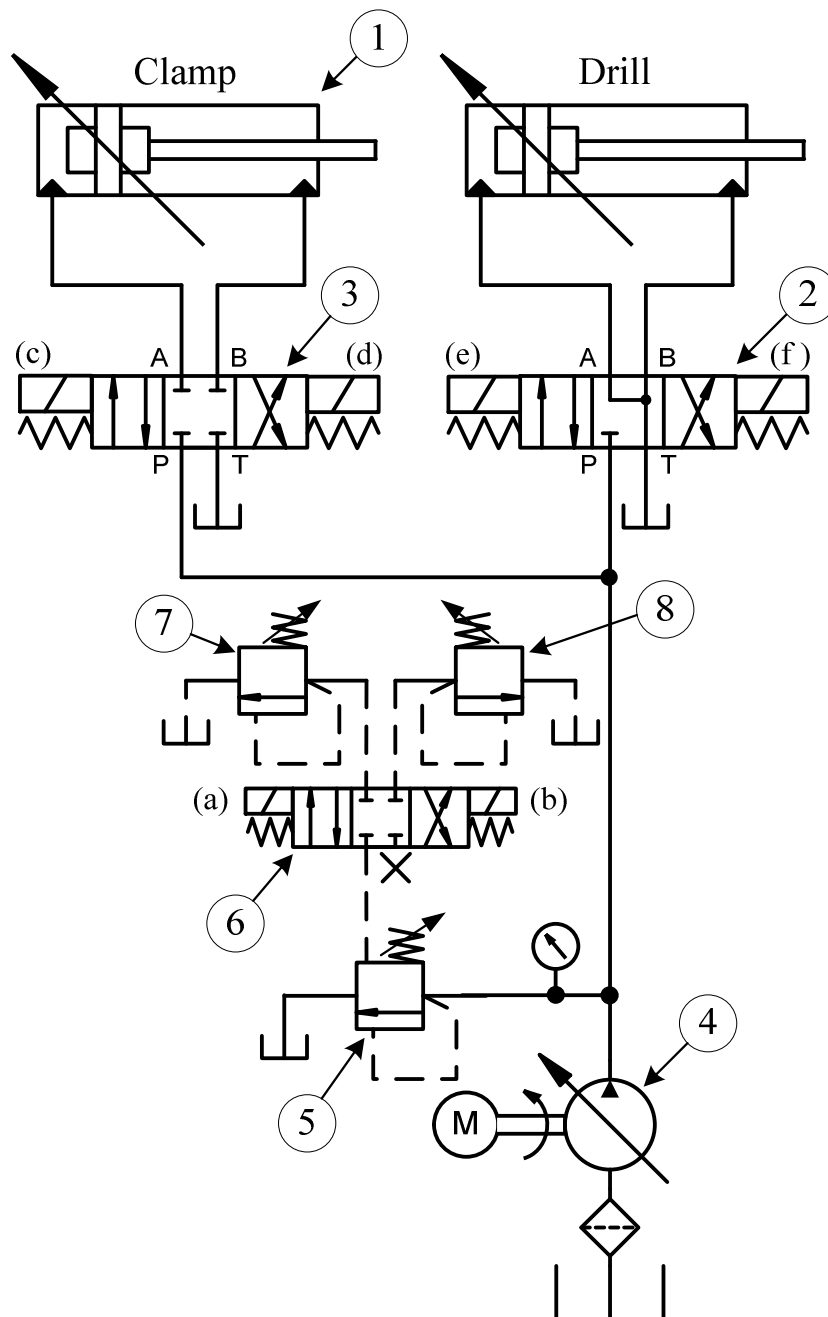
_____ When solenoid **a** is de-energized, the springs shift the DCV back to home position and the flow path is open from **P** to **T** and the ports **A** and **B** will be blocked; the cylinder stop at any position, the cylinder can't be moved from the actual position, and the pump will be unloaded through the DCV

1. _____ When the solenoid **a** is energized

_____ The load goes up under control

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Schematic Exercise #14



Choose from the master list to identify the components in Schematic Exercise #14

- | | | | |
|----------|----------|----------|----------|
| 1. _____ | 2. _____ | 3. _____ | 4. _____ |
| 5. _____ | 6. _____ | 7. _____ | 8. _____ |

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Choose the correct solenoid schedule to accomplish each individual operation for Schematic Exercise #14. (Note: all the solenoid schedules **will not** be used)

Five individual operations;

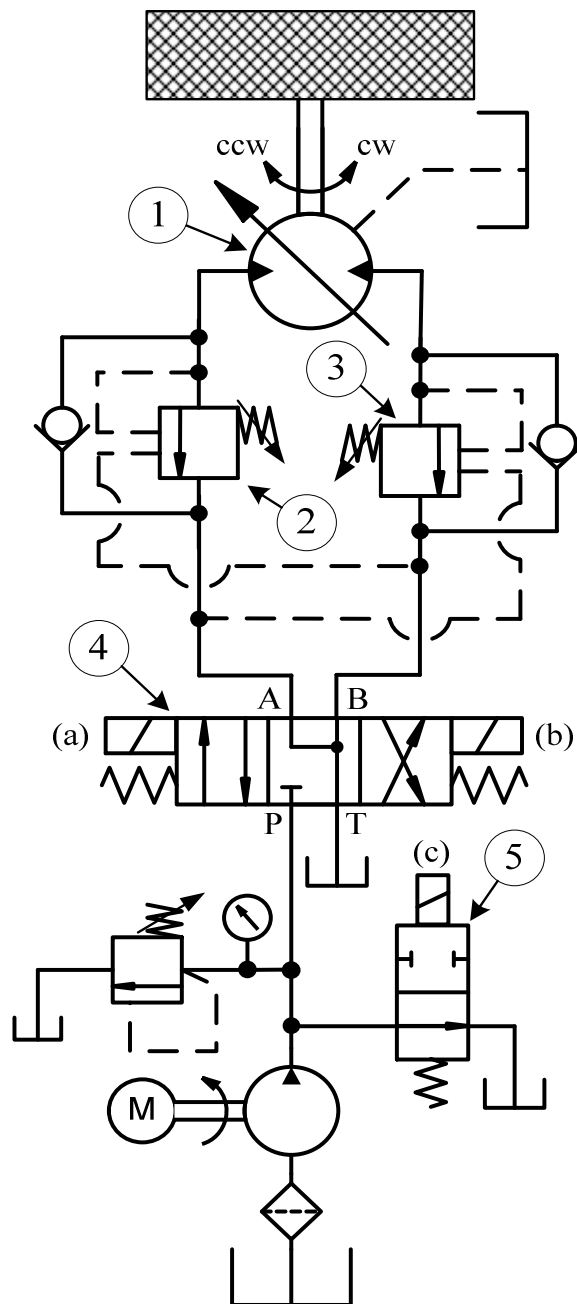
1. Clamp Extension at PCV5 Setting. _____
2. Drill Extension at PCV7 Setting. _____
3. Drill Retraction at PCV8 Setting. _____
4. Clamp Retraction at PCV7 Setting. _____
5. Pump Unloading at PCV8 Setting. _____

Solenoid schedules; Choose one schedule per individual operation.
(+ is energized; - is de-energized)

- A. a-, b-, c+, d-, e+, f-
- B. a+, b-, c-, d-, e+, f-
- C. a-, b-, c-, d-, e-, f+
- D. a+, b-, c-, d+, e-, f-
- E. a-, b-, c-, d+, e-, f-
- F. a-, b-, c+, d-, e-, f-
- G. a-, b+, c-, d-, e-, f+
- H. a-, b+, c+, d-, e-, f-
- I. a-, b+, c-, d-, e-, f-
- J. a+, b-, c+, d-, e-, f+

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Schematic Exercise #15



Choose from the master list to identify the components in Schematic Exercise #15

1. _____

2. _____

3. _____

4. _____

5. _____

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Number the statements in sequential order to describe the proper operation of Schematic Exercise #15. (Note: all the statements will be used)

- _____ When solenoid **a** and **c** are de-energized, the springs shift the DCV4 and DCV5 back to home position blocking the port **P** and connecting ports **A** and **B** to **T**; the PCV3 is close till its internal pilot pressure overcome the PCV3 setting
- _____ The flow path through the DCV5 is block, and the flow by-passes the PCV2
- _____ When the external pilot of the PCV3 overcome the PCV3 setting, the PCV3 open at low pressure
- _____ The motor rotates CCW
- _____ The motor rotates CW
- _____ The PCV3 open at high pressure and stop the motor. The pump is unloading through the DCV5
- _____ The flow path through the DCV5 is block, and the flow by-passes the PCV3
- _____ When the external pilot of the PCV2 overcome the PCV2 setting, the PCV 2 open at low pressure
- _____ When solenoid **b** and **c** are de-energized, the springs shift the DCV4 and DCV5 back to home position blocking the port **P** and connecting ports **A** and **B** to **T**; the PCV2 is close till its internal pilot pressure overcome the PCV2 setting and the PCV2 open at high pressure and stop the motor, the pump is unloading through the DCV5
- _____ When the solenoids **b** and **c** are energized, the DCV4 shifts and the flow path is open from **P** to **B** and from **A** to **T**
- _____ The DCV4 shifts and the flow paths are open from **P** to **A** and from **B** to **T**
- 1.** When the solenoids **a** and **c** are energized

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Schematic Application

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Hydraulic Formulas

Pressure, Force & Area

U. S. Units

$$\text{Pressure (PSI)} = \frac{\text{Force (lbs)}}{\text{Area (in}^2\text{)}} \quad \text{Force} = \text{Pressure} \times \text{Area} \quad \text{Area} = \frac{\text{Force}}{\text{Pressure}}$$

Metric Units

$$\text{Pressure (bar)} = \frac{\text{Force (kg)}}{\text{Area (cm}^2\text{)}} \quad \text{Force} = \text{Pressure} \times \text{Area} \quad \text{Area} = \frac{\text{Force}}{\text{Pressure}}$$

Hydraulic Cylinder Speed & Flow Rate

U. S. Units

$$\text{Cylinder Rod Speed (in / min)} = \frac{\text{GPM} \times 231}{\text{Piston Area (in}^2\text{)}}$$

$$\text{Cylinder Rod Speed (ft / min)} = \frac{\text{GPM} \times 19.25}{\text{Piston Area (in}^2\text{)}}$$

$$\text{GPM} = \frac{\text{Cylinder Rod Speed (in / min)} \times \text{Piston Area (in}^2\text{)}}{231}$$

Metric Units

$$\text{Cylinder Rod Speed (m / sec)} = \frac{\text{Flow (LPM)}}{6(\text{ factor}) \times \text{Piston Area (cm}^2\text{)}}$$

$$\text{Flow (LPM)} = 6 \times \text{Cylinder Rod Speed (m / sec)} \times \text{Piston Area (cm}^2\text{)}$$

Hydraulic (Circuit) Horsepower

U. S. Units

$$\text{Horsepower} = \text{GPM} \times \text{Pressure (psi)} \times 0.000583$$

Metric Units

$$\text{Horsepower (Kilowatt)} = \frac{\text{Flow Rate (LPM)} \times \text{Pressure (bar)}}{600}$$

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Hydraulic Formulas (Continued)

Hydraulic Motor Shaft Speed, Flow Rate, Torque & H/P

U. S. Units

$$RPM = \frac{GPM \times 231}{Displacement(in^3 / rev)}$$

$$GPM = \frac{RPM \times Displacement(in^3 / rev)}{231}$$

$$Torque(lb.in.) = \frac{Pressure(PSI) \times Displacement(in^3 / rev)}{2 \times \pi}$$

$$MotorHorsepower = \frac{RPM \times Torque(lb.in.)}{63025}$$

Metric Units

$$RPM = \frac{LPM \times 1000}{Displacement(cm^3 / rev)}$$

$$LPM = \frac{RPM \times Displacement(cm^3 / rev)}{1000}$$

$$Torque(N.m.) = \frac{15.9 \times Pressure(bar) \times Displacement(cm^3 / rev)}{1000}$$

$$MotorHorsepower(Kilowatt) = \frac{RPM \times Torque(N.m.)}{9543}$$

Hydraulic Pump Torque

U. S. Units

$$Torque(lb.in.) = \frac{Pressure(PSI) \times Displacement(in^3 / rev)}{2 \times \pi}$$

Metric Units

$$Torque(N.m.) = \frac{15.9 \times Pressure(bar) \times Displacement(cm^3 / rev)}{1000}$$

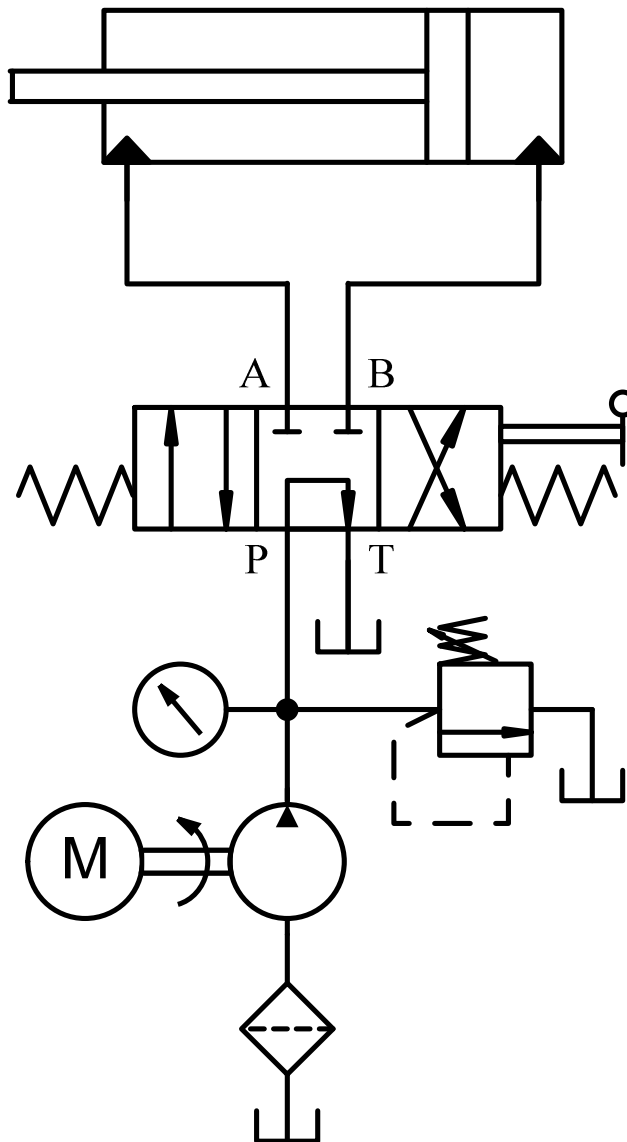
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Schematic Application Exercises

Schematic Application Exercise #1

Schematic #1 description: Hand lever operated dcv controls cylinder movement. When the hand lever is released cylinder movement stops and pump flow unloads to the tank.

1. Pipe this circuit on the trainer/simulator.



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2. Schematic #1 exercises: The cylinder size is a 5 inch (12.7 cm) bore, and 30 inch (76.2 cm) stroke. The cylinder speed is 20 feet/min (0.103 m/sec) and the maximum load being moved is 32,000 lbs (14514.92 kg).

a. How many gpm (l/min) are required to maintain the cylinder speed?

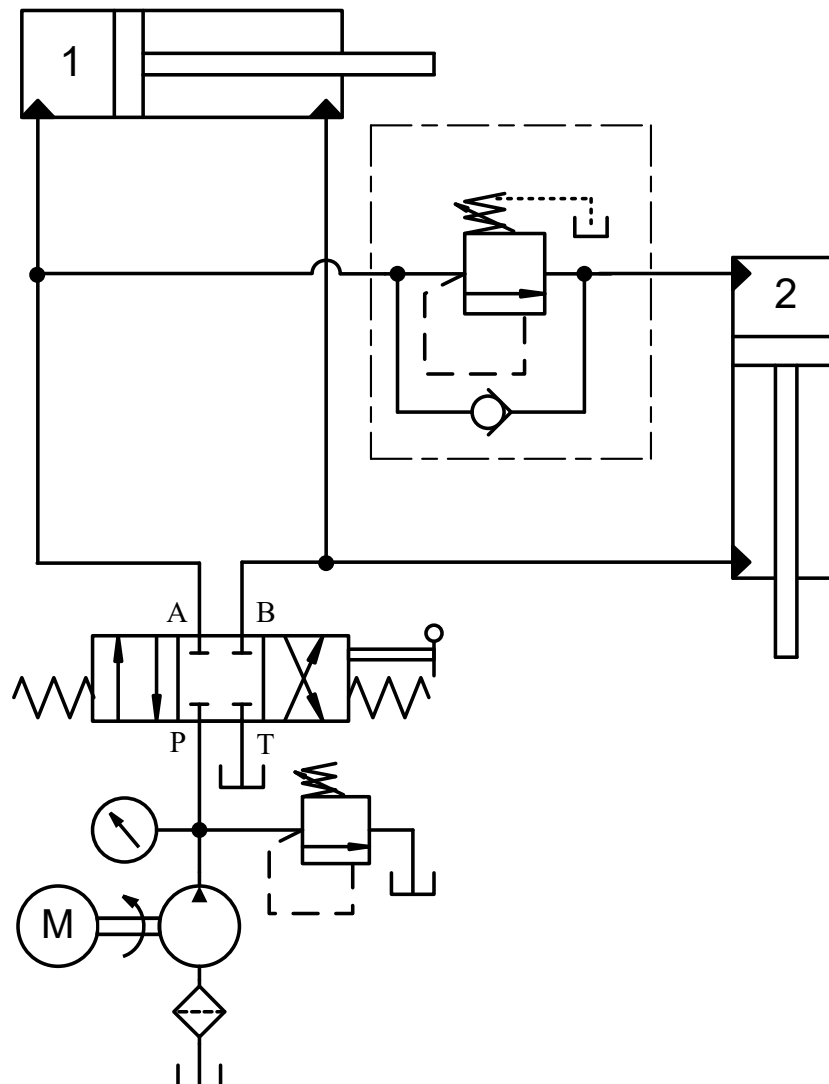
b. What is the required working pressure for the circuit?

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Schematic Application Exercise #2

Schematic #2 description: Hand lever operated dcv controls movement of two cylinders. During extension, cylinder 1 extends, when cylinder 1 has fully extended pressure increases and sequence valve ensures the extension of cylinder 2. The retraction strokes of both cylinders take place in an undetermined order.

1. Pipe this circuit on the trainer/simulator.



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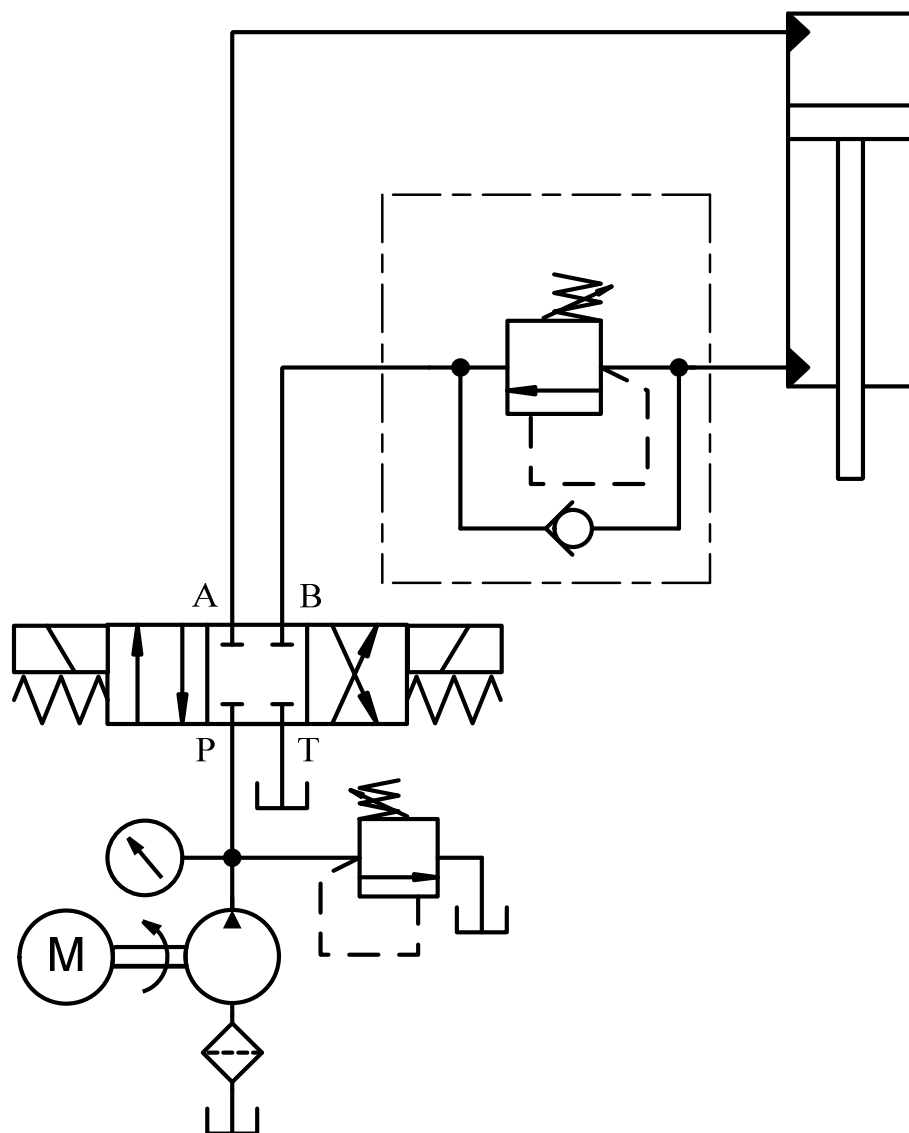
2. Schematic #2 exercises: Cylinder #1 has a 1.5 inch (3.81 cm) bore and the load is 500 lbs (226.8 kg).
 - a. What pressure would you set the sequence valve?
(Use 150 psi (10.55 b) for the cracking pressure.)

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Schematic Application Exercise #3

Schematic #3 description: A solenoid operated dcv controls the movement of a vertically mounted cylinder with an assisting load. The counter balance valve is used to prevent the load from free falling when the cylinder extends.

1. Pipe this circuit on the trainer/simulator.



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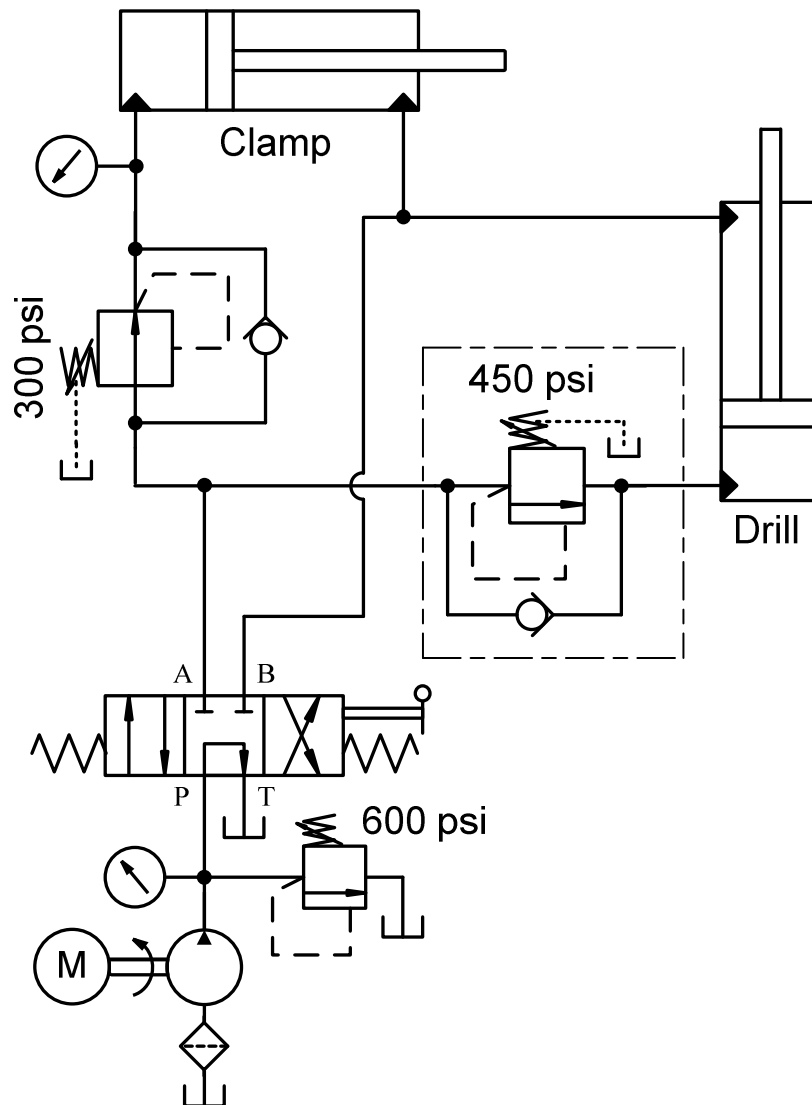
2. Schematic #3 exercises: The cylinder bore dimensions are 2.5 inches (6.35 cm), and the rod dimension is 1.375 inches (3.49 cm). With a hanging load of 1000 lbs (453.59 kg) and using internal pilot.
 - a. What pressure would you set the counter balance spring to ensure that the cylinder doesn't move until 250 psi (17.58 b) is achieved on the major side of the cylinder? (Use 150 psi (10.55 b) for the cracking pressure.)

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Schematic Application Exercise #4

Schematic #4 description: The clamp cylinder must extend first and clamp at a pressure of 300 psi (21 b). The relief valve setting is to be 600 psi (42 b). Assume that the drill cylinder will be mechanically restricted in its motion while drilling. This causes the relief valve to be cracked open (actuated). The sequence valve may be piloted internally or externally. The pressure-reducing valve may, or may not, need a by-pass check valve

1. Pipe this circuit on the trainer/simulator.



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2. Schematic #4 exercises: The drill cylinder has a 4 inch (10.16 cm) bore with a 1.75 inch (4.45 cm) rod, and the pump displacement is 11 gpm (41.64 l/min). When the machine is drilling 1.75 gpm (6.62 l/min) passes over the relief valve back to the reservoir. Pilot flow through the springs to tank on the sequence and reducing valves is approximately 0.1 gpm (0.38 l/min) for each valve.

a. How fast will the drill cylinder be extending?

b. Which cylinder will retract first?

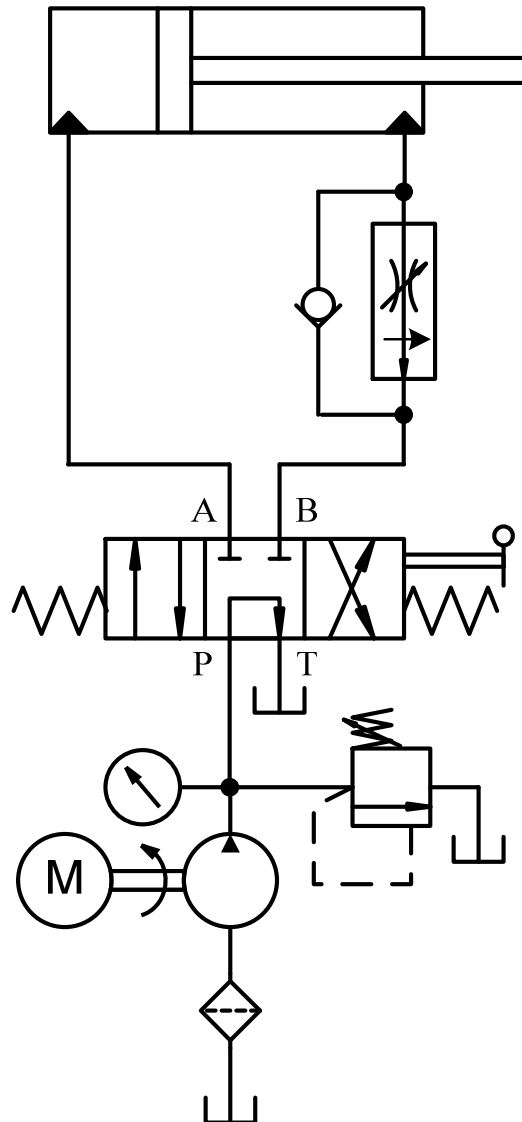
c. What is the total hydraulic horsepower of this circuit?

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Schematic Application Exercise #5

Schematic #5 description: Hand lever operated dcv controls cylinder movement. When the hand lever is released cylinder movement stops and pump flow unloads to the tank. The cylinder speed needs to be precise throughout the workday, even with changing loads on the cylinder.

1. Pipe this circuit on the trainer/simulator.



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2. Schematic #5 exercises:

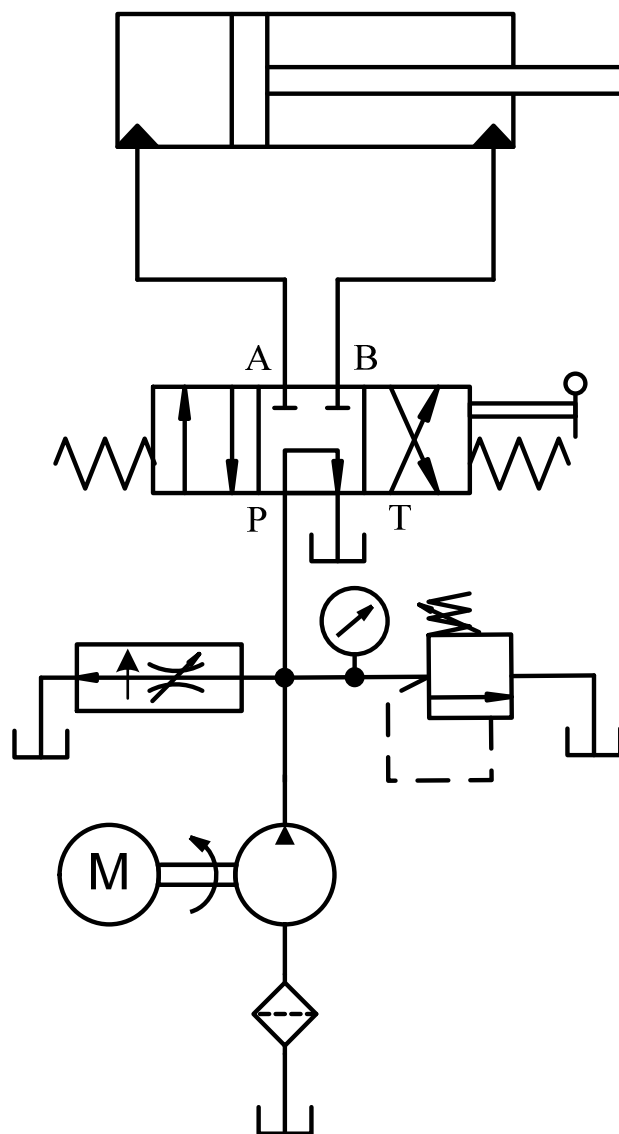
- a. What is the common name for this type of circuit?

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Schematic Application Exercise #6

Schematic #6 description: Hand lever operated dcv controls cylinder movement. When the hand lever is released cylinder movement stops and pump flow unloads to the tank. Another type of flow control circuit can be used to control the speed of a cylinder. Besides generating less heat than the previous circuit, it can also be more economical than either a meter-in or meter-out circuit.

1. Pipe this circuit on the trainer/simulator.



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2. Schematic #6 exercises:

a. What is the common name for this type of circuit?

b. Give a complete description of the directional control valve and what is the benefit of using this type of center condition?

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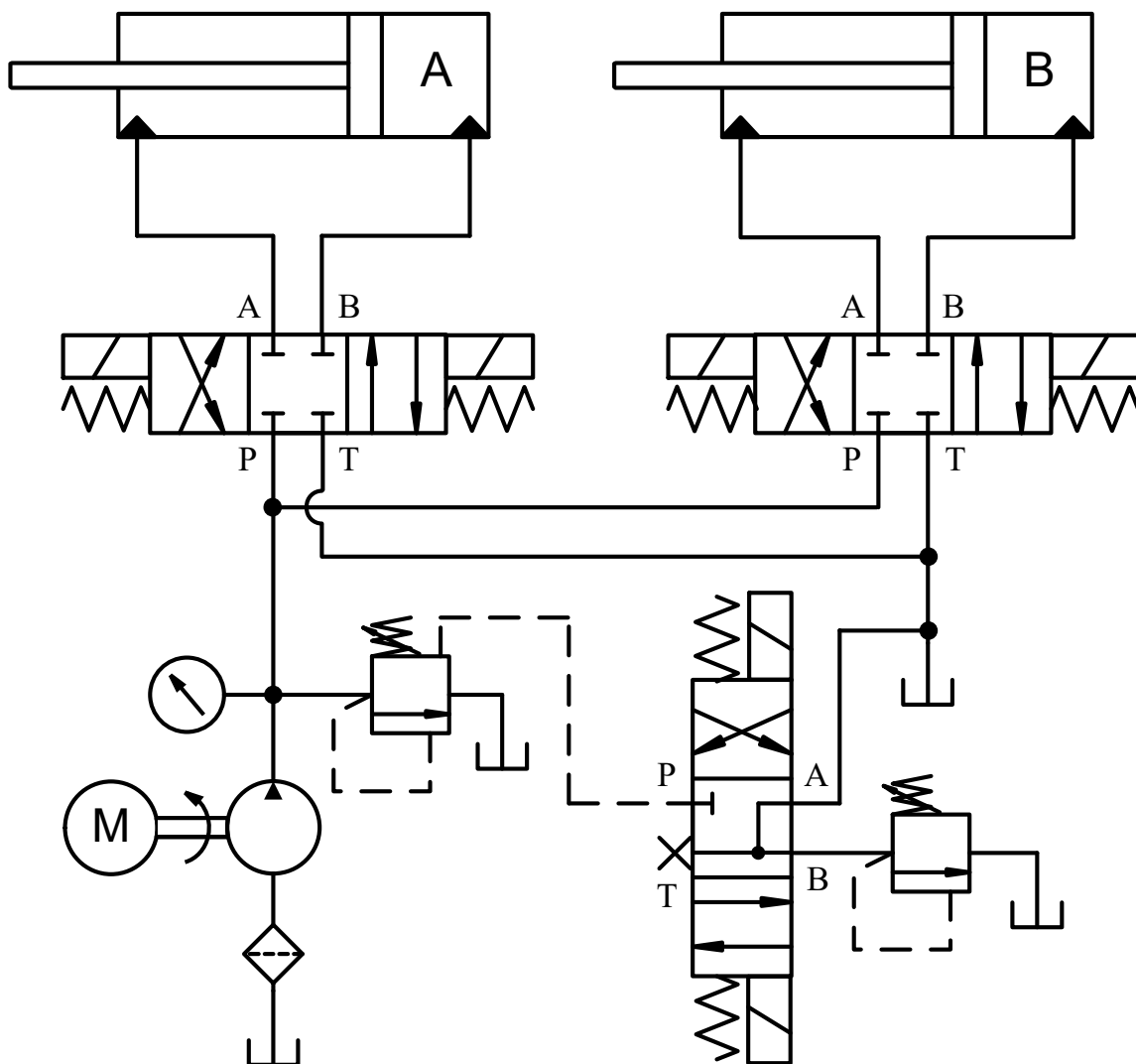
2. Schematic #7 exercises:

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Schematic Application Exercise #8

Schematic #8 description: Cylinder A must work at a maximum pressure of 2500 psi (175.77 b). Cylinder B should not exceed 1500 psi (105.46 b) when working. When the machine is idle and doing no work, the pump should be unloaded. Cylinder A and B operate independent of one another.

1. Pipe this circuit on the trainer/simulator.



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2. Schematic #8 exercises:

a. Describe in detail the type of pressure relief valve used.

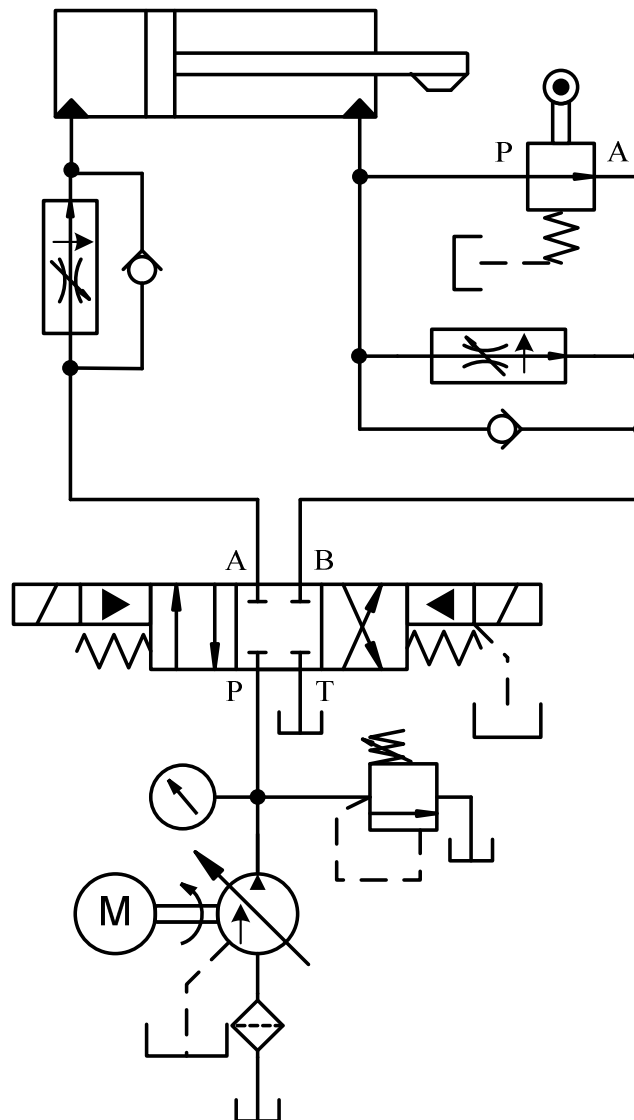
b. What is the best way to set the pressures of the relief valve?

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Schematic Application Exercise #9

Schematic #9 description: In this circuit, a rapid extension of 300 in/min (0.127 m/sec) is needed until the machining portion of the stroke is reached. Then a slow extension of 100 in/min (0.04 m/sec) is required. At the end of the machining process a rapid retraction of 500 in/min (0.212 m/sec) is needed to get into position for the next cycle.

1. Pipe this circuit on the trainer/simulator.



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2. Schematic #9 exercises: The cylinder bore is 4 inch (10.16 cm), and the rod diameter is 2 inch (5.08 cm).

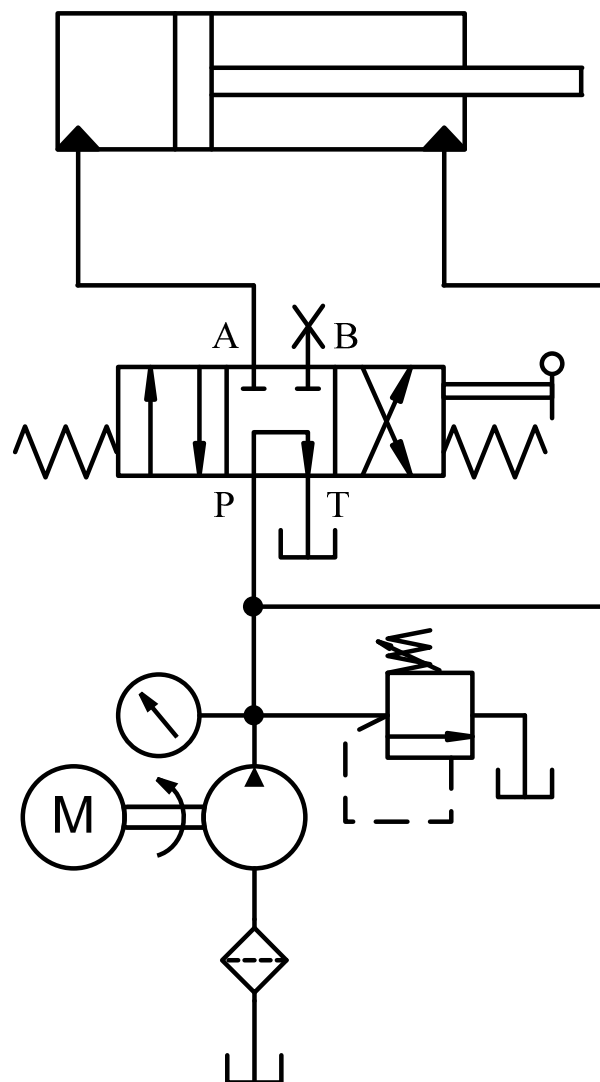
a. What is the gpm (l/min) requirement for the pump?

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Schematic Application Exercise #10

Schematic #10 description: Hand lever operated dcv controls cylinder movement. When the dcv is shifted to get **P to A**, a rapid extension is utilized. When the hand lever is released cylinder movement stops and pump flow unloads to the tank. Shifting the dcv to get **P to B** will provide a rapid retraction to position the cylinder for the next cycle.

1. Pipe this circuit on the trainer/simulator.



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2. Schematic #10 exercises: The piston surface area is 8 in² (51.613 cm²), and the rod surface area is 4 in². (25.806 cm²). The pump flow rate is 5 gpm (18.927 l/min). The pressure relief valve spring setting is 1000 psi (68.948 b).

a. What is the Rod Speed during the extension?

b. How much force can be created during the extension?

c. What is the Rod Speed during the retraction?

d. How much force can be created during the retraction?

e. What is the common name for this type of circuit?

f. What is the main purpose of this circuit?

g. what is sacrificed during extension?

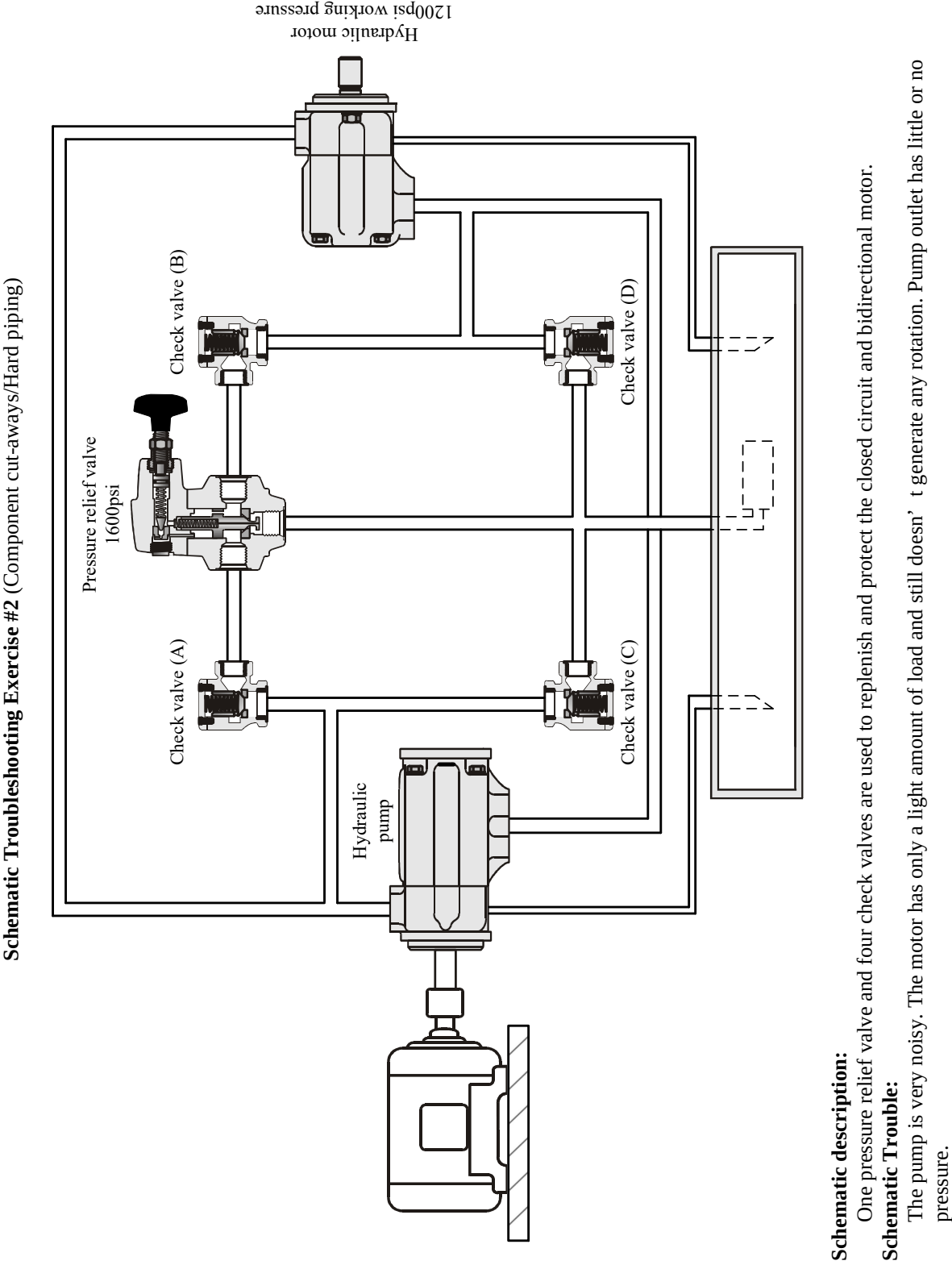
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Schematic Troubleshooting

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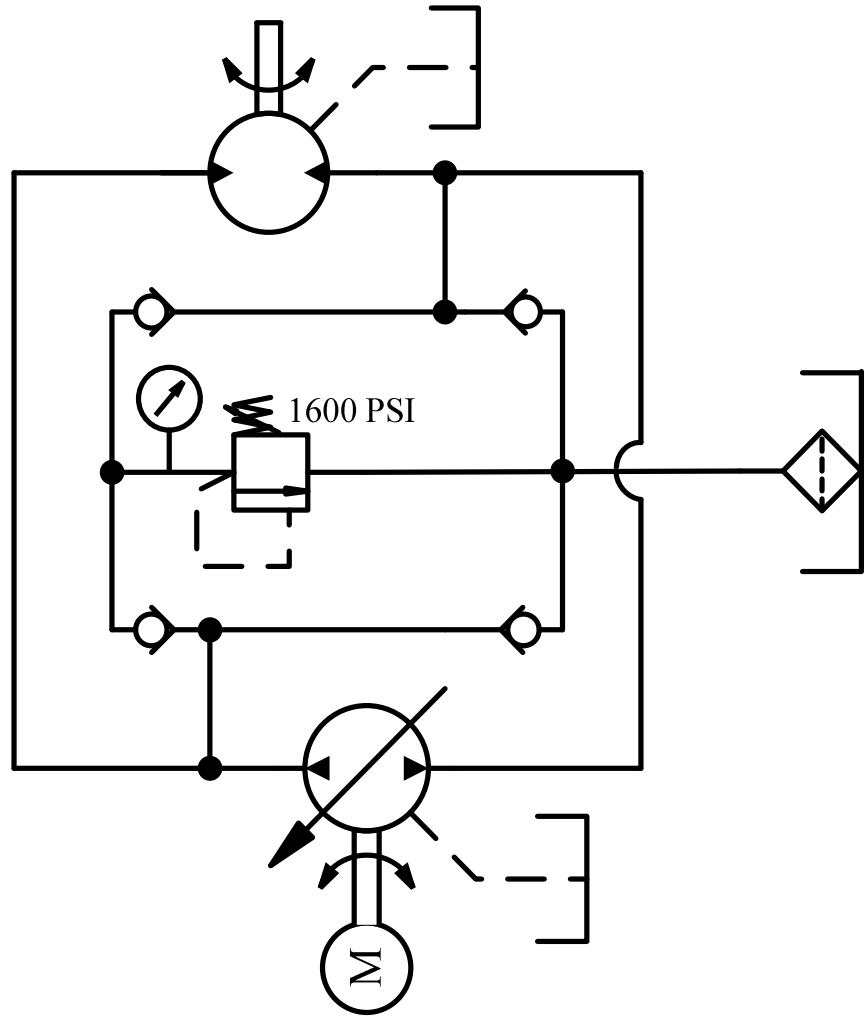
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Schematic Troubleshooting Exercise #2 (Blueprint schematic)



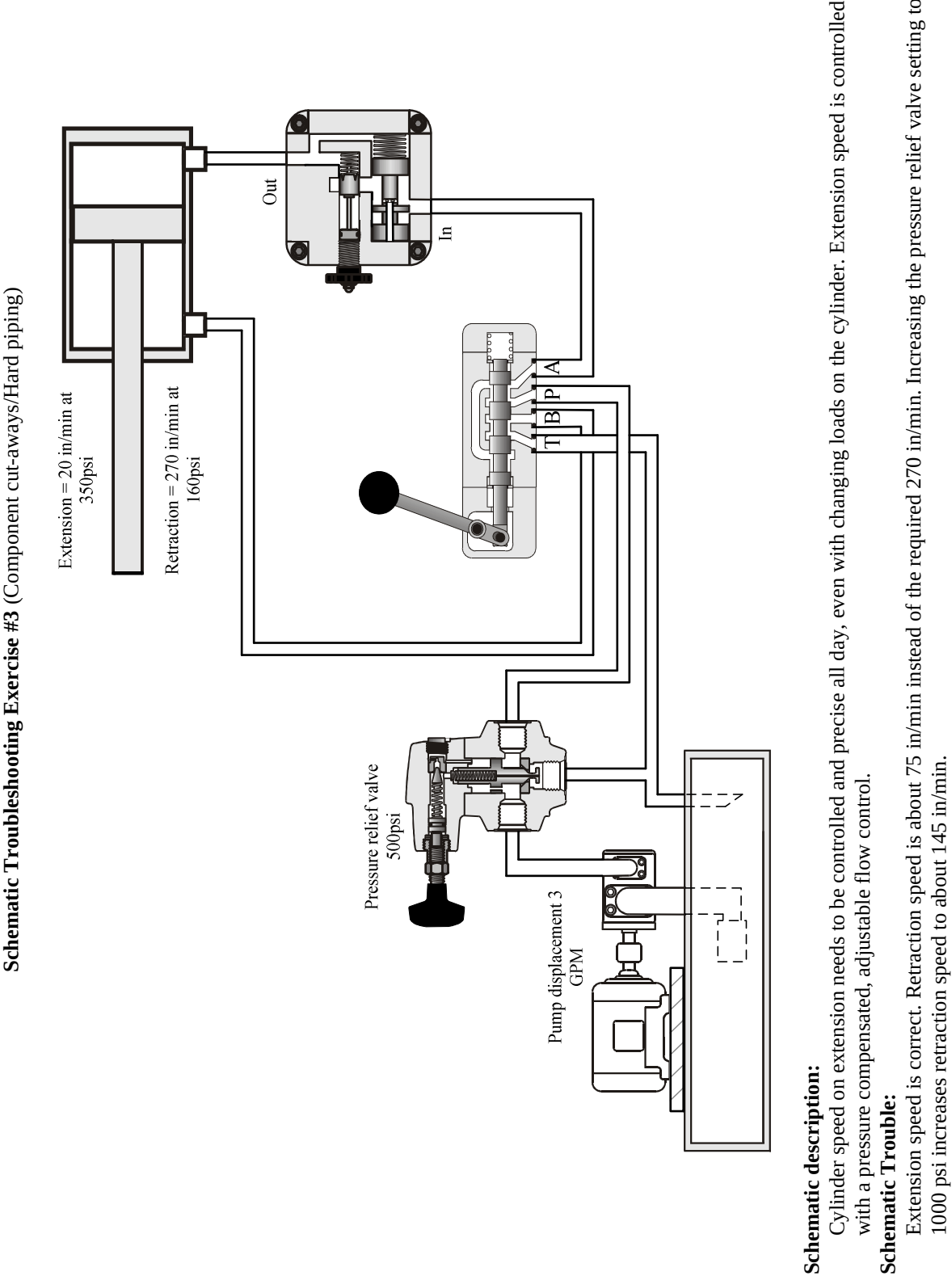
Schematic description:

One pressure relief valve and four check valves are used to replenish and protect the closed circuit and bidirectional motor.

Schematic Trouble:

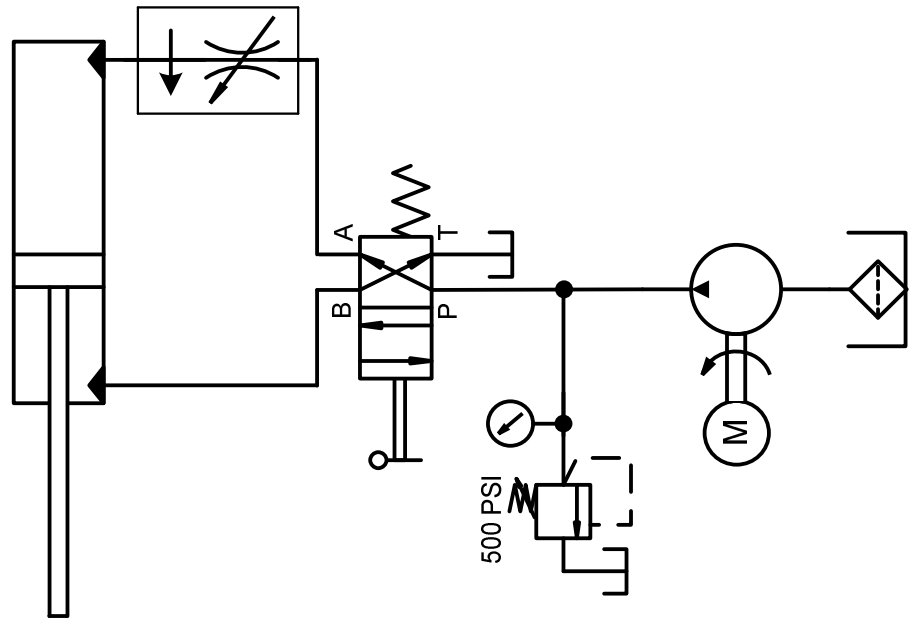
The pump is very noisy. The motor has only a light amount of load and still doesn't generate any rotation. Pump outlet has little or no pressure.

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Schematic Troubleshooting Exercise #3 (Blueprint schematic)



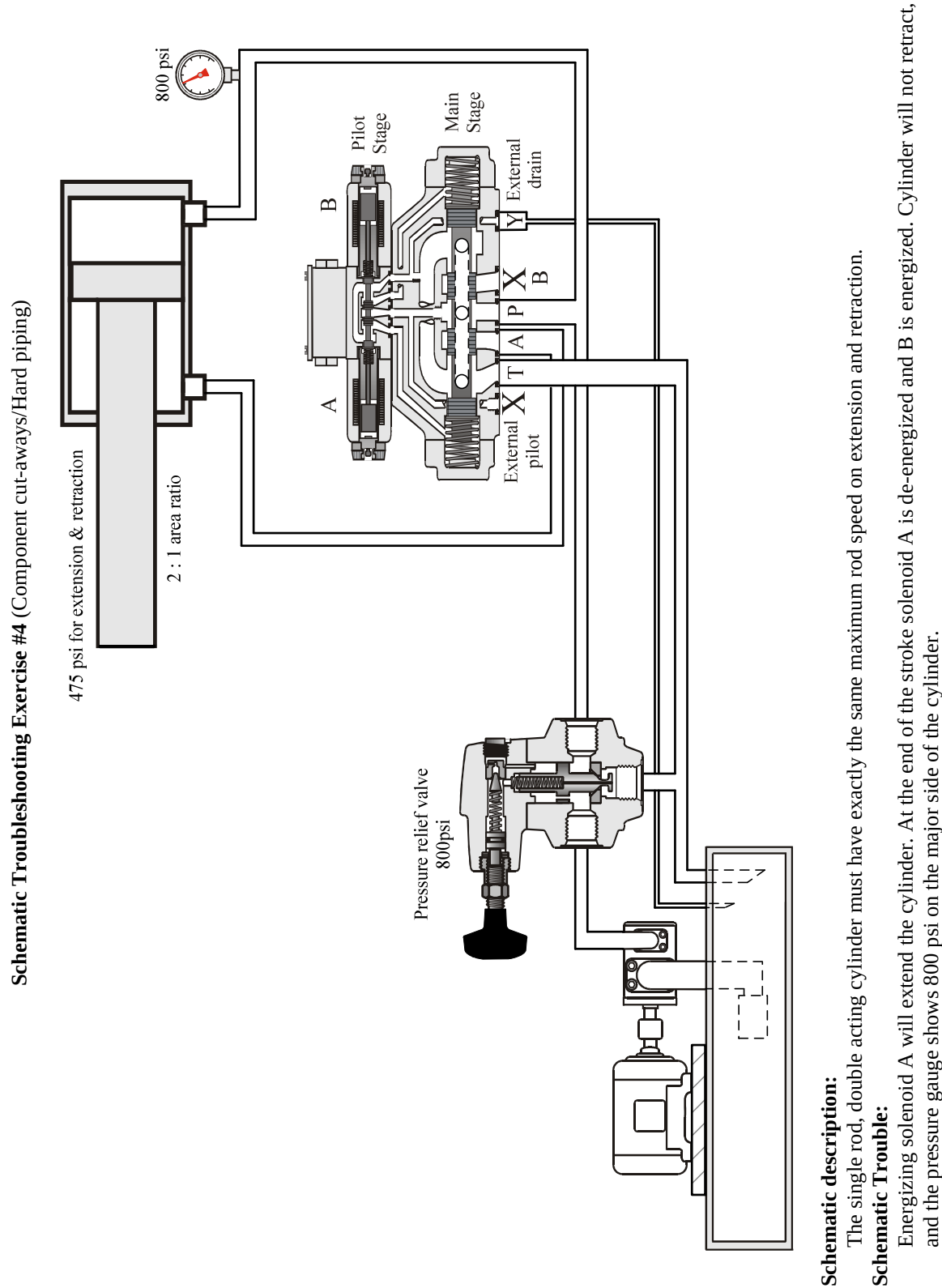
Schematic description:

Cylinder speed on extension needs to be controlled and precise all day, even with changing loads on the cylinder. Extension speed is controlled with a pressure compensated, adjustable flow control.

Schematic Trouble:

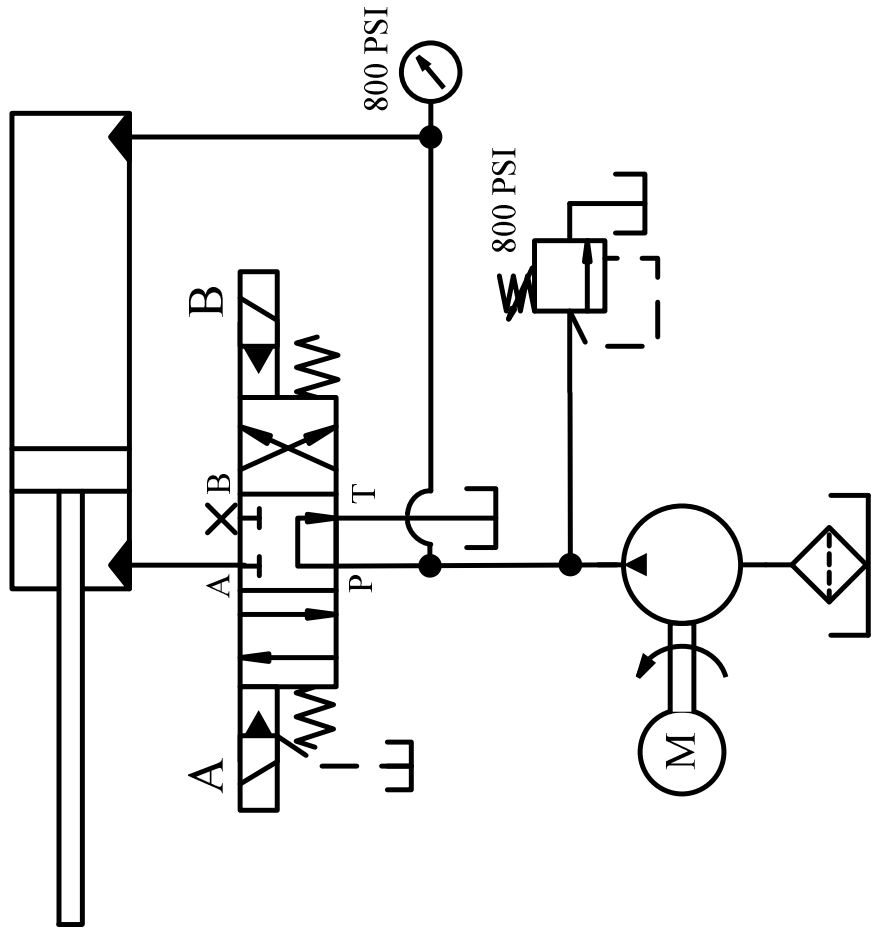
Extension speed is correct. Retraction speed is about 75 in/min instead of the required 270 in/min. Increasing the pressure relief valve setting to 1000 psi increases retraction speed to about 145 in/min.

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Schematic Troubleshooting Exercise #4 (Blueprint schematic)



Schematic description:

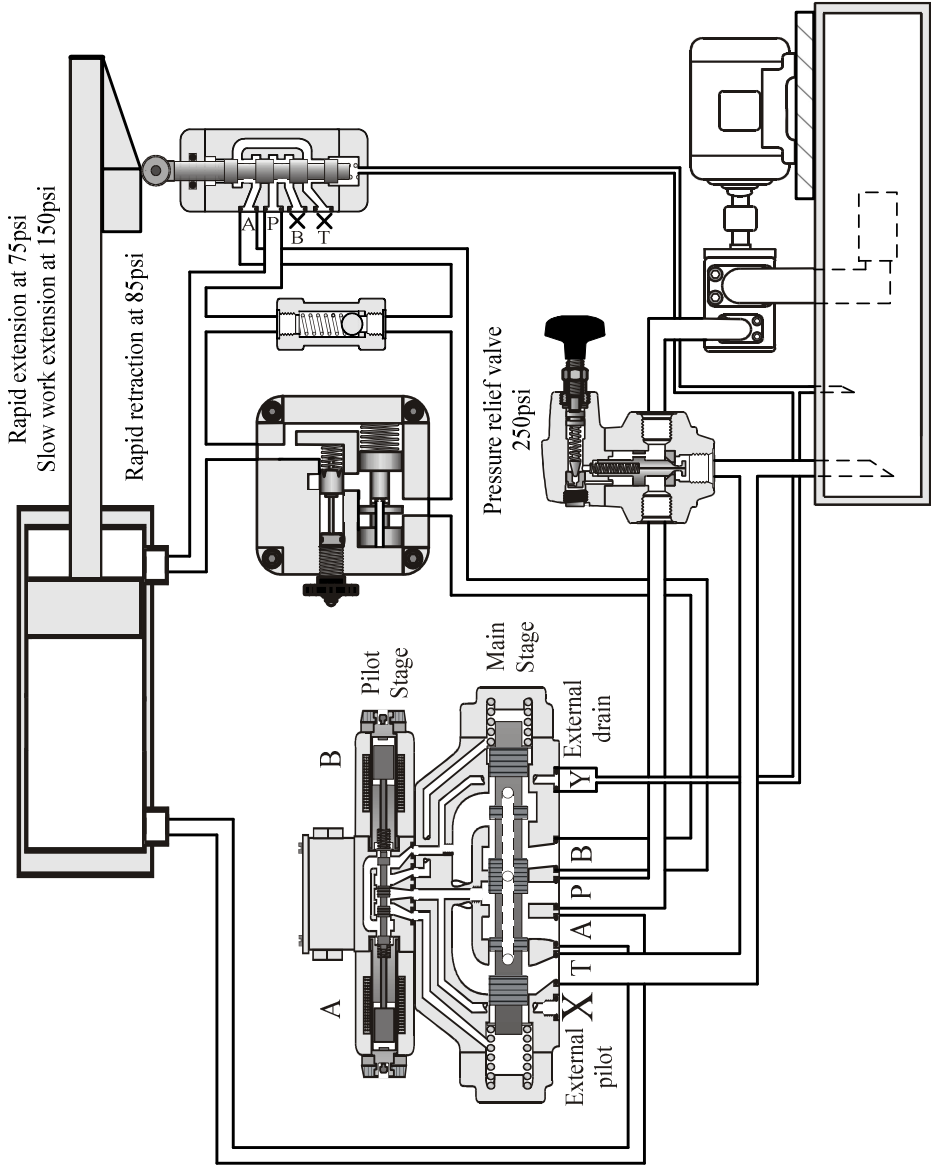
The single rod, double acting cylinder must have exactly the same maximum rod speed on extension and retraction.

Schematic Trouble:

Energizing solenoid A will extend the cylinder. At the end of the stroke solenoid A is de-energized and B is energized. Cylinder will not retract, and the pressure gauge shows 800 psi on the major side of the cylinder.

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Schematic Troubleshooting Exercise #5 (Component cut-aways/Hard piping)



Schematic description:

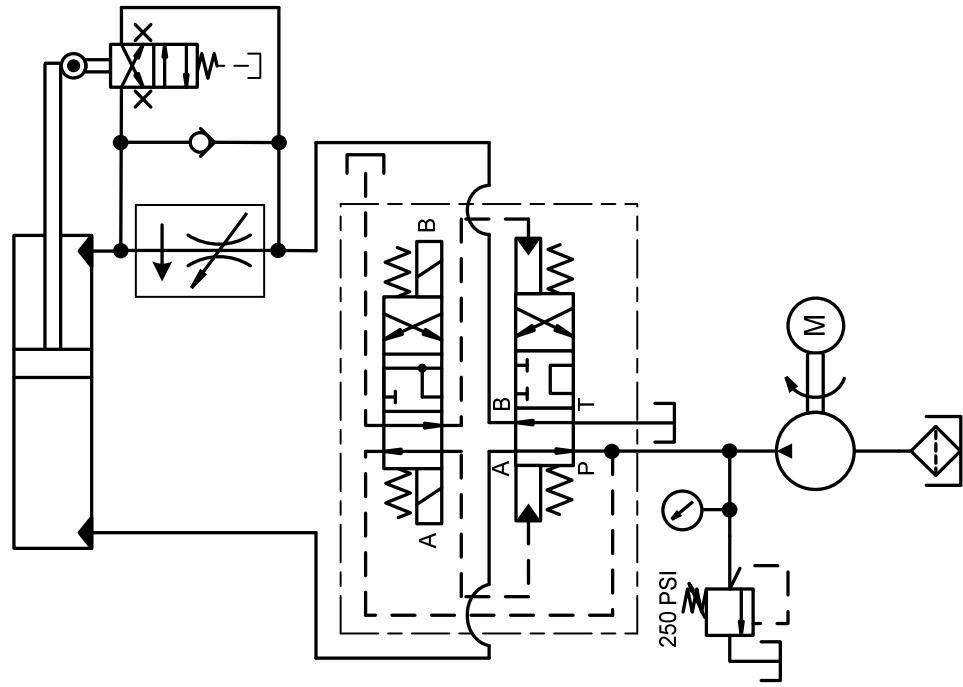
The cylinder will extend rapidly, until the deceleration valve is energized. Fluid is forced over the flow control and finishes extension as a slow work circuit. Cylinder will have rapid retraction for the entire return stroke.

Schematic Trouble:

During the slow work cycle the feed speed will vary and cannot get the feed as slow as required.

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Schematic Troubleshooting Exercise #5 (Blueprint schematic)



Schematic description:

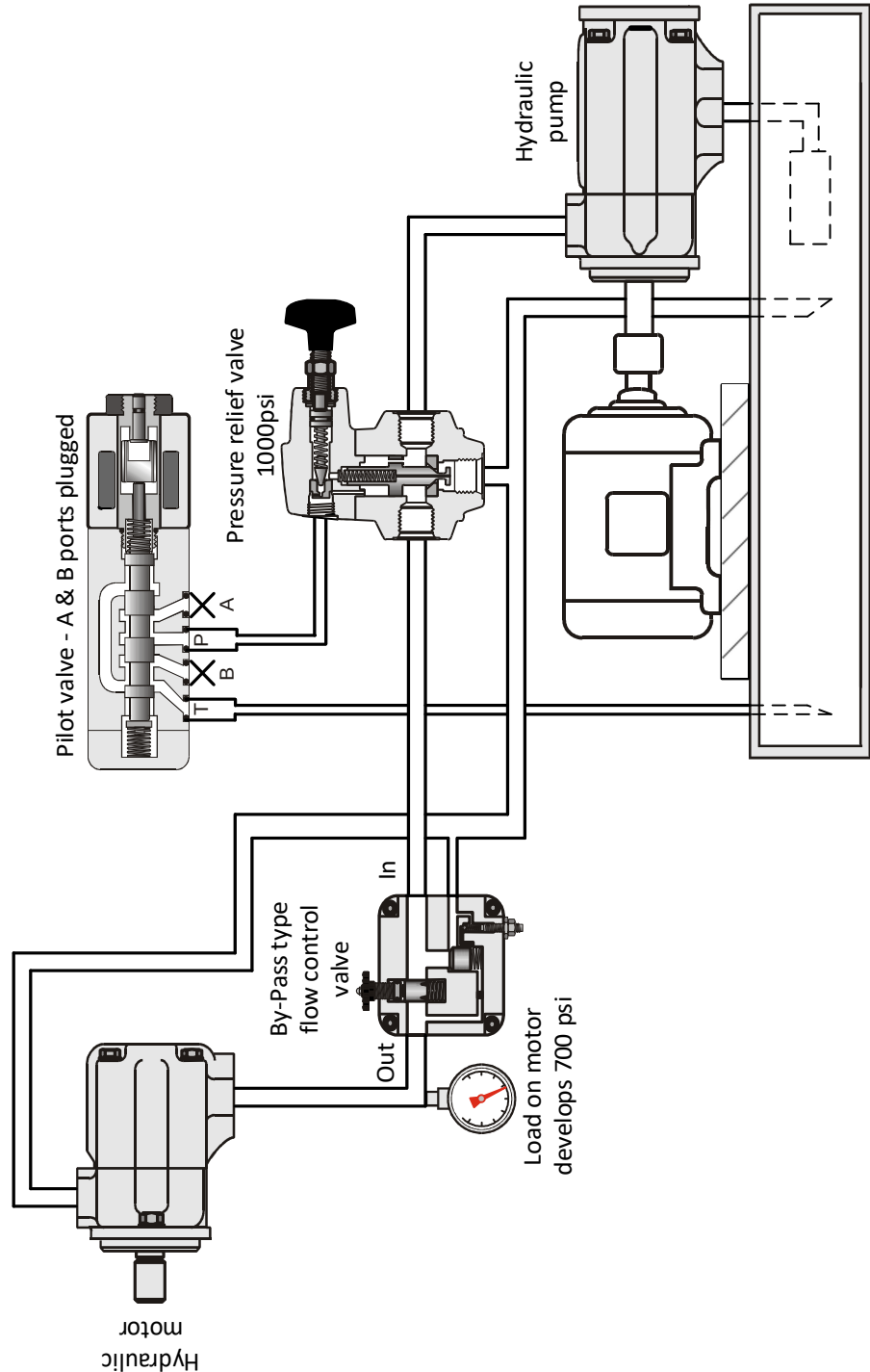
The cylinder will extend rapidly, until the deceleration valve is energized. Fluid is forced over the flow control and finishes extension as a slow work circuit. Cylinder will have rapid retraction for the entire return stroke.

Schematic Trouble:

During the slow work cycle the feed speed will vary and cannot get the feed as slow as required.

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Schematic Troubleshooting Exercise #6 (Component cut-aways/Hard piping)



Schematic description:

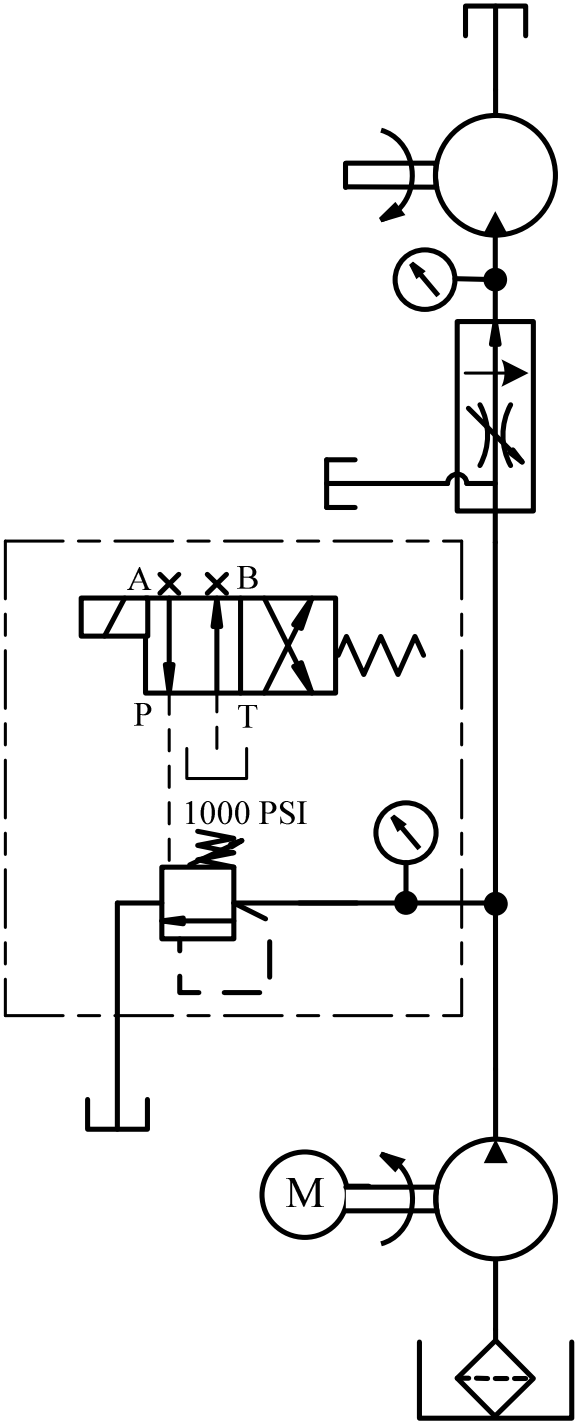
A variable speed hydraulic motor is required for the circuit. Speed is controlled by adjusting the flow control. De-energizing the pilot valve stops the motor by remote pilot operating the pressure relief to by-pass full pump flow to tank.

Schematic Trouble:

Energizing or de-energizing the solenoid on the pilot valve will not stop the motor.

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Schematic Troubleshooting Exercise #6 (Blueprint schematic)



Schematic description:

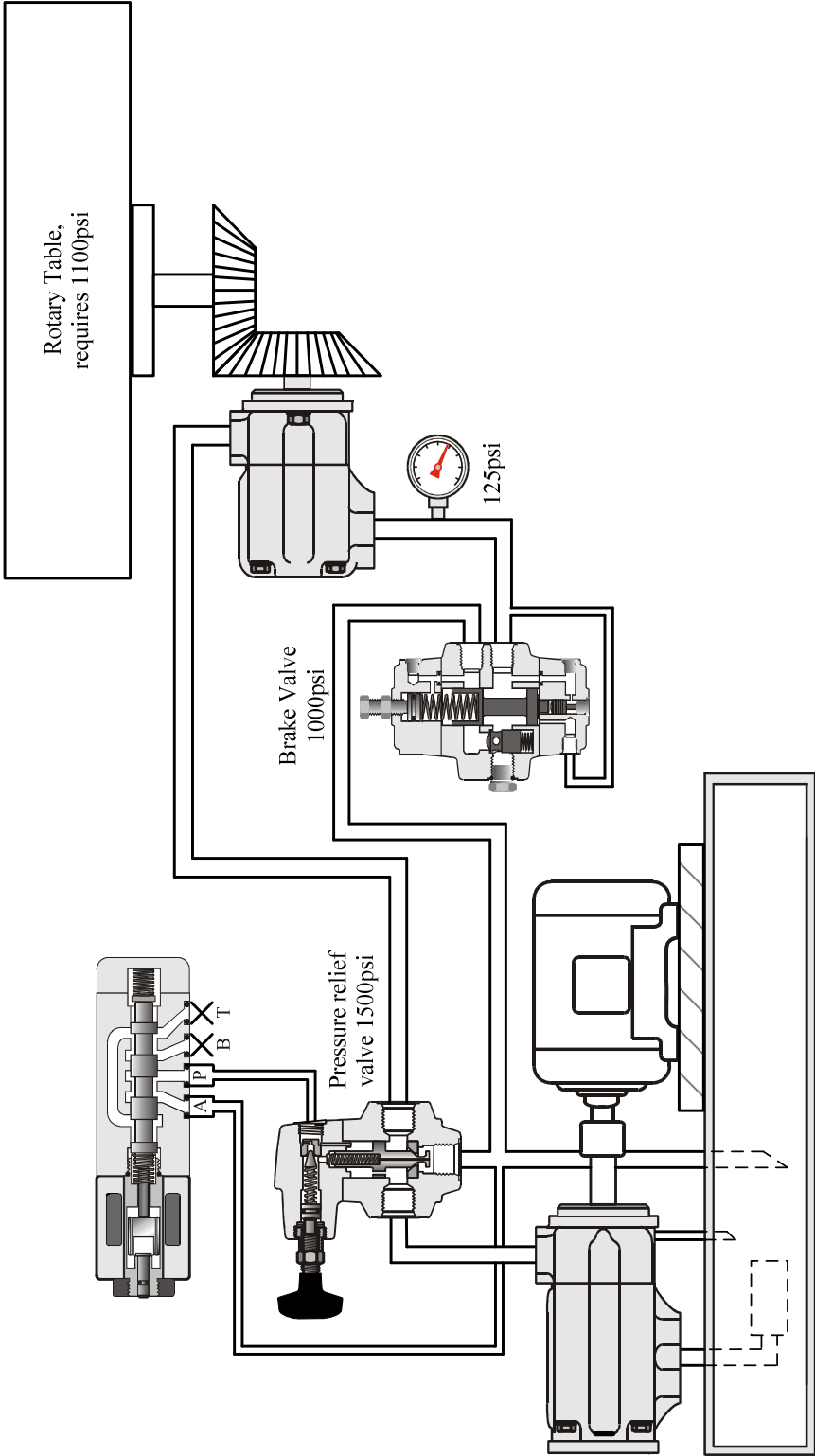
A variable speed hydraulic motor is required for the circuit. Speed is controlled by adjusting the flow control. De-energizing the pilot valve stops the motor by remote pilot operating the pressure relief to by-pass full pump flow to tank.

Schematic Trouble:

Energizing or de-energizing the solenoid on the pilot valve will not stop the motor.

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Schematic Troubleshooting Exercise #7 (Component cut-aways/Hard piping)



Schematic description:

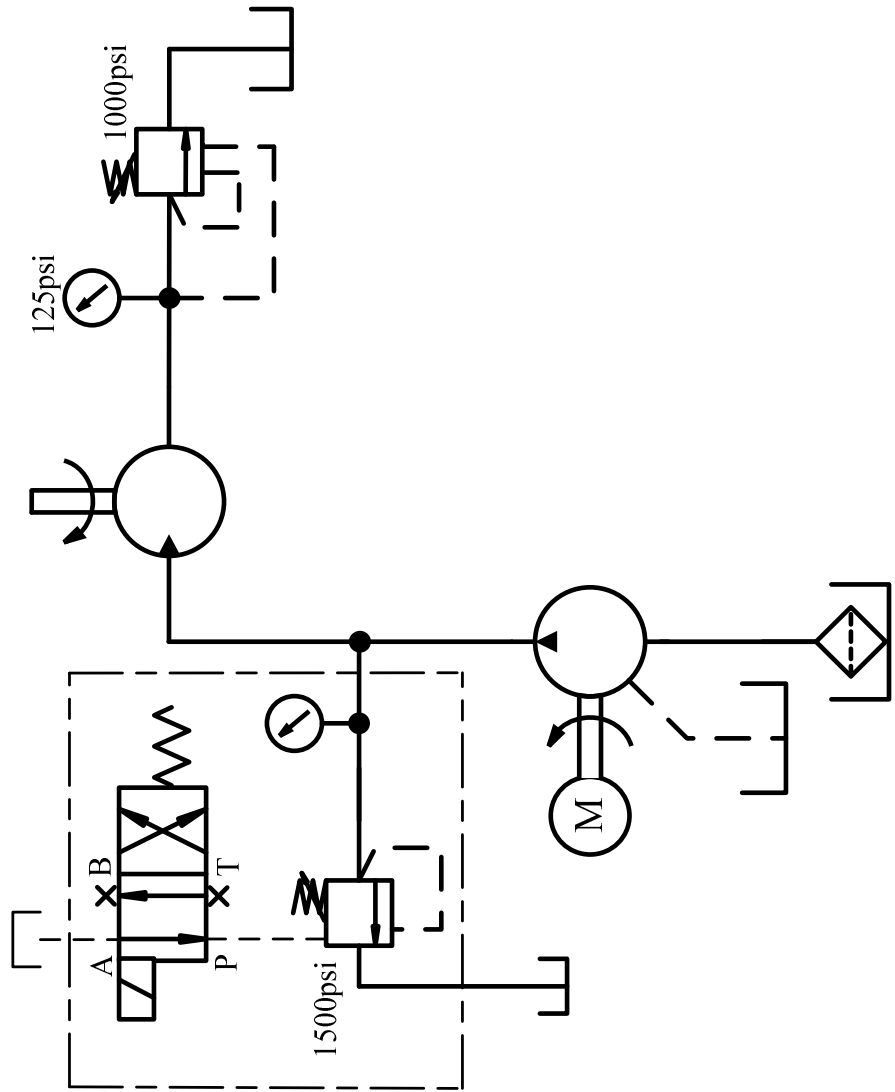
A Hydraulic motor rotates a heavy circular table. The coasting of the table is controlled by a brake valve. Energizing the DCV will unload the pressure relief valve spool and direct all pump flow to reservoir.

Schematic Trouble:

The table takes too much time to slow down after the DCV is energized to unload the pressure relief valve. The pressure gauge in the discharge side of the motor reads 125 psi during running and braking.

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Schematic Troubleshooting Exercise #7 (Blueprint schematic)



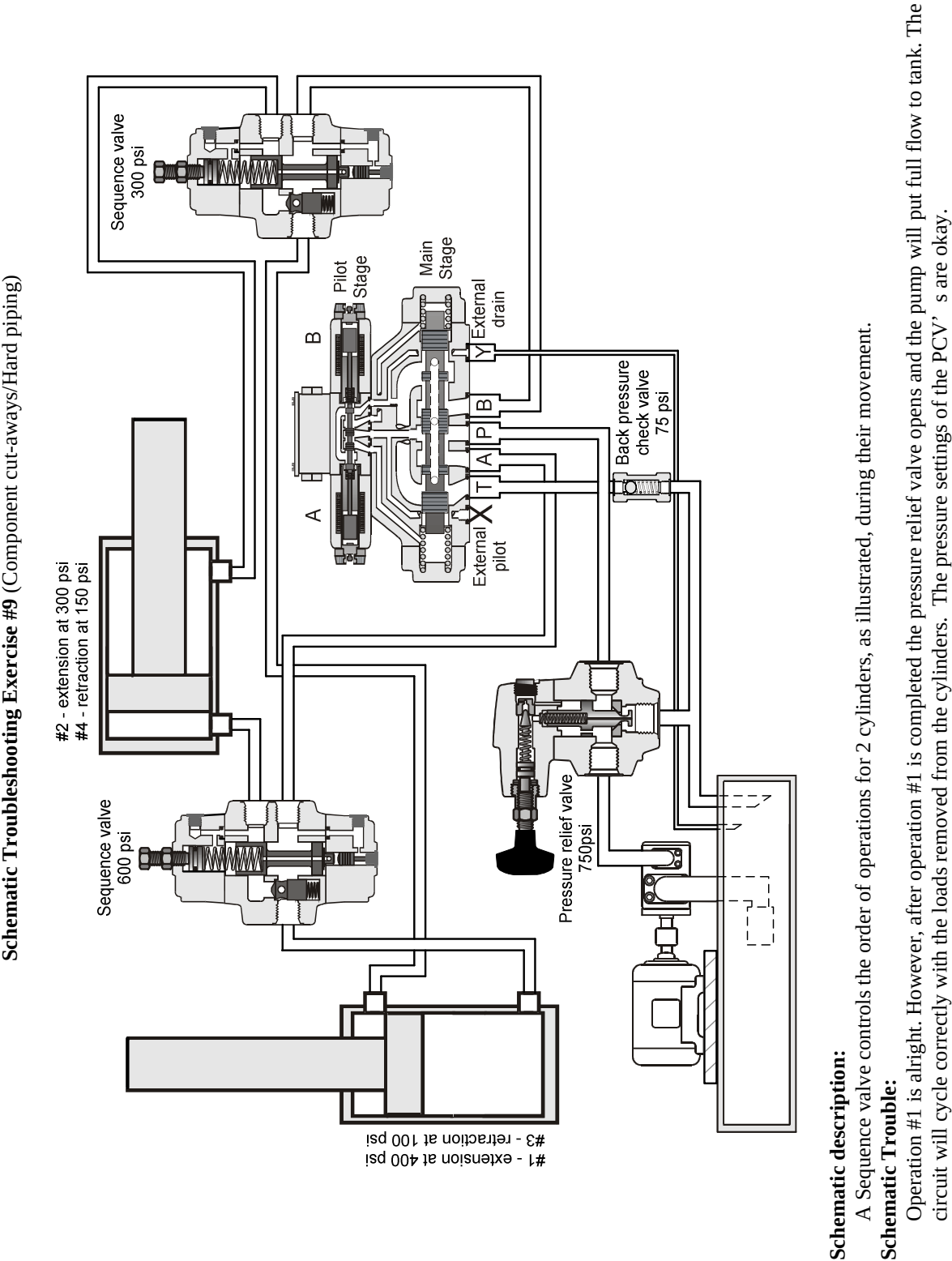
Schematic description:

A Hydraulic motor rotates a heavy circular table. The coasting of the table is controlled by a brake valve. Energizing the DCV will unload the pressure relief valve spool and direct all pump flow to reservoir.

Schematic Trouble:

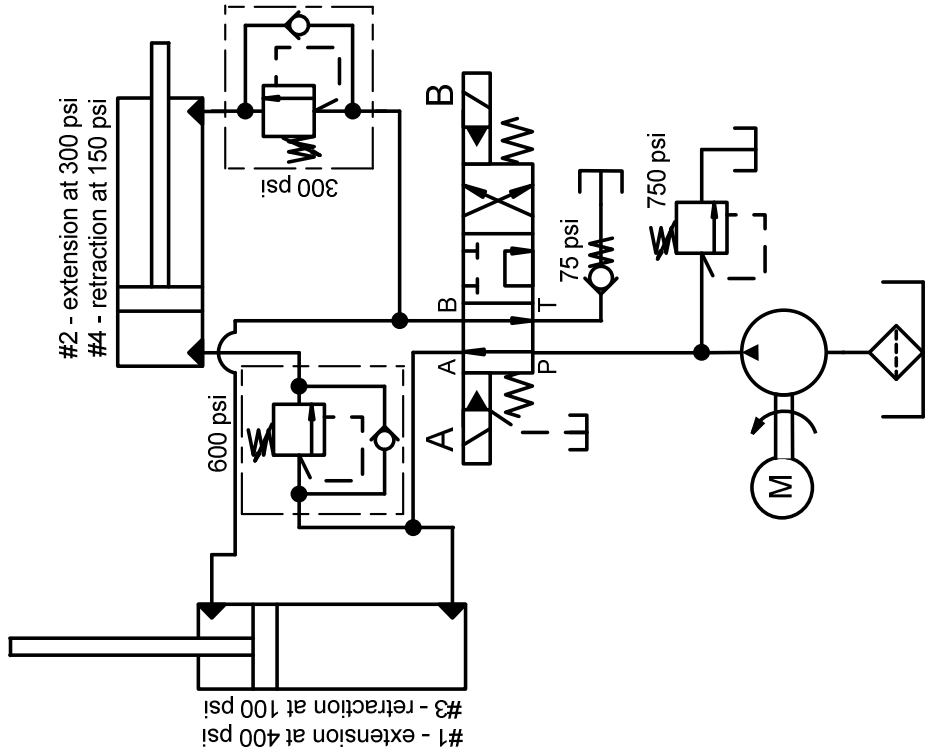
The table takes too much time to slow down after the DCV is energized to unload the pressure relief valve. The pressure gauge in the discharge side of the motor reads 125 psi during running and braking.

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Schematic Troubleshooting Exercise #9 (Blueprint schematic)



Schematic description:

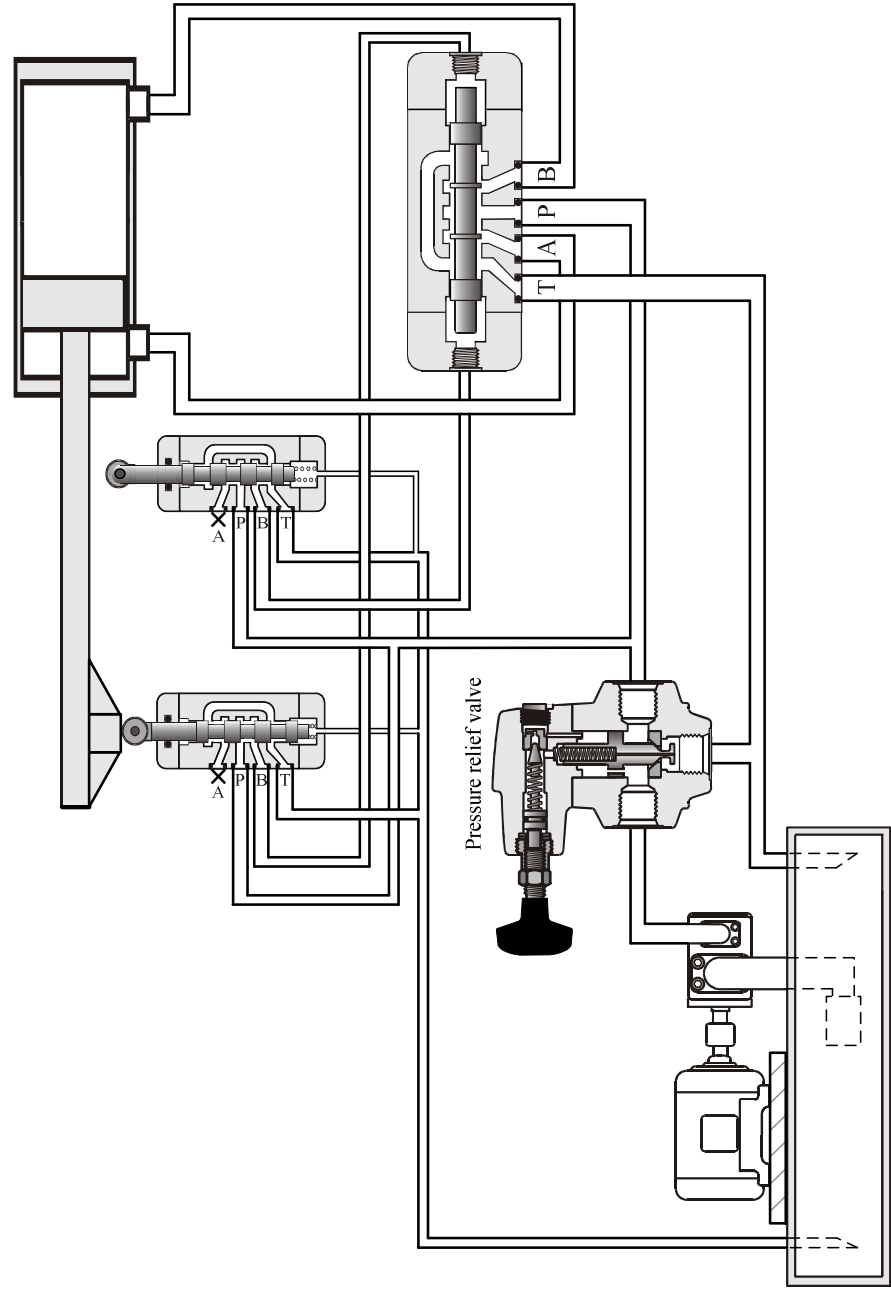
A Sequence valve controls the order of operations for 2 cylinders, as illustrated, during their movement.

Schematic Trouble:

Operation #1 is alright. However, after operation #1 is completed the pressure relief valve opens and the pump will put full flow to tank. The circuit will cycle correctly with the loads removed from the cylinders. The pressure settings of the PCV' s are okay.

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Schematic Troubleshooting Exercise #10 (Component cut-aways/Hard piping)



Schematic description:

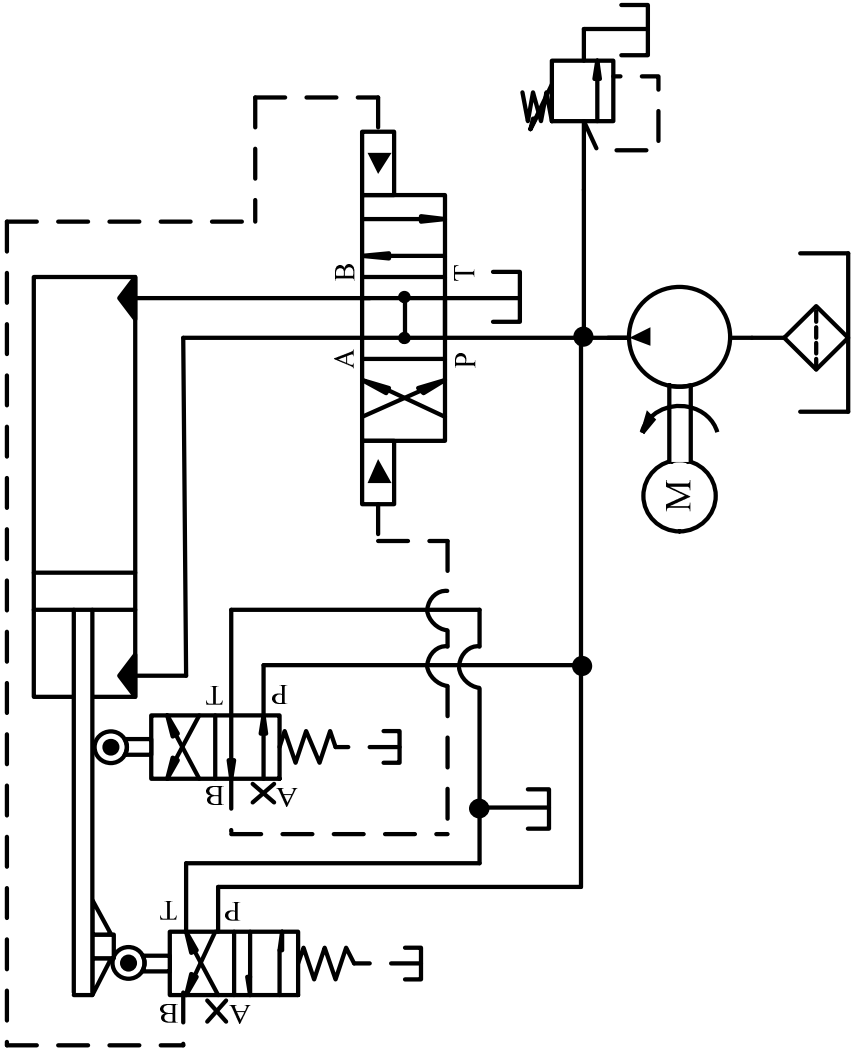
The cylinder will have an automatic extension and retraction type of cycle when the cylinder energizes the limit valves.

Schematic Trouble:

A repair was made on a PM, now when production tries to run the system it will not work. The pump is running, but the system will not work.

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Schematic Troubleshooting Exercise #10 (Blueprint schematic)



Schematic description:

The cylinder will have an automatic extension and retraction type of cycle when the cylinder energizes the limit valves.

Schematic Trouble:

A repair was made on a PM, now when production tries to run the system it will not work. The pump is running, but the system will not work.

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